Quantum Field Theory II

派蒙の素论派学习笔记

Author Be Like:

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University of Hamburg
Department of Physics

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Path integral method

1. Path integral in quantum mechanics

1.1 Functional method

Functional S maps a function to numbers, i.e.

$$S : x(t) \to S[x(t)]. \tag{1.1}$$

As we all know, the Hamilton operator in 1-dim is $H = \frac{p^2}{2m} + V(x)$, now let us consider the amplitude for motion of a particle from x_a to x_b. For this, we introduce the evolution operator $U(x_a, x_b; t)$ in Heisenburg picture, so we have

$$U(x_a, x_b; T) = \langle x_b| e^{-\mathrm{i}HT/\hbar} |x_a\rangle\,, \tag{1.2}$$

where the Hamiltonian **is already quantized**, but in the following calculation, the Hamiltonian that in the path integral **is classical** due to the path integral itself is the process of quantization. **So we don't need to quantize anything inside of the integration**. So the path integral is directly defined from this operator. The basic idea is to cut the space and time into small pieces, and add them together, i.e.,

$$U(x_a, x_b; T) = \int \mathcal{D}x(t) e^{\mathrm{i}S[x(t)]/\hbar}, \tag{1.3}$$

where $S[x(t)]$ is the action that depends on the paths. In QM, we can expand it as

$$S[x(t)] = \int_0^T \mathrm{d}t\, L = \int_0^T \mathrm{d}t\, \left[\frac{1}{2}mv^2 - V(x)\right]. \tag{1.4}$$

And the integration measure $\mathcal{D}x(t)$ accounts for all possible paths from x_a to x_b, i.e.,

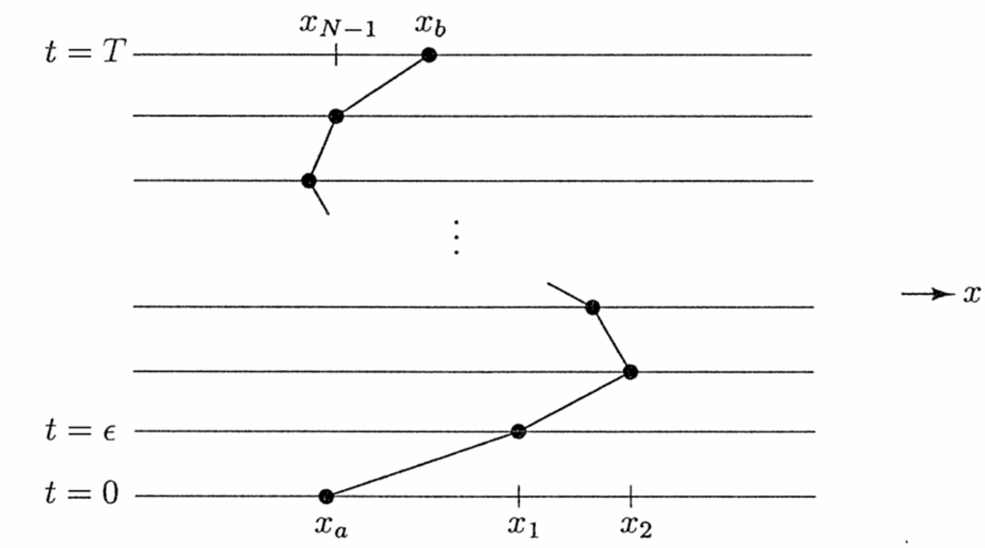
$$\sum_{\substack{\text{all paths from}\\ x_a \text{ to } x_b}} \longrightarrow \int \mathcal{D}x(t). \tag{1.5}$$

Now we need to prove that the definition we gave that can derive the same quantization result. Firstly, we discretize the time interval from 0 to T in steps $\Delta t = \varepsilon$:

$$S = \int_0^T \mathrm{d}t\, \left[\frac{1}{2}m\dot{x}^2 - V(x)\right] = \sum_k \left\{\frac{m}{2}\frac{(x_{k+1} - x_k)^2}{\varepsilon} - \varepsilon V\left(\frac{x_{k+1} + x_k}{2}\right)\right\}. \tag{1.6}$$

The path integral build up from N discrete time intervals

$$\mathcal{D}x(t) \equiv \frac{1}{c(\varepsilon)} \int \frac{\mathrm{d}x_1}{c(\varepsilon)} \frac{\mathrm{d}x_2}{c(\varepsilon)} \ldots \frac{\mathrm{d}x_N}{c(\varepsilon)} = \frac{1}{c(\varepsilon)} \prod_{k=1}^{N} \int_{-\infty}^{+\infty} \frac{\mathrm{d}x_k}{c(\varepsilon)}, \tag{1.7}$$



where $\frac{1}{c(\varepsilon)}$ is an undetermined normalization factor. Now we consider the evolution operator $U(x_a, x_b; T)$ evolve from $T-\varepsilon$ to T in the interval $[x_a, x_b]$, which is the last step in (1.7):

$$U(x_a, x_b; T) = \langle x_b| e^{-\mathrm{i}H[(T-\varepsilon)+\varepsilon]/\hbar} |x_a\rangle = \int \mathrm{d}x' \langle x_b| e^{-\mathrm{i}H\varepsilon/\hbar} |x'\rangle \langle x'| e^{-\mathrm{i}H(T-\varepsilon)/\hbar} |x_a\rangle$$
$$= \int_{-\infty}^{+\infty} \frac{\mathrm{d}x'}{c(\varepsilon)} \exp\left[\frac{\mathrm{i}}{\hbar}\frac{m(x_b-x')^2}{2\varepsilon} - \frac{\mathrm{i}}{\hbar}\varepsilon V\left(\frac{x_b+x'}{2}\right)\right] U(x_a, x'; T-\varepsilon) \tag{1.8}$$

Notice that we can expand $U(x_a, x'; T-\varepsilon)$ around x_b

$$U(x_a, x'; T-\varepsilon) = \left(1 + (x'-x_b)\frac{\partial}{\partial x_b} + \frac{1}{2}(x'-x_b)^2\frac{\partial^2}{\partial x_b^2} + ...\right) U(x_a, x_b; T-\varepsilon) \tag{1.9}$$

and for $\varepsilon \to 0$, the potential term can also expand as

$$\exp\left\{\frac{\mathrm{i}}{\hbar}\varepsilon V\left(\frac{x_b - x_{x'}}{2}\right)\right\} \overset{x'\to x_b}{=} e^{\frac{\mathrm{i}}{\hbar}\varepsilon V(x_b)} = 1 - \frac{i}{\hbar}\varepsilon V(x_b) + ... \tag{1.10}$$

Thus, (1.9) can be rewritten as

$$U(x_a, x_b; T) = \int_{-\infty}^{+\infty} \frac{\mathrm{d}x'}{c(\varepsilon)} \exp\left\{\frac{\mathrm{i}}{\hbar}\frac{m(x_b-x')^2}{\varepsilon}\right\} \left[1 - \frac{\mathrm{i}\varepsilon}{\hbar}V(x_b) + \dots\right]$$
$$\times \left[1 + (x'-x_b)\frac{\partial}{\partial x_b} + \frac{1}{2}(x'-x_b)^2\frac{\partial^2}{\partial x_b^2} + \dots\right] U(x_a, x_b; T-\varepsilon) \tag{1.11}$$

To do this integration the Gauss integration formulas are very useful in the next:

$$\int_{-\infty}^{+\infty} \mathrm{d}\xi\, e^{-b\xi^2} = \sqrt{\frac{\pi}{b}}; \tag{1.12}$$
$$\int_{-\infty}^{+\infty} \mathrm{d}\xi\, \xi e^{-b\xi^2} = 0; \tag{1.13}$$
$$\int_{-\infty}^{+\infty} \mathrm{d}\xi\, \xi^2 e^{-b\xi^2} = \frac{1}{2b}\sqrt{\frac{\pi}{b}}. \tag{1.14}$$

Thus,

$$U(x_a, x_b; T) = \frac{1}{c(\varepsilon)}\sqrt{\frac{\pi}{b}}\left[1 - \frac{\mathrm{i}\varepsilon}{\hbar}V(x_b) + \frac{1}{4b}\frac{\partial^2}{\partial x_b^2} + \mathcal{O}(\varepsilon^2)\right] U(x_a, x_b; T-\varepsilon), \tag{1.15}$$

where $b = \frac{-im}{2\varepsilon\hbar}$. When $\varepsilon \to 0$,

$$U(x_a, x_b; T) = \overbrace{\frac{1}{c(\varepsilon)}\sqrt{\frac{\pi}{b}}}^{=1} U(x_a, x_b; T-\varepsilon)$$
$$\Rightarrow c(\varepsilon) = \sqrt{\frac{\pi}{b}}. \tag{1.16}$$

So we have the Schrödinger equation for the evolution operator

$$\mathrm{i}\hbar \frac{U(x_a, x_b; T) - U(x_a, x_b; T-\varepsilon)}{\varepsilon} = \left(V(x_b) - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial x_b^2}\right) U(x_a, x_b; T-\varepsilon). \tag{1.17}$$

So we have derived the familiar differential equation with the canonical quantization did. Now we also need to consider the boundary condition. For $T \to 0$, canonical quantization requires $U(x_a, x_b; 0) = \langle x_b| \mathbb{1} |x_a\rangle \delta(x_a - x_b)$. For path integral, we firstly take a piece of time slice, since $T \to 0$ there is no difference for labeling the space interval, so we can just take $\delta x = x_b - x_a$:

$$\lim_{\varepsilon\to 0} \frac{1}{c(\varepsilon)} \exp\left[\frac{im}{\hbar}\frac{(x_b - x_a)^2}{\varepsilon}\right] = \delta(x_b - x_a) \tag{1.18}$$

Thus,

$$\lim_{\varepsilon\to 0}\int \mathcal{D}x(t) e^{\mathrm{i}S[x(t)]/\hbar} = \lim_{\varepsilon\to 0} \frac{1}{c(\varepsilon)} \exp\left[\frac{im}{\hbar}\frac{(x_b - x_a)^2}{\varepsilon}\right] \to \delta(x_a - x_b) \tag{1.19}$$

- Path integral formulas quantize system through summation/integration over all possible paths; weighted by the phase $e^{-\mathrm{i}S[x(t)]/\hbar}$.
- In the limit $\hbar \to 0$ (classical physics) path integral is dominated by classical path with $\delta S = 0$.

1.2 Generalization to quantum system with coordinates $\{q_i\}$ and conjugate momenta $\{p_i\}$

Now we consider the general case for a quantum system with Hamiltonian $H(q,p)$, and evolution operator $U(q_a, q_b; T) = \langle q_b| e^{-\mathrm{i}HT} |q_a\rangle$. Split up $f(q,p) = e^{-\mathrm{i}HT}$ in N time intervals $\Delta t = \varepsilon$

$$e^{-\mathrm{i}HT} = \prod_{k=1}^{N} e^{-\mathrm{i}H\varepsilon} = e^{-\mathrm{i}H\varepsilon}\mathbb{1}e^{-\mathrm{i}H\varepsilon}\mathbb{1}\cdots\mathbb{1}e^{-\mathrm{i}H\varepsilon}, \tag{1.20}$$

with $\mathbb{1} = \left(\prod_i \int \mathrm{d}q_k^i\right) |q_k\rangle \langle q_k|$. At k-th position,

$$\langle q_k + 1| f(q) |q_k\rangle \xrightarrow{\varepsilon=0} \langle q_k + 1| 1 - i\varepsilon H + \mathcal{O}(\varepsilon^2) |q_k\rangle \tag{1.21}$$

For a function that its variable is only q, we have

$$\langle q_{k+1}| f(q) |q_k\rangle = f(q_k)\prod_i \delta(q_k^i - q_{k+1}^i) = f\left(\frac{q_k + q_{k+1}}{2}\right)\left(\prod_i \int \frac{\mathrm{d}p_i}{2\pi}\right)\exp\left\{\mathrm{i}\sum_i p_k^i(q_{k+1}^i - q_k^i)\right\}, \tag{1.22}$$

and for a function that its variable is only p, we have

$$\langle q_{k+1}| f(p) |q_k\rangle = \left(\prod_i \int \frac{\mathrm{d}p_k^i}{2\pi}\right) f(p_k)\exp\left\{\mathrm{i}\sum_i p_k^i(q_{k+1}^i - q_k^i)\right\}. \tag{1.23}$$

In general, we normally face the more complicated case that using the Weyl ordering is convenient to keep the properties we have in the above, i.e.,

$$\langle q_{k+1}| \frac{1}{4}\left(q^2p^2 + 2qp^2q + p^2q^2\right)|q_k\rangle = \left(\frac{q_{k+1}+q_k}{2}\right)^2 \langle q_k + 1| p^2 |q_k\rangle\,, \tag{1.24}$$

where the Weyl order is defined as

$$W(p,q) = \frac{1}{2}\{x,p\}. \tag{1.25}$$

Assume $H(p,q)$ is Weyl oerdered, i.e.,

$$\langle q_{k+1}| H(p,q) |q_k\rangle = \left(\prod_i \int \frac{\mathrm{d}p_k^i}{2\pi}\right) H\left(\frac{q_{k+1}+q_k}{2}, p_k\right) \exp\left\{\mathrm{i}\sum_i p_k^i(q_{k+1}^i - q_k^i)\right\}. \tag{1.26}$$

Thus

$$\langle q_{k+1}| e^{-\mathrm{i}H\varepsilon} |q_k\rangle = \left(\prod_i \int \frac{\mathrm{d}p_k^i}{2\pi}\right) \exp\left\{-\mathrm{i}\varepsilon H\left(\frac{q_{k+1}+q_k}{2}, p_k\right)\right\} \exp\left\{\mathrm{i}\sum_i p_k^i(q_{k+1}^i - q_k^i)\right\}. \tag{1.27}$$

Take $q_0 = q_a$ and $q_N = q_b$, the integration are

$$U(q_0, q_N; T) = \left(\prod_{i,k} \int \mathrm{d}q_k^i \int \frac{\mathrm{d}p_k^i}{2\pi}\right) \exp\left\{\mathrm{i}\sum_k \left[\sum_i p_k^i(q_{k+1}^i - q_k^i) - \varepsilon H\left(\frac{q_{k+1}+q_k}{2}, p_k\right)\right]\right\}. \tag{1.28}$$

In path integral notation

$$U(q_0, q_N; T) = \left(\prod_{i,k} \int \mathcal{D}q^i(t) \int \mathcal{D}p^i(t)\right) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \left[\sum_i p^i\dot{q}^i - H(p,q)\right]\right\}. \tag{1.29}$$

2. Path integral in the Scalar field

For a scalar field $\phi(\vec{x},t)$, we can simply change the variables $\vec{q}_i(t) \longmapsto \phi(\vec{x},t)$, the evolution operator is then became:

$$U(\phi_a, \phi_b; T) = \langle \phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle. \tag{1.30}$$

The action S and Hamiltonian H for thw scalar field are defined as:

$$S = \int \mathrm{d}^4x \left(\frac{1}{2}(\partial\cdot\phi)^2 - V(\phi)\right), \tag{1.31}$$

$$H = \int \mathrm{d}^3x \left(\frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi)\right). \tag{1.32}$$

The path integral form can then be simply seen from (1.29) as

$$\langle \phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle = \int_{\phi(0,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x) \int \mathcal{D}\pi(x) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left(\pi\dot{\phi} - \frac{1}{2}\pi^2 - \frac{1}{2}(\nabla\phi)^2 - V(\phi)\right)\right\} \tag{1.33}$$

$$= \int \mathcal{D}\phi(x) \int \mathcal{D}(\pi(x) - \dot{\phi}) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left[-\frac{1}{2}\left(-2\pi\dot{\phi} + \pi^2 + \dot{\phi}^2 - \dot{\phi}^2\right) - \frac{1}{2}(\nabla\phi)^2 - V(\phi)\right]\right\} \tag{1.34}$$

To cancel the $\int \mathcal{D}\pi(x)$, we change the variable $\mathcal{D}\pi(x) \to \mathcal{D}(\pi(x) - \partial_0\phi)$, i.e.,

$$\langle \phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle = \int \mathcal{D}\phi(x) \int \mathcal{D}(\pi(x) - \dot{\phi}) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left[-\frac{1}{2}\left(\pi - \dot{\phi}\right)^2 + \frac{1}{2}(\partial\phi)^2 - V(\phi)\right]\right\} \tag{1.35}$$

$$= \int \mathcal{D}\phi(x) \int \mathcal{D}\tilde{\pi}(x) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left[-\frac{1}{2}\tilde{\pi}(x)^2 + \frac{1}{2}(\partial\phi)^2 - V(\phi)\right]\right\}. \tag{1.36}$$

Thus

$$\langle \phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle = \mathcal{C}\int \mathcal{D}\phi(x) \exp\left\{\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \left[\frac{1}{2}(\partial\phi)^2 - V(\phi)\right]\right\}, \tag{1.37}$$

where

$$\mathcal{C} = \int \mathcal{D}\tilde{\pi}(x) \exp\left\{-\mathrm{i}\int_0^T \mathrm{d}t \int \mathrm{d}^3x \, \frac{1}{2}\tilde{\pi}(x)^2\right\} \tag{1.38}$$

So we can directly see it is

$$\langle \phi_b(\vec{x})|e^{-\mathrm{i}\hat{H}T}|\phi_a(\vec{x})\rangle = \mathcal{C}\int_{\phi(0,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x) \exp\left\{\mathrm{i}\int \mathrm{d}^4x \, \mathcal{L}\right\} \tag{1.39}$$

For some reason that we will consider later, the normalization constant $\mathcal{C}$ is ignored in the next calculation.

2.1 Correlation Function

Now we consider the two-point correlation function. For that, we firstly consider the structure

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x)\, \phi(x_1)\phi(x_2) \exp\left\{\mathrm{i}\int_{-T}^T \mathrm{d}^4x \, \mathcal{L}\right\}, \tag{1.40}$$

where we assume $x_2^0 > x_1^0$. Now the boundary condition changes from $-T$ to T. So we can separate the integration in three area $-T \to x_1^0$, $x_1^0 \to x_2^0$ and $x_2^0 \to T$ such that we can fix the field at x_1^0 and x_2^0 as $\phi_1(\vec{x})$ and $\phi_2(\vec{x})$ respectively. This operation means no matter how the field evolve from $-T$ to T, we can always separate the integration with 2 steps: First, fixing the field at these two time point and integrate the three area. Second, we integrate all possible situation of these two field.

$$\int \mathcal{D}\phi(x) = \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \int\limits_{\substack{\phi(x_1^0,\vec{x})=\phi_1(\vec{x}) \\ \phi(x_2^0,\vec{x})=\phi_2(\vec{x})}} \mathcal{D}\phi(x). \tag{1.41}$$

So we have

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x)\, \phi(x_1)\phi(x_2) \exp\left\{\mathrm{i}\int_{-T}^T \mathrm{d}^4x \, \mathcal{L}\right\}$$

$$= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x}) \int \mathcal{D}\phi(x)\, \phi(x_1)\phi(x_2) \exp\left\{\mathrm{i}\int_{-T}^T \mathrm{d}^4x \, \mathcal{L}\right\}. \tag{1.42}$$

Because we have fixed the time point x_1^0 and x_2^0, so $\phi(x_1)$ and $\phi(x_2)$ can be directly picked from the integration $\int \mathcal{D}\phi$ and becomes $\phi_1(\vec{x_1})$ and $\phi_2(\vec{x_2})$, i.e.,

$$\begin{aligned}
&\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x)\,\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\,\mathcal{L}\right\} \\
&\quad= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x})\,\phi_1(\vec{x}_1)\phi_2(\vec{x}_2)\left[\int \mathcal{D}\phi(x)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\,\mathcal{L}\right\}\right] \qquad (1.43)\\
&\quad= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x})\,\phi_1(\vec{x}_1)\phi_2(\vec{x}_2)\left[\langle\phi_b|e^{-\mathrm{i}\hat{H}(2T)}|\phi_a\rangle\right] \qquad (1.44)\\
&\quad= \int \mathcal{D}\phi_1(\vec{x}) \int \mathcal{D}\phi_2(\vec{x})\,\phi_1(\vec{x}_1)\phi_2(\vec{x}_2)\langle\phi_b|e^{-\mathrm{i}\hat{H}(T-x_2^0)}|\phi_2\rangle\langle\phi_2|e^{-\mathrm{i}\hat{H}(x_2^0-x_1^0)}|\phi_1\rangle\langle\phi_1|e^{-\mathrm{i}\hat{H}(x_1^0+T)}|\phi_a\rangle. \qquad (1.45)
\end{aligned}$$

Using $\hat{\phi}_S(\vec{x})|\phi_i\rangle = \phi_i(\vec{x})|\phi_i\rangle$ and $\int \mathcal{D}\phi_i|\phi_i\rangle\langle\phi_i| = \mathbb{1}$:

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x)\,\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\,\mathcal{L}\right\} = \langle\phi_b|e^{-\mathrm{i}\hat{H}(T-x_2^0)}\hat{\phi}_S(\vec{x}_2)e^{-\mathrm{i}\hat{H}(x_2^0-x_1^0)}\hat{\phi}_S(\vec{x}_1)e^{-\mathrm{i}\hat{H}(x_1^0+T)}|\phi_a\rangle. \qquad (1.46)$$

Defining the Heisenberg picture operator $\hat{\phi}_H(x) = e^{\mathrm{i}\hat{H}x^0}\hat{\phi}_S(\vec{x})e^{-\mathrm{i}\hat{H}x^0}$:

$$\int_{\phi(-T,\vec{x})=\phi_a}^{\phi(T,\vec{x})=\phi_b} \mathcal{D}\phi(x)\,\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\,\mathcal{L}\right\} = \langle\phi_b|e^{-\mathrm{i}\hat{H}T}\hat{\phi}_H(x_2)\hat{\phi}_H(x_1)e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle. \qquad (1.47)$$

For a general time ordering $x_2^0 > x_1^0$, this becomes the time-ordered product:

$$\langle\phi_b|e^{-\mathrm{i}\hat{H}T}T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle. \qquad (1.48)$$

Using the ground state projection

$$\lim_{T\to\infty(1-i\epsilon)} e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle = \lim_{T\to\infty(1-i\epsilon)} \sum_n e^{-\mathrm{i}E_nT}\,|n\rangle\,\langle n|\phi_a\rangle = \langle\Omega|\phi_a\rangle e^{-\mathrm{i}E_0\infty(1-i\epsilon)}|\Omega\rangle, \qquad (1.49)$$

and similarly,

$$\lim_{T\to\infty(1-i\epsilon)} \langle\phi_b|\,e^{-\mathrm{i}\hat{H}T} = \lim_{T\to\infty(1-i\epsilon)} \sum_n \langle\phi_b|n\rangle\,\langle n|\,e^{-\mathrm{i}E_nT} = \langle\phi_b|\Omega\rangle\,\langle\Omega|\,e^{-\mathrm{i}E_0\infty(1-i\epsilon)}. \qquad (1.50)$$

So we have

$$\langle\phi_b|e^{-\mathrm{i}\hat{H}T}T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle = \langle\phi_b|\phi_a\rangle\langle\Omega|T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}|\Omega\rangle. \qquad (1.51)$$

Thus, we can finally reach the correlation function:

$$\langle\Omega|T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}|\Omega\rangle = \frac{\langle\phi_b|e^{-\mathrm{i}\hat{H}T}T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}e^{-\mathrm{i}\hat{H}T}|\phi_a\rangle}{\langle\phi_b|\phi_a\rangle}. \qquad (1.52)$$

So in the end, we have

$$\langle\Omega|T\{\hat{\phi}_H(x_1)\hat{\phi}_H(x_2)\}|\Omega\rangle = \lim_{T\to\infty(1-\mathrm{i}\epsilon)} \frac{\int \mathcal{D}\phi\,\phi(x_1)\phi(x_2)e^{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\mathcal{L}}}{\int \mathcal{D}\phi\,e^{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\mathcal{L}}}. \qquad (1.53)$$

2.2 Feynman Rules

Now we want to get the Feynman rules from (1.53), so we need to calculate the numerator and the denominator respectively. For that, we firstly consider the free action of the scalar field:

$$S_0 = \int \mathrm{d}^4x\, \mathcal{L}_0 = \int \mathrm{d}^4x \left(\frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2\right). \tag{1.54}$$

The general Gaussian integration formulas could be useful in the next:

$$\int \mathrm{d}^n x\; e^{-x_i A_{ij} x_j} = \text{const.} \cdot (\det A)^{-1/2}; \tag{1.55}$$

$$\int \mathcal{D}\phi\, e^{\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\, \phi(\partial^2 - m^2)\phi} = \text{const.} \cdot \left(\det\left(\partial^2 - m^2\right)\right)^{-1/2}. \tag{1.56}$$

The idea is firstly discretizing the path integral and then take the limit to get it back to the continuous form. So we work in the lattice $\{x_i\}$, where we can define the discrete field to replace the continuous variable:

$$\mathcal{D}\phi = \prod_i \mathrm{d}\phi\,(x_i). \tag{1.57}$$

And the field can be expressed by the Fourier series:

$$\phi(x_i) = \frac{1}{V}\sum_n e^{-\mathrm{i}k_n\cdot x_i}\phi(k_n); \quad k_n^\mu = \frac{2\pi n^\mu}{L}, \tag{1.58}$$

where n^μ is integers and $|k^\mu| < \pi/\varepsilon$, ε is the distance between two close lattice points. $V = L^4$ is the volume of the space. Because of $\phi(x_n)$ is real, so

$$\begin{aligned} \phi^*(x_i) &= \frac{1}{V}\sum_n e^{+\mathrm{i}k_n^*\cdot x_i}\phi^*(k_n) \\ &= \frac{1}{V}\sum_n e^{-\mathrm{i}(-k_n)\cdot x_i}\phi(-k_n), \end{aligned} \tag{1.59}$$

which means $\phi^*(k_n) = \phi(-k_n)$. So the integral measurement becomes

$$\mathcal{D}\phi(x) = \prod_i \mathrm{d}\phi\,(x_i) = |\det(M)| \prod_n \mathrm{d}\phi\,(k_n) = \prod_{k_n^0>0} \mathrm{d}\,\mathrm{Re}\,\phi(k_n)\,\mathrm{d}\,\mathrm{Im}\,\phi(k_n), \tag{1.60}$$

where M is the Fourier transformation matrix, so the Jacobian determinant is 1. And the last equality is because $\mathrm{d}\phi\,(k_n)$ is an area measurement in the complex plane, so this small piece of area measurement is $\mathrm{d}\,\mathrm{Re}\,\phi(k_n)\,\mathrm{d}\,\mathrm{Im}\,\phi(k_n)$. And the free action in momentum space can be rewritten as

$$S_0 = -\frac{1}{V}\sum_{k_n^0>0}\frac{1}{2}(m^2 - k_n^2)|\phi_n|^2 = -\frac{1}{V}\sum_{k_n^0>0}(m^2 - k_n^2)\left[(\mathrm{Re}\ \phi_n)^2 + (\mathrm{Im}\ \phi_n)^2\right]. \tag{1.61}$$

Therefore, we can together all these parts to express the denominator of (1.53), then using the Gaussian integration formula to calculate the integral:

$$\int \mathcal{D}\phi e^{\mathrm{i}S_0} = \prod_{k_n^0>0} \left(\int \mathrm{d}\,\mathrm{Re}\,\phi_n \exp\left[-\frac{\mathrm{i}}{V}(m^2 - k_n^2)(\mathrm{Re}\,\phi_n)^2 \right] \right) \left(\int \mathrm{d}\,\mathrm{Im}\,\phi_n \exp\left[-\frac{\mathrm{i}}{V}(m^2 - k_n^2)(\mathrm{Im}\,\phi_n)^2 \right] \right) \tag{1.62}$$

$$= \prod_{k_n^0>0} \sqrt{\frac{-\mathrm{i}\pi V}{m^2 - k_n^2}} \sqrt{\frac{-\mathrm{i}\pi V}{m^2 - k_n^2}} \tag{1.63}$$

$$= \prod_{k_n^0>0} \sqrt{\frac{-\mathrm{i}\pi V}{m^2 - k_n^2}} \sqrt{\frac{-\mathrm{i}\pi V}{m^2 - (-k_n)^2}} \tag{1.64}$$

$$= \prod_{\text{all } k_n} \sqrt{\frac{-\mathrm{i}\pi V}{m^2 - k_n^2}}, \tag{1.65}$$

where the last step is counting from the upper half plane to the whole complex k_n plane. And to avoid the divergence, we take $k^0 \to k^0(1+\mathrm{i}\epsilon)$ in the denominator. For the expression for the propagator, we still need to calculating $\phi(x_1)\phi(x_2)$ for the numerator of (1.53), so we have

$$\phi(x_1)\phi(x_2) = \frac{1}{V^2} \sum_m e^{-\mathrm{i}k_m x_1}\phi_m \sum_l e^{-\mathrm{i}k_l x_2}\phi_l. \tag{1.66}$$

So the numerator becomes

$$\begin{aligned} \int \mathcal{D}\phi\, \phi(x_1)\phi(x_2) e^{\mathrm{i}\int_{-T}^{T} \mathrm{d}^4x \mathcal{L}} = \frac{1}{V^2} \sum_{m,l} e^{-\mathrm{i}(k_m x_1 + k_l x_2)} & \left(\prod_{n|k_n^0>0} \int \mathrm{d}\,\mathrm{Re}\,\phi_n\, \mathrm{d}\,\mathrm{Im}\,\phi_n \right) \\ & \times (\mathrm{Re}\,\phi_m + \mathrm{i}\,\mathrm{Im}\,\phi_m)(\mathrm{Re}\,\phi_l + \mathrm{i}\,\mathrm{Im}\,\phi_l) \\ & \times \exp\left\{ -\frac{\mathrm{i}}{V} \sum_{n|k_n^0>0} (m^2 - k_n^2)\left[(\mathrm{Re}\,\phi_n)^2 + (\mathrm{Im}\,\phi_n)^2 \right] \right\}. \end{aligned} \tag{1.67}$$

Notice that for $m \neq l$ and $m \neq -l$, the term

$$(\mathrm{Re}\,\phi_m + \mathrm{i}\,\mathrm{Im}\,\phi_m)(\mathrm{Re}\,\phi_l + \mathrm{i}\,\mathrm{Im}\,\phi_l) \exp\left\{ (\mathrm{Re}\,\phi_n)^2 + (\mathrm{Im}\,\phi_n)^2 \right\} \tag{1.68}$$

is (**odd function**)$\times$(**even function**) = (**odd function**), so the integral is 0. And for $m = -l$, because of $\phi(-k) = \phi^*(k)$, which means $\phi_l = \phi_{-m} = \phi_m^*$, the production term is then

$$(\phi_m) \cdot (\phi_l) = \phi_m \cdot \phi_m^* = |\phi_m|^2 = (\mathrm{Re}\,\phi_m)^2 + (\mathrm{Im}\,\phi_m)^2. \tag{1.69}$$

So the integration is $\sim \int x^2 e^{-ax^2}$, which is non-zero. Thus only the terms with $k_m = -k_l$ contribute to the integral. Finally, we apply the Gaussian integration formula to (1.67), then we have

$$\int \mathcal{D}\phi\, \phi(x_1)\phi(x_2) e^{\mathrm{i}\int_{-T}^{T} \mathrm{d}^4x \mathcal{L}} = \frac{1}{V^2} \sum_m e^{-\mathrm{i}k_m(x_1 - x_2)} \left(\prod_{k_n^0>0} \frac{-\mathrm{i}\pi V}{m^2 - k_n^2} \right) \frac{-\mathrm{i}V}{m^2 - k_m^2 - \mathrm{i}\varepsilon} \tag{1.70}$$

$$= \frac{1}{V^2} \sum_m e^{-\mathrm{i}k_m(x_1 - x_2)} \left(\prod_{\text{all } k_n} \sqrt{\frac{-\mathrm{i}\pi V}{m^2 - k_n^2}} \right) \frac{-\mathrm{i}V}{m^2 - k_m^2 - \mathrm{i}\varepsilon} \tag{1.71}$$

Finally, we got the Feynman propagator through dividing (1.71) by (1.65):

$$\langle 0|T\{\phi(x_1)\phi(x_2)\}|0\rangle = \frac{\int \mathcal{D}\phi\, \phi(x_1)\phi(x_2) e^{\mathrm{i}\int_{-T}^{T} \mathrm{d}^4x \mathcal{L}}}{\int \mathcal{D}\phi\, e^{\mathrm{i}\int_{-T}^{T} \mathrm{d}^4x \mathcal{L}}} \tag{1.72}$$

$$= \frac{\frac{1}{V^2}\sum_m e^{-\mathrm{i}k_m(x_1-x_2)} \left(\prod_{\text{all } k_n} \sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}}\right) \frac{-\mathrm{i}V}{m^2-k_m^2-\mathrm{i}\varepsilon}}{\prod_{\text{all } k_n} \sqrt{\frac{-\mathrm{i}\pi V}{m^2-k_n^2}}} \tag{1.73}$$

$$= \sum_m \frac{1}{V} \frac{\mathrm{i}}{k_m^2 - m^2 - \mathrm{i}\varepsilon} e^{-\mathrm{i}k_m(x_1-x_2)} \tag{1.74}$$

$$= \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\varepsilon} e^{-\mathrm{i}k(x_1-x_2)} = D_F(x_1 - x_2) \tag{1.75}$$

2.3 Generating Functional

Now, we consider a more convenient and general method to calculate the Feynman rules. But it might be not very direct to understanding. First, we introduce $J(x)$ as a source in the outside to define the generating functional. For this, it's convenient to review some properties of the functional derivative:

$$\frac{\delta}{\delta J(x)} J(y) = \delta^{(4)}(x-y) \implies \frac{\delta}{\delta J(x)} \int \mathrm{d}^4y\, J(y)\phi(y) = \phi(x). \tag{1.76}$$

And some basic examples of the functional derivative:

1. Exponential function:

$$\frac{\delta}{\delta J(x)} \exp\left\{\mathrm{i}\int \mathrm{d}^4y\, J(y)\phi(y)\right\} = \mathrm{i}\phi(x) \exp\left\{\mathrm{i}\int \mathrm{d}^4y\, J(y)\phi(y)\right\}. \tag{1.77}$$

2. Source terms containing derivatives (obtained through integration by parts):

$$\frac{\delta}{\delta J(x)} \int \mathrm{d}^4y\, \partial_\mu J(y) V^\mu(y) = -\frac{\delta}{\delta J(x)} \int \mathrm{d}^4y\, J(y)\partial_\mu V^\mu(y) = -\partial_\mu V^\mu(x). \tag{1.78}$$

3. Functional composite differentiation (chain rule):

$$\frac{\delta F[\phi]}{\delta J(x)} = \int \mathrm{d}^4z \frac{\delta F[\phi]}{\delta\phi(z)} \frac{\delta\phi(z)}{\delta J(x)} \quad \left(\text{Generalization of } \frac{\partial x_i(z_k)}{\partial y_j} = \sum_k \frac{\partial x_i}{\partial z_k}\frac{\partial z_k}{\partial y_j}\right). \tag{1.79}$$

Define the **Generating Functional** $Z[J]$ as:

$$Z[J] = \int \mathcal{D}\phi \exp\left\{\mathrm{i}\int \mathrm{d}^4x\, (\mathcal{L} + J(x)\phi(x))\right\}. \tag{1.80}$$

Using $Z[J]$, we can conveniently calculate the n-point correlation function:

$$\langle\Omega|\, T\{\phi(x_1)\phi(x_2)\}\, |\Omega\rangle = \frac{1}{Z[0]} \left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right) \left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right) Z[J]\bigg|_{J=0} \tag{1.81}$$

2.4 Feynman Rules for the Free Theory

For a free scalar field, the Lagrangian is

$$\mathcal{L}_0 = \frac{1}{2}\left[(\partial_\mu\phi)(\partial^\mu\phi) + (-m^2 + \mathrm{i}\epsilon)\phi^2\right] \tag{1.82}$$
$$= \frac{1}{2}\left[\partial_\mu(\phi\,\partial^\mu\phi) - (\phi\,\partial^2\phi) + (-m^2 + \mathrm{i}\epsilon)\phi^2\right] \tag{1.83}$$
$$= \frac{1}{2}\phi(-\partial^2 - m^2 + \mathrm{i}\epsilon)\phi. \tag{1.84}$$

Define the operator $K_x = -\partial^2 - m^2 + \mathrm{i}\epsilon$, and the action with source can be written as:

$$S_J = \int \mathrm{d}^4x\,(\mathcal{L}_0 + J\phi) = \int \mathrm{d}^4x\left(\frac{1}{2}\phi K_x\phi + J\phi\right) \tag{1.85}$$
$$= \int \mathrm{d}^4x\,\mathrm{d}^4y\left(\frac{1}{2}\phi(x)K(x,y)\phi(y)\right) + \int \mathrm{d}^4x\,J(x)\phi(x), \tag{1.86}$$

where $K(x,y) = \delta^{(4)}(x-y)K_y$. Now the generating functional is:

$$Z_0[J] = \int \mathcal{D}\phi\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(\frac{1}{2}\phi K\phi + J\phi\right)\right\}. \tag{1.87}$$

Now we change the variable $\phi \to \phi' = \phi + K^{-1}J$, so the measurement keeps unchanged $\mathcal{D}\phi' = \mathcal{D}\phi$:

$$Z_0[J] = \int \mathcal{D}\phi'\left|\det\frac{\delta\phi'}{\delta\phi}\right|\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left[\frac{1}{2}(\phi' - K^{-1}J)K(\phi' - K^{-1}J) + J(\phi' - K^{-1}J)\right]\right\} \tag{1.88}$$
$$= \left(\int \mathcal{D}\phi'\exp\left\{\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\phi' K\phi'\right\}\right)\exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\,J(x)K^{-1}(x,y)J(y)\right\} \tag{1.89}$$
$$= Z_0[J=0]\exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\,J(x)K^{-1}(x,y)J(y)\right\}, \tag{1.90}$$

where $\left|\det\frac{\delta\phi'}{\delta\phi}\right| = 1$. And notice for

$$J(y) = K(x,y)\left(K^{-1}(x,y)J(y)\right) \tag{1.91}$$
$$= \left(-\partial_x^2 - m^2 + \mathrm{i}\epsilon\right)K^{-1}(x,y)\delta^{(4)}(x-y)J(y) \tag{1.92}$$
$$= \left(-\partial_x^2 - m^2 + \mathrm{i}\epsilon\right)K^{-1}(x,y)J(x). \tag{1.93}$$

Thus, K^{-1} satisfies the Green's function (except an $-\mathrm{i}$ factor in the right hand side):

$$(-\partial_x^2 - m^2 + \mathrm{i}\epsilon)K^{-1}(x,y) = \delta^{(4)}(x-y). \tag{1.94}$$

Take a sharp look! So we can directly realize that the Feynman propagator $-\mathrm{i}D_F(x-y)$ also satisfies this equation, so we realize that $K^{-1}(x-y)$ is nothing but $-\mathrm{i}D_F(x-y)$. So the **generating functional in free theory** can be then expressed as

$$Z_0[J] = Z_0[0]\exp\left\{-\frac{1}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\,J(x)D_F(x-y)J(y)\right\}. \tag{1.95}$$

So the correlation function can be obtained from (1.81):

$$\langle 0|\, T\{\phi(x_1)\phi(x_2)\}\, |0\rangle = \frac{1}{Z_0[0]}\left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right) Z_0[J]\bigg|_{J=0} \tag{1.96}$$

$$= \left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right)\exp\left\{-\frac{1}{2}\int \mathrm{d}^4x\, \mathrm{d}^4y\, J(x)D_F(x-y)J(y)\right\}\bigg|_{J=0} \tag{1.97}$$

$$= \left(-\frac{\delta}{\delta J(x_1)}\right)\left[-\frac{1}{2}\left(\int \mathrm{d}^4y\, D_F(x_2-y)J(y) + \int \mathrm{d}^4x\, J(x)D_F(x-x_2)\right)\right] \times \exp\left\{-\frac{1}{2}\int \mathrm{d}^4x\, \mathrm{d}^4y\, J(x)D_F(x-y)J(y)\right\}\bigg|_{J=0} \tag{1.98}$$

$$= \frac{\delta}{\delta J(x_1)}\left[\int \mathrm{d}^4y\, D_F(y-x_2)J(y)\right]\frac{Z_0[J]}{Z_0[0]}\bigg|_{J=0} \tag{1.99}$$

$$= D_F(x_1-x_2)\,\frac{Z_0[J]}{Z_0[0]}\bigg|_{J=0} + \left[\int \mathrm{d}^4y\, D_F(y-x_2)J(y)\right]\left[\int \mathrm{d}^4y\, D_F(y-x_1)J(y)\right]\frac{Z_0[J]}{Z_0[0]}\bigg|_{J=0} \tag{1.100}$$

$$= D_F(x_1-x_2). \tag{1.101}$$

Technically, the last step we do, which impose the boundary condition to cancel the second term is not very strict. In general, one can not impose the boundary condition before actually calculating the integration. More precisely, the reason why we have confidence to do so is based on the property that we can let $f \in L^2(\mathbb{R}^{1,3})$ be fixed, and define a functional

$$F_f : L^2(\mathbb{R}^{1,3}) \to \mathbb{C}, \qquad F_f[J] := \int_{\mathbb{R}^{1,3}} f(x)\, J(x)\, dx. \tag{1.102}$$

Then F_f is a continuous linear functional on $L^2(\mathbb{R}^{1,3})$. Indeed, by the Cauchy–Schwarz inequality,

$$|F_f[J]| \le \|f\|_{L^2}\, \|J\|_{L^2}, \tag{1.103}$$

so F_f is bounded and hence continuous. In particular, for any sequence (or net) $J_n \to 0$ in $L^2(\mathbb{R}^{1,3})$, one has

$$\lim_{n\to\infty} F_f[J_n] = F_f[0] = 0. \tag{1.104}$$

Therefore, the notation$F_f[J]\,|_{J=0}$ is to be understood as the evaluation of the functional F_f at the zero element of $L^2(\mathbb{R}^{1,3})$, i.e.

$$F_f[J]\,|_{J=0} := F_f(0). \tag{1.105}$$

Moreover, by continuity, this evaluation coincides with the limit along any sequence $J_n \to 0$ in the L^2 topology:

$$F_f(0) = \lim_{J\to 0} F_f[J], \tag{1.106}$$

where the limit is taken with respect to the L^2 norm.

2.5 Interaction theory: ϕ^4 theory as an example

For theories involving interactions, such as $\mathcal{L} = \mathcal{L}_0 - \frac{\lambda}{4!}\phi^4$ and $\mathcal{L}_0 = \frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2$, the generating functional can be written as:

$$Z[J] = \int \mathcal{D}\phi \exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4 + \mathcal{L}_0 + J(x)\phi(x)\right)\right\}. \tag{1.107}$$

Notice that

$$\begin{aligned}\frac{\delta}{\delta J(x)} Z_0[J] &= \frac{\delta}{\delta J(x)} \int \mathcal{D}\phi \exp\left\{-\mathrm{i}\int \mathrm{d}^4x\, \mathcal{L}_0 + J(x)\phi(x)\right\} \\ &= \phi(x) \int \mathcal{D}\phi \exp\left\{-\mathrm{i}\int \mathrm{d}^4x\, \mathcal{L}_0 + J(x)\phi(x)\right\} \\ &= \phi(x) Z_0[J]. \end{aligned} \tag{1.108}$$

So we have

$$\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4 Z_0[J] = \phi^4 Z_0[J], \tag{1.109}$$

which means

$$Z[J] = \int \mathcal{D}\phi \left\{1 + \left(-\mathrm{i}\int \mathrm{d}^4x\, \frac{\lambda}{4!}\phi^4\right) + \frac{1}{2!}\left(-\mathrm{i}\int \mathrm{d}^4x\, \frac{\lambda}{4!}\phi^4\right)^2 + \ldots\right\} \exp\left\{\mathrm{i}\int \mathrm{d}^4x\, [\mathcal{L}_0 + J(x)\phi(x)]\right\} \tag{1.110}$$

$$= \int \mathcal{D}\phi \left\{1 + \left[-\mathrm{i}\int \mathrm{d}^4x\, \frac{\lambda}{4!}\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4\right] + \frac{1}{2!}\left[-\mathrm{i}\int \mathrm{d}^4x\, \frac{\lambda}{4!}\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4\right]^2 + \ldots\right\} \times \exp\left\{\mathrm{i}\int \mathrm{d}^4x\, [\mathcal{L}_0 + J(x)\phi(x)]\right\} \tag{1.111}$$

$$= \exp\left\{-\mathrm{i}\int \mathrm{d}^4x\, \frac{\lambda}{4!}\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4\right\} \int \mathcal{D}\phi \exp\left\{\mathrm{i}\int \mathrm{d}^4x\, [\mathcal{L}_0 + J(x)\phi(x)]\right\} \tag{1.112}$$

$$= \exp\left\{-\mathrm{i}\int \mathrm{d}^4x\, \frac{\lambda}{4!}\left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4\right\} Z_0[J], \tag{1.113}$$

where we expanded the exponential term of interaction part as the perturbed series, and using the derivative relation (1.109). So we link the interaction theory generating functional with the free theory generating functional:

$$Z[J] = \left(1 - \frac{\mathrm{i}\lambda}{4!}\int \mathrm{d}^4x \left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4 + \ldots\right) Z_0[J]. \tag{1.114}$$

So the correlation function can be expanded by the free propagator as

$$\langle\Omega| T\{\phi(x_1)\phi(x_2)\} |\Omega\rangle = \frac{1}{Z[0]}\left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right) Z[J]\Big|_{J=0} \tag{1.115}$$

$$= \frac{1}{Z[0]}\left(-\mathrm{i}\frac{\delta}{\delta J(x_1)}\right)\left(-\mathrm{i}\frac{\delta}{\delta J(x_2)}\right)\left(1 - \frac{\mathrm{i}\lambda}{4!}\int \mathrm{d}^4x \left(-\mathrm{i}\frac{\delta}{\delta J(x)}\right)^4 + \ldots\right) Z_0[J]\Big|_{J=0} \tag{1.116}$$

$$= \frac{1}{Z[0]}\left[(-\mathrm{i})^2 \frac{\delta^2 Z_0[J]}{\delta J(x_1)\delta J(x_2)} + \left(-\frac{\mathrm{i}\lambda}{4!}\right)\int \mathrm{d}^4x\, (-\mathrm{i})^6 \frac{\delta^6 Z_0[J]}{\delta J(x)^4 \delta J(x_1)\delta J(x_2)} + \ldots\right]\Bigg|_{J=0} \tag{1.117}$$

$$= \frac{1}{Z[0]}\langle 0| T\{\phi(x_1)\phi(x_2)\} |0\rangle + \frac{1}{Z[0]}\left(-\mathrm{i}\frac{\lambda}{4!}\right)\int \mathrm{d}^4x\, \langle 0| T\{\phi^4(x)\phi(x_1)\phi(x_2)\} |0\rangle + \ldots \tag{1.118}$$

$$= \frac{1}{Z[0]}\langle 0| T\left\{\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int \mathrm{d}^4x \left(-\frac{\lambda}{4!}\phi^4\right)\right\}\right\} |0\rangle. \tag{1.119}$$

Comparing with the result we got in the canonical quantization

$$\langle\Omega| T\{\phi(x_1)\phi(x_2)\} |\Omega\rangle = \frac{\langle 0| T\left\{\phi(x_1)\phi(x_2)\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4\right)\right\}\right\}|0\rangle}{\langle 0| T\left\{\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4\right)\right\}\right\}|0\rangle}, \tag{1.120}$$

we find in the interaction case, $Z[0]$ has different meaning comparing with $Z_0[0]$ in the free case, i.e.,

$$Z[0] = T\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(-\frac{\lambda}{4!}\phi^4\right)\right\}, \tag{1.121}$$

which included all the **Vacuum Bubbles**.

2.6 Other Type of Generating functional

We define the **Energy Functional** for the connected graphs $W[J]$ as

$$W[J] = \mathrm{i}\ln Z[J]. \tag{1.122}$$

So we have

$$\frac{\delta W[J]}{\delta J(x)} = \frac{\mathrm{i}}{Z[0]}\frac{\delta Z[J]}{\delta J(x)} = -\frac{\langle\Omega|\phi(x)|\Omega\rangle}{\langle\Omega|\Omega\rangle} = -\phi_{\text{classical}}(x), \tag{1.123}$$

where the second step is because $\delta Z[J] = \langle\Omega|\,\mathrm{i}\phi(x)\,|\Omega\rangle$. And in physics, it is defined as the **Classical Field**. And the definition of $W[J]$ can be easily seen that it's very similar with the free energy in classical statistical mechanics with the partition function $Z[J]$. So using the Legendre transformation, we can define the **Effective Action**, which looks like the Gibbs free energy $\Gamma[\phi_{cl}]$ as

$$\Gamma[\phi_{cl}] = -W[J] - \int \mathrm{d}^4x\, J(x)\phi_{cl}(x). \tag{1.124}$$

Base on this, we can see there is a symmetric relation:

$$\frac{\delta\Gamma[\phi_{cl}]}{\delta\phi_{cl}(x)} = -J(x); \quad \frac{\delta W[J]}{\delta J(x)} = -\phi_{cl}(x). \tag{1.125}$$

Taking another functional derivative with respect to $J(y)$ on both sides yields:

$$\Rightarrow \frac{\delta}{\delta J(y)}\frac{\delta}{\delta\Phi_{\mathrm{cl}}(x)}\Gamma[\Phi_{\mathrm{cl}}] = -\frac{\delta J(x)}{\delta J(y)} = -\delta(x-y). \tag{1.126}$$

Using the chain rule for functional differentiation $\frac{\delta}{\delta J(y)} = \int \mathrm{d}^4z\,\frac{\delta\Phi_{\mathrm{cl}}(z)}{\delta J(y)}\frac{\delta}{\delta\Phi_{\mathrm{cl}}(z)}$, we obtain the following identity:

$$-\delta(x-y) = \int \mathrm{d}^4z\,\frac{\delta\Phi_{\mathrm{cl}}(z)}{\delta J(y)}\frac{\delta^2\Gamma[\Phi_{\mathrm{cl}}]}{\delta\Phi_{\mathrm{cl}}(z)\delta\Phi_{\mathrm{cl}}(x)}. \tag{1.127}$$

We can rewrite the first term as $\frac{\delta\Phi_{\mathrm{cl}}(z)}{\delta J(y)} = \frac{\delta^2 W[J]}{\delta J(y)\delta J(z)}$, where $W[J]$ is the generating functional of connected Green's functions:

$$-\delta(x-y) = \int \mathrm{d}^4z\,\underbrace{\frac{\delta^2 W[J]}{\delta J(y)\delta J(z)}}_{-\mathrm{i}G(y-z)}\underbrace{\frac{\delta^2\Gamma[\Phi_{\mathrm{cl}}]}{\delta\Phi_{\mathrm{cl}}(z)\delta\Phi_{\mathrm{cl}}(x)}}_{\mathrm{i}G^{-1}(z-x)}. \tag{1.128}$$

Here, the first part corresponds to the connected two-point function $-\mathrm{i}G(y-z)$, and the second part corresponds to the inverse propagator $\mathrm{i}G^{-1}(z-x)$. Thus, the equation simplifies to:

$$\delta(x-y) = \int \mathrm{d}^4 z \, [-\mathrm{i}G(y-z)] \left[\mathrm{i}G^{-1}(z-x)\right]. \tag{1.129}$$

The n-th derivative of $W[J]$ gives the connected correlation function

$$\left.\frac{\delta^n W[J]}{\delta J(x_1)\dots\delta J(x_n)}\right|_{J=0} = (\mathrm{i})^{n+1} \langle\Omega| \, T\{\phi(x_1)\dots\phi(x_n)\} \, |\Omega\rangle_{connect}. \tag{1.130}$$

And the n-th derivative of $\Gamma[\phi_{cl}]$ gives the correlation function for 1PI graphs

$$\frac{\delta^n \Gamma[\phi_{cl}]}{\delta\phi_{cl}(x_1)\dots\delta\phi_{cl}(x_n)} = -\mathrm{i} \, \langle\Omega| \, T\{\phi(x_1)\dots\phi(x_n)\} \, |\Omega\rangle_{1PI}. \tag{1.131}$$

3. Quantization of EM Field in Path Integral Formalism

The classical action for the electromagnetic field is given by:

$$S[A] = -\frac{1}{4}\int \mathrm{d}^4 x \, F_{\mu\nu}F^{\mu\nu}, \tag{1.132}$$

where $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$. Integrating by parts, we can rewrite the action as:

$$S[A] = \frac{1}{2}\int \mathrm{d}^4 x \, A_\mu D^{\mu\nu} A_\nu, \quad \text{with} \quad D^{\mu\nu} = g^{\mu\nu}\Box - \partial^\mu\partial^\nu. \tag{1.133}$$

In the path integral formulation, the vacuum-to-vacuum transition amplitude is given by:

$$\int \mathcal{D}A \, e^{\mathrm{i}S[A]}. \tag{1.134}$$

But the problem the measure $\mathcal{D}A = \mathcal{D}A^0\mathcal{D}A^1\mathcal{D}A^2\mathcal{D}A^3$ is **not** well defined. The integral is ill-defined because of the redundancy due to infinite gauge copies. Under a local gauge transformation:

$$A_\mu \to A_\mu^\alpha = A_\mu + \frac{1}{e}\partial_\mu\alpha(x), \tag{1.135}$$

where $\alpha(x)$ is an arbitrary scalar function. For a pure gauge choice where $A_\mu = \frac{1}{e}\partial_\mu\alpha$, we find that the field strength $F_{\mu\nu} = 0$, leading to:

$$S[A] = 0 \quad \Rightarrow \quad e^{\mathrm{i}S[A]} = 1. \tag{1.136}$$

This implies that we have an unbounded path integral because the integrand does not decay along the gauge orbits.

3.1 Faddeev-Popov Quantization

Solution: We need to divide out the gauge degrees of freedom in the measure $\mathcal{D}A$. This procedure is known as the **Faddeev-Popov quantization**.

Let $G(A)$ be a function that fixes the gauge (for example, $G(A) = \partial_\mu A^\mu = 0$ for the Lorentz gauge). We want to define $\int \mathcal{D}A \, e^{\mathrm{i}S[A]}$ with a constraint such that the path integral only sums over field configurations fulfilling $G(A) = 0$.

To motivate this, recall the standard Gauss integration with matrix M (rank N):

$$(\det M)^{-1/2} = (2\pi)^{-N/2} \int_{-\infty}^{+\infty} \mathrm{d}x_1 \dots \mathrm{d}x_N \, e^{-\frac{1}{2}x^T M x}. \tag{1.137}$$

Generalizing this to a functional determinant (Path integral):

$$(\det M)^{-1/2} = \int \mathcal{D}\phi \, e^{-\frac{1}{2}\int \mathrm{d}^4x \phi(x) M(x) \phi(x)}, \tag{1.138}$$

where M is some differential operator, e.g., $M(x) = \Box + m_\phi^2$.

Next, consider the Dirac δ-distribution on an N-dimensional space for an arbitrary function $\vec{g}(x_1, \dots, x_N)$:

$$\int_{-\infty}^{+\infty} \mathrm{d}x_1 \dots \mathrm{d}x_N \; \delta^{(N)} \left[\vec{g}(x_1, \dots, x_N)\right] \det \left[\frac{\partial g_i}{\partial x_j}\right] = 1 \tag{1.139}$$

In 1-dimension, this reduces to the property:

$$\int \mathrm{d}x \, \delta(ax) a = \int \mathrm{d}x \, \frac{1}{a} \delta(x) a = 1. \tag{1.140}$$

Extending this to a functional Dirac distribution for gauge fields, we have the Faddeev-Popov identity:

$$\int \mathcal{D}\alpha(x) \underbrace{\delta\left[G(A^\alpha)\right] \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right]}_{\text{Faddeev-Popov det.}} = 1. \tag{1.141}$$

Quantization of electrodynamics proceeds by inserting this identity "1" into the path integral:

$$I = \int \mathcal{D}A \, e^{\mathrm{i}S[A]} = \int \mathcal{D}A (1) e^{\mathrm{i}S[A]} = \int \mathcal{D}A \left(\int \mathcal{D}\alpha \, \delta\left[G(A^\alpha)\right] \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] \right) e^{\mathrm{i}S[A]}. \tag{1.142}$$

If $G(A^\alpha)$ is the Lorentz gauge condition, we have:

$$G(A^\alpha) = \partial_\mu A^\mu + \frac{1}{e} \Box \alpha. \tag{1.143}$$

Then the functional derivative becomes:

$$\det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] = \det \left(\frac{\Box}{e}\right). \tag{1.144}$$

Notice that in quantum electrodynamics, this determinant is invariant under gauge transformations and independent of A_μ.

Due to gauge invariance, we have $\mathcal{D}A = \mathcal{D}A^\alpha$ and $S[A] = S[A^\alpha]$. Therefore, the path integral can be rewritten as:

$$\begin{aligned} I &= \int \mathcal{D}\alpha \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] \int \mathcal{D}A \, \delta\left[G(A^\alpha)\right] e^{\mathrm{i}S[A]} \\ &= \int \mathcal{D}\alpha \left(\det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] \int \mathcal{D}A^\alpha \, \delta\left[G(A^\alpha)\right] e^{\mathrm{i}S[A^\alpha]} \right) \\ &= \int \mathcal{D}\alpha \, \tilde{I}. \end{aligned} \tag{1.145}$$

By dropping the infinite volume factor of the gauge group $\int \mathcal{D}\alpha$, we define the physical path integral for QED as:

$$\tilde{I} = \det \left[\frac{\delta G(A^\alpha)}{\delta \alpha}\right] \int \mathcal{D}A^\alpha \, \delta\left[G(A^\alpha)\right] e^{\mathrm{i}S[A]}. \tag{1.146}$$

3.2 Gauge Fixing and the Photon Propagator

Our next task is to absorb the Dirac constraint $\delta[G(A^\alpha)]$ into a modified action. Let us generalize the gauge fixing condition by setting:

$$G(A^\alpha) = \partial^\mu A_\mu(x) - \omega(x), \tag{1.147}$$

where $\omega(x)$ is an arbitrary scalar function. We also set the determinant remains independent of ω as the form:

$$\det\left[\frac{\delta G(A^\alpha)}{\delta\alpha}\right] = \det\left(\frac{\Box}{e}\right). \tag{1.148}$$

So the functional integration $\tilde{I}$ becomes

$$\int \mathcal{D}A e^{-\mathrm{i}S[A]} = \det\left(\frac{\Box}{e}\right)\int \mathcal{D}A^\alpha\, \delta\left[\partial^\mu A_\mu - \omega(x)\right] e^{\mathrm{i}S[A]}. \tag{1.149}$$

Since this formula is satisfied for any $\omega(x)$, so we can integrating over all possible functions $\omega(x)$ with a Gaussian weight centered at $\omega = 0$:

$$I' = N(\xi)\int \mathcal{D}\omega\, e^{-\mathrm{i}\int \mathrm{d}^4x\frac{\omega(x)^2}{2\xi}}\,\tilde{I}, \tag{1.150}$$

where $N(\xi)$ is a normalization factor. Then we substitute $\tilde{I}$ and explicitly show the delta function:

$$\begin{aligned}
I' &= N(\xi)\det\left(\frac{\Box}{e}\right)\int \mathcal{D}A\, e^{\mathrm{i}S[A]}\int \mathcal{D}\omega\, e^{-\mathrm{i}\int \mathrm{d}^4x\frac{\omega^2}{2\xi}}\underbrace{\delta[\partial^\mu A_\mu - \omega]}_{\text{linear covariant gauge}}\\
&= N(\xi)\det\left(\frac{\Box}{e}\right)\int \mathcal{D}A\, e^{\mathrm{i}S[A]}e^{-\frac{\mathrm{i}}{2\xi}\int \mathrm{d}^4x(\partial^\mu A_\mu)^2}\\
&= N(\xi)\det\left(\frac{\Box}{e}\right)\int \mathcal{D}A\, e^{\mathrm{i}S'[A]}.
\end{aligned} \tag{1.151}$$

The modified action $S'[A]$ becomes:

$$S'[A] = \frac{1}{2}\int \mathrm{d}^4x\left[A_\mu D^{\mu\nu}A_\nu - \frac{1}{\xi}(\partial^\mu A_\mu)^2\right] = \frac{1}{2}\int \mathrm{d}^4x\, A_\mu\tilde{D}^{\mu\nu}A_\nu, \tag{1.152}$$

which implies the new differential operator:

$$\tilde{D}^{\mu\nu} = g^{\mu\nu}\Box - \left(1 - \frac{1}{\xi}\right)\partial^\mu\partial^\nu. \tag{1.153}$$

From this modified action, we can derive the generating functional as

$$Z[J_\mu] = \int \mathcal{D}A\exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(\frac{1}{2}A_\mu\tilde{D}^{\mu\nu}A_\nu + J^\mu(x)A_\mu(x)\right)\right\}. \tag{1.154}$$

Then transform the field as $A_\mu \to A'_\mu = A_\mu + \tilde{D}^{-1}_\mu{}^\nu J_\nu$, we have

$$Z[J_\mu] = \int \mathcal{D}A\ \exp\left\{\mathrm{i}\int \mathrm{d}^4x\left(\frac{1}{2}\left(A'_\mu - \tilde{D}^{-1}_\mu{}^\lambda J_\lambda\right)\tilde{D}^{\mu\nu}\left(A'_\nu - \tilde{D}^{-1}_\nu{}^\lambda J_\lambda\right) + J^\mu(x)\left(A'_\mu - \tilde{D}^{-1}_\mu{}^\lambda J_\lambda\right)\right)\right\} \tag{1.155}$$

$$= \left(\int \mathcal{D}A'\ \exp\left\{\frac{\mathrm{i}}{2}A'_\mu\tilde{D}^{\mu\nu}A'_\nu\right\}\right)\exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\left(J_\mu(x)\tilde{D}^{-1\mu\nu}J_\nu(y)\right)\right\} \tag{1.156}$$

$$= Z[0]\exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\left(J_\mu(x)\tilde{D}^{-1\mu\nu}J_\nu(y)\right)\right\} \tag{1.157}$$

$$= Z[0]\exp\left\{-\frac{\mathrm{i}}{2}\int \mathrm{d}^4x\,\mathrm{d}^4y\left(J_\mu(x)\tilde{G}^{\mu\nu}J_\nu(y)\right)\right\}, \tag{1.158}$$

where we have ignored the determinant due to $\det\left(\frac{\delta A'}{\delta A}\right) = 1$. And the photon propagator is the inverse of $\tilde{D}^{\mu\nu}$ in momentum space:

$$G^{\mu\nu}(x-y) = -\mathrm{i}\int \frac{\mathrm{d}^4k}{(2\pi)^4} e^{-\mathrm{i}k(x-y)} \frac{g^{\mu\nu} - (1-\xi)\frac{k^\mu k^\nu}{k^2}}{k^2 + \mathrm{i}\epsilon}. \tag{1.159}$$

The effective action then can be constructed as same as (1.124)

$$\Gamma[A^\mu_{cl}] = -\mathrm{i}Z[J_\mu] - \int \mathrm{d}^4x\, J_\mu(x) A^\mu_{cl}. \tag{1.160}$$

$S'[A]$ with $\tilde{D}^{\mu\nu}$ now has no problems with the longitudinal degrees of freedom for photons.

3.3 Standard Gauge Choices

Depending on the choice of the parameter ξ and conditions, we have different gauges in the linear covariant gauge family:

- $\xi = 0$: **Landau gauge**, corresponding to the condition $\partial^\mu A_\mu = 0$.
- $\xi = 1$: **Feynman gauge**.
- $i = 1, 2, 3, \quad \partial^i A_i = 0$: **Coulomb gauge**.
- $n^\mu A_\mu = 0$: **Axial gauge**.

Finally, the path integral in QED for calculating an n-point correlator is written as:

$$\langle\Omega|T\,\hat{\mathcal{O}}(A)|\Omega\rangle = \lim_{T\to\infty(1-\mathrm{i}\epsilon)} \frac{\int \mathcal{D}A\,\hat{\mathcal{O}}(A)\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\left[\mathcal{L}(x) - \frac{1}{2\xi}(\partial^\mu A_\mu)^2\right]\right\}}{\int \mathcal{D}A\,\exp\left\{\mathrm{i}\int_{-T}^{T}\mathrm{d}^4x\left[\mathcal{L}(x) - \frac{1}{2\xi}(\partial^\mu A_\mu)^2\right]\right\}}. \tag{1.161}$$

4. Path Integral for Fermions and Dirac Spinors

For a scalar field the path integral is an ordinary Gaussian integral repeated at every spacetime point. For a fermion field this cannot be right, because the quantum operators obey anti-commutation relations. The path-integral version of this anti-commuting nature is to let the classical integration variables themselves be Grassmann numbers. Thus in the free Dirac theory we formally write

$$Z_0 = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi\,\exp\left\{\mathrm{i}\int \mathrm{d}^4x\;\bar{\psi}(x)\big(\mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon\big)\psi(x)\right\}. \tag{1.162}$$

Here ψ and $\bar{\psi}$ are treated as independent Grassmann-valued integration variables. The relation $\bar{\psi} = \psi^\dagger\gamma^0$ is the operator relation in canonical quantization; in the functional integral the two variables are integrated independently and the correct conjugation properties are recovered in correlation functions.

4.1 Grassmann algebra

A set of Grassmann numbers θ_i is defined by

$$\theta_i\theta_j = -\theta_j\theta_i, \qquad \theta_i^2 = 0. \tag{1.163}$$

Because every variable squares to zero, a function of one Grassmann variable stops after the linear term:

$$f(\theta) = a + b\theta. \tag{1.164}$$

We use the left derivative convention. The basic rule is

$$\frac{\partial}{\partial\theta}\theta = 1, \qquad \frac{\partial}{\partial\theta}(AB) = \frac{\partial A}{\partial\theta}B + (-1)^{|A|}A\frac{\partial B}{\partial\theta}, \tag{1.165}$$

where $|A| = 0$ if A is Grassmann-even and $|A| = 1$ if A is Grassmann-odd. For example, if b is Grassmann-odd, then

$$\frac{\partial}{\partial\theta}(b\theta) = -b. \tag{1.166}$$

The minus sign is not optional; it is the same sign that will later produce the minus signs of fermion loops and fermion contractions.

Berezin integration is defined so that it is the same operation as differentiation:

$$\int \mathrm{d}\theta\ 1 = 0, \qquad \int \mathrm{d}\theta\ \theta = 1. \tag{1.167}$$

For many variables the order matters. With the convention that the rightmost integration is performed first,

$$\int \mathrm{d}\theta_1\, \mathrm{d}\theta_2\ \theta_1\theta_2 = \int \mathrm{d}\theta_1 \left(\int \mathrm{d}\theta_2\ \theta_1\theta_2\right) = \int \mathrm{d}\theta_1\, (-\theta_1) = -1. \tag{1.168}$$

This simple example is often the safest way to remember that Grassmann integration is sensitive to ordering.

The finite-dimensional Gaussian integral is the key formula. For one pair $\eta, \bar\eta$,

$$\int \mathrm{d}\bar\eta\, \mathrm{d}\eta\ \exp(-\bar\eta b\eta) = \int \mathrm{d}\bar\eta\, \mathrm{d}\eta\ (1 - \bar\eta b\eta) = \int \mathrm{d}\bar\eta\, \mathrm{d}\eta\ (1 + \eta\bar\eta b) = b. \tag{1.169}$$

With an insertion one obtains

$$\int \mathrm{d}\bar\eta\, \mathrm{d}\eta\ \eta\bar\eta\ \exp(-\bar\eta b\eta) = 1 = b^{-1}\det b. \tag{1.170}$$

For N pairs this becomes

$$\int \prod_{i=1}^{N} \mathrm{d}\bar\eta_i\, \mathrm{d}\eta_i\ \exp\left(-\bar\eta_i A_{ij}\eta_j\right) = \det A, \tag{1.171}$$

and the two-point insertion gives the inverse matrix:

$$\int \prod_{i=1}^{N} \mathrm{d}\bar\eta_i\, \mathrm{d}\eta_i\ \eta_j\bar\eta_k\ \exp\left(-\bar\eta_i A_{i\ell}\eta_\ell\right) = (A^{-1})_{jk}\det A. \tag{1.172}$$

This is the fermionic analogue of the ordinary bosonic Gaussian, but the determinant appears in the numerator rather than in the denominator. In parallel,

$$\int \mathrm{d}^N x\ \exp\left(-\frac{1}{2}x_i A_{ij}x_j + J_i x_i\right) = \text{const.}\,(\det A)^{-1/2}\exp\left(\frac{1}{2}J_i(A^{-1})_{ij}J_j\right), \tag{1.173}$$

$$\int \prod_i \mathrm{d}\bar\eta_i\, \mathrm{d}\eta_i\ \exp\left(-\bar\eta_i A_{ij}\eta_j + \bar\xi_i\eta_i + \bar\eta_i\xi_i\right) = \det A\ \exp\left(\bar\xi_i(A^{-1})_{ij}\xi_j\right). \tag{1.174}$$

The inverse power in the bosonic case and the direct determinant in the fermionic case are not cosmetic differences; this is why closed fermion loops contribute an extra minus sign and a trace over spinor indices.

The change of variables also goes oppositely to the bosonic case. If

$$\theta_i = U_{ij}\theta'_j, \tag{1.175}$$

then the product of all Grassmann variables transforms as

$$\theta_1\theta_2\cdots\theta_N = (\det U)\,\theta'_1\theta'_2\cdots\theta'_N. \tag{1.176}$$

Since the integral of the top product must remain normalized to one, the measure transforms as

$$\mathrm{d}\theta_N\cdots\mathrm{d}\theta_1 = (\det U)^{-1}\,\mathrm{d}\theta'_N\cdots\mathrm{d}\theta'_1\,. \tag{1.177}$$

This inverse Jacobian is the finite-dimensional origin of fermionic functional determinants in field theory.

4.2 Fermion field propagator

Now apply the same Gaussian formula to a Dirac field. A Dirac spinor is a four-component column of Grassmann-valued fields,

$$\psi(x) = \begin{pmatrix}\psi_1(x)\\ \psi_2(x)\\ \psi_3(x)\\ \psi_4(x)\end{pmatrix}, \qquad \bar\psi(x) = \psi^\dagger(x)\gamma^0, \tag{1.178}$$

and the free Dirac Lagrangian is

$$\mathcal{L}_{\text{Dirac}} = \bar\psi_a(x)(\mathrm{i}\gamma^\mu_{ab}\partial_\mu - m\delta_{ab})\psi_b(x). \tag{1.179}$$

After discretizing spacetime, the labels (x,a) play the role of the finite-dimensional index i. Therefore the free vacuum functional is the continuum version of a Grassmann Gaussian:

$$Z_0[0,0] = \int \mathcal{D}\bar\psi\mathcal{D}\psi\ \exp\left[\mathrm{i}\int \mathrm{d}^4x\ \bar\psi(\mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon)\psi\right] = \text{const.}\ \det\bigl(\mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon\bigr). \tag{1.180}$$

The constant depends on the precise normalization of the measure and will cancel in normalized correlation functions.

To compute correlation functions, introduce Grassmann sources $\eta, \bar\eta$:

$$Z_0[\eta,\bar\eta] = \int \mathcal{D}\bar\psi\mathcal{D}\psi\ \exp\left\{\mathrm{i}\int \mathrm{d}^4x\ \left[\bar\psi(\mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon)\psi + \bar\psi\eta + \bar\eta\psi\right]\right\}. \tag{1.181}$$

Let

$$D = \mathrm{i}\partial\!\!\!/ - m + \mathrm{i}\epsilon. \tag{1.182}$$

Completing the square gives

$$\bar\psi D\psi + \bar\psi\eta + \bar\eta\psi = \bigl(\bar\psi + \bar\eta D^{-1}\bigr)D\bigl(\psi + D^{-1}\eta\bigr) - \bar\eta D^{-1}\eta. \tag{1.183}$$

The shift of Grassmann integration variables has unit Jacobian, so

$$Z_0[\eta,\bar{\eta}] = Z_0[0,0]\,\exp\left[-\mathrm{i}\int \mathrm{d}^4x\,\mathrm{d}^4y\;\bar{\eta}(x)D^{-1}(x,y)\eta(y)\right]. \tag{1.184}$$

It is conventional to define the Feynman propagator by

$$S_F(x-y) = \mathrm{i}D^{-1}(x,y), \tag{1.185}$$

or equivalently

$$\big(\mathrm{i}\partial\!\!\!/_x - m + \mathrm{i}\epsilon\big)S_F(x-y) = \mathrm{i}\delta^{(4)}(x-y). \tag{1.186}$$

Hence the generating functional can be written in the compact form

$$Z_0[\eta,\bar{\eta}] = Z_0[0,0]\,\exp\left[-\int \mathrm{d}^4x\,\mathrm{d}^4y\;\bar{\eta}(x)S_F(x-y)\eta(y)\right]. \tag{1.187}$$

In momentum space this propagator is

$$S_F(p) = \frac{\mathrm{i}}{p\!\!\!/ - m + \mathrm{i}\epsilon} = \frac{\mathrm{i}(p\!\!\!/ + m)}{p^2 - m^2 + \mathrm{i}\epsilon}. \tag{1.188}$$

Finally, the time-ordered two-point function is obtained by differentiating with respect to the Grassmann sources. With the derivative convention used above,

$$\big\langle 0|T\{\psi_a(x)\bar{\psi}_b(y)\}|0\big\rangle = -\frac{1}{Z_0[0,0]}\,\frac{\delta^2 Z_0[\eta,\bar{\eta}]}{\delta\bar{\eta}_a(x)\,\delta\eta_b(y)}\bigg|_{\eta=\bar{\eta}=0} = S_{F,ab}(x-y). \tag{1.189}$$

Thus the whole fermionic path integral is fixed by the same idea as the scalar Gaussian integral, but with Grassmann variables: the Gaussian gives a determinant, and insertions give the inverse Dirac operator, namely the fermion propagator.

5. Symmetries in Path Integral Formalism

The path integral approach provides a direct link between symmetries and physical observables through the following hierarchy: **Variational calculus** $\to$ **Schwinger-Dyson equations** $\to$ **Noether's theorem** $\to$ **Ward identities in QED**.

5.1 Schwinger-Dyson Equations

For simplicity, we still consider the scalar field theory as an example. By doing the partial integration, the action can be rewritten as:

$$S[\phi] = \int \mathrm{d}^4x\,\mathcal{L}[\phi] = \int \mathrm{d}^4x\left\{\frac{1}{2}(\partial_\mu\phi)^2 - \frac{1}{2}m^2\phi^2\right\} = -\frac{1}{2}\int \mathrm{d}^4x\,\phi\left(\Box + m^2\right)\phi. \tag{1.190}$$

The n-point correlator is given by:

$$\langle\Omega|T\{\phi(x_1)\ldots\phi(x_n)\}|\Omega\rangle = \frac{\int\mathcal{D}\phi\,\phi_1\ldots\phi_n e^{\mathrm{i}S[\phi]}}{\int\mathcal{D}\phi\,e^{\mathrm{i}S[\phi]}}. \tag{1.191}$$

We derive classical equations of motion from a variation of fields: $\phi(x) \to \phi'(x) = \phi(x) + \varepsilon(x)$, i.e., $\delta\phi(x) = \varepsilon(x)$. The integration measurement is invariant due to ε is infinitesimal: $\mathcal{D}\phi' = \mathcal{D}\phi$. And the action changes as: $S[\phi'] = S[\phi + \varepsilon] = S[\phi] + \delta S$. So we have

$$\int \mathcal{D}\phi\, \phi_1 \ldots \phi_n e^{\mathrm{i}S[\phi]} = \int \mathcal{D}\phi'\, \phi'_1 \ldots \phi'_n e^{\mathrm{i}S[\phi+\varepsilon]} \tag{1.192}$$

$$= \int \mathcal{D}\phi\, e^{\mathrm{i}(S[\phi]+\delta S)}(\phi_1 + \varepsilon_1)\ldots(\phi_n + \varepsilon_n), \tag{1.193}$$

where we have denoted that $\phi_i \equiv \phi(x_i)$ and $\varepsilon_i \equiv \varepsilon(x_i)$. Then we expand ε to the first order, the exponential is $e^{\mathrm{i}(S+\delta S)} \simeq e^{\mathrm{i}S}(1 + \mathrm{i}\delta S)$. And for δS, we have

$$\delta S = \int \mathrm{d}^4 y \left(\frac{\partial \mathcal{L}}{\partial \phi}\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}\partial_\mu \delta\phi \right) \tag{1.194}$$

$$= \int \mathrm{d}^4 y \left[-m^2 \phi \varepsilon + \partial^\mu \phi \partial_\mu \varepsilon \right] \tag{1.195}$$

$$= -\int \mathrm{d}^4 y \left[\varepsilon(y) \left(\Box_y + m^2 \right) \phi(y) \right]. \tag{1.196}$$

Thus, put these result into (1.193), we have:

$$\begin{aligned}
&\int \mathcal{D}\phi\, e^{\mathrm{i}(S[\phi]+\delta S)}(\phi_1 + \varepsilon_1)\ldots(\phi_n + \varepsilon_n) \\
&= \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left[1 - \mathrm{i} \int \mathrm{d}^4 y\, \varepsilon(y)(\Box_y + m^2)\phi(y) \right] \left[(\phi_1 \ldots \phi_n) + (\varepsilon_1 \phi_2 \ldots \phi_n) + \cdots + (\phi_1 \ldots \varepsilon_n) \right]
\end{aligned} \tag{1.197}$$

$$\begin{aligned}
&= \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left[(\phi_1 \ldots \phi_n) + (\varepsilon_1 \phi_2 \ldots \phi_n) + \cdots + (\phi_1 \ldots \varepsilon_n) \right] \\
&\qquad - \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \int \mathrm{d}^4 y\, \mathrm{i} \left[\varepsilon(y)(\Box_y + m^2)\phi(y) \right] (\phi_1 \ldots \phi_n) + \mathcal{O}(\varepsilon^2).
\end{aligned} \tag{1.198}$$

Subtracting the first term $\int \mathcal{D}\phi\, \phi_1 \ldots \phi_n e^{\mathrm{i}S[\phi]}$ with (1.192), we have

$$0 = -\mathrm{i} \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left\{ \int \mathrm{d}^4 y\ \varepsilon(y) \left[(\Box_y + m^2)\phi(y) \right] \phi_1 \ldots \phi_n + \mathrm{i} \sum_{i=1}^{n} \phi_1 \ldots \epsilon_i \ldots \phi_n \right\}, \tag{1.199}$$

where ε_i will replace the corresponding ϕ_j when $i = j$ in the summation. And we can notice that $\varepsilon(x_i) = \int \mathrm{d}^4 y\ \varepsilon(y)\delta^{(4)}(x_i - y)$, so we have

$$\int \mathrm{d}^4 y\ \varepsilon(y) \left\{ -\mathrm{i} \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left[\left[(\Box_y + m^2)\phi(y) \right] \phi_1 \ldots \phi_n + \mathrm{i} \sum_{i=1}^{n} \phi_1 \ldots \delta^{(4)}(y - x_i) \ldots \phi_n \right] \right\} = 0. \tag{1.200}$$

For arbitrary ε holds this relation, the only possible case is

$$\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} \left[\left[(\Box_y + m^2)\phi(y) \right] \phi_1 \ldots \phi_n + \mathrm{i} \sum_{i=1}^{n} \phi_1 \ldots \delta^{(4)}(y - x_i) \ldots \phi_n \right] = 0. \tag{1.201}$$

Example: Consider two-point correlator, i.e., $n = 1$:

$$(\Box_y + m^2) \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}\phi(y)\phi(x_1) = -\mathrm{i}\delta^{(4)}(y - x_1) \int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]} = -\mathrm{i}\delta^{(4)}(y - x_1) Z[J = 0] \tag{1.202}$$

Putting it back in the expression for the correlator:

$$(\Box_y + m^2)\underbrace{\langle\Omega|T\{\phi(y)\phi(x_1)\}|\Omega\rangle}_{G_F(y-x_1)} = -\mathrm{i}\delta^{(4)}(y-x_1). \tag{1.203}$$

Thus,

$$G_F(y-x_1) = -\mathrm{i}\delta^{(4)}(y-x_1)\left(\Box_y + m^2\right)^{-1}. \tag{1.204}$$

Similarly, we would like to generalize this form into n-point correlator. From (1.201) we have

$$\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}(\Box_y + m^2)\phi(y)\,(\phi_1\ldots\phi_n) = -\mathrm{i}\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}\left(\sum_{i=1}^{n}\phi_1\ldots\delta^{(4)}(y-x_i)\ldots\phi_n\right). \tag{1.205}$$

Notice $-(\Box_y + m^2)\phi(y)$ in the left-hand side is nothing but the variation of the action:

$$\frac{\delta S[\phi]}{\delta\phi(y)} = \int \mathrm{d}^4x\left(\frac{\delta\mathcal{L}}{\delta\phi}\delta(x-y) + \frac{\delta\mathcal{L}}{\delta\partial_\mu\phi}\partial_\mu\delta(x-y)\right) = \frac{\delta\mathcal{L}}{\delta\phi(y)} - \partial_\mu\frac{\delta\mathcal{L}}{\delta\partial_\mu\phi(y)}, \tag{1.206}$$

the result of this variation always the equation of motion (or the Euler-Lagrange equation, if you like). So Without loss of generality, we can replace $-(\Box_y + m^2)\phi(y)$ by $\frac{\delta S[\phi]}{\delta\phi(y)}$ in (1.205) and divide $Z[0]$ as

$$\frac{\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}\left(\frac{\delta S[\phi]}{\delta\phi(y)}\phi_1\ldots\phi_n\right)}{\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}} = \frac{\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}\left(\mathrm{i}\sum_{i=1}^{n}\phi_1\ldots\delta^{(4)}(y-x_i)\ldots\phi_n\right)}{\int \mathcal{D}\phi\, e^{\mathrm{i}S[\phi]}}. \tag{1.207}$$

We then get the **Schwinger-Dyson equation** for n-point functions:

$$\left\langle\Omega\left|\frac{\delta S[\phi]}{\delta\phi(y)}T\{\phi_1\ldots\phi_n\}\right|\Omega\right\rangle = \mathrm{i}\sum_{i=1}^{n}\left\langle\Omega\left|T\{\phi_1\ldots\delta^{(4)}(y-x_i)\ldots\phi_n\}\right|\Omega\right\rangle. \tag{1.208}$$

We can simply denote it as:

$$\left\langle\frac{\delta S[\phi]}{\delta\phi(y)}T\{\phi_1\ldots\phi_n\}\right\rangle = \mathrm{i}\sum_{i=1}^{n}\left\langle T\{\phi_1\ldots\delta^{(4)}(y-x_i)\ldots\phi_n\}\right\rangle. \tag{1.209}$$

5.2 Conservation Laws and Noether's Theorem

Consider a complex scalar field $\mathcal{L} = \partial_\mu\phi\partial^\mu\phi^* - m^2\phi\phi^*$. Under the global $U(1)$ transformation $\phi \to \phi' = e^{\mathrm{i}\alpha}\phi$, the variation is:

$$\delta\phi = \mathrm{i}\alpha\phi; \qquad \text{for } \alpha \in \mathbb{R}, \tag{1.210}$$

with the unchanged Lagrangian. For a *local* transformation $\alpha \to \alpha(x)$, we have $\delta\phi(x) = \mathrm{i}\alpha(x)\phi(x)$, so the variation for the Lagrangian is

$$\begin{aligned}\delta\mathcal{L} &= \mathcal{L}[\phi'] - \mathcal{L}[\phi]\\ &= \partial_\mu\phi\partial^\mu\phi^* + \mathrm{i}\partial_\mu\alpha(x)\left(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi\right) + \mathcal{O}(\alpha^2) - m^2\phi\phi^* - \partial_\mu\phi\partial^\mu\phi^* + m^2\phi\phi^*\\ &= \partial_\mu\alpha(x)j^\mu(x),\end{aligned} \tag{1.211}$$

where $j^\mu(x) = \mathrm{i}\left(\phi\partial^\mu\phi^* - \phi^*\partial^\mu\phi\right)$. An n-point correlator is invariant under this transformation

$$\int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}(\phi_1\ldots\phi_{n/2})(\phi_1^*\ldots\phi_{n/2}^*) = \int \mathcal{D}\phi'\mathcal{D}\phi'^*\, e^{\mathrm{i}S'[\phi',\phi'^*]}(\phi_1'\ldots\phi_{n/2}')(\phi_1'^*\ldots\phi_{n/2}'^*), \tag{1.212}$$

Notice that $e^{\mathrm{i}S'[\phi',\phi'^*]} = e^{\mathrm{i}(S+\delta S)} \simeq e^{\mathrm{i}S}(1+\mathrm{i}\delta S)$. Expand the right-hand side to order $\alpha(x)$, and subtracting with the left-hand side, we have

$$0 = \int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\,(1+\mathrm{i}\delta S)\left((1+\mathrm{i}\alpha)\phi_1\dots(1+\mathrm{i}\alpha)\phi_n^*\right) \tag{1.213}$$

$$= \int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\Big\{\mathrm{i}\delta S(\phi_1\dots\phi_n^*) + [(\mathrm{i}\alpha\phi_1)\dots\phi_n^*] + \cdots + [\phi_1\cdots(\pm\mathrm{i}\alpha\phi_i^{(*)})\dots\phi_n^*] + \cdots + [\phi_1\dots(-\mathrm{i}\alpha\phi_n^*)]\Big\} \tag{1.214}$$

$$= \int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\left\{\mathrm{i}\int \mathrm{d}^4y\,\partial_\mu\alpha(y)j^\mu(y)(\phi_1\dots\phi_n^*) + \sum_{i=1}^n \phi_1\cdots(\pm\mathrm{i}\alpha\phi_i^{(*)})\dots\phi_n^*\right\}, \tag{1.215}$$

where $\pm\mathrm{i}\phi^{(*)_i}$ denote that for ϕ_i^* taking $-\mathrm{i}$ and for ϕ_i taking $+\mathrm{i}$ .After partial integration to factor out $\alpha(y)$:

$$0 = \int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\left\{\int \mathrm{d}^4y\,\alpha(y)\partial_\mu j^\mu(y)\phi_1\dots\phi_n + \mathrm{i}\sum_{i=1}^n \phi_1\dots(\pm\mathrm{i})\left[\int \mathrm{d}^4y\,\delta(y-x_i)\alpha(y)\right]\phi_i^{(*)}\dots\phi_n\right\} \tag{1.216}$$

$$= \int \mathrm{d}^4y\,\alpha(y)\int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\left\{-\mathrm{i}\partial_\mu j^\mu(y)\phi_1\dots\phi_n + \mathrm{i}\sum_{i=1}^n \phi_1\dots(\pm\mathrm{i})\delta(y-x_i)\phi_i^{(*)}\dots\phi_n\right\}. \tag{1.217}$$

Similarly, for any infinitesimally α holding this relation, so the only possible way is

$$\int \mathcal{D}\phi\mathcal{D}\phi^*\, e^{\mathrm{i}S[\phi,\phi^*]}\left\{-\mathrm{i}\partial_\mu j^\mu(y)\phi_1\dots\phi_n + \mathrm{i}\sum_{i=1}^n \phi_1\dots(\pm\mathrm{i})\delta(y-x_i)\phi_i^{(*)}\dots\phi_n\right\} = 0. \tag{1.218}$$

We can also use the correlator notation:

$$\langle\partial_\mu j^\mu(y)\phi(x_1)\dots\phi(x_n)\rangle = -\mathrm{i}\sum_{i=1}^n\langle\phi(x_1)\dots(\pm\mathrm{i})\delta(y-x_i)\phi_i^{(*)}\dots\phi(x_n)\rangle. \tag{1.219}$$

Now we generalize this relation to a set of real fields ϕ_a. For a local transformation $\phi_a(x) \to \phi'_a(x) = \phi_a(x) + \varepsilon\Delta\phi_a(x)$, if the global transformation gives $\mathcal{L} \to \mathcal{L} + \varepsilon\partial_\mu J^\mu$, then for a local $\varepsilon(x)$ we have

$$\partial_\mu\phi'_a = \partial_\mu(\phi_a + \varepsilon\Delta\phi_a) = \partial_\mu\phi_a + \varepsilon\partial_\mu(\Delta\phi_a) + (\partial_\mu\varepsilon)\Delta\phi_a. \tag{1.220}$$

So when we expand the Lagrangian, we have:

$$\mathcal{L}(\phi',\partial\phi') = \mathcal{L}(\phi,\partial\phi) + \sum_a\left(\frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\delta(\partial_\mu\phi_a)\right), \tag{1.221}$$

then plug the variation $\delta\phi_a = \varepsilon\Delta\phi_a$ into it and using the Euler-Lagrange equation:

$$\delta\mathcal{L} = \sum_a\left[\frac{\partial\mathcal{L}}{\partial\phi_a}(\epsilon\Delta\phi_a) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\left(\epsilon\partial_\mu(\Delta\phi_a) + (\partial_\mu\epsilon)\Delta\phi_a\right)\right] \tag{1.222}$$

$$= \sum_a\left[\partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}(\epsilon\Delta\phi_a) + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\left(\epsilon\partial_\mu(\Delta\phi_a) + (\partial_\mu\epsilon)\Delta\phi_a\right)\right] \tag{1.223}$$

$$= \sum_a\left[\varepsilon\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\Delta\phi_a\right) + (\partial_\mu\epsilon)\Delta\phi_a\right]. \tag{1.224}$$

Then we get the variation for the Lagrangian:

$$\mathcal{L}(\phi_a) \to \mathcal{L}(\phi'_a) = \mathcal{L}(\phi_a) + \varepsilon(x)\partial_\mu J^\mu + \partial_\mu\varepsilon(x)\sum_a \Delta\phi_a \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)} \tag{1.225}$$

$$= \mathcal{L}(\phi_a) + \varepsilon(x)\partial_\mu J^\mu - \varepsilon(x)\partial_\mu\left(\sum_a \Delta\phi_a \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\right) \tag{1.226}$$

$$= \mathcal{L}(\phi_a) - \varepsilon(x)\partial_\mu\left(\sum_a \Delta\phi_a \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)} - J^\mu\right), \tag{1.227}$$

which appears an extra term due to the infinitesimal transformation rely on the coordinate. Therefore, the Noether current j^μ can be defined from the variation for the action

$$S[\phi'_a] = \int \mathrm{d}^4y\, [\mathcal{L}[\phi_a] - \varepsilon(y)\partial_\mu j^\mu(y)] = S[\phi_a] - \int \mathrm{d}^4y\, \varepsilon(y)\partial_\mu j^\mu(y). \tag{1.228}$$

Thus, the Noether current j^μ is

$$j^\mu = \sum_a \Delta\phi_a \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)} - J^\mu. \tag{1.229}$$

Put this current back to (1.219), we get the conserved symmetry Schwinger-Dyson equation as:

$$\langle\partial_\mu j^\mu(y)\phi_{a_1}(x_1)\ldots\phi_{a_n}(x_n)\rangle = -\mathrm{i}\sum_{i=1}^{n}\langle\phi_{a_1}(x_1)\ldots\delta(y-x_i)\Delta\phi_{a_i}\ldots\phi_{a_n}(x_n)\rangle \tag{1.230}$$

5.3 Ward-Takahashi Identity in QED

Consider the QED Lagrangian with fermions:

$$\mathcal{L}[\psi,\bar\psi,A_\mu] = \bar\psi(\mathrm{i}\gamma^\mu D_\mu - m)\psi + A_\mu j^\mu, \tag{1.231}$$

where $j^\mu = e\bar\psi\gamma^\mu\psi$. The global $U(1)$ transformation gives

$$\psi \to e^{\mathrm{i}\alpha}\psi \implies \delta\psi = \mathrm{i}\alpha\psi. \tag{1.232}$$

So the local transformation gives

$$\mathcal{L}[\psi',\bar\psi'] = \mathcal{L}[\psi,\bar\psi] - (\partial_\mu\alpha)j^\mu. \tag{1.233}$$

Applying Schwinger-Dyson equation with two fields, we have

$$\partial_\mu\langle 0|T\{j^\mu(y)\psi(x_1)\bar\psi(x_2)\}|0\rangle = \delta^{(4)}(y-x_1)\langle 0|T\{\psi(x_1)\bar\psi(x_2)\}|0\rangle - \delta^{(4)}(y-x_2)\langle 0|T\{\psi(x_1)\bar\psi(x_2)\}|0\rangle. \tag{1.234}$$

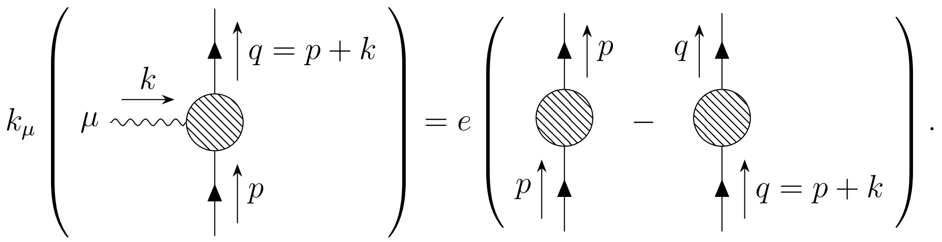
The minus sign shows that ψ follows **Fermi statistics**. Take a Fourier transform to momentum space:

$$k_\mu\langle 0|T\{j^\mu(k)\psi(q)\bar\psi(p)\}|0\rangle = e\left(\langle 0|T\{\psi(q-k)\bar\psi(p)\}|0\rangle - \langle 0|T\{\psi(q)\bar\psi(p+k)\}|0\rangle\right), \tag{1.235}$$

which is nothing but the **Ward-Takahashi identity** in QED that we are familiar with:

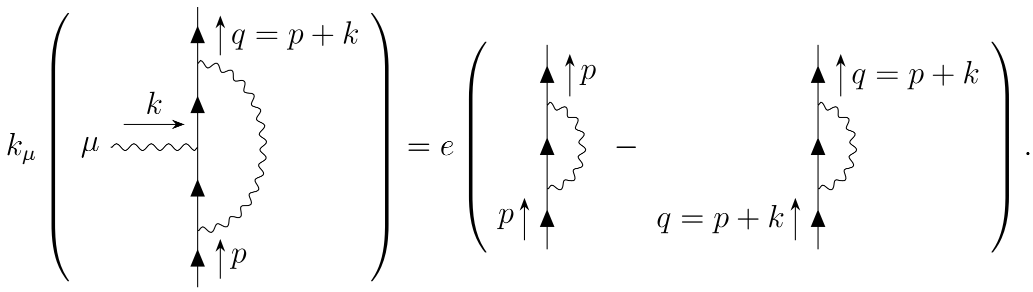
$$k_\mu\mathcal{M}^\mu(k;p,q=p+k) = e\mathcal{M}_0(p) - e\mathcal{M}_0(q=p+k). \tag{1.236}$$

It can also be expressed by the Feynman diagram. Notice that the interaction $e\bar{\psi}\gamma^\mu\psi A_\mu$ contains one ψ and one $\bar{\psi}$, but this does not mean that the two fermion segments in an electron scattering diagram should be drawn as two disconnected particles. They are the two ends of one continuous fermion line. With the photon momentum k flowing into the current insertion, momentum conservation gives $q = p + k$:



(1.237)

Especially for the one-loop vertex correction, the same convention must be used. The fermion line is continuous; the upper external momentum is therefore $q = p + k$, not $p - k$:



(1.238)

The lower and upper segments in the first diagram are not separate particle species. They are the incoming and outgoing parts of the same fermion line. Algebraically this corresponds to the two fields $\bar{\psi}$ and ψ at the QED vertex; graphically it is represented by one continuous fermion arrow.

Renormalization

1. Renormalization in ϕ^3-theory

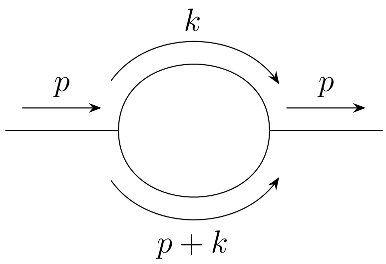
One loop of self energy

Simply speaking, renormalization is to solve the problem of divergent when we try to solve the loop graphs by subtracting the divergent part in the expression. In the following, we will use ϕ^3-theory as an example to discuss the renormalization.

Firstly, the Lagrangian of ϕ^3-theory is:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{g}{3!}\phi^3. \tag{2.1}$$

As the simplest case, the 1-loop correlation for the propagator is (notice that in this graph ,there are **all inner lines**, i.e., propagators):



where the momentum k is **off-shelled**, which means k doesn't follow the relation $k^2+E_k^2=m^2$, so technically k could be very large. But we don't need to worry about it, because k only appears in the loop, which means that it must follows the 4-momentum conservation. Thus we need to sum over all possible k values, which finally will contribute a integral measurement when we calculate the loop part. Thus, generally, we can write the whole graph as

$$\frac{\mathrm{i}}{p^2-m^2+\mathrm{i}\epsilon}\left[-\mathrm{i}\Sigma_1(p^2)\right]\frac{\mathrm{i}}{p^2-m^2+\mathrm{i}\epsilon}, \tag{2.2}$$

where $-\mathrm{i}\Sigma_1(p^2)$ represent the loop part, i.e.,

$$-\mathrm{i}\Sigma_1(p^2) = \frac{g^2}{2}\int\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{(p+k)^2-m^2+\mathrm{i}\epsilon}, \tag{2.3}$$

where we will note the integral measurement $\frac{\mathrm{d}^4k}{(2\pi)^4}\to\mathrm{d}^4k$ for simple. Next, we will focus on this integral as an example to talk about how to solve this type of integral.

1.1 Wick rotation

The integral in (2.3) over the loop energy can be also written as

$$I(p,\vec{k}) = \int_{-\infty}^{+\infty}\mathrm{d}k^0\,\frac{\mathrm{i}}{(k^0)^2-E_k^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{(p^0+k^0)^2-E_{p+k}^2+\mathrm{i}\epsilon}, \tag{2.4}$$

where

$$E_k = \sqrt{\vec{k}^{\,2} + m^2}, \qquad E_{p+k} = \sqrt{(\vec{p} + \vec{k})^2 + m^2}. \tag{2.5}$$

The variable k^0 is now regarded as a complex variable. The four poles are

$$k^0 = +E_k - \mathrm{i}\epsilon, \qquad k^0 = -E_k + \mathrm{i}\epsilon, \tag{2.6}$$

$$k^0 = -p^0 + E_{p+k} - \mathrm{i}\epsilon, \qquad k^0 = -p^0 - E_{p+k} + \mathrm{i}\epsilon. \tag{2.7}$$

The small $\mathrm{i}\epsilon$ is important: it tells us which poles sit slightly above or slightly below the real axis. Wick rotation means changing the integration path in the complex k^0 plane. Originally the integral is along the real axis,

$$C_{\mathrm{M}}: \quad k^0 \in (-\infty, +\infty).$$

In Euclidean variables we use

$$k^0 = \mathrm{i}\omega, \qquad \mathrm{d}k^0 = \mathrm{i}\,\mathrm{d}\omega,$$

so the new path is the imaginary axis,

$$C_{\mathrm{E}}: \quad k^0 = \mathrm{i}\omega, \quad \omega \in (-\infty, +\infty).$$

In the figures below, the blue path is the real-axis Minkowski contour C_{M}, the green path is the imaginary-axis Euclidean contour C_{E}, and the crosses are poles.

There is one orientation point which is very easy to mix up. The Euclidean contour itself is oriented upward, because

$$\omega : -\infty \to +\infty \quad \Longrightarrow \quad k^0 = \mathrm{i}\omega : -\mathrm{i}\infty \to +\mathrm{i}\infty.$$

But when we prove the Wick rotation by closing the contour, the closed contour is

$$C_{\text{closed}} = C_{\mathrm{M}} + A_+ - C_{\mathrm{E}} + A_-,$$

where A_+ runs from $+R$ to $+\mathrm{i}R$ in the first quadrant, while A_- runs from $-\mathrm{i}R$ back to $-R$ in the third quadrant. Thus the imaginary-axis part of the closed contour is the downward path $-C_{\mathrm{E}}$. After moving it to the other side of the equation, the final Euclidean integral is again along C_{E}, upward. The orange dashed arrows in the figures show these auxiliary closing arcs, not the direction of the final integral along C_{E}.

The phrase "change the contour" means the following simple thing: we slide the integration line from one path to another in the complex k^0 plane. If the sliding line does not hit any pole, the integral is unchanged, apart from the Jacobian $\mathrm{d}k^0 = \mathrm{i}\,\mathrm{d}\omega$. If the line hits a pole, then we must bend the path around that pole, and the difference is a residue term. Symbolically,

$$\int_{C_{\text{old}}} f(k^0)\,\mathrm{d}k^0 - \int_{C_{\text{new}}} f(k^0)\,\mathrm{d}k^0 = 2\pi\mathrm{i} \sum_{\text{poles swept by the contour}} \operatorname{Res} f, \tag{2.8}$$

up to the sign determined by the orientation of the contour.

Case 1: p^0 is imaginary.

Take, for example, $p^0 = -\mathrm{i}P$ with $P > 0$. Then $-p^0 = +\mathrm{i}P$, so the two poles from the second propagator are shifted upward:

$$k^0 = \mathrm{i}P + E_{p+k} - \mathrm{i}\epsilon, \qquad k^0 = \mathrm{i}P - E_{p+k} + \mathrm{i}\epsilon.$$

This is the Euclidean starting point. The current integration path is C_E, not C_M.

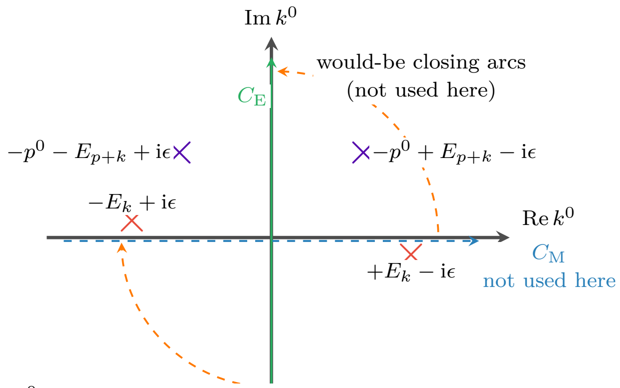


Fig. 1: p^0 imaginary. The first-quadrant pole is harmless, because the contour currently being used is only the green line C_E.

The point which is easy to miss is this: the pole in the first quadrant does not automatically give a residue. A residue appears only when the contour is actually moved through that pole, or when the pole moves through the contour. In this first figure we are not rotating C_E into C_M while keeping p^0 imaginary. We are simply defining the Euclidean integral on the green line.

Case 2: p^0 is real and $|p^0| < m$.

Now keep the green contour C_E fixed, and analytically continue p^0 from imaginary value back to a real value. The poles move because their positions contain p^0. If $|p^0| < m$, then

$$E_{p+k} \geq m > |p^0|,$$

so

$$-p^0 + E_{p+k} > 0, \qquad -p^0 - E_{p+k} < 0.$$

Thus the second pair of poles ends up in the same pattern as the first pair: one pole is slightly below the real axis on the right, and one pole is slightly above the real axis on the left.

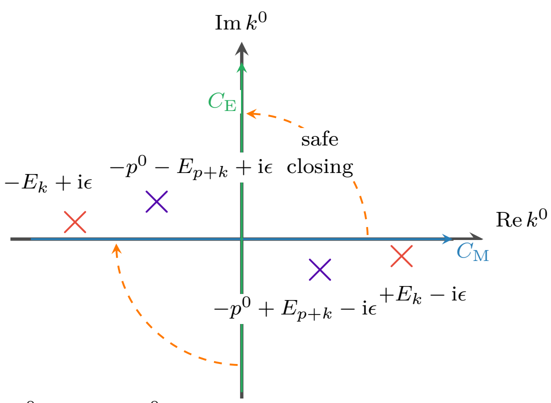


Fig. 2: p^0 real and $|p^0| < m$. No pole lies in the region swept when the blue contour is rotated into the green contour.

Therefore, for $|p^0| < m$, the real-axis integral and the imaginary-axis integral are related by an ordinary Wick rotation. On the green contour,

$$k^0 = \mathrm{i}\omega, \qquad k^2 = (k^0)^2 - \vec{k}^{\,2} = -(\omega^2 + \vec{k}^{\,2}) < 0,$$

and similarly $(p+k)^2 < 0$ in the large Euclidean region. This is why power counting becomes Euclidean.

What does "a pole crosses a contour" mean?

A pole is not just a fixed dot: its position depends on the external energy p^0. For example, if $p^0 = -P$ with $P > 0$, then one pole is

$$k^0 = -p^0 - E_{p+k} + \mathrm{i}\epsilon = P - E_{p+k} + \mathrm{i}\epsilon. \tag{2.9}$$

As P increases, this pole moves horizontally. If $P < E_{p+k}$, it is on the left side of the imaginary axis. If $P = E_{p+k}$, it sits on the imaginary-axis contour. If $P > E_{p+k}$, it has moved to the right side of the imaginary axis. This is what we mean by saying that the pole has crossed C_{E}.

In formula language, crossing the imaginary axis means

$$\mathrm{Re}\, k^0_{\text{pole}} \quad \text{changes sign.}$$

Crossing the real axis would similarly mean

$$\mathrm{Im}\, k^0_{\text{pole}} \quad \text{changes sign.}$$

But only crossing the contour that we are actually using produces a residue.

Case 3: p^0 is real and $|p^0| > m$.

Take the negative-energy example $p^0 = -P$ with $P > m$, which is the case drawn in the book. The pole

$$k^0 = P - E_{p+k} + \mathrm{i}\epsilon$$

crosses the green contour whenever

$$P > E_{p+k} \quad \Longleftrightarrow \quad (p^0)^2 > m^2 + (\vec{p} + \vec{k})^2.$$

When this happens, the contour cannot pass straight through the pole. It must detour around it, and this detour gives an extra residue, usually called the pole term.

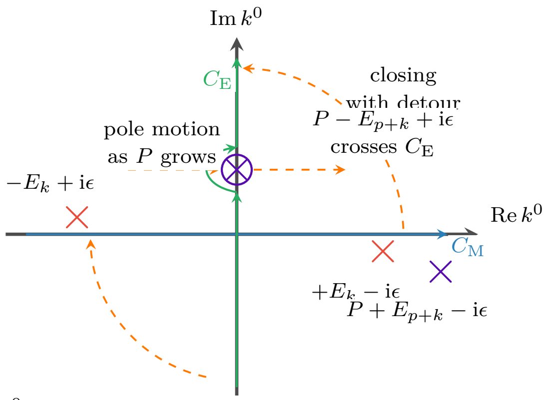


Fig. 3: $p^0 = -P$ and $P > m$. A pole crosses the green contour, so the Wick-rotated answer contains an extra residue term.

However, this pole term can occur only when

$$(p^0)^2 > m^2 + (\vec{p} + \vec{k})^2.$$

For very large $|\vec{k}|$, the right-hand side is very large, so the inequality cannot hold. Therefore the pole term comes only from a finite region of $\vec{k}$, not from the UV region $|\vec{k}| \to \infty$. The UV divergence is still captured by the Euclidean large-momentum integral along C_{E}.

So the divergence in Eq.(2.3) is called the UV divergence, which can be seen by firstly taking the polar coordinate: $\mathrm{d}^4 k_E = k_E^3 \mathrm{d}k_E \mathrm{d}\Omega$, thus in large k

$$\Sigma_1 \sim \int \mathrm{d}^d k \frac{1}{k^4} \sim \int \mathrm{d}k \mathrm{d}\Omega \frac{1}{k} \sim \int \frac{\mathrm{d}k}{k} \sim \ln \Lambda \to \infty. \tag{2.10}$$

1.2 On-shell renormalization

Renormalization begins by separating a loop integral into two parts: a finite, momentum-dependent part and a divergent part that is local. For the one-loop self-energy in ϕ^3 theory,

$$\Sigma_1(p^2, m^2) = \frac{\mathrm{i}g^2}{32\pi^4} \int \mathrm{d}^4 k \, \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{2.11}$$

The integral is logarithmically divergent in the ultraviolet. A convenient way to display the divergent piece is to add and subtract a simple reference integral. We introduce an arbitrary finite mass scale μ and write

$$\begin{aligned} \Sigma_1 = \frac{\mathrm{i}g^2}{32\pi^4} \Bigg[\int \mathrm{d}^4 k \bigg\{ \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)} - \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} \bigg\} \\ + \int \mathrm{d}^4 k \, \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} \Bigg]. \end{aligned} \tag{2.12}$$

This is just adding zero, so no physics has been changed. The reason for this particular subtraction is that the first integral now falls faster at large loop momentum:

$$\frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)} - \frac{1}{(k^2 - \mu^2 + \mathrm{i}\epsilon)^2} = O\left(\frac{1}{k^6}\right). \tag{2.13}$$

In four dimensions, $\mathrm{d}^4 k \sim k^3 \mathrm{d}k$, and therefore

$$\int^{\infty} \mathrm{d}^4 k \, O\left(\frac{1}{k^6}\right) \sim \int^{\infty} \mathrm{d}k \, \frac{1}{k^3}, \tag{2.14}$$

which is ultraviolet finite. Thus the logarithmic divergence has been isolated in the second integral. Since that second integral has no dependence on the external momentum p, its divergent part can be canceled by a local mass counterterm.

To see the divergence explicitly, regulate the theory on a spacetime lattice with lattice spacing a. The lattice spacing acts as a UV cutoff of order $1/a$. If $S_F(k; m, a)$ denotes the regulated free propagator, then

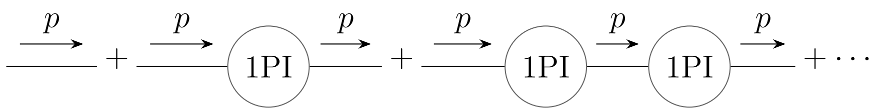
$$\begin{aligned} \Sigma_1 &= \frac{\mathrm{i}g^2}{32\pi^4} \left[\int \mathrm{d}^4 k \, \{ S_F(k; m, a) S_F(p+k; m, a) - S_F^2(k; \mu, a) \} + \int \mathrm{d}^4 k \, S_F^2(k; \mu, a) \right] \\ &\equiv \Sigma_{1,\mathrm{fin}}(p, m, \mu, a) + \Sigma_{1,\mathrm{div}}(\mu, a). \end{aligned} \tag{2.15}$$

The divergent reference part is

$$
\begin{aligned}
\Sigma_{1,\mathrm{div}}(\mu, a) &= \frac{\mathrm{i}g^2}{32\pi^4} \int \mathrm{d}^4k\, S_F^2(k;\mu,a) \\
&= -\frac{\mathrm{i}g^2}{32\pi^4} \int_{\omega^2+\vec{k}^{\,2}<1/a^2} \mathrm{d}\omega\, \mathrm{d}^3\vec{k}\, \frac{1}{(\omega^2+\vec{k}^{\,2}+\mu^2)^2} \\
&= -\frac{\mathrm{i}g^2}{16\pi^2} \int_0^{1/a} \mathrm{d}K\, \frac{K^3}{(K^2+\mu^2)^2} \\
&= -\frac{\mathrm{i}g^2}{16\pi^2} \ln\frac{1}{a} + \text{finite terms}, \qquad a \to 0.
\end{aligned}
\tag{2.16}
$$

This agrees with the power-counting estimate in Eq. (2.10): the one-loop two-point graph in four-dimensional ϕ^3 theory is logarithmically divergent.

The same divergence can be understood from the full propagator. Let $-\mathrm{i}\Sigma(p^2)$ denote the sum of all one-particle-irreducible two-point insertions. Repeated insertions give

$G_2(p^2) =$ 

$$
\begin{aligned}
&= \frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2} + \frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2}(-\mathrm{i}\Sigma)\frac{\mathrm{i}}{p^2-m^2} + \cdots \\
&= \frac{\mathrm{i}}{p^2-m^2}\left[1 + \frac{\Sigma(p^2)}{p^2-m^2} + \left(\frac{\Sigma(p^2)}{p^2-m^2}\right)^2 + \cdots\right] \\
&= \frac{\mathrm{i}}{p^2-m^2-\Sigma(p^2)}.
\end{aligned}
\tag{2.17}
$$

The physical mass is defined by the pole of the full propagator:

$$
m_{\mathrm{ph}}^2 - m^2 - \Sigma(m_{\mathrm{ph}}^2) = 0, \qquad m_{\mathrm{ph}}^2 = m^2 + \Sigma(m_{\mathrm{ph}}^2). \tag{2.18}
$$

This equation fixes only the pole position. Away from the pole, the full propagator is not simply $\mathrm{i}/(p^2-m_{\mathrm{ph}}^2)$; the remaining momentum dependence of $\Sigma(p^2)$ still contributes.

In the on-shell scheme the renormalized mass parameter is chosen to be the physical pole mass. Thus the loop integral is written with m_{ph}:

$$
\Sigma_1(p^2) = \frac{\mathrm{i}g^2}{32\pi^4} \int \mathrm{d}^4k\, \frac{1}{(k^2-m_{\mathrm{ph}}^2+\mathrm{i}\epsilon)((p+k)^2-m_{\mathrm{ph}}^2+\mathrm{i}\epsilon)}. \tag{2.19}
$$

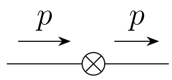
The bare mass is then split as

$$
m_0^2 = m_{\mathrm{ph}}^2 + \delta m^2. \tag{2.20}
$$

In the Lagrangian this means

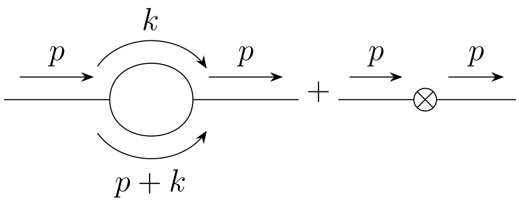
$$
-\frac{1}{2}m_0^2\phi^2 = -\frac{1}{2}m_{\mathrm{ph}}^2\phi^2 - \frac{1}{2}\delta m^2\phi^2. \tag{2.21}
$$

The counterterm vertex is therefore

 $= -\mathrm{i}\,\delta m^2.$ (2.22)

The renormalized one-loop self-energy is the sum of the ordinary bubble and the counterterm graph:

$$-\mathrm{i}\Sigma_{1R}(p^2) =$$



$$= -\mathrm{i}\Sigma_1(p^2) - \mathrm{i}\delta m^2. \tag{2.23}$$

Equivalently,

$$\Sigma_{1R}(p^2) = \Sigma_1(p^2) + \delta m^2 = \Sigma_{1,\mathrm{fin}}(p^2) + \big(\Sigma_{1,\mathrm{div}} + \delta m^2\big). \tag{2.24}$$

Choosing

$$\delta m^2 = -\Sigma_{1,\mathrm{div}} \tag{2.25}$$

cancels the divergent part. In the on-shell scheme, the remaining finite part is fixed by the condition that the pole sits at

$$p^2 = m_{\mathrm{ph}}^2. \tag{2.26}$$

A useful way to compute the finite answer is to differentiate with respect to p^2. The counterterm and the subtracted reference integral do not depend on p, so the derivative acts only on the original loop integral:

$$\begin{aligned}\frac{\partial \Sigma_{1R}}{\partial p^2} &= \frac{p^\mu}{2p^2}\frac{\partial \Sigma_1}{\partial p^\mu} \\ &= -\frac{\mathrm{i}g^2}{32\pi^4 p^2}\int \mathrm{d}^4k\, \frac{p\cdot(p+k)}{(k^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon)((p+k)^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon)^2}.\end{aligned} \tag{2.27}$$

To evaluate it, use

$$\frac{1}{AB^n} = \int_0^1 \mathrm{d}x\, \frac{nx^{n-1}}{[xB + (1-x)A]^{n+1}}, \tag{2.28}$$

with

$$A = k^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon, \qquad B = (p+k)^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon, \qquad n = 2. \tag{2.29}$$

Then

$$\begin{aligned}\frac{\partial \Sigma_{1R}}{\partial p^2} &= -\frac{\mathrm{i}g^2}{32\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k\, \frac{2x\, p\cdot(p+k)}{\big[k^2 + 2x\, p\cdot k + xp^2 - m_{\mathrm{ph}}^2 + \mathrm{i}\epsilon\big]^3} \\ &= \frac{\mathrm{i}g^2}{16\pi^4 p^2}\int_0^1 \mathrm{d}x \int \mathrm{d}^4k\, \frac{x\, p\cdot(p+k)}{\big[m_{\mathrm{ph}}^2 - k^2 - 2x\, p\cdot k - xp^2 - \mathrm{i}\epsilon\big]^3}.\end{aligned} \tag{2.30}$$

Complete the square by shifting the loop momentum:

$$k' = k + xp, \qquad \Delta(x) = m_{\mathrm{ph}}^2 - p^2 x(1-x). \tag{2.31}$$

The numerator becomes

$$x\, p\cdot(p+k) = x\big[p^2(1-x) + p\cdot k'\big]. \tag{2.32}$$

The term proportional to $p\cdot k'$ is odd in the shifted integration variable and vanishes. Therefore

$$\frac{\partial \Sigma_{1R}}{\partial p^2} = \frac{\mathrm{i}g^2}{16\pi^4}\int_0^1 \mathrm{d}x\, x(1-x) \int \mathrm{d}^4k'\, \frac{1}{[\Delta(x) - k'^2 - \mathrm{i}\epsilon]^3}. \tag{2.33}$$

Using

$$\int \frac{\mathrm{d}^d l}{(2\pi)^d} \frac{1}{(l^2 - \Delta + \mathrm{i}\epsilon)^m} = \frac{\mathrm{i}(-1)^m}{(4\pi)^{d/2}} \frac{\Gamma(m - d/2)}{\Gamma(m)} \Delta^{d/2-m}, \tag{2.34}$$

with $d = 4$ and $m = 3$, gives

$$\begin{aligned} \frac{\partial \Sigma_{1R}}{\partial p^2} &= -\frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \frac{x(1-x)}{m_{\mathrm{ph}}^2 - p^2 x(1-x)} \\ &= \frac{g^2}{32\pi^2} \frac{\partial}{\partial p^2} \int_0^1 \mathrm{d}x \, \ln\left[m_{\mathrm{ph}}^2 - p^2 x(1-x)\right]. \end{aligned} \tag{2.35}$$

Integrating back gives

$$\Sigma_{1R}(p^2) = \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln\left[m_{\mathrm{ph}}^2 - p^2 x(1-x)\right] + C. \tag{2.36}$$

The on-shell condition $\Sigma_{1R}(m_{\mathrm{ph}}^2) = 0$ fixes the constant:

$$C = -\frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln\left[m_{\mathrm{ph}}^2 - m_{\mathrm{ph}}^2 x(1-x)\right]. \tag{2.37}$$

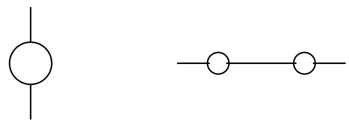
Thus

$$\Sigma_{1R}(p^2) = \frac{g^2}{32\pi^2} \int_0^1 \mathrm{d}x \, \ln\left[\frac{m_{\mathrm{ph}}^2 - p^2 x(1-x)}{m_{\mathrm{ph}}^2 (1 - x + x^2)}\right]. \tag{2.38}$$

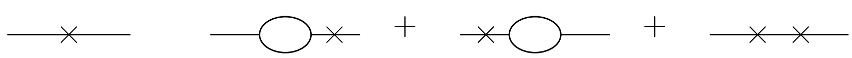
This expression is finite and vanishes at $p^2 = m_{\mathrm{ph}}^2$ by construction.

At higher orders, the same logic must be applied to divergent subgraphs inside larger graphs. If a one-loop bubble appears as a subgraph, the renormalized graph is obtained by adding the corresponding local counterterm graph:

1. The original larger graph may contain a bubble subgraph:



2. Each divergent bubble must be accompanied by a counterterm insertion. The cross denotes the local counterterm:



3. The sum of the original graph and all required counterterm graphs is finite. In this sense the divergent bubble subgraph is replaced by its finite renormalized value $-\mathrm{i}\Sigma_{1R}$.

The important point is locality: whenever a divergent short-distance subgraph appears, its divergent part has the same form as a local counterterm with the same external legs.

1.3 Degree of divergence and required counterterms

The superficial degree of divergence is a quick estimate of how a Feynman integral behaves at large loop momentum. For the one-loop self-energy in d spacetime dimensions,

$$\Sigma_1(p^2, m^2; d) = \frac{\mathrm{i}g^2}{2(2\pi)^d} \int \mathrm{d}^d k \, \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{2.39}$$

At large k, each propagator behaves as $1/k^2$, while the measure contributes $k^{d-1}\mathrm{d}k$. Therefore

$$\mathrm{d}^d k \, \frac{1}{k^2 k^2} \sim \mathrm{d}k \, k^{d-5}. \tag{2.40}$$

Equivalently, before the final radial integral, the graph has superficial degree

$$D = d - 4. \tag{2.41}$$

Thus

$$d < 4 \implies \text{the graph is superficially convergent,} \tag{2.42}$$
$$d = 4 \implies \text{the graph is logarithmically divergent,} \tag{2.43}$$
$$d > 4 \implies \text{the graph is power divergent by superficial counting.} \tag{2.44}$$

External-momentum derivatives reduce the degree of divergence. Differentiating once gives

$$\frac{\partial \Sigma_1}{\partial p^\mu} = -\frac{\mathrm{i}g^2}{(2\pi)^d} \int \mathrm{d}^d k \, \frac{(p+k)_\mu}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)^2}. \tag{2.45}$$

At large k,

$$\mathrm{d}^d k \, \frac{k}{k^2 (k^2)^2} \sim \mathrm{d}k \, k^{d-6}, \tag{2.46}$$

so the degree has been lowered from $d-4$ to $d-5$. If $d=5$, naive power counting gives a logarithm. The leading logarithmic piece is odd under symmetric integration of the loop momentum, so it does not produce a Lorentz-invariant divergence in the self-energy.

Differentiating a second time gives

$$\frac{\partial^2 \Sigma_1}{\partial p^\mu \partial p^\nu} = \frac{\mathrm{i}g^2}{(2\pi)^d} \int \mathrm{d}^d k \, \frac{2(p+k)_\mu (p+k)_\nu - g_{\mu\nu}((p+k)^2 - m^2)}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)^3}. \tag{2.47}$$

Its superficial degree is

$$D_{\partial^2 \Sigma} = d - 6. \tag{2.48}$$

In four dimensions, this is finite. When we integrate twice to reconstruct Σ_1, the most general missing polynomial is

$$A + B_\mu p^\mu. \tag{2.49}$$

Since the self-energy is a Lorentz scalar, the linear term is forbidden, so $B_\mu = 0$. The only local ambiguity in $d=4$ is the constant A, which is precisely a mass counterterm.

In six dimensions, the original graph has $D = 2$. After enough derivatives make the integral finite, integrating back allows a quadratic local polynomial:

$$A + Bp^2 + C_\mu p^\mu. \tag{2.50}$$

Lorentz invariance again sets $C_\mu = 0$. Therefore in $d = 6$ the two-point counterterm must contain both a mass counterterm and a wave-function counterterm. With the sign convention used below, their contribution to the self-energy is

$$\Sigma_{\mathrm{ct}}(p^2) = \delta m^2 - \delta Z \, p^2. \tag{2.51}$$

The corresponding local Lagrangian terms are

$$\mathcal{L}_{\mathrm{ct}} = -\frac{1}{2}\delta Z\, \partial_\mu\phi\, \partial^\mu\phi - \frac{1}{2}\delta m^2\phi^2. \tag{2.52}$$

The δZ term is called wave-function or field-strength renormalization.

The field-strength factor relates the bare field to the renormalized field:

$$\phi_0 = Z^{1/2}\phi, \qquad \phi = Z^{-1/2}\phi_0. \tag{2.53}$$

Therefore

$$\begin{aligned} G_{2(0)}(p) &= \langle 0|\, \mathrm{T}\{\phi_0(p)\phi_0(-p)\}\, |0\rangle \\ &= Z\, \langle 0|\, \mathrm{T}\{\phi(p)\phi(-p)\}\, |0\rangle = Z G_2(p). \end{aligned} \tag{2.54}$$

The full bare propagator is

$$G_{2(0)}(p^2) = \frac{\mathrm{i}}{p^2 - m_0^2 - \Sigma_{1(0)}(p^2) + \mathrm{i}\epsilon}. \tag{2.55}$$

The physical mass is the pole position:

$$m_{\mathrm{ph}}^2 - m_0^2 - \Sigma_{1(0)}(m_{\mathrm{ph}}^2) = 0, \qquad \Sigma_{1(0)}(m_{\mathrm{ph}}^2) = m_{\mathrm{ph}}^2 - m_0^2. \tag{2.56}$$

Hence

$$p^2 - m_0^2 - \Sigma_{1(0)}(p^2) = p^2 - m_{\mathrm{ph}}^2 - \big[\Sigma_{1(0)}(p^2) - \Sigma_{1(0)}(m_{\mathrm{ph}}^2)\big]. \tag{2.57}$$

When only mass renormalization is needed, this motivates

$$\Sigma_{1R}(p^2) = \Sigma_{1(0)}(p^2) - \Sigma_{1(0)}(m_{\mathrm{ph}}^2). \tag{2.58}$$

With a wave-function counterterm included, the general form is

$$\Sigma_{1R}(p^2) = \Sigma_{1(0)}(p^2) + \delta m^2 - \delta Z\, p^2. \tag{2.59}$$

Then the bare and renormalized propagators are related by

$$G_{2(0)}(p^2) = \frac{\mathrm{i}Z}{p^2 - m_{\mathrm{ph}}^2 - \Sigma_{1R}(p^2) + \mathrm{i}\epsilon}. \tag{2.60}$$

The on-shell scheme imposes two conditions. The first fixes the pole:

$$\Sigma_{1R}(m_{\mathrm{ph}}^2) = 0. \tag{2.61}$$

The second fixes the pole residue. Expanding near $p^2 = m_{\mathrm{ph}}^2$,

$$\begin{aligned} \Sigma_{1R}(p^2) &= \Sigma_{1R}(m_{\mathrm{ph}}^2) + (p^2 - m_{\mathrm{ph}}^2)\Sigma'_{1R}(m_{\mathrm{ph}}^2) + O\big((p^2 - m_{\mathrm{ph}}^2)^2\big) \\ &= (p^2 - m_{\mathrm{ph}}^2)\Sigma'_{1R}(m_{\mathrm{ph}}^2) + O\big((p^2 - m_{\mathrm{ph}}^2)^2\big). \end{aligned} \tag{2.62}$$

Thus

$$G_2(p^2) \simeq \frac{\mathrm{i}}{(p^2 - m_{\mathrm{ph}}^2)\big[1 - \Sigma'_{1R}(m_{\mathrm{ph}}^2)\big]}. \tag{2.63}$$

Requiring unit residue gives

$$\Sigma'_{1R}(m_{\rm ph}^2) = 0. \tag{2.64}$$

For the bare propagator,

$$\begin{aligned} G_{2(0)}(p^2) &\simeq \frac{\mathrm{i}}{(p^2 - m_{\rm ph}^2)\Big[1 - \Sigma'_{1(0)}(m_{\rm ph}^2)\Big]} \\ &\equiv \frac{\mathrm{i}R_{(0)}}{p^2 - m_{\rm ph}^2}, \end{aligned} \tag{2.65}$$

so

$$R_{(0)} = \frac{1}{1 - \Sigma'_{1(0)}(m_{\rm ph}^2)}. \tag{2.66}$$

Since $G_{2(0)} = ZG_2$, choosing the renormalized propagator to have unit residue means

$$Z = R_{(0)} = \frac{1}{1 - \Sigma'_{1(0)}(m_{\rm ph}^2)}. \tag{2.67}$$

Finally, rewrite the bare ϕ^3 Lagrangian:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi_0\,\partial^\mu\phi_0 - \frac{1}{2}m_0^2\phi_0^2 - \frac{g_0}{3!}\phi_0^3. \tag{2.68}$$

Using $\phi_0 = Z^{1/2}\phi$,

$$\mathcal{L} = \frac{1}{2}Z\,\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}Zm_0^2\phi^2 - \frac{g_0}{3!}Z^{3/2}\phi^3. \tag{2.69}$$

Define

$$Z = 1 - \delta Z, \qquad Zm_0^2 = m_{\rm ph}^2 + \delta m^2, \qquad Z^{3/2}g_0 = g_{\rm ph} + \delta g. \tag{2.70}$$

Then

$$\begin{aligned} \mathcal{L} = {} & \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m_{\rm ph}^2\phi^2 - \frac{g_{\rm ph}}{3!}\phi^3 \\ & - \frac{1}{2}\delta Z\,\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}\delta m^2\phi^2 - \frac{\delta g}{3!}\phi^3. \end{aligned} \tag{2.71}$$

The first line supplies the renormalized propagator and vertex. The second line supplies the counterterm vertices needed to cancel divergences order by order.

1.4 Arbitrariness in a renormalization graph

Counterterms are required by finiteness, but finiteness does not uniquely determine their finite parts. This freedom is the origin of different renormalization schemes.

In four-dimensional ϕ^3 theory at one loop, the two-point graph is logarithmically divergent, so a mass counterterm is enough and we may set

$$\delta Z = 0 \tag{2.72}$$

at this order. The renormalized self-energy can then be written as

$$\Sigma_{1R}(p^2) = \Sigma_{1,\rm fin}(p^2) + \big(\Sigma_{1,\rm div} + \delta m^2\big). \tag{2.73}$$

The divergent part of δm^2 must cancel $\Sigma_{1,\rm div}$, but the finite part is still a convention.

In the mass-shell scheme, the convention is fixed by requiring the pole of the full propagator to be at $p^2 = m_{\text{ph}}^2$:

$$\Sigma_{1R}(m_{\text{ph}}^2) = 0. \tag{2.74}$$

Thus

$$0 = \Sigma_{1,\text{fin}}(m_{\text{ph}}^2) + \Sigma_{1,\text{div}} + \delta m^2, \tag{2.75}$$

and hence

$$\delta m_{\text{OS}}^2 = -\Sigma_{1,\text{div}} - \Sigma_{1,\text{fin}}(m_{\text{ph}}^2). \tag{2.76}$$

The on-shell renormalized self-energy is

$$\Sigma_{1R}^{\text{OS}}(p^2) = \Sigma_{1,\text{fin}}(p^2) - \Sigma_{1,\text{fin}}(m_{\text{ph}}^2). \tag{2.77}$$

Therefore, through order g^2,

$$G_2(p^2) = \frac{\mathrm{i}}{p^2 - m_{\text{ph}}^2 - \Sigma_{1,\text{fin}}(p^2) + \Sigma_{1,\text{fin}}(m_{\text{ph}}^2) + O(g^4)}. \tag{2.78}$$

The bare mass is

$$m_0^2 = m_{\text{ph}}^2 + \delta m_{\text{OS}}^2 = m_{\text{ph}}^2 - \Sigma_{1,\text{div}} - \Sigma_{1,\text{fin}}(m_{\text{ph}}^2) + O(g^4). \tag{2.79}$$

Now choose a different finite part. The only requirement for finiteness is

$$\delta m^2 = -\Sigma_{1,\text{div}} + \text{finite}. \tag{2.80}$$

For example, choose the finite part to be $-g^2 C$:

$$\delta m_C^2 = -\Sigma_{1,\text{div}} - g^2 C, \tag{2.81}$$

where C is an arbitrary finite constant. The renormalized self-energy is then

$$\Sigma_{1R}^{(C)}(p^2, m^2) = \Sigma_{1,\text{fin}}(p^2, m^2) - g^2 C. \tag{2.82}$$

Here m is the renormalized mass parameter in this scheme. It is not automatically the physical pole mass. The physical pole must be found from

$$p^2 - m^2 - \Sigma_{1R}^{(C)}(p^2, m^2) = 0 \tag{2.83}$$

order by order in perturbation theory. The propagator is

$$G_2^{(C)}(p^2) = \frac{\mathrm{i}}{p^2 - m^2 - \Sigma_{1,\text{fin}}(p^2, m^2) + g^2 C + O(g^4)}. \tag{2.84}$$

The bare mass in this scheme is

$$m_0^2 = m^2 + \delta m_C^2 = m^2 - \Sigma_{1,\text{div}} - g^2 C + O(g^4). \tag{2.85}$$

The bare mass is the same object in every scheme. Equating the on-shell expression and the C-scheme expression gives

$$m^2 - \Sigma_{1,\text{div}} - g^2 C = m_{\text{ph}}^2 - \Sigma_{1,\text{div}} - \Sigma_{1,\text{fin}}(m_{\text{ph}}^2) + O(g^4). \tag{2.86}$$

The divergent pieces cancel, leaving

$$m^2 = m_{\text{ph}}^2 + g^2 C - \Sigma_{1,\text{fin}}(m_{\text{ph}}^2) + O(g^4). \tag{2.87}$$

This is a finite redefinition between two renormalization schemes. The parameter m is scheme dependent, while m_{ph} is defined by the pole of the propagator and is observable. Changing the finite part of the counterterm changes the numerical value assigned to the renormalized parameter, but it does not change physical predictions when all quantities are converted consistently.

1.5 Dimensional regularization

Dimensional regularization is another systematic way to handle ultraviolet divergences. Instead of putting a hard cutoff Λ on the loop momentum, we temporarily continue the number of spacetime dimensions from 4 to a complex variable d. The integral is first evaluated in a region of d where it converges, and the result is then analytically continued back to the physical value. In this language, UV divergences appear as poles of Gamma functions.

For the same one-loop two-point function in ϕ^3 theory, we write

$$\Sigma_1(p^2, m^2; d) = \frac{\mathrm{i}g^2\mu^{4-d}}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{(k^2 - m^2 + \mathrm{i}\epsilon)((p+k)^2 - m^2 + \mathrm{i}\epsilon)}. \tag{2.88}$$

The factor μ^{4-d} is inserted so that the coupling g keeps its four-dimensional mass dimension while d is moved away from 4.

The basic Gaussian integral is

$$\int \mathrm{d}^n x\, \mathrm{e}^{-x^2} = (\sqrt{\pi})^n = \pi^{n/2}. \tag{2.89}$$

After Wick rotation, $k^0 = \mathrm{i}k_E^0$, the Minkowski square becomes $k^2 = -k_E^2$, and the measure contributes one factor of i:

$$\begin{aligned}\int \mathrm{d}^d k\, \mathrm{e}^{k^2} &= \mathrm{i} \int \mathrm{d}^d k_E\, \mathrm{e}^{-k_E^2} \\ &= \mathrm{i}\left(\int \mathrm{d}k_E^0\, \mathrm{e}^{-(k_E^0)^2}\right)\left(\int \mathrm{d}^{d-1}\vec{k}_E\, \mathrm{e}^{-\vec{k}_E^2}\right) \\ &= \mathrm{i}\sqrt{\pi}\,\pi^{(d-1)/2} = \mathrm{i}\pi^{d/2}. \end{aligned} \tag{2.90}$$

One way to derive the dimensionally regularized answer is to use Schwinger parameters. The propagator can be represented as

$$\frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} = \int_0^\infty \mathrm{d}a\, \mathrm{e}^{-\mathrm{i}a(m^2 - k^2 - \mathrm{i}\epsilon)}. \tag{2.91}$$

Using one parameter for each propagator gives

$$\begin{aligned}\Sigma_1 &= \frac{\mathrm{i}g^2\mu^{4-d}}{2(2\pi)^d} \int \mathrm{d}^d k \int_0^\infty \mathrm{d}a \int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}a(m^2-k^2-\mathrm{i}\epsilon)} \mathrm{e}^{-\mathrm{i}b(m^2-(p+k)^2-\mathrm{i}\epsilon)} \\ &= \frac{\mathrm{i}g^2\mu^{4-d}}{2(2\pi)^d} \int \mathrm{d}^d k \int_0^\infty \mathrm{d}a \int_0^\infty \mathrm{d}b\, \mathrm{e}^{-\mathrm{i}[(a+b)m^2 - (a+b)k^2 - bp^2 - 2b\,p\cdot k - \mathrm{i}\epsilon(a+b)]}. \end{aligned} \tag{2.92}$$

Complete the square by defining

$$k' = k + \frac{b}{a+b}p. \tag{2.93}$$

Then

$$(a+b)k^2 + 2b\,p\cdot k + bp^2 = (a+b)k'^2 + \frac{ab}{a+b}p^2. \tag{2.94}$$

It is convenient to change variables from (a, b) to

$$z = a+b, \qquad x = \frac{a}{a+b}, \qquad a = xz, \qquad b = (1-x)z. \tag{2.95}$$

The Jacobian is

$$da\,db = \begin{vmatrix} \frac{\partial a}{\partial x} & \frac{\partial a}{\partial z} \\ \frac{\partial b}{\partial x} & \frac{\partial b}{\partial z} \end{vmatrix} dx\,dz = \begin{vmatrix} z & x \\ -z & 1-x \end{vmatrix} dx\,dz = z\,dx\,dz. \tag{2.96}$$

Thus $z \in (0,\infty)$ and $x \in (0,1)$, and the exponent contains the standard combination

$$\Delta(x) = m^2 - p^2 x(1-x). \tag{2.97}$$

After the Gaussian loop integral and the z integral, one obtains

$$\Sigma_1(p^2, m^2; d) = -\frac{g^2\mu^{4-d}}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 dx\,\left[m^2 - p^2x(1-x)\right]^{d/2-2}. \tag{2.98}$$

The same result follows more directly from the Feynman-parameter identity

$$\frac{1}{AB} = \int_0^1 dx\,\frac{1}{[xB+(1-x)A]^2}. \tag{2.99}$$

With

$$A = k^2 - m^2 + i\epsilon, \qquad B = (p+k)^2 - m^2 + i\epsilon, \tag{2.100}$$

we get

$$\begin{aligned}\Sigma_1 &= \frac{ig^2\mu^{4-d}}{2}\int_0^1 dx \int \frac{d^dk}{(2\pi)^d}\frac{1}{[(k+xp)^2 - \Delta(x) + i\epsilon]^2} \\ &= -\frac{g^2\mu^{4-d}}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 dx\,\Delta(x)^{d/2-2}.\end{aligned} \tag{2.101}$$

The divergence is now entirely in $\Gamma(2-d/2)$. Near $d=4$, set

$$d = 4-\epsilon. \tag{2.102}$$

Then

$$\Gamma\left(2-\frac{d}{2}\right) = \Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \tag{2.103}$$

and

$$\Delta(x)^{d/2-2} = \Delta(x)^{-\epsilon/2} = 1 - \frac{\epsilon}{2}\ln\Delta(x) + O(\epsilon^2). \tag{2.104}$$

Using also

$$\mu^\epsilon(4\pi)^{\epsilon/2} = 1 + \epsilon\ln\mu + \frac{\epsilon}{2}\ln 4\pi + O(\epsilon^2), \tag{2.105}$$

the unrenormalized self-energy becomes

$$\Sigma_1 = -\frac{g^2}{32\pi^2}\int_0^1 dx\left[\frac{2}{\epsilon} - \gamma_E + \ln 4\pi + \ln\mu^2 - \ln\Delta(x)\right] + O(\epsilon). \tag{2.106}$$

1.6 Minimal subtraction scheme

As we introduced above, we may have different schemes to choose for renormalization. The on-shell subtraction is especially transparent in dimensional regularization. Define

$$\Sigma_{1R}^{\rm OS}(p^2) = \Sigma_1(p^2, m_{\rm ph}^2; d) - \Sigma_1(m_{\rm ph}^2, m_{\rm ph}^2; d). \tag{2.107}$$

The two terms have the same pole, so the pole cancels before taking $\epsilon \to 0$. Since

$$\Delta_p(x) = m_{\rm ph}^2 - p^2 x(1-x), \qquad \Delta_{\rm OS}(x) = m_{\rm ph}^2 - m_{\rm ph}^2 x(1-x) = m_{\rm ph}^2(1-x+x^2), \tag{2.108}$$

we obtain

$$\Sigma_{1R}^{\rm OS}(p^2) = \frac{g^2}{32\pi^2}\int_0^1 {\rm d}x\, \ln \frac{m_{\rm ph}^2 - p^2 x(1-x)}{m_{\rm ph}^2(1-x+x^2)}. \tag{2.109}$$

This is exactly the finite on-shell result obtained earlier by differentiating with respect to p^2.

But on-shell scheme would have problems when we are dealing with the massless case. So we introduce the **Minimal Subtraction (MS)**, which is a different prescription. It subtracts only the pole part at the physical dimension. In $d = 4 - \epsilon$, the mass counterterm in the MS scheme is chosen as

$$\delta m_{\rm MS}^2 = \frac{g^2}{32\pi^2}\frac{2}{\epsilon}. \tag{2.110}$$

Therefore

$$\begin{aligned}\Sigma_{1R}^{\rm MS}(p^2) &= \Sigma_1(p^2) + \delta m_{\rm MS}^2 \\ &= \frac{g^2}{32\pi^2}\int_0^1 {\rm d}x \left[\ln\left(\frac{\Delta(x)}{4\pi\mu^2}\right) + \gamma_E\right] + O(\epsilon).\end{aligned} \tag{2.111}$$

This expression is finite, but it is not the on-shell scheme result. In particular, $\Sigma_{1R}^{\rm MS}(m^2)$ is not required to vanish. The MS scheme removes only the pole; the finite part is chosen by convention.

The scale μ appears because the mass dimension of the coupling changes when the dimension is continued. In d dimensions,

$$[\mathcal{L}] = [m]^d, \qquad [\phi] = [m]^{(d-2)/2}, \tag{2.112}$$

and the interaction

$$\frac{g}{3!}\phi^3 \tag{2.113}$$

implies

$$[g] = [m]^{(6-d)/2}. \tag{2.114}$$

Around four dimensions, g has dimension one, and writing $g_0 = \mu^{(4-d)/2} g$ keeps the renormalized coupling normalized as a four-dimensional parameter. Around six dimensions, where ϕ^3 is classically marginal, the corresponding continuation is

$$g_0 = \mu^{3-d/2}(g + \delta g). \tag{2.115}$$

Now consider the same two-point graph near six dimensions. Set

$$d = 6 - \epsilon, \qquad \Delta(x) = m^2 - p^2 x(1-x). \tag{2.116}$$

The dimensionally regularized result is

$$\Sigma_1 = -\frac{g^2\mu^{6-d}}{2(4\pi)^{d/2}}\Gamma\left(2 - \frac{d}{2}\right)\int_0^1 {\rm d}x\, \Delta(x)^{d/2-2}. \tag{2.117}$$

Here

$$\Gamma\left(2-\frac{d}{2}\right) = \Gamma\left(-1+\frac{\epsilon}{2}\right) = -\frac{2}{\epsilon} + \gamma_E - 1 + O(\epsilon). \tag{2.118}$$

Expanding the remaining factors gives

$$\Sigma_1 = -\frac{g^2}{128\pi^3}\int_0^1 \mathrm{d}x\,\Delta(x)\left[-\frac{2}{\epsilon}+\gamma_E-1+\ln\left(\frac{\Delta(x)}{4\pi\mu^2}\right)\right]+O(\epsilon). \tag{2.119}$$

Since

$$\int_0^1 \mathrm{d}x\,\Delta(x) = m^2-\frac{p^2}{6}, \tag{2.120}$$

the pole part is

$$\Sigma_{1,\text{pole}} = \frac{g^2}{64\pi^3}\frac{1}{\epsilon}\left(m^2-\frac{p^2}{6}\right). \tag{2.121}$$

Thus a mass counterterm alone is no longer sufficient: the divergence contains both a constant and a p^2 term. Writing

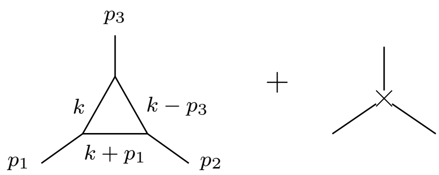
$$\Sigma_{1R} = \Sigma_1+\delta m^2-\delta Z\,p^2, \tag{2.122}$$

minimal subtraction at $d=6-\epsilon$ requires

$$\delta m^2 = -\frac{g^2m^2}{64\pi^3}\frac{1}{\epsilon} = \frac{g^2m^2}{64\pi^3}\frac{1}{d-6}, \tag{2.123}$$

$$\delta Z = -\frac{g^2}{6\cdot 64\pi^3}\frac{1}{\epsilon} = \frac{g^2}{6\cdot 64\pi^3}\frac{1}{d-6}. \tag{2.124}$$

In six dimensions the three-point vertex is also logarithmically divergent. The tree vertex is corrected by the one-loop triangle, and the divergence is canceled by a local three-leg counterterm:



To extract the UV pole, set the external momenta to zero after noting that the superficial divergence is logarithmic. Momentum-dependent terms are less divergent and affect only finite parts. The triangle contribution is then

$$\begin{aligned}\Gamma_1^{(3)} &= (-\mathrm{i}g\mu^{3-d/2})^3\int\frac{\mathrm{d}^dk}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{(k+p_1)^2-m^2+\mathrm{i}\epsilon}\frac{\mathrm{i}}{(k-p_2)^2-m^2+\mathrm{i}\epsilon}\\ &\xrightarrow{p_i\to 0}(-\mathrm{i}g\mu^{3-d/2})^3\int\frac{\mathrm{d}^dk}{(2\pi)^d}\frac{\mathrm{i}^3}{(k^2-m^2+\mathrm{i}\epsilon)^3}\\ &= -\frac{\mathrm{i}g^3}{2(4\pi)^3}\left(\frac{2}{\epsilon}+\text{finite}\right).\end{aligned} \tag{2.125}$$

The counterterm vertex is $-\mathrm{i}\delta g$. Requiring the pole to cancel gives

$$-\mathrm{i}\delta g+\left[-\frac{\mathrm{i}g^3}{2(4\pi)^3}\frac{2}{\epsilon}\right]=0, \tag{2.126}$$

so

$$\delta g = -\frac{g^3}{64\pi^3}\frac{1}{\epsilon}\,\mu^{3-d/2} = \frac{g^3}{64\pi^3}\frac{1}{d-6}\,\mu^{3-d/2}. \tag{2.127}$$

This is the vertex counterterm in minimal subtraction near six dimensions.

1.7 Two loop graph in massless ϕ^3 theory

The on-shell scheme is natural for a massive particle, because the physical mass is defined by an isolated pole at $p^2 = m_{\rm ph}^2$. For a strictly massless interacting theory this prescription becomes delicate. The reason is not a new ultraviolet problem; it is an infrared problem caused by taking the subtraction point to the massless pole.

Recall the one-loop on-shell mass counterterm in dimensional regularization $d = 4 - \epsilon$:

$$\delta m_{\rm OS}^2 = -\Sigma_1(m_{\rm ph}^2, m_{\rm ph}^2) = \frac{g^2\mu^\epsilon}{2(4\pi)^{d/2}}\Gamma\left(2 - \frac{d}{2}\right)\int_0^1 {\rm d}x\,\left[m_{\rm ph}^2(1 - x + x^2)\right]^{d/2-2}, \tag{2.128}$$

with the approximation formulas:

$$\Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \qquad A^{-\epsilon/2} = 1 - \frac{\epsilon}{2}\ln A + O(\epsilon^2). \tag{2.129}$$

These gives

$$\delta m_{\rm OS}^2 = \frac{g^2}{32\pi^2}\left[\frac{2}{\epsilon} - \gamma_E + \ln 4\pi + \ln\mu^2 - \ln m_{\rm ph}^2 - \int_0^1 {\rm d}x\,\ln\big(1 - x + x^2\big)\right] + O(\epsilon). \tag{2.130}$$

The pole $2/\epsilon$ is the ultraviolet divergence. The separate term

$$-\ln m_{\rm ph}^2 \tag{2.131}$$

blows up when $m_{\rm ph} \to 0$. This is an infrared divergence. It appears because the on-shell subtraction point has been moved to $p^2 = 0$, where a massless field has long-range propagation.

Minimal subtraction does not require an on-shell subtraction point. Setting $m = 0$ after MS subtraction gives

$$\begin{aligned}\Sigma_{1R}^{\rm MS}(p^2)\big|_{m=0} &= \frac{g^2}{32\pi^2}\int_0^1 {\rm d}x\left[\ln\left(\frac{-p^2x(1-x)}{4\pi\mu^2}\right) + \gamma_E\right] \\ &= \frac{g^2}{32\pi^2}\left[\ln\left(\frac{-p^2}{4\pi\mu^2}\right) + \gamma_E - 2\right].\end{aligned} \tag{2.132}$$

This expression is meaningful for $p^2 \neq 0$, but it is still singular as $p^2 \to 0$. Therefore the failure of the mass-shell scheme in the strictly massless case is an infrared statement: near the massless point, the full propagator is not described by the same simple isolated massive-particle pole.

For perturbation theory it is useful to separate the Lagrangian into the free part, the basic interaction, and the counterterms:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_b + \mathcal{L}_{ct} \tag{2.133}$$

with

$$\mathcal{L}_0 = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}m^2\phi^2, \tag{2.134}$$

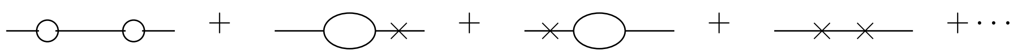
$$\mathcal{L}_b = -\frac{g}{3!}\phi^3, \tag{2.135}$$

and

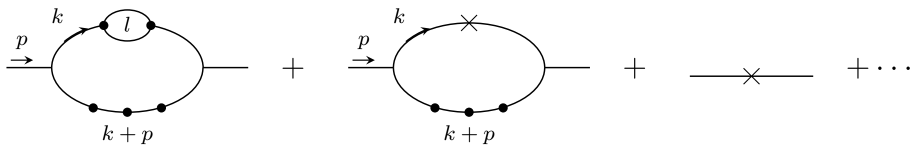
$$\mathcal{L}_{ct} = -\frac{1}{2}\delta Z\,\partial_\mu\phi\,\partial^\mu\phi - \frac{1}{2}\delta m^2\phi^2 - \frac{\delta g}{3!}\phi^3 - \delta h\,\phi. \tag{2.136}$$

The term δh cancels tadpoles, namely one-point divergences. At order g^4 we must draw not only the genuine two-loop graphs. We must also draw graphs in which a lower-order divergent subgraph has already been replaced by its local counterterm. The relevant classes are:

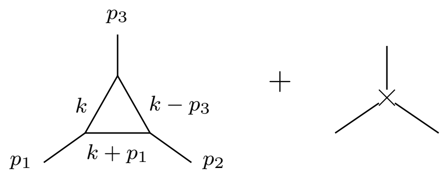
1. Iterated one-loop self-energy insertions on the same propagator line:



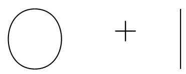
2. A two-loop self-energy graph with a one-loop two-point subdivergence:



3. A one-loop three-point vertex correction and its vertex counterterm:

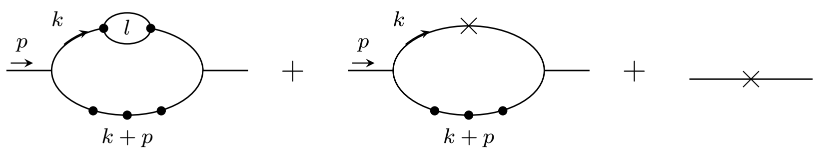


Each cross stands for a local counterterm. The rule is simple but important: every divergent subgraph needs its own subtraction before the remaining overall divergence can be removed. Tadpoles are not shown in the main list, because they are local one-point divergences and are canceled by the $\delta h\, \phi$ term:



A pole multiplying a non-polynomial function such as $\ln(-p^2)$ is called a non-local subdivergence. It cannot be canceled by a local term in the Lagrangian. After all subdivergences have been subtracted, the remaining overall divergence is local: in momentum space it is a polynomial in the external momenta.

Now we consider the two-loop self-energy in $d = 6$. The diagram that matters for subdivergences is the graph in which a one-loop two-point subgraph is inserted into an internal line. We must include three contributions:



(a) (b) (c)

Graph (a) is the genuine two-loop graph. Graph (b) replaces the divergent one-loop subgraph inside (a) by the lower-order counterterm. Graph (c) is the remaining overall local two-point counterterm. For a two-point function the cross abbreviates the local insertion $\delta m^2 - \delta Z\, p^2$; in the massless calculation below the relevant overall counterterm is the wave-function part. The sum of all three graphs is finite.

$$
\begin{aligned}
\Sigma_{2a} = (\mathrm{i} g \mu^{3-d/2})^4 \frac{1}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{d}^d l}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{l^2 - m^2 + \mathrm{i}\epsilon} \\
\times \frac{\mathrm{i}}{(k-l)^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(k+p)^2 - m^2 + \mathrm{i}\epsilon}.
\end{aligned}
\tag{2.137}
$$

The two factors of $1/(k^2 - m^2)$ are the two propagator segments on the line that contains the one-loop subgraph. For convenience, we now take the massless case:

$$\Sigma_{2a} = \frac{1}{2} g^4 \mu^{12-2d} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{d}^d l}{(2\pi)^d} \frac{1}{(k^2 + \mathrm{i}\epsilon)^2 (l^2 + \mathrm{i}\epsilon)((k-l)^2 + \mathrm{i}\epsilon)((k+p)^2 + \mathrm{i}\epsilon)}. \tag{2.138}$$

After Wick rotation, we have

$$\begin{aligned}
\Sigma_{2a} &= -\frac{1}{2} g^4 \mu^{12-2d} \int \frac{\mathrm{d}^d k_E}{(2\pi)^d} \frac{\mathrm{d}^d l_E}{(2\pi)^d} \frac{1}{[l_E^2 (k_E - l_E)^2][(k_E^2)^2 (k_E + p_E)^2]} \\
&= -\frac{1}{2} g^4 \mu^{12-2d} \int \frac{\mathrm{d}^d k_E}{(2\pi)^d} \frac{1}{(k_E^2)^2 (k_E + p_E)^2} \left(\int \frac{\mathrm{d}^d l_E}{(2\pi)^d} \frac{1}{l_E^2 (k_E - l_E)^2} \right) \\
&= -\frac{1}{2} g^4 \mu^{12-2d} \int \frac{\mathrm{d}^d k_E}{(2\pi)^d} \frac{1}{(k_E^2)^2 (k_E + p_E)^2} \mathrm{i}\pi^{d/2} \frac{\Gamma(2 - d/2)}{(2\pi)^d} \int_0^1 \mathrm{d}x \, [-k_E^2 x(1-x)]^{d/2-2} \\
&= -\frac{1}{2} g^4 \mu^{12-2d} \int \frac{\mathrm{d}^d k_E}{(2\pi)^d} \frac{1}{(k_E^2)^2 (k_E + p_E)^2} \mathrm{i}\pi^{d/2} \Gamma(2 - d/2) \frac{\Gamma(\frac{d}{2} - 1)^2}{\Gamma(d-2)} (-k_E^2)^{d/2-2} \\
&= \frac{\mathrm{i} g^4}{2^{13} \pi^9} (16\pi^3 \mu^4)^{3-d/2} \frac{\Gamma(2 - d/2)\Gamma(\frac{d}{2} - 1)^2}{\Gamma(d-2)} \int \mathrm{d}^d k_E \frac{1}{(-k_E^2)^{4-d/2} (k_E + p_E)^2}.
\end{aligned} \tag{2.139}$$

For the integral, one can introduce a Feynman parameter

$$\frac{1}{A^{4-d/2} B} = \frac{\Gamma(5 - d/2)}{\Gamma(4 - d/2)} \int_0^1 \mathrm{d}x \, \frac{x^{3-d/2}}{[Ax + B(1-x)]^{5-d/2}}. \tag{2.140}$$

Thus, the result is

$$\Sigma_{2a} = \left(\frac{g^2}{64\pi^3} \right)^2 \frac{p^2}{2} \left(\frac{-p^2}{4\pi\mu^2} \right)^{d-6} \frac{\Gamma(2 - d/2)\Gamma(5-d)\Gamma(\frac{d}{2} - 1)^3 \Gamma(d-4)}{\Gamma(d-2)\Gamma(4 - d/2)\Gamma(3d/2 - 5)} \tag{2.141}$$

$$\equiv \left(\frac{g^2}{64\pi^3} \right)^2 \frac{p^2}{2} \left(\frac{-p^2}{4\pi\mu^2} \right)^{d-6} \Gamma(2 - d/2)\Gamma(5-d) K(d) \tag{2.142}$$

Here $\Gamma(2 - d/2)$ is the pole of the one-loop subgraph, while $\Gamma(5-d)$ is the remaining overall pole. The massless formula also has an infrared singularity as $p^2 \to 0$; this is separate from the ultraviolet subdivergence.

Now expand near six dimensions. To keep track of the non-local momentum dependence, define

$$d = 6 - \epsilon, \qquad L_p = \ln \left(\frac{-p^2}{4\pi\mu^2} \right). \tag{2.143}$$

The singular factors are

$$\Gamma\left(2 - \frac{d}{2}\right) = \Gamma\left(-1 + \frac{\epsilon}{2}\right) = -\frac{2}{\epsilon} + O(1), \tag{2.144}$$

$$\Gamma(5-d) = \Gamma(-1+\epsilon) = -\frac{1}{\epsilon} + O(1), \tag{2.145}$$

$$\left(\frac{-p^2}{4\pi\mu^2} \right)^{d-6} = 1 - \epsilon L_p + \frac{\epsilon^2}{2} L_p^2 + O(\epsilon^3). \tag{2.146}$$

Therefore graph (a) has the pole structure

$$\Sigma_{2a} = \left(\frac{g^2}{64\pi^3} \right)^2 \frac{p^2}{36} \left[\frac{1}{(d-6)^2} + \frac{L_p + c_a}{d-6} + \frac{1}{2} L_p^2 + c_a' L_p + c_a'' + O(d-6) \right]. \tag{2.147}$$

The constants c_a, c'_a, c''_a are not important for the present point. The important term is $L_p/(d-6)$: it is a pole multiplying a non-polynomial function of the external momentum. A local counterterm cannot cancel such a term directly. It must be canceled by graph (b), which subtracts the divergent one-loop subgraph before the remaining overall divergence is removed.

For $m = 0$, the one-loop two-point counterterm in the subgraph is the wave-function counterterm

$$\delta_1 Z = \frac{g^2}{6\cdot 64\pi^3}\frac{1}{d-6} + O(g^4). \tag{2.148}$$

Because the counterterm in graph (b) sits on the line carrying momentum k, its Feynman rule contributes a factor proportional to $\delta_1 Z\, k^2$:

$$\begin{aligned}
\Sigma_{2b} &= (-\mathrm{i}g\mu^{3-d/2})^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{(k+p)^2-m^2}\delta_1 Z\cdot k^2 && (2.149)\\
&= \mathrm{i}g^2\mu^{6-d}\delta_1 Z\int\frac{\mathrm{d}^d k_E}{(2\pi)^d}\frac{k_E^2}{(k_E^2)^2(k_E+p_E)^2} && (2.150)\\
&= \mathrm{i}g^2\mu^{6-d}\delta_1 Z\frac{\mathrm{i}\pi^{d/2}}{(2\pi)^d}\Gamma(2-d/2)\frac{\Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}(p^2)^{d/2-2} && (2.151)\\
&= \left(\frac{g^2}{64\pi^3}\right)\left(\frac{p^2}{6}\right)\frac{\Gamma(2-d/2)\Gamma^2(\frac{d}{2}-1)}{(d-6)\Gamma(d-2)}\left(\frac{-p^2}{4\pi\mu^2}\right)^{d/2-3} && (2.152)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\left(\frac{p^2}{36}\right)\left(-\frac{2}{\epsilon}+\gamma_E-1\right)\frac{1}{\epsilon}\left(1-\frac{\epsilon}{2}L_p+\frac{\epsilon^2}{8}L_p^2\right) && (2.153)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\left(\frac{p^2}{36}\right)\left[-\frac{2}{\epsilon^2}+\frac{\gamma_E-1+L_p}{\epsilon}-\frac{1}{2}(\gamma_E-1)L_p-\frac{1}{4}L_p^2\right] && (2.154)\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[-\frac{2}{(d-6)^2}+\frac{-L_p+\text{const.}}{d-6}-\frac{1}{4}L_p^2+\text{const.}\,L_p+O(d-6)\right]. && (2.155)
\end{aligned}$$

Adding graphs (a) and (b), the pole multiplying L_p cancels:

$$\begin{aligned}
&\Sigma_{2a}+\Sigma_{2b}\\
&= \left(\frac{g^2}{64\pi^3}\right)^2\frac{p^2}{36}\left[\frac{-1}{(d-6)^2}+\frac{1}{d-6}\text{Const.}+\frac{1}{4}\ln^2\frac{-p^2}{4\pi\mu^2}+\text{Const.}\ln\frac{-p^2}{4\pi\mu^2}+\text{const.}+O(d-6)\right]
\end{aligned} \tag{2.156}$$

The non-local pole has disappeared:

$$\frac{1}{d-6}\left(\ln\frac{-p^2}{4\pi\mu^2}+\text{Const.}\right)\to\frac{1}{d-6}\text{const.} \tag{2.157}$$

If this term survived, canceling it would require a non-local counterterm schematically of the form

$$\phi(x)\ln(\Box)\phi(x) \tag{2.158}$$

for the subdivergence $\ln(p^2)/(d-6)$, which is unacceptable in a local renormalizable Lagrangian. Since the non-local pole has been canceled by graph (b), the remaining divergent part is local and can be canceled by graph (c). The two-loop wave-function counterterm is fixed by

$$-p^2\delta_2 Z = -(\Sigma_{2a}+\Sigma_{2b})_{\text{local pole}}. \tag{2.159}$$

Equivalently, in MS notation,

$$\delta_2 Z = \left(\frac{g^2}{64\pi^3}\right)^2 \frac{1}{36}\left[-\frac{1}{(d-6)^2} + \frac{c_Z}{d-6}\right], \tag{2.160}$$

where c_Z depends on the precise subtraction convention.

Thus, the renormalized result is

$$\begin{aligned} \Sigma_{2R}^{\mathrm{MS}} &= \Sigma_{2a} + \Sigma_{2b} - p^2\delta_2 Z \\ &= \left(\frac{g^2}{64\pi^3}\right)^2 \frac{p^2}{36}\left[\frac{1}{4}L_p^2 + c_1 L_p + c_0\right]. \end{aligned} \tag{2.161}$$

1.8 External-momentum derivatives and locality of counterterms

We now explain why the remaining overall counterterm of a 1PI graph must be a polynomial in the external momenta. This is the basic locality statement behind renormalization: after all subdivergences have been subtracted, the only ultraviolet divergence left in a graph is local.

For the two-loop graph above in $d = 6$, the superficial degree of the overall divergence is $D = 2$. Each derivative with respect to the external momentum lowers the degree by one. Therefore three external-momentum derivatives should remove the overall divergence. To make the bookkeeping clean, use the shorthand

$$\partial^p_\mu \equiv \frac{\partial}{\partial p^\mu}. \tag{2.162}$$

We also isolate the one-loop subgraph as

$$\Pi(k) = \int \mathrm{d}^d l \,\frac{1}{l^2(k-l)^2}. \tag{2.163}$$

Then the two-loop graph has the schematic form

$$\Sigma_{2a}(p) \sim \int \mathrm{d}^d k\, \Pi(k)\, \frac{1}{(k^2)^2\,(k+p)^2}, \tag{2.164}$$

$$\partial^p_\mu\partial^p_\nu\partial^p_\rho \Sigma_{2a}(p) \sim \int \mathrm{d}^d k\, \Pi(k)\, \partial^p_\mu\partial^p_\nu\partial^p_\rho\left[\frac{1}{(k^2)^2\,(k+p)^2}\right]. \tag{2.165}$$

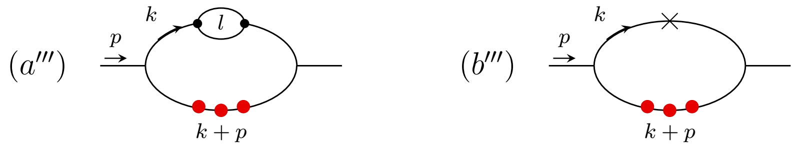
The derivative acts only on propagators through which the external momentum flows. In the graph above this is the line with momentum $k + p$:

$$\partial^p_\mu \frac{1}{(k+p)^2 - m^2} = -\frac{2(k+p)_\mu}{[(k+p)^2 - m^2]^2}. \tag{2.166}$$

Thus differentiating a propagator produces one extra propagator power and a numerator proportional to the momentum through that line. In graph language we denote this bookkeeping operation by adding a marked insertion on the differentiated line:

$$\partial^p_\mu(\underline{\hspace{3em}}) \quad \longrightarrow \quad \underline{\hspace{3em}}. \tag{2.167}$$

The three red dots below mean that the external-momentum dependent propagator has been differentiated three times. They are not new interaction vertices; they only remind us where the powers of momentum in the numerator come from. The differentiated versions of graphs (a) and (b) are shown below.



The claim is slightly subtle. The graph (a''') by itself still contains the same one-loop subdivergence in the l loop. The finite object is the subtracted combination

$$(a''') + (b'''). \tag{2.168}$$

After this subtraction, the third derivative of the full graph is ultraviolet finite. Equivalently, the divergent part of the un-differentiated graph must obey

$$\partial^p_\mu \partial^p_\nu \partial^p_\rho \Sigma_{2,\mathrm{div}}(p) = 0. \tag{2.169}$$

Therefore $\Sigma_{2,\mathrm{div}}(p)$ can be at most a quadratic polynomial in the external momentum:

$$\Sigma_{2,\mathrm{div}}(p) = A(d)p^2 + B_\mu(d)p^\mu + C(d), \tag{2.170}$$

$$B_\mu(d) = 0 \qquad \text{by Lorentz invariance,} \tag{2.171}$$

$$\Sigma_{2,\mathrm{div}}(p) = A(d)p^2 + C(d). \tag{2.172}$$

The term $A(d)p^2$ is canceled by a wave-function counterterm. The constant $C(d)$ is a mass counterterm; in the massless MS example discussed above, the displayed overall divergence is proportional to p^2, so the relevant graph (c) is the δZ counterterm.

Let us check explicitly why the subtracted third derivative is finite. There are three ultraviolet regions to consider. If k and l become large together, the original overall degree is $D = 2$, and three derivatives lower it to $D = -1$, so this region is convergent. If $l \to \infty$ while k is fixed, the divergence is precisely the one-loop subgraph and is canceled by (b'''). The only region that is easy to miss is the nested region

$$l \gg k \gg |p|. \tag{2.173}$$

In this region the loop momentum l is much larger than the momentum k entering the subgraph, so the inner loop can be expanded in powers of k/l. Define

$$I(k) = \int \mathrm{d}^6 l \, \frac{1}{l^2(l+k)^2}. \tag{2.174}$$

Here we keep $d = 6$, because this is the case where the two-loop graph has the subdivergence we want to subtract. The useful expansion is

$$(l+k)^2 = l^2 + 2l\cdot k + k^2 = l^2\left(1 + \frac{2l\cdot k + k^2}{l^2}\right), \tag{2.175}$$

$$\frac{1}{(l+k)^2} = \frac{1}{l^2}\left[1 - \frac{2l\cdot k + k^2}{l^2} + \left(\frac{2l\cdot k + k^2}{l^2}\right)^2 + \cdots\right]. \tag{2.176}$$

This is the same as the Taylor expansion of the subgraph in its external momentum k:

$$I(k) = I(0) + k^\mu \left.\frac{\partial I}{\partial k^\mu}\right|_{k=0} + \frac{1}{2}k^\mu k^\nu \left.\frac{\partial^2 I}{\partial k^\mu \partial k^\nu}\right|_{k=0} + \cdots. \tag{2.177}$$

Now look term by term. In six dimensions $\mathrm{d}^6 l = \Omega_5 l^5 \mathrm{d}l$, and angular symmetry gives $\langle (l\cdot k)^2\rangle = l^2k^2/6$. The first three terms behave as

$$I_{(0)}(k) = \int \mathrm{d}^6 l\, \frac{1}{l^4} \sim \Omega_5 \int^\Lambda \mathrm{d}l\, l \sim \Lambda^2, \tag{2.178}$$

$$I_{(1)}(k) = -\int \mathrm{d}^6 l\, \frac{2l\cdot k}{l^6} = -2k_\mu \int \mathrm{d}^6 l\, \frac{l^\mu}{l^6} = 0, \tag{2.179}$$

$$\begin{aligned} I_{(2)}(k) &= \int \mathrm{d}^6 l \left(-\frac{k^2}{l^6} + \frac{4(l\cdot k)^2}{l^8} \right) \\ &\sim \Omega_5 \int^\Lambda \mathrm{d}l\, l^5 \left(-\frac{k^2}{l^6} + \frac{4}{l^8}\frac{l^2k^2}{6} \right) \\ &= -\frac{\Omega_5 k^2}{3} \int^\Lambda \frac{\mathrm{d}l}{l} \sim -\frac{\Omega_5 k^2}{3} \ln\Lambda. \end{aligned} \tag{2.180}$$

Thus $I_{(0)}$ is quadratically divergent, $I_{(1)}$ vanishes by oddness, and $I_{(2)}$ is logarithmically divergent.

Thus the divergent part of the subgraph is local in k. More explicitly, the divergent part can be written as a polynomial in k; the appearance of that polynomial depends on the regulator:

$$\begin{aligned} I^{(\Lambda)}_{\mathrm{div}}(k) &= a_0\Lambda^2 + a_2 k^2 \ln\frac{\Lambda^2}{\mu^2}, && \text{hard cutoff,} \\ I^{(\mathrm{DR})}_{\mathrm{div}}(k) &= \frac{b_2 k^2}{d-6}, && \text{dimensional regularization near } d=6. \end{aligned} \tag{2.181}$$

The constants a_0, a_2, b_2 are regulator dependent. What matters is not their numerical value here, but the fact that the divergent part is a local polynomial: a constant plus a term proportional to k^2. The counterterm in graph (b) removes exactly these local divergent pieces. After subtraction, the large-k behavior of the inner subgraph is at worst

$$I_R(k) = I(k) - I_{\mathrm{ct}}(k) = O\big(k^2 \ln k^2\big). \tag{2.182}$$

Terms of order k^3 and higher in the expansion of the subgraph are already finite in the l loop, so they do not introduce new subdivergences.

Finally combine this with the differentiated outer line. For $k \gg |p|$,

$$\partial^p_\mu \partial^p_\nu \partial^p_\rho \frac{1}{(k+p)^2} = O\left(\frac{1}{k^5}\right). \tag{2.183}$$

The worst large-k behavior of the subtracted differentiated two-loop graph is therefore obtained by multiplying the six-dimensional measure, the renormalized subgraph, the two k^2 propagators on the upper line, and the differentiated $k+p$ propagator:

$$\int^\infty \mathrm{d}k\, k^5 \left(k^2 \ln k^2\right) \frac{1}{(k^2)^2} \frac{1}{k^5} \sim \int^\infty \mathrm{d}k\, \frac{\ln k}{k^2} < \infty. \tag{2.184}$$

which is finite. This is the concrete reason why the non-local subdivergence must cancel between graphs (a) and (b), leaving only a local polynomial counterterm.

1.9 Composite operators

The new object in this subsection is a product of fields at nearby points, for example $\phi(x)\phi(y)$ when $x-y$ is small. Such a product is more singular than an ordinary field, because two quantum fields are being multiplied at almost the same spacetime point. The useful statement is that, at short distances, this product can be replaced by a sum of local operators at one point:

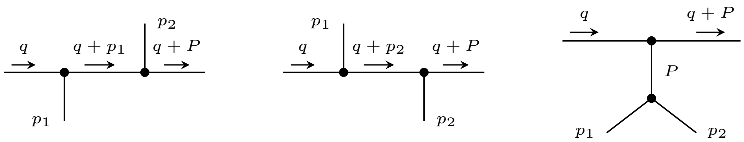
$$\phi(x)\phi(y) \sim C_1(x-y)\mathbb{1}+C_{\phi^2}(x-y)\left[\frac{1}{2}\phi^2(y)\right]+C_\phi(x-y)[\phi(y)]+C^\mu_{\partial\phi}(x-y)[\partial_\mu\phi(y)]+\dots \tag{2.185}$$

This is the operator product expansion, or OPE. The bracket $[A]$ means that the local operator has been renormalized. The functions C_i are ordinary functions, not operators; they are called Wilson coefficients. A Wilson coefficient contains the short-distance, high-momentum part of a graph, while the matrix element of the local operator contains the soft long-distance part.

We now see this mechanism in ϕ^3 theory. The large momentum q flows through the pair of fields that are being brought close together, whereas p_1 and p_2 are held fixed. In the labels of the figures,

$$P = p_1 + p_2, \qquad q^2 \gg p_1^2,\ p_2^2,\ P^2,\ m^2. \tag{2.186}$$

The relevant contributions to the four-point function are



(a) hard (b) hard (c) soft

The words "hard" and "soft" only describe which subgraph carries the large momentum. In graphs (a) and (b), the three propagators along the upper line are all hard. In graph (c), only the two upper propagators are hard; the lower part, which carries momentum P, remains soft and will be identified with a matrix element of a local operator.

The expansion used repeatedly below is the large-q expansion of a propagator. If p is fixed while q is taken large, then

$$\frac{1}{(q+p)^2-m^2} = \frac{1}{q^2}\left[1-\frac{2q\cdot p+p^2-m^2}{q^2}+O\left(\frac{1}{q^2}\right)\right]. \tag{2.187}$$

Equivalently, the small parameters are p/q and m/q. Only propagators carrying the hard momentum are expanded. The soft denominators are kept exact:

$$D_1 = p_1^2 - m^2, \qquad D_2 = p_2^2 - m^2, \qquad D_P = P^2 - m^2. \tag{2.188}$$

$$(a) = (-\mathrm{i}g)^2\frac{\mathrm{i}}{q^2-m^2}\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{(q+p_1)^2-m^2}\frac{\mathrm{i}}{(q+P)^2-m^2}\frac{\mathrm{i}}{D_2} \tag{2.189}$$

$$\xrightarrow{q\to\infty} \mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3}+O\left(\frac{1}{q^7}\right), \tag{2.190}$$

$$(b) \xrightarrow{q\to\infty} \mathrm{i}g^2\left(\frac{\mathrm{i}}{D_1}\frac{\mathrm{i}}{D_2}\right)\frac{1}{(q^2)^3}+O\left(\frac{1}{q^7}\right), \tag{2.191}$$

$$(a) + (b) \simeq 2\mathrm{i}g^2 \left(\frac{\mathrm{i}}{D_1} \frac{\mathrm{i}}{D_2} \right) \frac{1}{(q^2)^3}, \tag{2.192}$$

$$(c) = (-\mathrm{i}g)^2 \frac{\mathrm{i}}{q^2 - m^2} \frac{\mathrm{i}}{(q+P)^2 - m^2} \frac{\mathrm{i}}{D_P} \frac{\mathrm{i}}{D_1} \frac{\mathrm{i}}{D_2}. \tag{2.193}$$

For graph (c), the hard expansion contains two hard propagators instead of three:

$$(c) = -\frac{\mathrm{i}g^2}{(q^2)^2} \left[1 - \frac{2q\cdot P}{q^2} + \frac{2m^2 - P^2}{q^2} + \frac{4(q\cdot P)^2}{(q^2)^2} + \cdots \right] \times \frac{1}{(P^2 - m^2)(p_1^2 - m^2)(p_2^2 - m^2)}, \qquad P = p_1 + p_2. \tag{2.194}$$

The first important point is that the soft factor in (a)+(b) is exactly the lowest-order matrix element of the local operator $\frac{1}{2}\phi^2(0)$:

$$M_{\phi^2}(p_1, p_2) \equiv \langle 0|\, \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\, |0\rangle_{0,\mathrm{conn}} = \frac{\mathrm{i}}{p_1^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{p_2^2 - m^2 + \mathrm{i}\epsilon}. \tag{2.195}$$

The factor $\frac{1}{2}$ removes the double counting of the two identical contractions. In the connected part,

$$\langle 0|\, \mathrm{T}()\{\phi(x)\phi(y)\frac{1}{2}\phi^2(0)\}\, |0\rangle_{0,\mathrm{conn}} = \frac{1}{2}D(x)D(y) + \frac{1}{2}D(y)D(x) = D(x)D(y). \tag{2.196}$$

The contraction in which the two fields inside $\phi^2(0)$ contract with each other gives a tadpole proportional to $D(0)$; it is not part of the connected graph displayed here and is removed by operator renormalization. In momentum space:

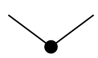
$$\int \mathrm{d}^4x\, \mathrm{d}^4y\, \mathrm{e}^{-\mathrm{i}p_1\cdot x - \mathrm{i}p_2\cdot y} D(x)D(y) = \frac{\mathrm{i}}{p_1^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{p_2^2 - m^2 + \mathrm{i}\epsilon}, \tag{2.197}$$

$$M_{\phi^2}(p_1, p_2) = \int \mathrm{d}^4x\, \mathrm{d}^4y\, \mathrm{e}^{-\mathrm{i}p_1\cdot x - \mathrm{i}p_2\cdot y} \langle 0|\, \mathrm{T}()\{\phi(x)\phi(y)\frac{1}{2}\phi^2(0)\}\, |0\rangle_{0,\mathrm{conn}}\,. \tag{2.198}$$

Here $\phi(0)$ is not Fourier transformed, because it is the local operator inserted at the origin. The same rule follows directly from the path integral:

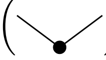
$$\langle 0|\, \mathrm{T}()\{\phi(x)\phi(y)\frac{1}{2}\phi^2(0)\}\, |0\rangle = N \int [\mathrm{d}A]\, A(x)A(y)\frac{1}{2}A^2(0)\mathrm{e}^{\mathrm{i}S[A]}. \tag{2.199}$$

Here ϕ denotes the quantum operator, while A is the classical field inside the path integral. At lowest order, the insertion of $\frac{1}{2}\phi^2(0)$ therefore gives one local black-dot vertex with two scalar lines:



This insertion $\frac{1}{2}\phi^2(0)$ is our first example of a composite operator. Therefore

$$\begin{aligned}
(a)+(b) &\simeq \underbrace{\frac{2\mathrm{i}g^2}{(q^2)^3}}_{C_{\phi^2}(q)} \underbrace{M_{\phi^2}(p_1,p_2)}_{\text{soft matrix element}} \\
&= \frac{2\mathrm{i}g^2}{(q^2)^3} \langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle_{0,\mathrm{conn}} \\
&= \frac{2\mathrm{i}g^2}{(q^2)^3}\Big(\quad \Big). \qquad (2.200)
\end{aligned}$$



Graph (c) is slightly different. The soft part is not the matrix element of $\frac{1}{2}\phi^2$; it is the matrix element of the elementary field ϕ between the vacuum and two external scalar legs. This matrix element first appears at order g:

$$\begin{aligned}
&\langle 0|\,\mathrm{T}()\{\phi(x)\phi(y)\phi(z)\}\,|0\rangle_{O(g)} \\
&\quad = \langle 0|\,\mathrm{T}()\{\phi(x)\phi(y)\phi(z)\int \mathrm{d}^4w\,\frac{-\mathrm{i}g}{3!}\phi^3(w)\}\,|0\rangle_0 \\
&\quad = -\mathrm{i}g\int \mathrm{d}^4w\,D(x-w)D(y-w)D(z-w). \qquad (2.201)
\end{aligned}$$

We use

$$D(u-v)=\int\frac{\mathrm{d}^4k}{(2\pi)^4}\mathrm{e}^{\mathrm{i}k\cdot(u-v)}\frac{\mathrm{i}}{k^2-m^2+\mathrm{i}\epsilon}. \qquad (2.202)$$

In momentum space, with $P=p_1+p_2$, define

$$\begin{aligned}
M_\phi(x;p_1,p_2) &\equiv \langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(x)\}\,|0\rangle_{O(g)} \\
&= \int \mathrm{d}^4y\mathrm{d}^4z\,\mathrm{e}^{-\mathrm{i}p_1\cdot y-\mathrm{i}p_2\cdot z}\,\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\phi(z)\}\,|0\rangle \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\int \mathrm{d}^4w\,\mathrm{e}^{-\mathrm{i}P\cdot w}D(x-w) \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\int\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{i}\,\mathrm{e}^{\mathrm{i}k\cdot x}}{k^2-m^2+\mathrm{i}\epsilon}\int \mathrm{d}^4w\,\mathrm{e}^{-\mathrm{i}(P+k)\cdot w} \\
&= -\mathrm{i}g\left(\frac{\mathrm{i}}{p_1^2-m^2+\mathrm{i}\epsilon}\right)\left(\frac{\mathrm{i}}{p_2^2-m^2+\mathrm{i}\epsilon}\right)\frac{\mathrm{i}}{P^2-m^2+\mathrm{i}\epsilon}\mathrm{e}^{-\mathrm{i}P\cdot x}. \qquad (2.203)
\end{aligned}$$

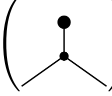
The factor $\mathrm{e}^{-\mathrm{i}P\cdot x}$ is the important part. Once the soft subgraph is represented by a local operator at $x=0$, powers of P in the hard coefficient become derivatives acting on that local operator:

$$\partial_\mu M_\phi(x;p_1,p_2)\big|_{x=0}=-\mathrm{i}P_\mu M_\phi(0;p_1,p_2), \qquad P_\mu\to\mathrm{i}\partial_\mu, \qquad P^2\to-\Box. \qquad (2.204)$$

$$(c)\sim\frac{\mathrm{i}g}{(q^2)^2}\left[1-\frac{2q\cdot P}{q^2}+\frac{2m^2-P^2}{q^2}+\frac{4(q\cdot P)^2}{(q^2)^2}+\cdots\right]M_\phi(0;p_1,p_2) \qquad (2.205)$$

$$=\frac{\mathrm{i}g}{(q^2)^2}\left[1-\frac{2\mathrm{i}q^\mu\partial_\mu}{q^2}+\frac{2m^2+\Box}{q^2}-\frac{4q^\mu q^\nu\partial_\mu\partial_\nu}{(q^2)^2}+\cdots\right]\langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(x)\}\,|0\rangle\Big|_{x=0} \qquad (2.206)$$

$$=\frac{\mathrm{i}g}{(q^2)^2}\sum_n c_n(q,\partial)\Big(\quad \Big). \qquad (2.207)$$



Putting the two types of terms together gives the OPE form of this four-point function:

$$\begin{aligned}&\langle 0|\,\mathrm{T}()\{\tilde{\phi}(q)\phi(0)\tilde{\phi}(p_1)\tilde{\phi}(p_2)\}\,|0\rangle\\ &\quad\sim\sum_i C_i(q)\,\langle 0|\,\mathrm{T}()\{O_i(0)\tilde{\phi}(p_1)\tilde{\phi}(p_2)\}\,|0\rangle\,.\end{aligned}\tag{2.208}$$

In this example the first few local operators are

$$O_1=\frac{1}{2}\phi^2,\qquad \text{from graphs (a) and (b),}\tag{2.209}$$

$$O_2=\phi,\qquad \text{from the leading term of graph (c),}\tag{2.210}$$

$$O_3=\partial_\mu\phi,\ \Box\phi,\ldots,\qquad \text{from higher terms in graph (c).}\tag{2.211}$$

The lesson is the separation of scales: the large momentum q is stored in the Wilson coefficients $C_i(q)$, while the dependence on the soft external momenta is stored in matrix elements of local operators.

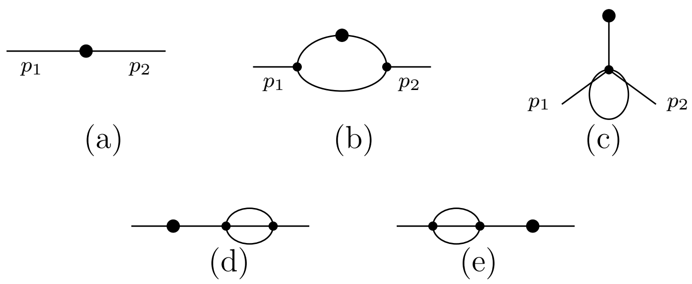
1.10 Renormalization of composite operators

The previous subsection used $\frac{1}{2}\phi^2(0)$ as a local insertion. This is not yet a complete definition of the operator. When the fields inside ϕ^2 approach interaction vertices, new ultraviolet divergences can occur at the insertion point itself. These divergences are not removed by renormalizing only the parameters in the Lagrangian; they require a renormalization of the composite operator.

Consider ϕ^3 theory in $d=6-\epsilon$ dimensions and the Green function with one insertion of $\frac{1}{2}\phi^2$:

$$\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(z)\}\,|0\rangle\tag{2.212}$$

The connected graphs up to order g^2 are shown below. The black dot is the insertion of $\frac{1}{2}\phi^2$. A crossed mark, when it appears later, denotes a local counterterm.



We now Fourier transform the two ordinary external fields, while the composite operator remains localized at the point z:

$$\begin{aligned}G&=\langle 0|\,\mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(z)\}\,|0\rangle\\ &=\int \mathrm{d}^dx\,\mathrm{d}^dy\,\mathrm{e}^{\mathrm{i}(p_1\cdot x+p_2\cdot y)}\,\langle 0|\,\mathrm{T}\{\phi(x)\phi(y)\frac{1}{2}\phi^2(z)\}\,|0\rangle\,.\end{aligned}\tag{2.213}$$

The lowest-order graph (a) is just the insertion vertex connected to the two external propagators. We use the shorthand

$$S_i=\frac{\mathrm{i}}{p_i^2-m^2+\mathrm{i}\epsilon},\qquad S_{12}=S_1S_2.\tag{2.214}$$

$$G_a = S_{12}. \tag{2.215}$$

For the order-g^2 graphs, the large-k behavior is

$$G_b \sim \int \mathrm{d}^6 k \, \frac{1}{k^2}\frac{1}{k^2}\frac{1}{k^2} \sim \int^{\Lambda} \frac{\mathrm{d}k}{k}, \qquad \text{logarithmically divergent,} \tag{2.216}$$

$$G_c,\ G_d,\ G_e \sim \int \mathrm{d}^6 k \, \frac{1}{k^4} \sim \int^{\Lambda} \mathrm{d}k\, k, \qquad \text{quadratically divergent by power counting.} \tag{2.217}$$

Graphs (d) and (e) are ordinary self-energy corrections on the external lines. Their divergences are cancelled by the usual wave-function and mass counterterms. After ordinary field and parameter renormalization has been performed, they do not give a new counterterm for $[\phi^2]$.

Graphs (b) and (c) are different. Their short-distance divergence sits at the composite-operator insertion itself, so the bare operator ϕ^2 is not enough. We must add local counterterms to define the finite operator $[\phi^2]$.

$$G_b = S_{12}(-\mathrm{i}g)^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(p_1 - k)^2 - m^2 + \mathrm{i}\epsilon} \frac{\mathrm{i}}{(p_2 + k)^2 - m^2 + \mathrm{i}\epsilon}. \tag{2.218}$$

After the continuation $g \to g\mu^{(6-d)/2}$, the loop part is

$$\begin{aligned} G_b &= S_{12}\left(\frac{\mathrm{i}g^2\mu^{6-d}}{(2\pi)^d}\right) \\ &\quad \times \int \mathrm{d}^d k \, \frac{1}{(k^2 - m^2)[(p_1 - k)^2 - m^2][(p_2 + k)^2 - m^2]}. \end{aligned} \tag{2.219}$$

$$\begin{aligned} &\frac{1}{(k^2 - m^2)[(p_1 - k)^2 - m^2][(p_2 + k)^2 - m^2]} \\ &\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[(1-x-y)(k^2 - m^2) + x((p_1 - k)^2 - m^2) + y((p_2 + k)^2 - m^2)\right]^{-3} \end{aligned} \tag{2.220}$$

$$\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[k^2 - m^2 - 2(xp_1 - yp_2)\cdot k + xp_1^2 + yp_2^2\right]^{-3}. \tag{2.221}$$

Completing the square with $k' = k - xp_1 + yp_2$ and $P = p_1 + p_2$ gives

$$\begin{aligned} &2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[(k - xp_1 + yp_2)^2 - (xp_1 - yp_2)^2 + xp_1^2 + yp_2^2 - m^2\right]^{-3} \\ &\quad = 2\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[k'^2 - \Delta_b(x,y)\right]^{-3}, \end{aligned} \tag{2.222}$$

$$\Delta_b(x,y) = m^2 - p_1^2 x(1-x-y) - p_2^2 y(1-x-y) - P^2 xy. \tag{2.223}$$

After Wick rotation, the momentum integral is

$$\begin{aligned} &\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \int \mathrm{d}^d k_E \, \frac{2}{(k_E^2 + \Delta_b)^3} \\ &\quad = \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \, \frac{(2\pi)^d}{(4\pi)^{d/2}} \Gamma\left(3 - \frac{d}{2}\right) \Delta_b^{d/2-3}. \end{aligned} \tag{2.224}$$

Therefore

$$G_b = S_{12}\frac{g^2}{64\pi^3}\Gamma\left(3-\frac{d}{2}\right) \times \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left(\frac{\Delta_b(x,y)}{4\pi\mu^2}\right)^{d/2-3}. \tag{2.225}$$

The pole is especially transparent if we write $d = 6-\epsilon$. Since

$$\Gamma\left(3-\frac{d}{2}\right) = \Gamma\left(\frac{\epsilon}{2}\right) = \frac{2}{\epsilon} - \gamma_E + O(\epsilon), \qquad \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y = \frac{1}{2}, \tag{2.226}$$

the divergent part of graph (b) is

$$G_b^{\text{pole}} = S_{12}\frac{g^2}{64\pi^3}\frac{1}{\epsilon} = -S_{12}\frac{g^2}{64\pi^3(d-6)}. \tag{2.227}$$

Thus the operator counterterm that cancels this pole is local and proportional to the same two-leg insertion $\frac{1}{2}\phi^2$.

For graph (c), use the same P and set

$$D_{12P} = (p_1^2 - m^2)(p_2^2 - m^2)(P^2 - m^2). \tag{2.228}$$

Then

$$G_c = (-\mathrm{i}g\mu^{\frac{6-d}{2}})^2 \frac{-\mathrm{i}}{D_{12P}} \frac{1}{2!} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{i}}{k^2 - m^2} \frac{\mathrm{i}}{(k+P)^2 - m^2} \tag{2.229}$$

$$= \frac{-\mathrm{i}g^2\mu^{6-d}}{D_{12P}} \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d k}{(2\pi)^d} \left[(k+xP)^2 - \Delta_c(x)\right]^{-2},$$

$$\Delta_c(x) = m^2 - P^2 x(1-x) \tag{2.230}$$

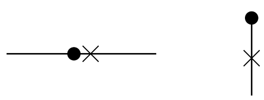
$$= \frac{\frac{1}{2}g^2\mu^{6-d}}{D_{12P}} \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d k_E}{(2\pi)^d} \frac{1}{(k_E^2 + \Delta_c(x))^2} \tag{2.231}$$

$$= \frac{\frac{1}{2}g^2(\mu^2)^{3-d/2}}{D_{12P}} \frac{1}{(4\pi)^{d/2}} \Gamma\left(2-\frac{d}{2}\right) \int_0^1 \mathrm{d}x\, \Delta_c(x)^{d/2-2} \tag{2.232}$$

$$= \frac{g^2}{D_{12P}} \frac{1}{128\pi^3} \Gamma\left(2-\frac{d}{2}\right) \int_0^1 \mathrm{d}x \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}} \tag{2.233}$$

$$G_c = \frac{-g\mu^{3-d/2}}{D_{12P}} \frac{-g\mu^{d/2-3}}{128\pi^3} \Gamma\left(2-\frac{d}{2}\right) \times \int_0^1 \mathrm{d}x \frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}} \tag{2.234}$$

G_b and G_c both diverge, so the bare composite operator ϕ^2 does not define finite Green functions by itself. For the OPE we need a local operator with the same quantum numbers and a finite insertion. In the MS scheme, the required counterterms are obtained by replacing each divergent short-distance loop by a local vertex:



For graph (b), the pole part is

$$G_b = S_{12}\frac{g^2}{64\pi^3}\Gamma\left(3-\frac{d}{2}\right)\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left(\frac{\Delta_b(x,y)}{4\pi\mu^2}\right)^{d/2-3} \tag{2.235}$$

$$= S_{12}\frac{g^2}{64\pi^3}\int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y \left[\frac{2}{\epsilon} - \gamma_E - \ln\frac{\Delta_b(x,y)}{4\pi\mu^2}\right] + O(\epsilon) \tag{2.236}$$

$$= -S_{12}\frac{g^2}{64\pi^3(d-6)} + \text{finite}. \tag{2.237}$$

Therefore the corresponding MS counterterm insertion is

$$G_{b,\text{ct}} = S_{12}\frac{g^2}{64\pi^3(d-6)}. \tag{2.238}$$

For graph (c), the pole part is

$$G_c = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{128\pi^3}\Gamma\left(2-\frac{d}{2}\right)\int_0^1 \mathrm{d}x\,\frac{\Delta_c(x)^{d/2-2}}{(4\pi\mu^2)^{d/2-3}}, \tag{2.239}$$

where $\Delta_c(x) = m^2 - P^2x(1-x)$, $\qquad P = p_1 + p_2$, so

$$G_c = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{128\pi^3}\left(-\frac{2}{\epsilon}\right)\int_0^1 \mathrm{d}x\,\Delta_c(x) + \text{finite} \tag{2.240}$$

$$= \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right) + \text{finite}. \tag{2.241}$$

Thus

$$G_{c,\text{ct}} = \frac{-g\mu^{3-d/2}}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\left\{\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2-\frac{P^2}{6}\right)\right\}. \tag{2.242}$$

This is exactly what is produced by inserting the local operator

$$\frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2+\frac{1}{6}\Box\right)\phi. \tag{2.243}$$

Indeed, in momentum space $\Box$ acting on the local field gives $-P^2$, so $m^2+\Box/6$ becomes $m^2-P^2/6$. Throughout this subsection $d = 6-\epsilon$, so $1/(d-6) = -1/\epsilon$; this is the source of the signs in the MS counterterms.

Thus, the renormalized value at $d = 6$ is

$$R(G_b) = \frac{\mathrm{i}^2}{(p_1^2-m^2)(p_2^2-m^2)}\frac{g^2}{64\pi^3}\left[-\frac{\gamma_E}{2} - \int_0^1 \mathrm{d}x \int_0^{1-x} \mathrm{d}y\,\ln\frac{\Delta_b(x,y)}{4\pi\mu^2}\right], \tag{2.244}$$

$$R(G_c) = \frac{-g}{(p_1^2-m^2)(p_2^2-m^2)(P^2-m^2)}\frac{-g}{128\pi^3}\left[(\gamma_E-1)\left(m^2-\frac{P^2}{6}\right) + \int_0^1 \mathrm{d}x\,\Delta_c(x)\ln\frac{\Delta_c(x)}{4\pi\mu^2}\right]. \tag{2.245}$$

The counterterms can now be read as local operator insertions. First, the two-leg insertion satisfies

$$\langle 0|\,\mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\,|0\rangle_{0,\text{conn}} = S_{12}. \tag{2.246}$$

Therefore the counterterm from graph (b) is equivalently

$$G_{b,\mathrm{ct}} = \frac{g^2}{64\pi^3(d-6)} \langle 0|\, \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\, |0\rangle_{0,\mathrm{conn}} . \tag{2.247}$$

Second, the one-leg operator insertion satisfies

$$\langle 0|\, \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\phi(0)\}\, |0\rangle_{O(g)} = -\frac{g}{D_{12P}}, \tag{2.248}$$

$$\langle 0|\, \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2 + \frac{1}{6}\Box\right)\phi(0)\}\, |0\rangle_{O(g)} = -\frac{g}{D_{12P}}\left(m^2 - \frac{P^2}{6}\right). \tag{2.249}$$

Therefore the counterterm from graph (c) is equivalently

$$G_{c,\mathrm{ct}} = \frac{g\mu^{d/2-3}}{64\pi^3(d-6)} \langle 0|\, \mathrm{T}()\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2 + \frac{1}{6}\Box\right)\phi(0)\}\, |0\rangle_{O(g)} . \tag{2.250}$$

Graphs (d) and (e) have already been handled by the usual field and mass counterterms. The remaining new information is therefore the renormalization of the insertion itself:

$$\begin{aligned} &G_a + R(G_b) + R(G_c) + R(G_d) + R(G_e) \\ &= \left(1 + \frac{g^2}{64\pi^3(d-6)}\right) \langle 0|\, \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\frac{1}{2}\phi^2(0)\}\, |0\rangle \\ &\quad + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)} \langle 0|\, \mathrm{T}\{\tilde{\phi}(p_1)\tilde{\phi}(p_2)\left(m^2 + \frac{1}{6}\Box\right)\phi(0)\}\, |0\rangle + \text{higher orders.} \end{aligned} \tag{2.251}$$

Thus the finite insertion is generated by the renormalized composite operator

$$\frac{1}{2}[\phi^2] \equiv \left(1 + \frac{g^2}{64\pi^3(d-6)}\right)\frac{1}{2}\phi^2 + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)}\left(m^2 + \frac{1}{6}\Box\right)\phi + \text{higher orders} \tag{2.252}$$

$$= \left(1 + \frac{g^2}{64\pi^3(d-6)}\right)\frac{1}{2}\phi^2 + \frac{g\mu^{d/2-3}}{64\pi^3(d-6)} m^2\phi + \frac{g\mu^{d/2-3}}{6\cdot 64\pi^3(d-6)}\Box\phi + \text{higher orders.} \tag{2.253}$$

The brackets remind us that this is a renormalized local operator, not the bare product of two fields at the same point. To all orders, the same structure is written as

$$\frac{1}{2}[\phi^2] = Z_a\frac{1}{2}\phi^2 + \mu^{d/2-3}Z_b m^2\phi + \mu^{d/2-3}Z_c\Box\phi. \tag{2.254}$$

The constants Z_a, Z_b, Z_c contain only short-distance information; in minimal subtraction they are series in g and poles in $d-6$. The reason that only these operators appear is the same power-counting logic used before: counterterms must be local, and an operator can mix only with operators whose dimension is not larger than its own and whose quantum numbers match. In six-dimensional ϕ^3 theory, the allowed operators with the right dimension and quantum numbers are ϕ^2, $m^2\phi$, and $\Box\phi$. After ordinary field and mass renormalization has been performed, the remaining operator mixing can be summarized as

$$\begin{pmatrix} \frac{1}{2}[\phi^2] \\ \phi \\ \Box\phi \end{pmatrix} = \begin{pmatrix} Z_a & \mu^{d/2-3}Z_b m^2 & \mu^{d/2-3}Z_c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2}\phi^2 \\ \phi \\ \Box\phi \end{pmatrix}. \tag{2.255}$$

2. Renormalization in ϕ^4-theory

Firstly, we briefly recall some basic idea about renormalization. For counting the divergent of a theory, one may use the superficial degree of divergence:

$$D = \text{power of } k \text{ in numerator} - \text{power of } k \text{ in denominator}. \tag{2.256}$$

Suppose the Loop integrals in $d = 4$ space-time dimension have superficial degree of divergence D, which can be easily seen from the directly calculating:

$$\int^{\Lambda} \mathrm{d}^4k \frac{k^{a-4}}{k^b} = \int^{\Lambda} k^3 \,\mathrm{d}k \,\mathrm{d}\Omega \,\frac{k^{a-b}}{k^4} \sim \int^{\Lambda} \mathrm{d}k \,\frac{k^{a-b}}{k} = \begin{cases} \Lambda^D & \text{for } D = a - b \neq 0 \\ \ln \Lambda & \text{for } D = 0 \end{cases}, \tag{2.257}$$

where Λ is the cut-off. But we would like to use some more general form to think about it. As an example, considering the ϕ^n-theory

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{n!}\phi^n, \tag{2.258}$$

in which the propagators have the form of $(k^2 - m^2)^{-1}$, vertices are $\sim \lambda$ that connect n lines. So we can define So in this case, we can

1. $L = P - V + 1$
   - L: loops (loop integral with d^4k)
   - P: internal propagators (propagator with $(k^2)^{-1}$)
   - V: vertices (vertices with energy-momentum conservation $\delta(\sum p_i)$)
   - $+1$: global conservation of energy-momentum
2. $nV = 2P + N$
   - n: number of lines at each vertex
   - P: internal propagators
   - N: external lines (or external propagator)

So for each loop, the power of k in numerator is 4, which is contributed by an integral measurement $\sim \mathrm{d}^4k$, and the power of k in denominator is contributed by the propagator. Therefore, the superficial degree of divergence is

$$D = dL - 2P; \quad d = 4. \tag{2.259}$$

Thus, base on the relation we introduced above, we have

$$\begin{aligned} D &= 4(P - V + 1) - 2P \\ &= 2P - 4V + 4 \\ &= (n-4)V - N + 4. \end{aligned} \tag{2.260}$$

Thus, we find D is independent of P, but only depends on N for the case $n = 4$. But for a more general n in $\frac{\lambda}{n!}\phi^n$, we have

- $n = 2$: $\Longrightarrow$ free theory (included in the mass term);

- $n = 3$: $D = 4 - N - V \implies$ **super renormalizable**
- $n = 4$: $D = -N + 4 \implies$ **renormalizable**
- $n > 4$: $D = (n-4)V - N + 4 \implies$ **non-renormalizable**

For a theory that the signature in front of V is "$-$", we call it as **super renormalizable theory** because the divergence only appear in some leading order terms, with respect the order getting higher, the diagram will finally be convergent, in which $D < 0$. *So in the super renormalizable theory only contains finite number of divergent degree*, which means only need finite counterterms. Similarly, for a theory that calls **renormalizable**, it is because it has no relation with V, which means *no matter how many orders we are counting, the divergent degrees are the same!* Although in this case, it contains infinite divergent diagrams, we can still by re-define the "physical" parameter to include the divergence. In the end, it is quite obvious to understand why we call the last situation as **non-renormalizable**. Because the pre-signature of V is positive, which means with the order getting higher, the divergent will be keep increasing, *so we need infinite parameter to cancel the divergence*, which in most of cases are technically impossible.

Another case is QED, we have two type of propagators, but they all contribute $\sim k^{-2}$

- N_γ: external photon lines;
- N_e: external electron lines;
- P_e: electron propagators;
- P_γ: photon propagators;
- $V = 2P_\gamma + N_\gamma = \frac{1}{2}(2P_e + N_e)$: number of vertices;
- $L = P_e + P_\gamma - V + 1$: number of the loops.

$$\begin{aligned}
D &= 4L - P_e - 2P_\gamma \\
&= 4(P_e + P_\gamma - V + 1) - P_e - 2P_\gamma \\
&= 3P_e + 2P_\gamma - 4V + 4 \\
&= 4 - N_\gamma - \frac{3}{2}N_e.
\end{aligned} \tag{2.261}$$

So as we discussed above, since D is non-related with V, so QED is a renormalizable theory.

2.1 Analysis of dimensions in view of renormalization

Now we consider another way to analyze the renormalizability. We use the dimension of mass as a standard. So

$$[\text{Length}] = [\text{Time}] = [\text{Energy}]^{-1} = [\text{Mass}]^{-1}. \tag{2.262}$$

We require the dimension for the action $[S] = 0$, thus

$$[\mathrm{d}^4x] = -4, \quad [\mathcal{L}] = 4, \quad [\partial_\mu] = 1, \quad [\phi] = 1. \tag{2.263}$$

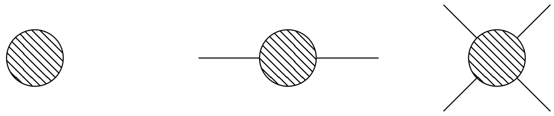
The last one is because $[\partial_\mu\phi\partial^\mu\phi] = 4$. The coupling constant $[\lambda]$ in $\frac{\lambda}{n!}\phi^n$ then has the dimension

$$[\lambda] = 4 - n. \tag{2.264}$$

Take a sharp look! This number, $4-n$, is nothing but the prefactor of superficial degree of divergence in ϕ^n theory, i.e.,

$$D = 4 - N - \underbrace{(4-n)}_{[\lambda]} V. \tag{2.265}$$

Thus, base on the discussion we did above, we find $[\lambda] < 0$ corresponds to a non-renormalizable theory. As an example, we consider the ϕ^4-theory. So $n = 4$ brings the superficial degree of divergence is $D = 4 - N \geq 0$ for degree up to $N = 4$, which is a renormalizable theory.



$D = 4$(vacuum) $D = 2$ (self energy) $D = 0$

We can also do the same thing for QED. For convenience, we rewrite the QED Lagrangian:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(\mathrm{i}\not{\partial} - m)\psi + eA_\mu \bar{\psi}\gamma^\mu\psi. \tag{2.266}$$

Similarly , since the action S is dimensionless, so

$$[\psi] = \frac{3}{2}, \quad [A_\mu] = 1. \tag{2.267}$$

Thus again, we can then see the coupling constant e is dimensionless, i.e., $[e] = 0$, which just corresponds to a renormalizable theory.

2.2 Renormalized perturbation theory

Now we still use ϕ^4-theory as an example to show the renormalization procedure. The bare Lagrangian is

$$\mathcal{L}_{\text{bare}} = \frac{1}{2}(\partial_\mu \phi_0)(\partial^\mu \phi_0) - \frac{1}{2} m_0^2 \phi_0^2 + \frac{1}{4!}\lambda_0 \phi_0^4 \tag{2.268}$$

where the corner mark "0" represent the bare parameters, i.e., λ_0, m_0, and ϕ_0. This Lagrangian is nothing special, it just the original Lagrangian that we used in perturbation theory. Now we just give it a different name to separate with the renormalized Lagrangian that we will discuss later. The name "bare" represent this quantity is not actually the physical quantity that we observe because the bare parameters are actually infinity. These parameters appears only for cancel the infinity in the loop integral. The rest of this canceling is then the physical quantity, which is the renormalized quantity. This explanation shows the core idea of why do we need renormalization, i.e., the physical part of graphs needs to be observable, so we have to find a well-defined quantity that suitable for any order. But in this case, we may have to accept that many "constants" we thought unchanged will start to running with the higher energy scale being considered. And this will be described by the renormalization group equation.

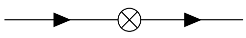
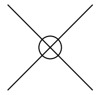
The relation of the renormalized quantities with the bare quantities are

$$\phi_0 = \sqrt{Z}\phi_r, \quad Z m_0^2 = m^2 + \delta m^2, \quad Z^2 \lambda_0 = \lambda + \delta\lambda, \tag{2.269}$$

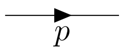
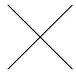
where $Z = 1 + \delta Z$. And δZ, δm^2 and $\delta\lambda$ are called the **counterterms**. So the bare Lagrangian can be written as

$$\begin{aligned}
\mathcal{L}_{bare} &= \mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{count}} \\
&= \underbrace{\frac{1}{2}(\partial_\mu \phi_r)(\partial^\mu \phi_r) - \frac{1}{2} m^2 \phi_r^2 + \frac{\lambda}{4!}\phi_r^4}_{\mathcal{L}_{\text{ren}}} + \underbrace{\frac{1}{2}\delta Z (\partial_\mu \phi_r)(\partial^\mu \phi_r) - \frac{1}{2}\delta m^2 \phi_r^2 + \frac{1}{4!}\delta\lambda \phi_r^4}_{\mathcal{L}_{\text{counter term}}}.
\end{aligned} \tag{2.270}$$

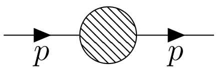
The counterterms also have corresponding Feynman rules

 $= \mathrm{i}(p^2\delta Z - \delta m^2);$  $= -\mathrm{i}\delta\lambda.$ (2.271)

And the Feynman rules of renormalized term is

 $= \dfrac{\mathrm{i}}{p^2 - m^2};$  $= -\mathrm{i}\lambda.$ (2.272)

The renormalization condition is usually called the **scheme**. The on-shell scheme for ϕ_r, m in two-point function is

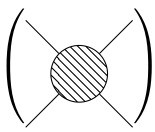


$$= \frac{i}{p^2 - m_0^2 - \Sigma(p^2)} \tag{2.273}$$

$$= \frac{\mathrm{i}}{p^2 - m_{\text{phy}}^2} + \underbrace{A + B(p^2 - m_{\text{phy}}^2) + C(p^2 - m_{\text{phy}}^2)^2 + \cdots}_{\text{Regular Terms}}, \tag{2.274}$$

where th $\Sigma(p^2)$ is the self-energy correction. The main point of this scheme is requiring the residue at the pole $p^2 = m_{\text{phys}}^2$ to be normalized to 1. The mission of the counterterms is to absorb UV divergences and ensure the self-energy $\Sigma(p^2)$ satisfies the specific renormalization conditions, rather than cancelling the regular terms.

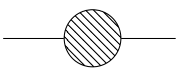
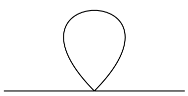
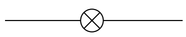
Now for the four-point functions, we impose a similar renormalization condition on the scattering amplitude:

 Amp $= \mathrm{i}\lambda,$ (2.275)

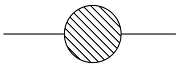
at the threshold $s = 4m^2$ and $t = u = 0$, which corresponds to the case of zero momentum transfer. The Mandelstam variables are defined as usual: $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, and $u = (p_1 - p_4)^2$.This condition, together with the two-point function conditions mentioned earlier, defines the on-shell renormalization scheme. To implement this in practice, we follow these steps:

1. Calculate the divergent diagrams: Evaluate all Feynman diagrams contributing to the n-point functions at a given order in perturbation theory. A suitable regularization scheme (e.g., dimensional regularization) must be chosen to handle the UV divergences.
2. Determine the counterterms: Extract the specific values of the counterterms δm^2, δZ, and $\delta\lambda$ by requiring that the renormalized Green's functions satisfy the chosen renormalization conditions (such as the one in (2.275)).

To conclude this section, we will briefly do the renormalization process for the 1-loop for ϕ^4-theory. Firstly, we expand the two point function until λ^2:

 = + ( + ) $+ O(\lambda^2),$ (2.276)

where the first term is just the normal propagator, the second and the third is the 1-loop graph and the counterterm. Thus, we have



$$= \frac{\mathrm{i}}{p^2-m^2} - \frac{\mathrm{i}}{2}\lambda \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{i}}{k^2-m^2} + \mathrm{i}(p^2\delta Z - \delta m^2) + O(\lambda^2). \tag{2.277}$$

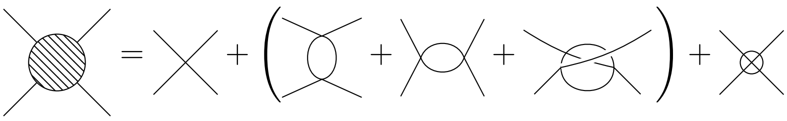
We can by directly calculating the tadpole integral to separate the divergent part, such that to determine the expression for the counterterms. As a result, we only shows the final expression that you should get:

$$\delta Z = 0 \tag{2.278}$$

$$\delta m^2 = -\frac{\lambda}{2}\frac{\Gamma(1-d/2)}{(4\pi)^{d/2}(m^2)^{1-d/2}}. \tag{2.279}$$

If we take $d = 4-2\epsilon$, the theory is renormalizable, so the form of the counterterms are suitable for any order.

The 4-point function also can be expanded into 1-loop order, but since there are two vertices, so the next to next leading order we have is λ^3:



$$+\,O(\lambda^3) \tag{2.280}$$

$$= -\mathrm{i}\lambda + (-\mathrm{i}\lambda)^2\left(\mathrm{i}V(s) + \mathrm{i}V(t) + \mathrm{i}V(u)\right) - \mathrm{i}\delta\lambda + O(\lambda^3) \tag{2.281}$$

The counterterm $\delta\lambda$ is determined by requiring the total scattering amplitude to satisfy the renormalization condition at the chosen subtraction point. According to the Feynman rules, the total amplitude at 1-loop order is the sum of the tree-level vertex, the loop corrections (s, t, and u channels), and the counterterm vertex. For each $\mathrm{i}V(p^2)$, we have

$$\mathrm{i}V(p^2) = \frac{\mathrm{i}}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2-m^2}\frac{\mathrm{i}}{(k+p)^2-m^2} \tag{2.282}$$

$$= -\frac{\mathrm{i}}{2}\frac{1}{(4\pi)^{d/2}}\int_0^1 \mathrm{d}x\, \frac{\Gamma(2-d/2)}{(m^2-x(1-x)p^2)^{2-d/2}}, \tag{2.283}$$

where x is the Feynman parameter. By taking $d = 4-2\epsilon$, we can separate the divergent part too:

$$\mathrm{i}V(p^2) = -\frac{\mathrm{i}}{32\pi^2}\int_0^1 \mathrm{d}x\left(\frac{1}{\epsilon} - \gamma_E + \ln(4\pi) - \ln\!\big(m^2 - x(1-x)p^2\big) + O(\epsilon)\right). \tag{2.284}$$

To satisfy the on-shell condition defined in (2.26), we require the amplitude to be exactly $-i\lambda$ at the threshold $s = 4m^2, t = u = 0$. This implies that the loop contributions and the counterterm must cancel each other out at this specific kinematic point:

$$(-i\lambda)^2\big[\mathrm{i}V(4m^2) + \mathrm{i}V(0) + \mathrm{i}V(0)\big] - \mathrm{i}\delta\lambda = 0 \tag{2.285}$$

Solving for $\delta\lambda$ yields the expression for the 1-loop counterterm in ϕ^4 theory:

$$\delta\lambda = -\lambda^2\left[V(4m^2) + 2V(0)\right], \tag{2.286}$$

Therefore, the divergence $1/\epsilon$ pole contained in $V(p^2)$ is exactly absorbed by $\delta\lambda$, rendering the final physical amplitude finite. This ensures that the coupling constant λ corresponds to the strength of the interaction measured at the specific energy scale $s = 4m^2$.

3. Renormalization of QED

3.1 The bare theory and the meaning of renormalization

Quantum electrodynamics is the quantum field theory of a Dirac field ψ interacting with a $U(1)$ gauge field A_μ. The classical gauge-invariant part of the Lagrangian is

$$\mathcal{L}_{\rm QED} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar\psi\big(\mathrm{i}\partial\!\!\!/ - m\big)\psi - e\bar\psi\gamma^\mu\psi A_\mu, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{2.287}$$

To quantize the photon field perturbatively we must fix a gauge. In a general covariant gauge,

$$\mathcal{L}_{\rm gf} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2. \tag{2.288}$$

For the explicit one-loop calculations below we use Feynman gauge, $\xi = 1$, because it gives the simplest photon propagator. Since QED is an abelian gauge theory, the Faddeev–Popov ghosts decouple and do not contribute to QED loop diagrams.

The bare Lagrangian is written in terms of bare fields and bare parameters:

$$\mathcal{L}_0 = -\frac{1}{4}F_{0\mu\nu}F_0^{\mu\nu} - \frac{1}{2\xi_0}(\partial_\mu A_0^\mu)^2 + \bar\psi_0\big(\mathrm{i}\partial\!\!\!/ - m_0\big)\psi_0 - e_0\bar\psi_0\gamma^\mu\psi_0 A_{0\mu}. \tag{2.289}$$

Dimensional regularization is used with

$$d = 4-\epsilon, \qquad \frac{1}{\bar\epsilon} \equiv \frac{2}{\epsilon} - \gamma_E + \ln 4\pi. \tag{2.290}$$

The factor $2/\epsilon$ appears because we use $d = 4-\epsilon$ rather than the alternative convention $d = 4-2\epsilon$.

The renormalized fields and parameters are defined by

$$\psi_0 = Z_2^{1/2}\psi, \qquad A_0^\mu = Z_3^{1/2}A^\mu, \qquad m_0 = Z_m m, \qquad e_0 = \mu^{\epsilon/2}Z_e e. \tag{2.291}$$

The scale μ is inserted so that the renormalized charge e remains dimensionless in $d \neq 4$. Substituting these definitions into the bare Lagrangian and separating the renormalized part from the counterterms gives

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \bar\psi\big(\mathrm{i}\partial\!\!\!/ - m\big)\psi - e\mu^{\epsilon/2}\bar\psi\gamma^\mu\psi A_\mu + \mathcal{L}_{\rm ct}, \tag{2.292}$$

$$\mathcal{L}_{\rm ct} = -\frac{1}{4}\delta_3 F_{\mu\nu}F^{\mu\nu} + \bar\psi\big(\mathrm{i}\delta_2\partial\!\!\!/ - \delta_m m\big)\psi - e\mu^{\epsilon/2}\delta_1\bar\psi\gamma^\mu\psi A_\mu. \tag{2.293}$$

Here

$$\delta_2 = Z_2 - 1, \qquad \delta_3 = Z_3 - 1, \qquad \delta_m = Z_2 Z_m - 1, \qquad \delta_1 = Z_1 - 1. \tag{2.294}$$

It is useful to introduce Z_1 by

$$Z_1 = Z_e Z_2 Z_3^{1/2}. \tag{2.295}$$

This notation means that the coefficient of the interaction $e\bar\psi\gamma^\mu\psi A_\mu$ is renormalized by Z_1, while the electron and photon fields are renormalized by Z_2 and Z_3.

The renormalized Feynman rules in Feynman gauge are

$$\text{electron propagator:} \qquad S_F(p) = \frac{\mathrm{i}(p\!\!\!/ + m)}{p^2 - m^2 + \mathrm{i}\epsilon}, \tag{2.296}$$

$$\text{photon propagator:} \qquad D_{\mu\nu}(k) = \frac{-\mathrm{i}g_{\mu\nu}}{k^2 + \mathrm{i}\epsilon}, \tag{2.297}$$

$$\text{QED vertex:} \qquad V^\mu = -\mathrm{i}e\mu^{\epsilon/2}\gamma^\mu. \tag{2.298}$$

The counterterm Feynman rules are

$$\text{electron two-point counterterm:} \qquad \mathrm{i}\big(\delta_2 \not{p} - \delta_m m\big), \tag{2.299}$$
$$\text{photon two-point counterterm:} \qquad \mathrm{i}\delta_3\big(q^2 g^{\mu\nu} - q^\mu q^\nu\big), \tag{2.300}$$
$$\text{vertex counterterm:} \qquad -\,\mathrm{i}e\mu^{\epsilon/2}\delta_1\gamma^\mu. \tag{2.301}$$

3.2 Power counting and the allowed counterterms

Before computing integrals, we should know which Green functions can be divergent. Let a QED graph have E_ψ external electron lines and E_A external photon lines. If I_ψ and I_A are the numbers of internal electron and photon propagators, and V is the number of interaction vertices, then

$$2I_\psi + E_\psi = 2V, \qquad 2I_A + E_A = V, \qquad L = I_\psi + I_A - V + 1. \tag{2.302}$$

Each loop integral contributes four powers of momentum, each internal electron propagator contributes one inverse power, and each photon propagator contributes two inverse powers. Therefore the superficial degree of divergence is

$$\begin{aligned} D &= 4L - I_\psi - 2I_A \\ &= 4(I_\psi + I_A - V + 1) - I_\psi - 2I_A \\ &= 3I_\psi + 2I_A - 4V + 4 \\ &= 4 - \frac{3}{2}E_\psi - E_A. \end{aligned} \tag{2.303}$$

The superficially divergent cases are therefore

$$E_\psi = 2,\ E_A = 0: \qquad D = 1, \qquad \text{electron self-energy,} \tag{2.304}$$
$$E_\psi = 0,\ E_A = 2: \qquad D = 2, \qquad \text{photon self-energy,} \tag{2.305}$$
$$E_\psi = 2,\ E_A = 1: \qquad D = 0, \qquad \text{electron-photon vertex.} \tag{2.306}$$

Other cases are either forbidden or made finite by symmetries. For example, diagrams with an odd number of external photons vanish by Furry's theorem. The four-photon amplitude has $D = 0$, but gauge invariance does not allow a local counterterm of the form A^4 in the QED Lagrangian, and the one-loop light-by-light amplitude is finite after the gauge-invariant sum of diagrams is taken.

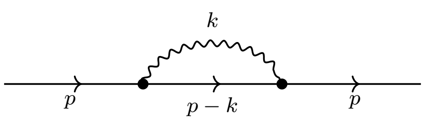
Thus the most general local counterterms needed for QED have exactly the same form as the terms already displayed in $\mathcal{L}_{\mathrm{ct}}$:

$$\bar\psi \mathrm{i}\not{\partial}\psi, \qquad m\bar\psi\psi, \qquad F_{\mu\nu}F^{\mu\nu}, \qquad e\bar\psi\gamma^\mu\psi A_\mu. \tag{2.307}$$

This is the concrete meaning of renormalizability in QED.

3.3 Electron self-energy

The one-loop correction to the electron propagator is



We define the electron self-energy $\Sigma(\not{p})$ by writing the full inverse propagator as

$$S^{-1}(p) = \not{p} - m - \Sigma(\not{p}). \tag{2.308}$$

In Feynman gauge the one-loop integral is

$$-\mathrm{i}\Sigma(\not{p}) = (-\mathrm{i}e\mu^{\epsilon/2})^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \gamma^\mu \frac{\mathrm{i}(\not{p} - \not{k} + m)}{(p-k)^2 - m^2 + \mathrm{i}\epsilon} \gamma^\nu \frac{-\mathrm{i}g_{\mu\nu}}{k^2 + \mathrm{i}\epsilon}. \tag{2.309}$$

The numerator simplifies by the Dirac identities

$$\gamma^\mu \gamma^\alpha \gamma_\mu = (2-d)\gamma^\alpha, \qquad \gamma^\mu \gamma_\mu = d. \tag{2.310}$$

Hence

$$\gamma^\mu(\not{p} - \not{k} + m)\gamma_\mu = (2-d)(\not{p} - \not{k}) + dm. \tag{2.311}$$

To combine the denominators, use

$$\frac{1}{k^2[(p-k)^2 - m^2]} = \int_0^1 \mathrm{d}x \, \frac{1}{\left[k^2 - 2xp\cdot k + x(p^2 - m^2)\right]^2}. \tag{2.312}$$

Complete the square with

$$\ell = k - xp, \qquad \Delta_e(x) = xm^2 - x(1-x)p^2. \tag{2.313}$$

Then the denominator becomes $(\ell^2 - \Delta_e)^2$, and the numerator is

$$(2-d)(\not{p} - \not{k}) + dm = (2-d)\big((1-x)\not{p} - \not{\ell}\big) + dm. \tag{2.314}$$

The term linear in ℓ integrates to zero by symmetry. The divergent part therefore comes from

$$\begin{aligned} \Sigma(\not{p})_{\text{div}} &= \frac{e^2}{16\pi^2}\frac{1}{\epsilon}\int_0^1 \mathrm{d}x \left[(2-d)(1-x)\not{p} + dm\right]_{d=4} \\ &= \frac{e^2}{16\pi^2}\frac{1}{\epsilon}\left[-\not{p} + 4m\right]. \end{aligned} \tag{2.315}$$

This form is important: the divergence is local and has only two possible Dirac structures, $\not{p}$ and m. Therefore it can be canceled by the two electron counterterms in $\mathcal{L}_{\text{ct}}$.

The counterterm contribution to the inverse propagator is

$$\mathrm{i}\big(\delta_2 \not{p} - \delta_m m\big). \tag{2.316}$$

Since the inverse propagator contains $-\Sigma(\not{p})$, the divergent part of

$$-\Sigma(\not{p}) + \delta_2\not{p} - \delta_m m \tag{2.317}$$

must vanish. This gives

$$\delta_2^{\text{MS}} = -\frac{e^2}{16\pi^2}\frac{1}{\epsilon}, \tag{2.318}$$

$$\delta_m^{\text{MS}} = -\frac{4e^2}{16\pi^2}\frac{1}{\epsilon}. \tag{2.319}$$

The parameter δ_m is the coefficient multiplying $m\bar{\psi}\psi$ inside $Z_2 Z_m$. The mass renormalization constant itself is obtained from

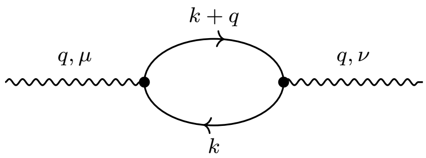
$$Z_2 Z_m = 1 + \delta_m, \qquad Z_2 = 1 + \delta_2. \tag{2.320}$$

To first order in e^2,

$$\begin{aligned} Z_m &= \frac{1+\delta_m}{1+\delta_2} = 1 + \delta_m - \delta_2 + O(e^4) \\ &= 1 - \frac{3e^2}{16\pi^2}\frac{1}{\bar{\epsilon}} + O(e^4). \end{aligned} \tag{2.321}$$

3.4 Photon self-energy and transversality

The one-loop photon self-energy is the fermion loop



The minus sign for a closed fermion loop is essential. The amplitude is

$$\mathrm{i}\Pi^{\mu\nu}(q) = -(-\mathrm{i}e\mu^{\epsilon/2})^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \operatorname{tr}\left[\gamma^\mu \frac{\mathrm{i}(\not{k}+m)}{k^2 - m^2 + \mathrm{i}\epsilon}\gamma^\nu \frac{\mathrm{i}(\not{k}+\not{q}+m)}{(k+q)^2 - m^2 + \mathrm{i}\epsilon}\right]. \tag{2.322}$$

Gauge invariance implies that the result must be transverse:

$$q_\mu \Pi^{\mu\nu}(q) = 0. \tag{2.323}$$

Equivalently, the tensor must have the form

$$\Pi^{\mu\nu}(q) = \big(q^\mu q^\nu - q^2 g^{\mu\nu}\big)\Pi(q^2). \tag{2.324}$$

This transversality is not an optional simplification; it is the reason QED does not generate a photon mass counterterm $m_\gamma^2 A_\mu A^\mu$.

To see how the divergence appears, combine the two fermion denominators:

$$\frac{1}{(k^2 - m^2)[(k+q)^2 - m^2]} = \int_0^1 \mathrm{d}x \, \frac{1}{[(\ell^2 - \Delta_\gamma(x))]^2}, \tag{2.325}$$

where

$$\ell = k + xq, \qquad \Delta_\gamma(x) = m^2 - x(1-x)q^2. \tag{2.326}$$

After the shift, terms odd in ℓ vanish. The tensor integrals are reduced by rotational symmetry:

$$\int \mathrm{d}^d\ell \, \ell^\mu \ell^\nu f(\ell^2) = \frac{g^{\mu\nu}}{d}\int \mathrm{d}^d\ell \, \ell^2 f(\ell^2). \tag{2.327}$$

Carrying out the Dirac trace and keeping only the ultraviolet pole gives

$$\Pi^{\mu\nu}_{\text{div}}(q) = \frac{e^2}{12\pi^2}\frac{1}{\bar{\epsilon}}\big(q^\mu q^\nu - q^2 g^{\mu\nu}\big). \tag{2.328}$$

The counterterm from $-\frac{1}{4}\delta_3 F_{\mu\nu}F^{\mu\nu}$ contributes

$$\mathrm{i}\Pi^{\mu\nu}_{\mathrm{ct}}(q) = \mathrm{i}\delta_3\big(q^2 g^{\mu\nu} - q^\mu q^\nu\big). \tag{2.329}$$

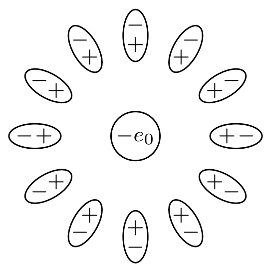
Thus the MS counterterm is

$$\delta_3^{\mathrm{MS}} = -\frac{e^2}{12\pi^2}\frac{1}{\epsilon}. \tag{2.330}$$

For N_f Dirac fermions with electric charges $Q_f e$, this generalizes to

$$\delta_3^{\mathrm{MS}} = -\frac{e^2}{12\pi^2}\frac{1}{\epsilon}\sum_{f=1}^{N_f} Q_f^2. \tag{2.331}$$

The same diagram also has a very direct physical meaning. The external photon creates a disturbance of the Dirac sea: for a short time, virtual electron-positron pairs are produced and then annihilated. In position space these pairs behave like tiny electric dipoles. Around a negative bare charge, the positive ends of the dipoles are pulled slightly inward and the negative ends are pushed slightly outward:



virtual e^+e^- dipoles screen the long-distance electric charge

This picture should not be taken as a literal classical cloud of particles. It is a useful way to remember the sign of the effect. The vacuum is polarized by the source charge. A probe very far away sees the source charge together with the induced dipoles, so the charge looks smaller. A probe at very short distance penetrates the polarization cloud and sees a larger effective charge.

The connection with the photon self-energy is as follows. In a static process the exchanged photon has $q^0 = 0$, so $q^2 = -\mathbf{q}^2$. The Dyson-resummed transverse photon propagator changes the Coulomb exchange from

$$\widetilde{V}_0(\mathbf{q}) = -\frac{4\pi\alpha}{\mathbf{q}^2} \tag{2.332}$$

to

$$\widetilde{V}(\mathbf{q}) = -\frac{4\pi\alpha}{\mathbf{q}^2}\frac{1}{1-\Pi_R(-\mathbf{q}^2)}. \tag{2.333}$$

Here Π_R is the renormalized scalar vacuum-polarization function, and the on-shell normalization convention is

$$\Pi_R(0) = 0. \tag{2.334}$$

This condition means that α is the electric charge measured at very long distance, for example in the Thomson limit. Expanding the denominator to first order gives

$$\widetilde{V}(\mathbf{q}) = -\frac{4\pi\alpha}{\mathbf{q}^2}\big[1 + \Pi_R(-\mathbf{q}^2) + O(\alpha^2)\big]. \tag{2.335}$$

Fourier transforming this correction gives the Uehling correction to the Coulomb potential. For an electron source and a positive test charge,

$$V(r) = -\frac{\alpha}{r}[1 + \delta_{\mathrm{VP}}(r)], \tag{2.336}$$

where

$$\delta_{\mathrm{VP}}(r) = \frac{2\alpha}{3\pi}\int_1^\infty \mathrm{d}s\, \mathrm{e}^{-2mrs}\left(1 + \frac{1}{2s^2}\right)\frac{\sqrt{s^2-1}}{s^2}. \tag{2.337}$$

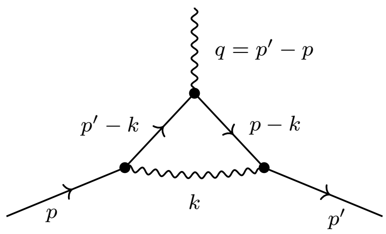
The factor e^{-2mr} reflects the threshold for making a virtual e^+e^- pair: the pair costs energy about $2m$. At distances much larger than the electron Compton wavelength, $r \gg m^{-1}$, the integral is dominated by $s \simeq 1$ and becomes

$$\delta_{\mathrm{VP}}(r) \simeq \frac{\alpha}{4\sqrt{\pi}}\frac{\mathrm{e}^{-2mr}}{(mr)^{3/2}}. \tag{2.338}$$

Thus the correction dies off exponentially at long distance. At shorter distances, or equivalently at larger momentum transfer $Q = \sqrt{-q^2}$, the probe resolves more of the unscreened charge. This is the physical origin of the running QED coupling.

3.5 Vertex correction and the Ward identity

The remaining superficially divergent one-loop graph is the correction to the electron-photon vertex:



Let the incoming electron momentum be p, the outgoing electron momentum be p', and the photon momentum be $q = p' - p$. We write the full renormalized vertex as

$$-\mathrm{i}e\mu^{\epsilon/2}\Gamma^\mu(p', p), \qquad \Gamma^\mu(p', p) = \gamma^\mu + \Lambda^\mu(p', p) + \delta_1\gamma^\mu. \tag{2.339}$$

The loop part is

$$\begin{aligned}-\mathrm{i}e\mu^{\epsilon/2}\Lambda^\mu(p', p) = (-\mathrm{i}e\mu^{\epsilon/2})^3 \int \frac{\mathrm{d}^d k}{(2\pi)^d}\gamma^\alpha \frac{\mathrm{i}(\not{p}' - \not{k} + m)}{(p'-k)^2 - m^2 + \mathrm{i}\epsilon}\gamma^\mu \\ \times \frac{\mathrm{i}(\not{p} - \not{k} + m)}{(p-k)^2 - m^2 + \mathrm{i}\epsilon}\gamma^\beta \frac{-\mathrm{i}g_{\alpha\beta}}{k^2 + \mathrm{i}\epsilon}.\end{aligned} \tag{2.340}$$

The ultraviolet divergent part is local, so it cannot depend on the external momenta except through terms that are suppressed by powers of the loop momentum. Therefore the divergent part must be proportional to γ^μ:

$$\Lambda^\mu_{\mathrm{div}}(p', p) = \frac{e^2}{16\pi^2}\frac{1}{\bar{\epsilon}}\gamma^\mu. \tag{2.341}$$

Finiteness of Γ^μ then gives

$$\delta_1^{\mathrm{MS}} = -\frac{e^2}{16\pi^2}\frac{1}{\bar{\epsilon}}. \tag{2.342}$$

This result is not independent of the electron self-energy. Gauge invariance implies the Ward–Takahashi identity

$$q_\mu \Gamma^\mu(p+q,p) = S^{-1}(p+q) - S^{-1}(p). \tag{2.343}$$

Taking the limit $q \to 0$ gives

$$\Gamma^\mu(p,p) = \frac{\partial S^{-1}(p)}{\partial p_\mu}. \tag{2.344}$$

Since

$$S^{-1}(p) = \not{p} - m - \Sigma(\not{p}), \tag{2.345}$$

the loop correction obeys

$$\Lambda^\mu(p,p) = -\frac{\partial \Sigma(\not{p})}{\partial p_\mu}. \tag{2.346}$$

Using the divergent part of the self-energy,

$$\Sigma_{\text{div}}(\not{p}) = \frac{e^2}{16\pi^2}\frac{1}{\bar{\epsilon}}\big(-\not{p} + 4m\big), \tag{2.347}$$

we find

$$-\frac{\partial \Sigma_{\text{div}}}{\partial p_\mu} = \frac{e^2}{16\pi^2}\frac{1}{\bar{\epsilon}}\gamma^\mu, \tag{2.348}$$

which is exactly the vertex divergence above. Therefore

$$\delta_1^{\text{MS}} = \delta_2^{\text{MS}}, \qquad Z_1 = Z_2. \tag{2.349}$$

This equality is the practical content of the Ward identity in QED.

3.6 Renormalization constants and the running charge

Collecting the one-loop MS counterterms for one Dirac fermion of unit charge, we have

$$Z_2 = 1 - \frac{e^2}{16\pi^2}\frac{1}{\bar{\epsilon}} + O(e^4), \tag{2.350}$$

$$Z_m = 1 - \frac{3e^2}{16\pi^2}\frac{1}{\bar{\epsilon}} + O(e^4), \tag{2.351}$$

$$Z_3 = 1 - \frac{e^2}{12\pi^2}\frac{1}{\bar{\epsilon}} + O(e^4), \tag{2.352}$$

$$Z_1 = Z_2 = 1 - \frac{e^2}{16\pi^2}\frac{1}{\bar{\epsilon}} + O(e^4). \tag{2.353}$$

Before extracting the running coupling, let us separate two ideas which are often compressed into one sentence. First, Z_3 is the photon field-strength renormalization:

$$A_0^\mu = Z_3^{1/2} A^\mu. \tag{2.354}$$

This means that the bare photon kinetic term becomes

$$\begin{aligned} -\frac{1}{4}F_{0\mu\nu}F_0^{\mu\nu} &= -\frac{1}{4}Z_3 F_{\mu\nu}F^{\mu\nu} \\ &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}\delta_3 F_{\mu\nu}F^{\mu\nu}. \end{aligned} \tag{2.355}$$

In momentum space the corresponding counterterm has the transverse form

$$\mathrm{i}\Pi^{\mu\nu}_{\text{ct}}(q) = \mathrm{i}\delta_3\big(q^2 g^{\mu\nu} - q^\mu q^\nu\big). \tag{2.356}$$

This is exactly the tensor structure produced by the divergent part of the vacuum-polarization loop. Therefore the divergence changes the normalization of the photon kinetic term, not the photon mass. A photon mass term $m_\gamma^2 A_\mu A^\mu$ would not be transverse and is forbidden by gauge invariance.

Second, the electric charge is the coefficient of the interaction between the photon and the conserved electron current. Starting from the bare interaction,

$$\begin{aligned} -e_0\bar{\psi}_0\gamma^\mu\psi_0 A_{0\mu} &= -\mu^{\epsilon/2} e\, Z_e Z_2 Z_3^{1/2}\bar{\psi}\gamma^\mu\psi A_\mu \\ &= -\mu^{\epsilon/2} e\, Z_1\bar{\psi}\gamma^\mu\psi A_\mu, \end{aligned} \tag{2.357}$$

so that

$$Z_1 = Z_e Z_2 Z_3^{1/2}. \tag{2.358}$$

Equivalently,

$$e_0 = \mu^{\epsilon/2} Z_e e, \qquad Z_e = \frac{Z_1}{Z_2 Z_3^{1/2}}. \tag{2.359}$$

The Ward identity gives

$$Z_1 = Z_2. \tag{2.360}$$

Thus the vertex renormalization and the electron field-strength renormalization cancel out of the electric charge. The result is

$$Z_e = Z_3^{-1/2}. \tag{2.361}$$

This is the algebraic form of the physical statement that QED charge renormalization is caused by vacuum polarization. The virtual charged pairs modify the photon propagator, and this modification changes the electric charge inferred from photon exchange.

The same statement can be seen directly from a current-current scattering amplitude. If two conserved external currents exchange a photon, then only the transverse photon propagator matters:

$$\mathrm{i}\mathcal{M}(q) = [-\mathrm{i}eJ_\mu(q)]D^{\mu\nu}_{\text{full}}(q)[-\mathrm{i}eJ'_\nu(-q)]. \tag{2.362}$$

Using

$$D^{\mu\nu}_{\text{full}}(q) = \frac{-\mathrm{i}}{q^2[1-\Pi_R(q^2)]}\left(g^{\mu\nu} - \frac{q^\mu q^\nu}{q^2}\right) + \text{longitudinal part}, \tag{2.363}$$

and current conservation,

$$q_\mu J^\mu(q) = 0, \qquad q_\nu J'^\nu(-q) = 0, \tag{2.364}$$

the longitudinal part drops out. Thus one can package the effect of vacuum polarization into an effective charge,

$$e^2_{\text{eff}}(q^2) = \frac{e^2}{1-\Pi_R(q^2)}. \tag{2.365}$$

For spacelike momentum transfer $q^2 = -Q^2$, this becomes

$$\alpha_{\text{eff}}(Q) = \frac{\alpha}{1-\Pi_R(-Q^2)}. \tag{2.366}$$

At large Q^2 compared with the electron mass, the one-loop vacuum-polarization function contains

$$\Pi_R(-Q^2) = \frac{\alpha}{3\pi} \ln \frac{Q^2}{m^2} + \text{scheme-dependent finite terms} + O(\alpha^2). \tag{2.367}$$

Therefore the effective charge grows when the probe momentum Q grows:

$$\alpha_{\text{eff}}(Q) \simeq \frac{\alpha}{1 - \dfrac{\alpha}{3\pi} \ln \dfrac{Q^2}{m^2}}. \tag{2.368}$$

This is the same physics as the real-space screening picture: long distances see a screened charge, while short distances see more of the unscreened charge.

At one loop,

$$\begin{aligned} Z_e &= \left(1 - \frac{e^2}{12\pi^2}\frac{1}{\bar{\epsilon}}\right)^{-1/2} + O(e^4) \\ &= 1 + \frac{e^2}{24\pi^2}\frac{1}{\bar{\epsilon}} + O(e^4). \end{aligned} \tag{2.369}$$

Since

$$\frac{1}{\bar{\epsilon}} = \frac{2}{\epsilon} - \gamma_E + \ln 4\pi, \tag{2.370}$$

the coefficient of the simple $1/\epsilon$ pole in Z_e is

$$Z_e = 1 + \frac{e^2}{12\pi^2}\frac{1}{\epsilon} + \text{finite MS convention terms} + O(e^4). \tag{2.371}$$

The bare charge does not depend on the arbitrary scale μ:

$$0 = \mu \frac{\mathrm{d}e_0}{\mathrm{d}\mu}. \tag{2.372}$$

Using

$$e_0 = \mu^{\epsilon/2} Z_e(e, \epsilon) e, \tag{2.373}$$

and extracting the coefficient of the simple pole in Z_e gives the one-loop beta function

$$\beta(e) \equiv \mu \frac{\mathrm{d}e}{\mathrm{d}\mu} = \frac{e^3}{12\pi^2} + O(e^5). \tag{2.374}$$

In terms of

$$\alpha = \frac{e^2}{4\pi}, \tag{2.375}$$

this is

$$\beta(\alpha) \equiv \mu \frac{\mathrm{d}\alpha}{\mathrm{d}\mu} = \frac{2}{3\pi}\alpha^2 + O(\alpha^3). \tag{2.376}$$

Integrating the one-loop equation between a reference scale μ and a momentum scale Q gives

$$\int_{\alpha(\mu)}^{\alpha(Q)} \frac{\mathrm{d}\alpha}{\alpha^2} = \frac{2}{3\pi} \int_{\mu}^{Q} \mathrm{d}\ln \mu', \tag{2.377}$$

$$\frac{1}{\alpha(Q)} = \frac{1}{\alpha(\mu)} - \frac{2}{3\pi} \ln \frac{Q}{\mu}. \tag{2.378}$$

Equivalently,

$$\alpha(Q) = \frac{\alpha(\mu)}{1 - \frac{2\alpha(\mu)}{3\pi}\ln\frac{Q}{\mu}} + O(\alpha^3). \tag{2.379}$$

Using $Q^2 = -q^2$ for spacelike momentum transfer, this is often written as

$$\alpha(Q) = \frac{\alpha(\mu)}{1 - \frac{\alpha(\mu)}{3\pi}\ln\frac{Q^2}{\mu^2}} + O(\alpha^3). \tag{2.380}$$

The positive sign of the QED beta function means that electric charge is screened in the infrared and becomes stronger in the ultraviolet. The one-loop formula would formally develop a Landau pole when the denominator vanishes, but that scale lies far outside the regime where perturbative QED alone should be trusted.

3.7 On-shell renormalization and physical interpretation

Minimal subtraction removes only ultraviolet poles. It is very convenient for deriving renormalization-group equations, but it does not define the parameters directly by physical measurements. In the on-shell scheme the renormalized mass is the pole of the electron propagator, the residue of that pole is one, and the electric charge is measured in the Thomson limit. These conditions are

$$S_R^{-1}(p)\big|_{\not p = m} = 0, \tag{2.381}$$

$$\frac{\partial S_R^{-1}(p)}{\partial \not p}\bigg|_{\not p = m} = 1, \tag{2.382}$$

$$\Pi_R(q^2 = 0) = 0, \tag{2.383}$$

$$\Gamma_R^\mu(p,p)\big|_{\not p = m} = \gamma^\mu. \tag{2.384}$$

The first two equations fix the mass and wave-function counterterms of the electron. The third equation fixes the normalization of the photon field so that the photon remains massless and the long-distance Coulomb law has the observed normalization. The last equation fixes the electric charge.

There is one extra subtlety in QED: the photon is massless, so on-shell Green functions can also contain infrared divergences. These are not UV renormalization divergences. They cancel in inclusive physical observables once real soft-photon emission is included. The UV counterterms derived above are local and are independent of that infrared physics.

The final picture is therefore simple. QED needs exactly four local renormalizations:

$$\psi \text{ field strength}, \qquad m \text{ mass}, \qquad A_\mu \text{ field strength}, \qquad e \text{ charge}. \tag{2.385}$$

The Ward identity reduces the number of independent charge-related renormalizations by enforcing $Z_1 = Z_2$, so the running of e is governed by Z_3, namely by vacuum polarization.

4. Wilson's approach to renormalization

4.1 Cut-off in high energy scale

The main idea of Wilson's renormalization is introducing a cut-off Λ and compute how the theory changes at different length scales. For that, we need to working with the generating functional with cut-off and successive integration over large momenta/small length scales:

$$Z = \int \mathcal{D}\phi_\Lambda \, e^{-S[\phi]}, \tag{2.386}$$

where we used Wick rotation changing to Euclidean space-time:

$$x^0 \to \mathrm{i}x^4, \quad \mathrm{d}^4x \to \mathrm{i}\mathrm{d}^4x_E. \tag{2.387}$$

Thus, the measurement is

$$\mathcal{D}\phi_\Lambda = \prod_{|k|<\Lambda} \mathrm{d}\phi(k). \tag{2.388}$$

In this case, the modes of ϕ are well defined in Euclidean space-time such that we can make sure that when $k \to \infty$ all the components of momentum also go to infinity. This step is to avoid in Minkowski space-time $k^2 = (k^0)^2 - (k^i)^2$. So when $k^2 \to \infty$, it could be $k^0 \to \infty$ and $k^i \to 0$, which doesn't suitable for our analysis.

Now, the action and Lagrangian become

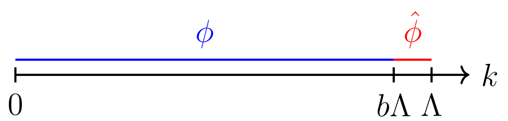
$$S[\phi] = \int \mathrm{d}^4x \mathcal{L}, \tag{2.389}$$

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4. \tag{2.390}$$

Suppose the mass $m \ll \Lambda$, and the coupling constant $\lambda \ll 1$, which means the last two terms in the Lagrangian can be treated as perturbation. For going to the perturbation theory, it's useful to split up fields $\phi \to \phi + \hat{\phi}$ as

$$\hat{\phi} = \begin{cases} \phi(k) & \text{for } b\Lambda \le |k| \le \Lambda \\ 0 & \text{for } |k| \le b\Lambda \end{cases}; \qquad \phi = \begin{cases} 0 & \text{for } b\Lambda \le |k| \le \Lambda \\ \phi(k) & \text{for } |k| \le b\Lambda \end{cases}, \tag{2.391}$$

where $(b\Lambda, \Lambda)$ is a very small interval controlled by parameter $b \le 1$



Then we can also split up the action as $S[\phi] \to S[\phi] + \hat{S}[\phi, \hat{\phi}]$, so we have

$$S[\phi] = \int \mathrm{d}^4x \frac{1}{2}\partial_\mu \left(\phi + \hat{\phi}\right) \partial^\mu \left(\phi + \hat{\phi}\right) + \frac{1}{2}m^2 \left(\phi + \hat{\phi}\right)^2 + \frac{\lambda}{4!}\left(\phi + \hat{\phi}\right)^4 \tag{2.392}$$

$$= \int \mathrm{d}^4x \frac{1}{2}\left(\partial_\mu\phi\partial^\mu\phi + 2\partial_\mu\phi\partial^\mu\hat{\phi} + \partial_\mu\hat{\phi}\partial^\mu\hat{\phi}\right) + \frac{1}{2}m^2\left(\phi^2 + \hat{\phi}^2 + 2\phi\hat{\phi}\right) \tag{2.393}$$

$$+ \frac{\lambda}{4!}\left(\phi^4 + \phi^2\hat{\phi}^2 + 2\phi^3\hat{\phi} + \hat{\phi}^2\phi^2 + \hat{\phi}^4 + 2\phi\hat{\phi}^3 + 2\phi^3\hat{\phi} + 2\phi\hat{\phi}^3 + 4\phi^2\hat{\phi}^2\right) \tag{2.394}$$

$$= \int \mathrm{d}^4x \left[\underbrace{\frac{1}{2}(\partial_\mu\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4}_{\mathcal{L}[\phi]} + \underbrace{\frac{1}{2}\left(\partial_\mu\hat{\phi}\right)^2 + \frac{1}{2}m^2\hat{\phi}^2 + \frac{\lambda}{4!}\left(6\phi^2\hat{\phi}^2 + 4\phi^3\hat{\phi} + 4\phi\hat{\phi}^3 + \hat{\phi}^4\right)}_{\hat{\mathcal{L}}[\phi,\hat{\phi}]}\right], \tag{2.395}$$

where we have dropped terms containing the products $\phi\hat{\phi}$ because Fourier components are orthogonal:

$$\int \frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\mathrm{d}^4k'}{(2\pi)^4}\phi(k)\hat{\phi}(k')e^{\mathrm{i}(k+k')\cdot x} \sim \delta^{(4)}(k+k') = 0 \tag{2.396}$$

Thus, we have

$$\hat{S}[\phi,\hat{\phi}] = \int \mathrm{d}^4x(\hat{\mathcal{L}}_0[\hat{\phi}] + \hat{\mathcal{L}}_{\text{int}}[\phi,\hat{\phi}]), \tag{2.397}$$

and

$$\hat{\mathcal{L}}_0[\hat{\phi}] = \frac{1}{2}\partial_\mu\hat{\phi}\partial^\mu\hat{\phi}, \tag{2.398}$$

$$\hat{\mathcal{L}}_{\text{int}}[\phi,\hat{\phi}] = \frac{1}{2}m^2\hat{\phi}^2 + \lambda\left(\frac{1}{3!}\phi\hat{\phi}^3 + \frac{1}{4}\phi^2\hat{\phi}^2 + \frac{1}{3!}\phi^3\hat{\phi} + \frac{1}{4!}\hat{\phi}^4\right). \tag{2.399}$$

Thus, the first part in $\hat{S}$ can be rewritten by the Fourier transformation as

$$\begin{aligned}\int \mathrm{d}^4x\,\hat{\mathcal{L}}_0[\hat{\phi}] &= \int \frac{\mathrm{d}^4k\mathrm{d}^4p}{(2\pi)^8}\left(\mathrm{i}k_\mu\hat{\phi}(k)\right)\left(\mathrm{i}p^\mu\hat{\phi}(p)\right)\int \mathrm{d}^4x\,e^{\mathrm{i}(k+p)\cdot x} && (2.400)\\ &= \int \frac{\mathrm{d}^4k\mathrm{d}^4p}{(2\pi)^8}(\mathrm{i}k_\mu)(\mathrm{i}p^\mu)\hat{\phi}(k)\hat{\phi}(p)\,(2\pi)^4\delta^{(4)}\,(k+p) && (2.401)\\ &= \frac{1}{2}\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\hat{\phi}(-k)k^2\hat{\phi}(k) && (2.402)\\ &= \frac{1}{2}\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\hat{\phi}^*(k)k^2\hat{\phi}(k). && (2.403)\end{aligned}$$

This Gaussian integral form remind us to write a propagator for $\hat{\phi}$ from Wick contraction of $\hat{\phi}$ fields:

$$\overbrace{\hat{\phi}(k)\hat{\phi}(p)} = \langle\hat{\phi}(k)\hat{\phi}(p)\rangle = \frac{\int \mathcal{D}\hat{\phi}\,\hat{\phi}(k)\hat{\phi}(p)e^{-S_0[\hat{\phi}]}}{\int \mathcal{D}\hat{\phi}e^{-S_0[\hat{\phi}]}} = \frac{1}{k^2}(2\pi)^4\delta^{(4)}(k+p)\Theta(k), \tag{2.404}$$

where

$$\Theta(k) = \begin{cases}1 & \text{for } b\Lambda < |k| < \Lambda\\ 0 & \text{else}\end{cases}. \tag{2.405}$$

4.2 Effective action and ideas about renormalization group

Therefore, we can separate the generating functional measurement with the **high energy modes** $\hat{\phi}$ and **low energy modes** $\phi_{b\Lambda}$ as $\mathcal{D}\phi_\Lambda = \mathcal{D}\phi_{b\Lambda}\mathcal{D}\hat{\phi}_\Lambda$. And we can integrate the high energy modes

$$\begin{aligned}Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]} = \int \mathcal{D}\phi_{b\Lambda}\mathcal{D}\hat{\phi}_\Lambda e^{-S[\phi]-\hat{S}[\phi,\hat{\phi}]} &= \int \mathcal{D}\phi_{b\Lambda}e^{-S[\phi]}\left(\int \mathcal{D}\hat{\phi}_\Lambda e^{-\hat{S}[\phi,\hat{\phi}]}\right) && (2.406)\\ = \int \mathcal{D}\phi_{b\Lambda}e^{-S_{\text{eff}}[\phi]}, &&& (2.407)\end{aligned}$$

where left a **effective action** defined as

$$S_{\text{eff}}[\phi] = S[\phi] + \delta S[\phi], \tag{2.408}$$

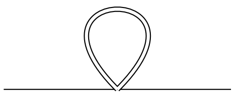
in which $\delta S[\phi]$ can be treated as perturbation, thus it can be expanded as

$$
\begin{aligned}
e^{-\delta S[\phi]} &= \int \mathcal{D}\hat{\phi}_\Lambda e^{-\hat{S}[\phi,\hat{\phi}]} = \int \mathcal{D}\hat{\phi}_\Lambda e^{-\int \mathrm{d}^4x \hat{\mathcal{L}}[\phi,\hat{\phi}]} && (2.409)\\
&\simeq \int \mathcal{D}\hat{\phi}_\Lambda \int \mathrm{d}^4x \left[1-\left(\hat{\mathcal{L}}_0+\frac{1}{2}m^2\hat{\phi}^2+\frac{\lambda}{3!}\phi\hat{\phi}^3+\frac{\lambda}{4}\phi^2\hat{\phi}^2+\frac{\lambda}{3!}\phi^3\hat{\phi}+\frac{\lambda}{4!}\hat{\phi}^4\right)+\mathcal{O}(\lambda^2)\right] && (2.410)\\
&= \int \mathcal{D}\hat{\phi}_\Lambda \int \mathrm{d}^4x \Bigg[1-\underbrace{\left(\hat{\mathcal{L}}_0+\frac{1}{2}m^2\hat{\phi}^2+\frac{\lambda}{4!}\hat{\phi}^4\right)}_{\substack{\text{Doesn't contain } \phi,\\ \text{which has no contribution for running}}} -\frac{\lambda}{4}\phi^2\hat{\phi}^2+\mathcal{O}(\lambda^2)\Bigg], && (2.411)
\end{aligned}
$$

where in the third step we have dropped all the odd terms which don't contribute to the integral. The introducing of $\delta S[\phi]$ will change the parameters in the original action $S[\phi]$, which will finally leading the parameters start running. To show this, we now need to work in the perturbation scenario. Since we have assumed that $m \ll \Lambda$ and $\lambda \ll 1$, so we can expand $e^{-S_{\text{eff}}[\phi]}$ with Dyson series in terms of λ. For that, we can generally write the effective Lagrangian $\mathcal{L}_{\text{eff}}$ for effective action $S_{\text{eff}}[\phi]$ as:

$$
\mathcal{L}_{\text{eff}}[\phi] = \frac{1}{2}(1+\Delta Z)\partial_\mu\phi\partial^\mu\phi+\frac{1}{2}(m^2+\Delta m^2)\phi^2+\frac{1}{4!}(\lambda+\Delta\lambda)\phi^4+(\text{higher dimensional operators}). \tag{2.412}
$$

Now the mission is to determine ΔZ, Δm^2 and $\Delta\lambda$. We firstly consider the 1-loop order $\mathcal{O}(\lambda)$ correction for the propagator, which corresponds to the term $\frac{1}{4}\lambda\phi^2\hat{\phi}^2$, the graph is :

$I_1 = \phi$  ϕ , (2.413)

$\mathcal{O}(\lambda)$

where the double line represent the contraction of $\hat{\phi}$, which doesn't contribute the real physical observable, thus it needs to be integrated. Thus, we have

$$
\begin{aligned}
I_1 &= -\frac{\lambda}{4}\int \mathrm{d}^4x \phi^2 \overbracket{\hat{\phi}\hat{\phi}} && (2.414)\\
&= -\frac{\lambda}{4}\int \mathrm{d}^4x \int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\cdots\frac{\mathrm{d}^4k_4}{(2\pi)^4} e^{\mathrm{i}(k_1+\cdots+k_4)\cdot x}\phi(k_1)\phi(k_2)\frac{1}{k_3^2}(2\pi)^4\delta^{(4)}(k_3+k_4)\Theta(k_3), && (2.415)
\end{aligned}
$$

integrate for x, k_4:

$$
\begin{aligned}
I_1 &= -\frac{\lambda}{4}\int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\cdots\frac{\mathrm{d}^4k_3}{(2\pi)^4}\phi(k_1)\phi(k_2)\frac{1}{k_3^2}(2\pi)^4\delta^{(4)}(k_1+k_2)\Theta(k_3) && (2.416)\\
&= -\frac{1}{2}\mu\int \frac{\mathrm{d}^4k_1}{(2\pi)^4}\phi(k_1)\phi(-k_1), && (2.417)
\end{aligned}
$$

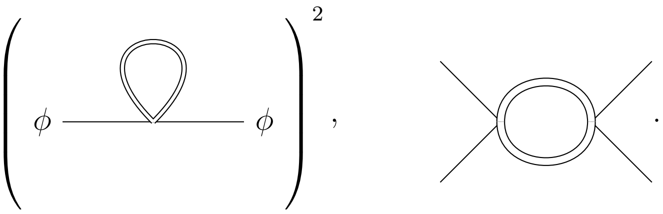
where

$$
\begin{aligned}
\mu &= \frac{\lambda}{2}\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\frac{\Theta(k)}{k^2} && (2.418)\\
&= \frac{\lambda}{2}\frac{1}{(2\pi)^4}\int \mathrm{d}\Omega\int_{b\Lambda}^{\Lambda} k\mathrm{d}k && (2.419)\\
&= \frac{\lambda}{32\pi^2}(1-b^2)\Lambda^2. && (2.420)
\end{aligned}
$$

I_1 has structure of mass correction, which can be put in exponent of effective action as

$$\exp\left\{-\int \mathrm{d}^4x \frac{1}{2}\mu\phi^2\right\}. \tag{2.421}$$

And the $\mathcal{O}(\lambda^2)$ order graphs therefore are:



For the first one, it's nothing but the disconnected graph $\sim \frac{1}{2}I_1^2$, which contribute $\mathcal{O}(\lambda^2)$ in the exponent. The second graph represent the correction for the ϕ^4 interaction vertex:

$$I_2 = -\frac{1}{4!}\xi \int \mathrm{d}^4x\phi^4(x), \tag{2.422}$$

factorizing the loop integral, then we have

$$\xi = -4! \cdot \underbrace{2}_{\substack{\text{2 possible}\\ \text{contractions}\\ \text{of } \phi^2(\hat{\phi}\hat{\phi})^2}} \cdot \underbrace{\frac{1}{2!}}_{\substack{\text{symmetry factor}\\ \text{of diagram}}} \cdot \left(\frac{\lambda}{4}\right)^2 \int_{b\Lambda<|k|<\Lambda} \frac{\mathrm{d}^4k}{(2\pi)^4}\left(\frac{1}{k^2}\right)^2 \Theta(k) = -\frac{3\lambda^2}{16\pi^2}\ln\frac{1}{b} \tag{2.423}$$

Thus, we find μ, ξ can be interpreted as combinations to $\Delta m^2, \Delta\lambda$.

Therefore, the generating functional $Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]}$ changing to $Z = \int \mathcal{D}\phi_{b\Lambda} e^{-S_{\text{eff}}[\phi]}$ is by rescaling S to effective action S_{eff} by changing the coordinates and momenta:

$$k' = \frac{k}{b} \quad , \quad x' = bx \tag{2.424}$$

Thus,

$$S_{\text{eff}} = \int \mathrm{d}^4x' \frac{1}{b^4}\left(\frac{1}{2}(1+\Delta Z)b^2\partial'_\mu\phi\partial'^\mu\phi + \frac{1}{2}(m^2+\Delta m^2)\phi^2 + \frac{1}{4!}(\lambda+\Delta\lambda)\phi^4 + \underbrace{\cdots + \frac{1}{n!}(\lambda_n+\Delta\lambda_n)\phi^n + \dots}_{\text{higher dim. operators}}\right) \tag{2.425}$$

$$= \int \mathrm{d}^4x' \left(\frac{1}{2}\partial'_\mu\phi'\partial'^\mu\phi' + \frac{1}{2}m'^2\phi'^2 + \frac{1}{4!}\lambda'\phi'^4 + \cdots + \frac{1}{n!}\lambda'_n(\phi')^n + \dots\right), \tag{2.426}$$

where

$$\phi' = \sqrt{(1+\Delta Z)b^{-2}}\phi \tag{2.427}$$
$$m'^2 = (m^2+\Delta m^2)(1+\Delta Z)^{-1}b^{-2} \tag{2.428}$$
$$\lambda' = (\lambda+\Delta\lambda)(1+\Delta Z)^{-2}b^0 \tag{2.429}$$
$$\lambda'_n = (\lambda_n+\Delta\lambda_n)(1+\Delta Z)^{-n/2}b^{n-4}. \tag{2.430}$$

This process, mapping $\phi' \to \phi$, $m'^2 \to m^2$, $\lambda' \to \lambda$, is exactly the **renormalization group transformations**. We can also explicitly show the relations for rescaling of parameters for mapping $x' = bx$ on $D = 4$ dimension:

$$m'^2 = b^{-2}\Delta m^2 \tag{2.431}$$
$$\lambda' = b^0\Delta\lambda \tag{2.432}$$
$$\lambda'_n = b^{n-4}\Delta\lambda_n \tag{2.433}$$

Generally, in D dimensions, we have:

$$m'^2 = b^{-2}\Delta m^2 \tag{2.434}$$
$$\lambda' = b^{D-4}\Delta\lambda \tag{2.435}$$
$$\lambda'_n = b^{D-4+n-4}\Delta\lambda_n \tag{2.436}$$

for $b < 1$ on $D = 4$:

- Super-renormalizable m^2 grows as $b^{-2} \longleftrightarrow$ relevant operator ϕ^2.
- Renormalizable λ scales as $b^0 \longleftrightarrow$ marginal operators $\lambda\phi^4$.
- Non-renormalizable λ_n vanishes as $b^{n-4} \longleftrightarrow$ irrelevant operators $\lambda_n\phi^n, n > 4$.

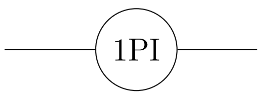
$$\lambda' = \lambda - \frac{3}{16\pi^2}\lambda^2 \ln\left(\frac{1}{b}\right) + O(\lambda^3) \tag{2.437}$$

5. Callan-Symanzik Equation

Goal: Quantify scale dependence of Green's functions $G^{(n)}(x_1 \dots x_n, M)$, where M sets the scale of external momenta, which leads to the Callan-Symanzik equation.

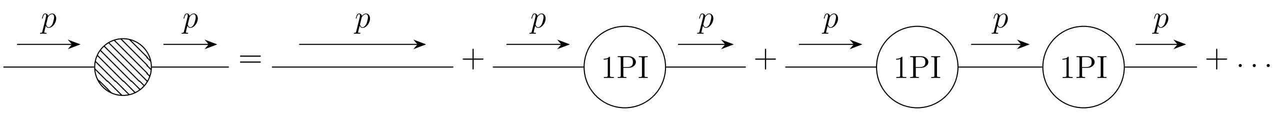
5.1 Alternative choice for renormalization in ϕ^4-theory

In ϕ^4-theory, the 1-PI graphs of propagator can be renormalized by taking the remormalization scheme. We simply recall this process again, firstly, taking



$$= -\mathrm{i}\Sigma(p^2), \tag{2.438}$$

for all 1-PI contributions to propagators. Thus, the full propagator goes like:

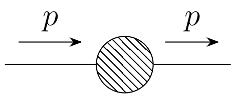


$$= \frac{\mathrm{i}}{p^2 - m^2 - \Sigma(p^2)}. \tag{2.439}$$

Here we took the on-shell scheme:

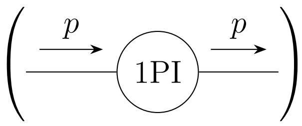
$$\Sigma(p^2)\big|_{p^2=m_{ph}^2} = 0, \quad \frac{\mathrm{d}}{\mathrm{d}p^2}\Sigma(p^2)\big|_{p^2=m_{ph}^2} = 0. \tag{2.440}$$

This scheme force the residue around the pole to be unity. Then we obtained the renormalized propagator:

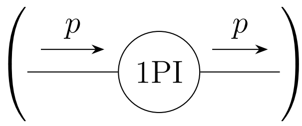


$$= \frac{\mathrm{i}}{p^2 - m_{ph}^2}. \tag{2.441}$$

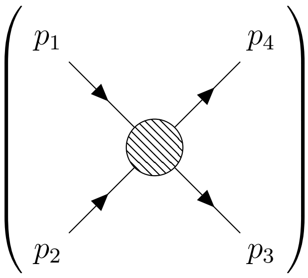
But this is not physical in many cases, say QCD and SM due to many of problems. For example, since the quarks are confined so normally we can hardly define the asymptotic state to imagine a free quark. And the massless field of SM in this scheme will meet IR-divergence, which will also lead us to many unnecessary problems. So we need to choose other better schemes for a more general model. One alternative is to define counterterms at arbitrary scale M, at the external space-like momenta $p^2 = -M^2 < 0$, we take



$$_{p^2=-M^2} = 0 \quad , \quad \frac{\mathrm{d}}{\mathrm{d}p^2}$$



$$_{p^2=-M^2} = 0,$$



$$_{s=t=u=-M^2} = \mathrm{i}\lambda. \tag{2.442}$$

where $(p_1+p_2)^2 = s$, $(p_1+p_3)^2 = t$, $(p_1+p_4)^2 = u$. The reason why we choose the space-like momentum is to avoid poles. If we choose a time-like momentum, it for sure corresponds to physical particle, however also having divergence. The first condition $\Sigma(-M^2) = 0$ requires the correction for mass Σ_{finite} is 0 at this non-fixing scale M, and all the divergence Σ_{div} will be subtracted by the counterterms. So basically it looks very similar with the on-shell condition, the only thing that we change is changing the fixed physical mass m^2 to the unphysical mass scale $-M^2$.

This arbitrary parameter M is called the **renormalization scaling**. These conditions defined the only certain vale at a fixed point, which canceled all the divergent. So we say **it defines the theory in scale $\boldsymbol{M}$**.

For seeing this scheme is indeed valid, we firstly look at the Green's function with renormalized fields $\phi = \sqrt{Z}\phi_r$ and coupling λ, and taking the renormalized scale as M, we have

$$\begin{aligned} G^{(n)}_{\text{ren}}(x_1 \dots x_n, \lambda, M) &= \langle \Omega | T\{\phi_r(x_1)\dots\phi_r(x_n)\}|\Omega\rangle \\ &= Z^{-n/2}\langle \Omega | T\{\phi(x_1)\dots\phi(x_n)\}|\Omega\rangle \\ &= Z^{-n/2} \underbrace{G^{(n)}_{\text{bare}}(x_1\dots x_n,\lambda,M)}_{\text{bare Green's fun. independent of } M} \quad . \end{aligned} \tag{2.443}$$

Now we change variables as $M \to M + \delta M$. This changing of our scaling will also change the other parameters to satisfy the renormalization condition. Thus it leads to $\lambda \to \lambda + \delta\lambda$ and $\phi_{\text{ren}} \to \phi'_{\text{ren}} = (1+\delta\eta)\phi_{\text{ren}}$, thus

$$(Z')^{-1/2}\phi_0 = Z^{-1/2}(1+\delta\eta)\phi_0 \implies Z' = Z(1+\delta\eta)^{-2} \simeq Z(1-2\delta\eta) \equiv Z(1+\delta Z), \tag{2.444}$$

where $\delta\eta = \frac{-\delta Z}{2}$. The latter gives the way of Green's function changing:

$$G^{(n)}_{\text{ren}} \to Z^{-n/2}(1+\delta Z)^{-n/2} G^{(n)}_{\text{bare}} \tag{2.445}$$
$$\simeq Z^{-n/2}(1-\frac{n}{2}\delta Z) G^{(n)}_{\text{bare}} \tag{2.446}$$
$$= Z^{-n/2}(1+n\delta\eta) G^{(n)}_{\text{bare}} \tag{2.447}$$
$$= (1+n\delta\eta) G^{(n)}_{\text{ren}}, \tag{2.448}$$

We will ignore the corner index "ren" in $G^{(n)}_{\text{ren}}$, **namely the renormalized Green's function denotes as $G^{(n)}$.** Thus from $\delta G_{\text{bare}} = 0$, we have

$$0 = \delta(Z^{n/2}G^{(n)}) = \underbrace{(\delta Z^{n/2})G^{(n)}}_{n\delta\eta Z^{n/2}} + Z^{n/2}\frac{\partial}{\partial M}G^{(n)}\delta M + Z^{n/2}\frac{\partial}{\partial\lambda}G^{(n)}\delta\lambda. \tag{2.449}$$

Introducing the new variables:

$$\beta = \frac{M}{\delta M}\delta\lambda, \quad \gamma = \frac{M}{\delta M}\delta\eta \tag{2.450}$$

Thus, dropping the arbitrary δM, we obtain **the Callan-Symanzik equation**:

$$\left(M\frac{\partial}{\partial M} + \beta\frac{\partial}{\partial\lambda} + n\gamma\right) G^{(n)}(x_1 \dots x_n, \lambda, M) = 0, \tag{2.451}$$

where β and γ are universal for a given theory, which is independent for particular Green's function. This also means that these parameters are independent of any UV-cut off (dimensionless).

$$\beta = \beta(\lambda), \qquad \gamma = \gamma(\lambda). \tag{2.452}$$

We can also directly compute $\mathrm{d}G^{(n)}$ as

$$\mathrm{d}G^{(n)} = (1 + n\delta\eta)G^{(n)} - G^{(n)} = n\delta\eta G^{(n)} = n\frac{\mathrm{d}M}{M}\left(M\frac{\mathrm{d}\eta}{\mathrm{d}M}\right)G^{(n)} \tag{2.453}$$

$$= \frac{\partial G^{(n)}}{\partial M}\mathrm{d}M + \frac{\partial G^{(n)}}{\partial\lambda}\mathrm{d}\lambda \tag{2.454}$$

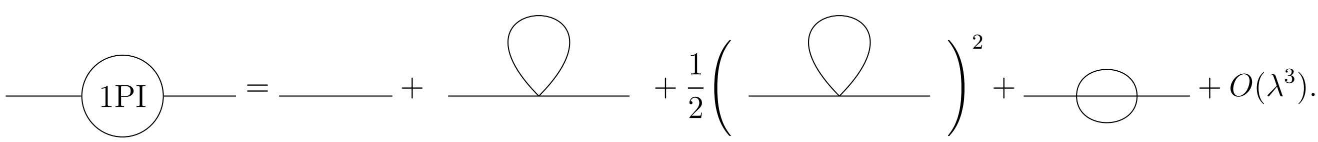
$$= \frac{\partial G^{(n)}}{\partial M}\mathrm{d}M + \left(M\frac{\mathrm{d}\lambda}{\mathrm{d}M}\right)\frac{\mathrm{d}M}{M}\frac{\partial G^{(n)}}{\partial\lambda} \tag{2.455}$$

$$= \frac{\mathrm{d}M}{M}\left[M\frac{\partial}{\partial M} + \left(M\frac{\mathrm{d}\lambda}{\mathrm{d}M}\right)\frac{\partial}{\partial\lambda}\right]G^{(n)}, \tag{2.456}$$

from this construction, we can also obtain the Callan-Symanzik equation.

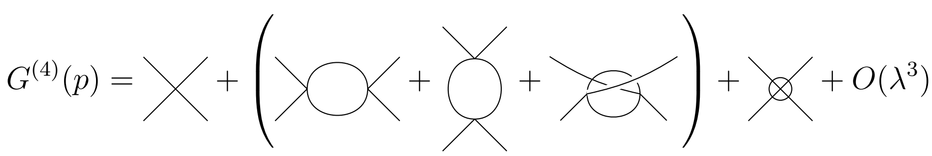
5.2 β-function in massless ϕ^4-theory

Now let's focus some actual cases to find the β-function and the anomalous dimension $\gamma(\lambda)$. For simplicity, we consider the 1-loop correction for **massless ϕ^4-theory**. The Callan-Symanzik equation for $G^{(2)}$ gives us $\gamma = 0 + O(\lambda^2)$, because in 1-loop order, there's no contribution to Z, but only to δm^2. Only until 2-loops or higher orders which contribute Z and also of course δm^2.



(2.457)

For $\beta(\lambda)$ at $O(\lambda)$, we need to consider the 4-point function correction:



(2.458)

$$= -\mathrm{i}\lambda + (-\mathrm{i}\lambda)^2(\mathrm{i}V(t) + \mathrm{i}V(s) + \mathrm{i}V(u)) - \mathrm{i}\delta\lambda + O(\lambda^3), \tag{2.459}$$

where the renormalization condition are taken in $t = s = u = -M^2$. So for each diagram vertex correction, for $m = 0$ we have:

$$V(p^2) = \frac{1}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{i}}{k^2}\frac{\mathrm{i}}{(p-k)^2}. \tag{2.460}$$

Counterterm $\delta\lambda$ determined from amputated Green's function

$$G^{(4)}_{\text{amp}}\big|_{s=t=u=-M^2} \to \delta\lambda = (-\mathrm{i}\lambda)^2 3V(-M^2). \tag{2.461}$$

Take $D = 4 - 2\epsilon$:

$$V(-M^2) = \lim_{\epsilon\to 0} \frac{1}{2}\frac{-1}{(4\pi)^{d/2}} \int_0^1 \mathrm{d}x \, \frac{\Gamma(\epsilon)}{(x(1-x)M^2)^\epsilon} \tag{2.462}$$

$$= -\frac{1}{32\pi^2}\left(\frac{1}{\epsilon} - \ln M^2 + O(1)\right). \tag{2.463}$$

Thus

$$\delta\lambda = \frac{3\lambda^2}{32\pi^2}\left(\frac{1}{\epsilon} - \ln M^2 + O(1)\right). \tag{2.464}$$

This will lead to

$$G^{(4)} = -\mathrm{i}\lambda + 3\mathrm{i}(\mathrm{i}\lambda)^2 V(p^2) - \mathrm{i}\delta\lambda + \mathcal{O}(\lambda^3) \tag{2.465}$$

$$= -\mathrm{i}\lambda + \frac{3\mathrm{i}\lambda^2}{32\pi^2}\left(\frac{1}{\epsilon} - \ln p^2 - \frac{1}{\epsilon} + \ln M^2\right) + \mathcal{O}(\lambda^3) \tag{2.466}$$

$$= -\mathrm{i}\lambda + \frac{3\mathrm{i}\lambda^2}{32\pi^2}\ln\left(\frac{M^2}{p^2}\right) + \mathcal{O}(\lambda^3) \tag{2.467}$$

Thus, we have

$$M\frac{\partial}{\partial M}G^{(4)} = \frac{3\mathrm{i}\lambda^2}{16\pi^2} + O(\lambda^3). \tag{2.468}$$

Thus, bring it into the Callan-Symanzik equation, we have

$$-\beta(\lambda)\frac{\partial}{\partial\lambda}G^{(4)} = \mathrm{i}\frac{3\lambda^2}{16\pi^2} + O(\lambda^3) \tag{2.469}$$

where $\gamma = 0 + O(\lambda^2)$. Since we only consider order $\mathcal{O}(\lambda^2)$, therefore we only need to expand $G^{(4)}$ into $\mathcal{O}(\lambda)$, i.e.,

$$\frac{\partial}{\partial\lambda}G^{(4)} = -\mathrm{i} + \frac{3\mathrm{i}\lambda}{16\pi^2}\ln\left(\frac{M^2}{p^2}\right) + \mathcal{O}(\lambda^2) \simeq -\mathrm{i} + \mathcal{O}(\lambda), \tag{2.470}$$

thus the β-function at 1-loop is:

$$\beta(\lambda) = \frac{3}{16\pi^2}\lambda^2 + O(\lambda^3). \tag{2.471}$$

5.3 Analogous analysis for massless QED

Now we consider the QED, which as a good example to calculate the anomalous dimension $\gamma(\lambda)$. In QED, we need to renormalize the vertex(Z_1) and the propagator for electron(Z_2) and photon(Z_3). These three parameters defined the electron charge renormalization:

$$e_0 = Z_1^{-1} Z_2 Z_3^{1/2} e. \tag{2.472}$$

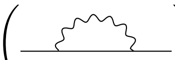
The Green's function $G^{(n_e, n_A)}(x_1 \ldots x_n, e, M)$ now contains two fields, so the Callan-Symanzik equation is

$$\left(M\frac{\partial}{\partial M} + \beta(e)\frac{\partial}{\partial e} + n_e\gamma_2 + n_A\gamma_3\right) G^{(n_e, n_A)}(x_1 \ldots x_n, e, M) = 0, \tag{2.473}$$

where n_e denote for the electron fields, n_A for photon fields. And γ_2, γ_3 are anomalous dimensions of electron and photon field respectively

$$\gamma_i = \frac{1}{2} M \frac{\partial}{\partial M} \delta_i, \quad i = 2, 3. \tag{2.474}$$

The γ_i comes from the field renormalization, which related with Z_2 and Z_3. And $Z_i = 1 + \delta_i$, where δ_i is the counterterms of fields. So for $i = 2$, i.e., the electron field

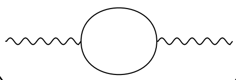
$$\delta_2 = \frac{\mathrm{d}}{\mathrm{d}\not{p}} \Sigma(p)\Big|_{p^2=-M^2} = \frac{\mathrm{d}}{\mathrm{d}\not{p}} \Bigg($$  $$\Bigg)$$

$$\begin{aligned} &= \frac{-e^2}{16\pi^2} \left(\frac{\Gamma(\epsilon)}{(M^2)^{\epsilon/2}} + O(1) \right) \simeq \frac{-e^2}{16\pi^2} \left(\frac{2}{\epsilon} - \gamma_E \right) (1 - \epsilon \ln M) \\ &= \frac{-e^2}{16\pi^2} \left(\frac{2}{\epsilon} - \gamma_E - 2 \ln M \right), \end{aligned} \tag{2.475}$$

which leads to

$$\gamma_2 = \frac{e^2}{16\pi^2} \equiv \frac{\alpha}{4\pi}. \tag{2.476}$$

Also $i = 3$ is for the photon field

$$\delta_3 = \Pi(p^2)\Big|_{p^2=-M^2} = \Bigg($$  $$\Bigg) = \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right) \Pi(p^2)$$

$$= \frac{-e^2}{12\pi^2} \left(\frac{\Gamma(\epsilon)}{(M^2)^{\epsilon/2}} + O(1) \right) \simeq \frac{-e^2}{12\pi^2} \left(\frac{2}{\epsilon} - \gamma_E - 2 \ln M \right). \tag{2.477}$$

Thus,

$$\gamma_3 = \frac{e^2}{12\pi^2} \equiv \frac{\alpha}{4\pi} \left(\frac{4}{3} \right). \tag{2.478}$$

For calculating the β-function, we first briefly review a very important solution in QED renormalization. Consider the Ward-Takahashi identity

$$(p - q)_\mu \Gamma^\mu(p, q) = S_{\mathrm{F}}^{-1}(p) - S_{\mathrm{F}}^{-1}(q), \tag{2.479}$$

when we take the limit that $p \to q$ and move it to the right, thus we have

$$\Gamma^\mu(p, p) = \frac{\partial S_{\mathrm{F}}^{-1}(p)}{\partial p_\mu}. \tag{2.480}$$

Notice $S_{\mathrm{F}}^{-1}(p) = \gamma^\mu p_\mu - m - \Sigma(p)$, thus we have

$$\Gamma^\mu(p, p) = \gamma^\mu - \frac{\partial \Sigma(p)}{\partial p_\mu}. \tag{2.481}$$

Beside the momentum p, the propagator can be renormalized as

$$S_{\mathrm{F}}^{-1}(p) = Z_2^{-1}(\gamma \cdot p - m_{\mathrm{ph}}) + \dots, \tag{2.482}$$

which means

$$\frac{\partial S_{\mathrm{F}}^{-1}}{\partial p_\mu} \approx Z_2^{-1} \gamma^\mu. \tag{2.483}$$

And for the renormalized vertex, we have

$$\Gamma^\mu = Z_1^{-1}\Gamma_R^\mu. \tag{2.484}$$

In low energy approximation, the vertex just $\Gamma_R^\mu \to \gamma^\mu$. Thus, comparing (2.481) and (2.484), we have

$$Z_1 = Z_2. \tag{2.485}$$

Thus, $\delta_1 = \delta_2$. Then we immediately have

$$0 = \frac{\partial e_0}{\partial M} = \frac{\partial}{\partial M}\left[e\left(1-\delta_1+\delta_2+\frac{1}{2}\delta_3\right)\right] \tag{2.486}$$

$$= e\frac{\partial}{\partial M}\left(1-\delta_1+\delta_2+\frac{1}{2}\delta_3\right) + \left(1-\delta_1+\delta_2+\frac{1}{2}\delta_3\right)\frac{\partial e}{\partial M} \tag{2.487}$$

$$\simeq e\frac{\partial}{\partial M}\left(-\delta_1+\delta_2+\frac{1}{2}\delta_3\right) + \frac{\partial e}{\partial M}. \tag{2.488}$$

Therefore, we get the final result for the β-function:

$$\begin{aligned}\beta(e) &= M\frac{\partial}{\partial M}e = -eM\frac{\partial}{\partial M}\left(\underbrace{-\delta_1+\delta_2}_{\delta_1=\delta_2}+\frac{1}{2}\delta_3\right)\\ &= \frac{e^3}{12\pi^2} \equiv e\frac{\alpha}{4\pi}\left(\frac{4}{3}\right).\end{aligned} \tag{2.489}$$

6. Evolution of coupling

6.1 Solution of Callan-Symanzik equation

Now we would like to solve the solutions of Callan-Symanzik equation. We firstly start with the 2-point function $G^{(2)}(p)$, which generally can be written as:

$$G^{(2)}(p) = \frac{\mathrm{i}}{p^2}f\left(\frac{-p^2}{M^2}\right), \tag{2.490}$$

where $f\left(\frac{-p^2}{M^2}\right)$ is a dimensionless function, which only depends on $\frac{-p^2}{M^2}$. In momentum space, without the amputation, which means we add the external lines to contribute the momentum p in the propagator. We have:

$$M\frac{\partial}{\partial M}G^{(2)}(p) = \frac{\mathrm{i}}{p^2}\left(\frac{2p^2}{M^2}\right)f'\left(\frac{-p^2}{M^2}\right), \tag{2.491}$$

$$p\frac{\partial}{\partial p}G^{(2)}(p) = -2G^{(2)}(p) - \frac{\mathrm{i}}{p^2}\left(\frac{2p^2}{M^2}\right)f'\left(\frac{-p^2}{M^2}\right). \tag{2.492}$$

Therefore, we have:

$$M\frac{\partial}{\partial M}G^{(2)}(p) = \left(-p\frac{\partial}{\partial p}-2\right)G^{(2)}(p). \tag{2.493}$$

Putting this results into the Callan-Symanzik equation for 2-point function, we then have:

$$\left(p\frac{\partial}{\partial p}-\beta\frac{\partial}{\partial\lambda}+2-2\gamma(\lambda)\right)G^{(2)}(p^2,\lambda,M) = 0. \tag{2.494}$$

This is an equation for a current. We can firstly write the solution in free theory, i.e. $\lambda = 0$ which means $\beta = \gamma = 0$:

$$G^{(2)}(p) = \frac{\mathrm{i}}{p^2}\bar{G}_0, \tag{2.495}$$

where $\bar{G}_0$ is a constant function in p^2. But one can not happy if we only have the free theory. Therefore, we need to solve the renormalization group equation with a more general form. For that, we firstly introduce some new variables to make the PDE simpler:

$$t = \ln\left(\frac{p}{M}\right), \tag{2.496}$$
$$v(x) = -\beta(\lambda), \tag{2.497}$$
$$\rho(x) = 2\gamma(\lambda) - 2, \tag{2.498}$$
$$x = \lambda. \tag{2.499}$$

Now we call the new function as $D(t,x)$, which is nothing but the $G^{(2)}$. So (2.494) becomes

$$\left(\frac{\partial}{\partial t} + v(x)\frac{\partial}{\partial x} - \rho(x)\right) D(t,x) = 0. \tag{2.500}$$

The solution of $D(t,x)$ is somehow like a current with the current density

$$j(t,x) = \rho(t,x)v(t,x). \tag{2.501}$$

So the solution of a current can be expressed as

$$D(t,x) = D_0(\bar{x}(t,x))\exp\left[\int_0^t \mathrm{d}t'\,\rho(\bar{x}(t',x))\right], \tag{2.502}$$

with the boundary condition and the initial condition:

$$\frac{\partial}{\partial t'}\bar{x}(t',x) = -v(\bar{x}), \qquad \bar{x}(0,x) = x, \tag{2.503}$$

where $\bar{x}(t,x)$ is the position of current volum element at t and x. So the Callan-Symanzik equation has the solution:

$$G^{(2)}(p^2,\lambda,M) = \bar{G}_0(\bar{\lambda}(p,\lambda))\exp\left[\int_{p'=M}^{p'=p} \mathrm{d}\ln\left(\frac{p'}{M}\right) 2\left[1 - \bar{\gamma}\left(\bar{\lambda}(p',\lambda)\right)\right]\right]. \tag{2.504}$$

It is important to note that $\bar{G}_0(\bar{\lambda}(p,\lambda))$ is an unknown function, which does not directly corresponds to the propagator. To fix this function, we firstly put (2.490) into (2.494):

$$0 = \left(p\frac{\partial}{\partial p} - \beta\frac{\partial}{\partial\lambda} + 2 - 2\gamma(\lambda)\right)\left[\frac{\mathrm{i}}{p^2} f\left(\frac{p^2}{M^2},\lambda\right)\right] \tag{2.505}$$
$$= \frac{\mathrm{i}}{p^2}\left(-2 + p\frac{\partial}{\partial p} - \beta\frac{\partial}{\partial\lambda} + 2 - 2\gamma(\lambda)\right) f\left(\frac{p^2}{M^2},\lambda\right) \tag{2.506}$$

Thus, we have

$$\left(p\frac{\partial}{\partial p} - \beta\frac{\partial}{\partial\lambda} - 2\gamma(\lambda)\right) f\left(\frac{p^2}{M^2},\lambda\right) = 0. \tag{2.507}$$

So the solution is similar with $G^{(2)}$:

$$f\left(\frac{p^2}{M^2},\lambda\right) = \bar{G}_0(\bar{\lambda}(p,\lambda))\exp\left[-2\int_M^p \mathrm{d}\ln\left(\frac{p'}{M}\right)\gamma(\bar{\lambda})\right]. \tag{2.508}$$

Therefore, we can take the another form of the solution by combine the solution of f and (2.490)

$$G^{(2)}(p^2,\lambda,M) = \frac{\mathrm{i}}{p^2}\bar{G}_0(\bar{\lambda}(p,\lambda))\exp\left[-2\int_M^p \mathrm{d}\ln\left(\frac{p'}{M}\right)\gamma(\bar{\lambda}(p',\lambda))\right]. \tag{2.509}$$

Now, we can take the boundary $p = M$, which all the integration part will vanish. So we have

$$\bar{G}_0(\lambda) = -\mathrm{i}M^2 \cdot G^{(2)}(M^2,\lambda,M). \tag{2.510}$$

We can then go to the specific case and use the perturbative expansion of propagator to get each order of $\bar{G}_0(\lambda)$.

6.2 Renormalization group equation

Now the boundary condition and the initial condition in (2.503) are changed to the so-called **renormalization group equation**:

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)}\bar{\lambda}(p,\lambda) = \beta(\bar{\lambda}), \qquad \bar{\lambda}(M,\lambda) = \lambda. \tag{2.511}$$

This equation shows that the flow of the running coupling constant $\bar{\lambda}(p)$ with the speed $\beta(\bar{\lambda})$. And we set the initial energy scale is M, and then set the value at this point is the original coupling constant λ. Now we expand $\bar{\lambda}$ to the first order and use (2.471):

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)}\bar{\lambda}(p,\lambda) = \beta(\bar{\lambda}) = \frac{3}{16\pi^2}\bar{\lambda}^2 + O(\bar{\lambda}^3). \tag{2.512}$$

So we actually are solving the ODE

$$\frac{\mathrm{d}\bar{\lambda}}{\mathrm{d}t} = A\bar{\lambda}^2 \tag{2.513}$$

It is easy to solve it by integrating

$$\int_{\bar{\lambda}(M)=\lambda}^{\bar{\lambda}(p)} \bar{\lambda}'^{-2}\,\mathrm{d}\bar{\lambda}' = \int_0^{\ln(p/M)} A\,\mathrm{d}t' \tag{2.514}$$

Thus, we have

$$\frac{1}{\bar{\lambda}(p)} = \frac{1}{\bar{\lambda}} - A\ln(p/M) \tag{2.515}$$

$$= \frac{1 - A\bar{\lambda}(M)\ln(p/M)}{\bar{\lambda}(M)}. \tag{2.516}$$

Put all the parameters back, we have

$$\ln\left(\frac{p}{M}\right) = \left(\frac{3}{16\pi^2}\right)^{-1}\left(\frac{1}{\bar{\lambda}(M)} - \frac{1}{\bar{\lambda}(p)}\right). \tag{2.517}$$

Take $\bar{\lambda}(M) = \lambda$, which leads to the expression for the running coupling $\bar{\lambda}$:

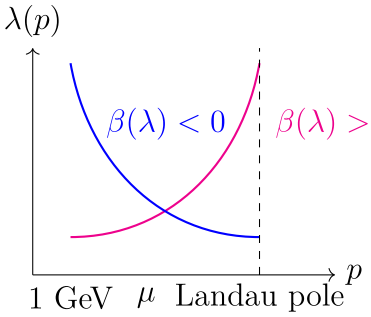
$$\bar{\lambda}(p) = \frac{\lambda}{1 - \frac{3}{16\pi^2}\lambda\ln(p/M)}. \tag{2.518}$$

So $\bar{\lambda}(p)$ grows for increasing scales p. And there is a pole when

$$1 - \frac{3}{16\pi^2}\lambda \ln\left(\frac{p}{M}\right) = 0, \tag{2.519}$$

which is called the **Landau pole**:

$$p = M \exp\left(\frac{16\pi^2}{3\lambda}\right). \tag{2.520}$$



For $p^2 \ll M^2$, theory becomes trivial:

$$\bar{\lambda}(p) \sim \frac{\bar{\lambda}(M)}{\lim\limits_{x\to 0}(1 - \ln x)} \to 0_+ \tag{2.521}$$

which corresponds to the free theory.

Generally, we have different cases for the β function, and that corresponds to the different background theory

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)}\bar{\lambda}(p,\lambda) = \beta(\bar{\lambda}) \begin{cases} > 0 & \text{IR free } (\phi^4\text{-theory, QED, ...}) \\ = 0 & \text{conformal theory } (N = 4 \text{ SYM}) \\ < 0 & \text{UV free/asymptotically free (QCD, SU(N) gauge theory).} \end{cases} \tag{2.522}$$

6.3 Higgs mass and fate of universe

As the second example, we consider the Higgs mechanism. The story begin with the Lagrangian:

$$\mathcal{L}_{\text{unbroken}} = \bar{\psi}(\mathrm{i}\gamma^\mu\partial_\mu)\psi + \frac{1}{2}(\partial^\mu\phi)(\partial_\mu\phi) - V(\phi) + y_\psi\phi\bar{\psi}\psi. \tag{2.523}$$

where $y_\psi\phi\bar{\psi}\psi$ is the Higgs-Yukawa interaction, and we dropped all the gauge terms. To get the spontaneous symmetry breaking, we take the potential as

$$V(\phi) = \mu^2\phi^2 + \lambda\phi^4. \tag{2.524}$$

The spontaneous symmetry breaking happens when

$$\phi = H + v. \tag{2.525}$$

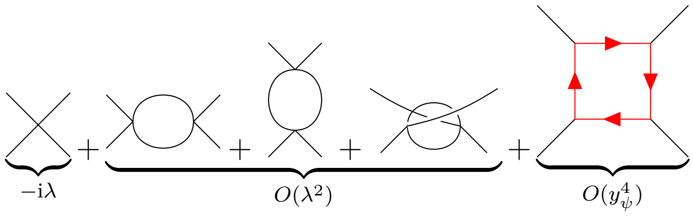
where H is the Higgs field, v is the vacuum expectation value, which in experiment measure are

$$v \simeq \frac{1}{\sqrt{\sqrt{2}G_F}} \simeq 246 \text{ GeV} \tag{2.526}$$

So plug (2.525) into the Lagrangian, we have the spontaneous symmetry breaking Lagrangian:

$$\mathcal{L}_{\text{broken}} = \bar{\psi}(\mathrm{i}\not{\partial})\psi + \frac{1}{2}(\partial_\mu H)(\partial^\mu H) - \underbrace{\lambda v^2}_{\frac{1}{2}M_H^2} H^2 - \lambda v H^3 - \frac{\lambda}{4}H^4 - \underbrace{\frac{1}{\sqrt{2}}y_\psi v}_{m_\psi}(1 + H/v)\bar{\psi}\psi \tag{2.527}$$

This mechanism turns out to the particle mass by the Higgs interact with itself and the other fields. So the coupling constant renormalization is very similar with ϕ^4 theory and Yukawa theory:



(2.528)

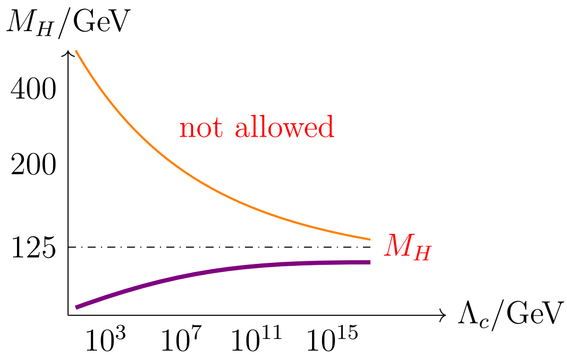
So the solution for $\lambda(Q)$ at scale $Q^2 \gg v$ is:

$$\lambda(Q) = \lambda(v)\frac{1}{1 - \frac{3}{16\pi^2}\lambda(v)\ln(Q/v)} \tag{2.529}$$

Cut off Λ_c corresponds to the Landau pole, up to which perturbative Higgs sector holds:

$$\Lambda_c = v\exp\left(\frac{16\pi^2}{3\lambda}\right) \simeq v\exp\left(\frac{16\pi^2 v^2}{M_H^2}\right) \tag{2.530}$$

where $M_H^2 = 2\lambda v^2$. So the area above the orange line are forbidden.



Now we consider fermion as the top quark, which is the heaviest fermion the standard model. It has contributions to $\beta(\lambda)$ from Higgs-Yukawa interaction, which the box diagram gives

$$Q\frac{\mathrm{d}}{\mathrm{d}Q}\lambda = \frac{1}{64\pi^2}(-3y_t^4 + 6\lambda y_t^2 + 12\lambda^2) + \dots. \tag{2.531}$$

It is interesting that the first term is negative, which means it evolves in the opposite way with the Higgs. Solving this equation for small λ ($\lambda \ll 1$), we have

$$\lambda(Q) = \lambda(v) + \frac{1}{32\pi^2}\left(-\frac{3m_t^4}{v^4}\right)\ln\left(\frac{Q}{v}\right) \tag{2.532}$$

since $m_t = y_t\frac{v}{\sqrt{2}}$, $m_t \sim 172$ GeV, so in principle, $\lambda(Q)$ can be negative at large Q! That means the vacuum will be super unstable. So to avoid that, we require a bound on Higgs mass:

$$M_H^2 \geq \frac{v^2}{16\pi^2}\left(-\frac{3m_t^4}{v^4}\right)\ln\left(\frac{Q}{v}\right), \tag{2.533}$$

such that we can keeping $\lambda(Q) > 0$.

7. Renormalization for local operators and evolution of mass parameter

We first introduce the idea of local operator. The local operator is multiple field operator compositing in the same space-time position, namely the production of fields. This will happen when the energy scale is much lower than the high energy scattering, e.g. weak decay.So to understand this, we can recall the beta decay in electroweak interaction.

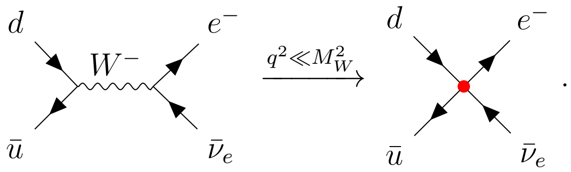
The electroweak interaction between left fermions and W-boson is:

$$\delta\mathcal{L} = \frac{g_{\text{ew}}}{\sqrt{2}} W_\mu \bar{\psi}\gamma^\mu(1-\gamma_5)\psi. \tag{2.534}$$

And the propagator of W-boson is

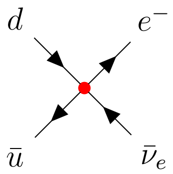
$$= \frac{-\mathrm{i}g^{\mu\nu}}{q^2 - M_W^2 + \mathrm{i}\epsilon} \tag{2.535}$$

In the low energy limit $q^2 \ll M_W^2$, the vertex and propagator of β-decay reduce to the 4-Fermi interaction vertex:



$$. \tag{2.536}$$

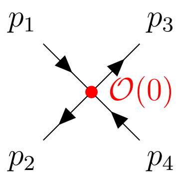
So we can reduce a process propagate from μ to ν to the process happens in a fixed point due to the reaction happening so fast, i.e.:



$$= \frac{g_{\text{ew}}^2}{2M_W^2}(\bar{\psi}\gamma^\mu(1-\gamma_5)\psi)_{\text{quark}}(\bar{\psi}\gamma_\mu(1-\gamma_5)\psi)_{\text{lep}} = \frac{g_{\text{ew}}^2}{2M_W^2}\mathcal{O}(x). \tag{2.537}$$

Generally, this special vertex can be seen as a Green's function:

$$\langle\psi(p_1)\bar{\psi}(p_2)\psi(p_3)\bar{\psi}(p_4)\mathcal{O}(0)\rangle = \tag{2.538}$$



7.1 Renormalized Green's functions with local operator $\mathcal{O}$ in ϕ^4-theory

Now we consider the renormalization of the local operator. Since the vertex is described by the Green's function, we can then consider a general form:

$$G^{(n,\mathcal{O})}(p_1\ldots p_n, k) = \langle\phi(p_1)\ldots\phi(p_n)\mathcal{O}(k)\rangle \tag{2.539}$$

$$= Z^{-n/2}(\mu)\underbrace{Z_{\mathcal{O}}^{-1}(M)}_{\substack{\text{operator}\\\text{renormalization}}}\langle\underbrace{\phi_0(p_1)\ldots\phi_0(p_n)}_{\text{bare fields}}\underbrace{\mathcal{O}_0(k)}_{\substack{\text{bare}\\\text{operator}}}\rangle \tag{2.540}$$

$$= Z^{-n/2}(\mu)Z_{\mathcal{O}}^{-1}(M)G_0^{(n,\mathcal{O})}(p_1\ldots p_n, k) \tag{2.541}$$

where $\mathcal{O} = \phi^2$ or ϕ^4 or $\ldots$ Tthe lower index "0" represent the bare quantity, and M is the renormalization scale. The bare operator and the renormalized operator have the relation:

$$\mathcal{O}_{\text{bare}} = Z_{\mathcal{O}}(M)\mathcal{O}_{\text{ren}}. \tag{2.542}$$

So we can derive the Callan-Symanzik equation by using

$$0 = M\frac{\mathrm{d}}{\mathrm{d}M}G_0^{(n,\mathcal{O}_0)} \tag{2.543}$$

$$= M\frac{\mathrm{d}}{\mathrm{d}M}\left(Z^{n/2}Z_{\mathcal{O}}G^{(n,\mathcal{O})}\right) \tag{2.544}$$

$$= \left(M\frac{\partial}{\partial M}Z^{n/2}\right)Z_{\mathcal{O}}G^{(n,\mathcal{O})} + Z^{n/2}\left(M\frac{\partial}{\partial M}Z_{\mathcal{O}}\right)G^{(n,\mathcal{O})} + Z^{n/2}Z_{\mathcal{O}}\left(M\frac{\partial}{\partial M}G^{(n,\mathcal{O})}\right), \tag{2.545}$$

dividing $Z^{n/2}Z_{\mathcal{O}}$ on both side, we have

$$0 = \left(\frac{M}{Z^{n/2}}\frac{\partial Z^{n/2}}{\partial M}\right)G^{(n,\mathcal{O})} + \left(\frac{M}{Z_{\mathcal{O}}}\frac{\partial Z_{\mathcal{O}}}{\partial M}\right)G^{(n,\mathcal{O})} + M\frac{\partial}{\partial M}G^{(n,\mathcal{O})} \tag{2.546}$$

$$= \left(M\frac{\partial}{\partial M}\ln Z^{n/2}\right)G^{(n,\mathcal{O})} + \left(M\frac{\partial}{\partial M}\ln Z_{\mathcal{O}}\right)G^{(n,\mathcal{O})} + M\frac{\partial}{\partial M}G^{(n,\mathcal{O})}, \tag{2.547}$$

where we can define the anomalous dimension respectively for field and local operator as:

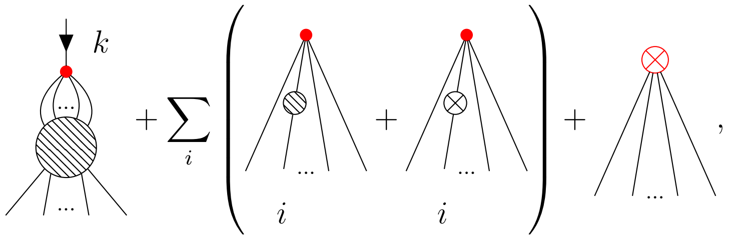
$$\gamma = \frac{M}{2}\frac{\partial}{\partial M}\ln Z_{\lambda}(M) \tag{2.548}$$

$$\gamma_{\mathcal{O}} = M\frac{\partial}{\partial M}\ln Z_{\mathcal{O}}(M). \tag{2.549}$$

Then we have the Callan-Symanzik equation:

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial\lambda} + n\gamma(\lambda) + \gamma_{\mathcal{O}}(\lambda)\right)G^{(n,\mathcal{O})} = 0, \tag{2.550}$$

Now we compute $\gamma_{\mathcal{O}}$ for N-point Green's function. The most general case can be expressed as

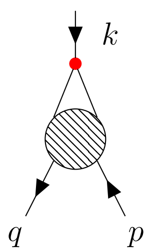


(2.551)

where the red point and red cross represent the operator vertex and the operator counterterm.

7.2 Renormalization for local operators in massless ϕ^4-theory

As an example to help us understanding this. We consider $\mathcal{O} = \phi^2$ inserted in two-point function:



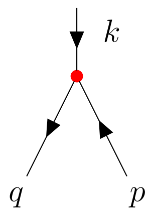
$$= \langle\Omega|\,\phi(q)\phi(p)\phi^2(k)\,|\Omega\rangle = G^{(2,\phi^2)}. \tag{2.552}$$

Thus, we can expand it as the normal Dyson series:

$$G^{(2,\phi^2)}(q,p;k) = \sum_{m=0}^{\infty} \frac{\mathrm{i}^m}{m!} \int \left[\prod_{a=1}^{m} \mathrm{d}^d x_a\right] \left(-\frac{\lambda}{4!}\right)^m \langle 0|T\{\phi(q)\phi(p)\phi^2(k)[\phi^4(x_1)\ldots\phi^4(x_m)]\}|0\rangle_{\text{connect}}. \tag{2.553}$$

The leading order, namely the tree-level non-amputated Green's function is just simply the contraction of the field and the operator:

$$G^{(2,\phi^2)}_{\text{tree}}(q,p;k) =$$



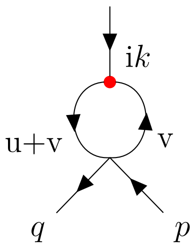
$$= \int \mathrm{d}^d x\, \mathrm{e}^{-\mathrm{i}k\cdot x} \langle 0|T\{\phi(q)\phi(p)\phi_A(x)\phi_B(x)\}|0\rangle \tag{2.554}$$

$$= \int \mathrm{d}^d x\, \mathrm{e}^{-\mathrm{i}k\cdot x} \left[\phi(q)\phi(p)\phi_A(x)\phi_B(x) + \phi(q)\phi(p)\phi_A(x)\phi_B(x)\right] \tag{2.555}$$

$$= \left(\frac{\mathrm{i}}{q^2}\right)\left(\frac{\mathrm{i}}{p^2}\right)\cdot 2. \tag{2.556}$$

To consider the renormalization for the operator, we need to compute 1-PI contribution at 1-loop (non-amputated):

$$G^{(2,\phi^2)}_{\text{1-loop}}(q,p;k) =$$



$$= \int \mathrm{d}^d x \mathrm{d}^d z\, \mathrm{e}^{-\mathrm{i}k\cdot x} \left\langle 0 \middle| T\left\{\phi(q)\phi(p)\left[\frac{-\mathrm{i}\lambda}{4!}\phi^4(z)\right]\left[\frac{1}{2}\phi_A(x)\phi_B(x)\right]\right\}\middle| 0\right\rangle \tag{2.557}$$

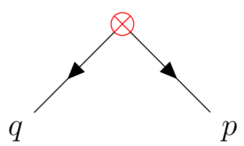
$$= \frac{-\mathrm{i}\lambda}{2} \int \mathrm{d}^d x \mathrm{d}^d z\, \mathrm{e}^{-\mathrm{i}k\cdot x} \left[\phi(q)\,\phi(p)\quad \phi(z)\,\phi(z)\,\phi(z)\,\phi(z)\quad \phi_A(x)\,\phi_B(x)\right] \tag{2.558}$$

$$= \left(\frac{\mathrm{i}}{q^2}\right)\left(\frac{\mathrm{i}}{p^2}\right) \int \frac{\mathrm{d}^d v}{(2\pi)^d} (-\mathrm{i}\lambda) \frac{\mathrm{i}}{v^2} \frac{\mathrm{i}}{(k+v)^2}, \tag{2.559}$$

with some calculation, one has:

$$G^{(2,\phi^2)}_{\text{1-loop}}(q,p;k) = \frac{\mathrm{i}}{p^2}\frac{\mathrm{i}}{q^2}\left(-\frac{\lambda}{(4\pi)^{d/2}}\frac{\Gamma(2-\frac{d}{2})}{\Delta^{2-d/2}}\right) \tag{2.560}$$

with the counterterm:



$$= \frac{\mathrm{i}}{p^2}\frac{\mathrm{i}}{q^2} 2\delta\phi^2 \tag{2.561}$$

Comparing of coefficients in these two graphs, we then can find the counterterm at scale $p^2 = -M^2$ should be:

$$\delta\phi^2 = \frac{\lambda}{2(4\pi)^{d/2}}\frac{\Gamma(2-\frac{d}{2})}{(M^2)^{2-d/2}}. \tag{2.562}$$

This implies the operator renormalization is

$$Z_{\phi^2} = 1 - \delta\phi^2 = 1 - \frac{\lambda}{2(4\pi)^{d/2}} \frac{\Gamma(2-\frac{d}{2})}{(M^2)^{2-d/2}} \tag{2.563}$$

$$\stackrel{d=4-\varepsilon}{=} 1 - \frac{\lambda}{(4\pi)^2}\frac{1}{\epsilon} + \text{finite terms.} \tag{2.564}$$

Using $\ln(1+x) \approx x$, we can obtain the anomalous dimension of the operator in $d = 4 - \varepsilon$:

$$\gamma_{\phi^2} = M\frac{\partial}{\partial M}\ln Z_{\phi^2} = \frac{-\lambda}{2(4\pi)^{d/2}}\Gamma\left(2-\frac{d}{2}\right)\cdot\Big[-(4-d)M^{-(4-d)}\Big] \tag{2.565}$$

$$\stackrel{d=4-\varepsilon}{=} \frac{-\lambda}{2(4\pi)^{2-\epsilon/2}}\Gamma\left(\frac{\epsilon}{2}\right)\cdot\Big[-\epsilon M^{-\epsilon}\Big] \tag{2.566}$$

$$= \frac{\lambda}{16\pi^2}. \tag{2.567}$$

More generally, if we insert the operator into an n-point Green's function, we then need to consider the field renormalization with the operator renormalization $Z_\phi^{n/2}Z_{\mathcal{O}}$, therefore the anomalous dimension becomes:

$$\gamma_{\mathcal{O}}(\lambda) = M\frac{\partial}{\partial M}\left(-\delta_{\mathcal{O}} + \frac{n}{2}\delta_Z\right), \tag{2.568}$$

with the approximation:

$$\ln\Big(Z_\phi^{n/2}Z_{\mathcal{O}}\Big) = \frac{n}{2}\ln Z_\phi + \ln Z_{\mathcal{O}} \approx \frac{n}{2}\delta_Z - \delta_{\mathcal{O}}. \tag{2.569}$$

This case can be treated as renormalized ϕ^4-theory with effective operator $-\frac{1}{2}m^2\phi^2$, where ϕ^2 is local operator inserted in Green's function. So the Callan-Symanzik equation with mass anomalous dimension $m(\mu)$ is

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial\lambda} + n\gamma(\lambda) + \gamma_{\phi^2}m^2\frac{\partial}{\partial m^2}\right)G^{(n,\phi^2)}(\{p_i\}, M, \lambda, m) = 0 \tag{2.570}$$

7.3 Evolution of mass parameter

Moreover, one can write the generalization to local operators $\mathcal{O}_i$ in ϕ^4-theory as:

$$\mathcal{L}(C_i) = \mathcal{L}_{\text{eff}} + \sum_i C_i\mathcal{O}_i(x) \tag{2.571}$$

where C_i is the coupling of local operators. So the Callan-Symanzik equation is

$$\left(M\frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial\lambda} + n\gamma(\lambda) + \sum_i\gamma_i(\lambda)C_i\frac{\partial}{\partial C_i}\right)G^{(n)}(M, \lambda, \{C_i\}) = 0 \tag{2.572}$$

To guarantee the dimension of Lagrangian is $[\text{Mass}]^4$ in 4-dimension spacetime, so we need to explicit mass dimension in coupling C_i as:

$$C_i = \rho_i M^{4-d_i} \tag{2.573}$$

where ρ_i is dimensionless, d_i is dimension of $\mathcal{O}_i$. Therefore, one can see that the variables change to $(M, \lambda, \{\rho_i\})$. To make the partial derivative independent to ρ_i, we have

$$M\frac{\partial}{\partial M}\Big|_{\rho_i} = M\frac{\partial}{\partial M}\Big|_{C_i} + \sum_i\left(M\frac{\partial C_i}{\partial M}\Big|_{\rho_i}\right)\frac{\partial}{\partial C_i} \tag{2.574}$$

Notice that

$$\rho_i \frac{\partial}{\partial \rho_i} = \rho_i M^{4-d_i} \frac{\partial}{\partial C_i} = C_i \frac{\partial}{\partial C_i} \tag{2.575}$$

so we have

$$M \frac{\partial}{\partial M}\Big|_{C_i} = M \frac{\partial}{\partial M}\Big|_{\rho_i} - \sum_i (4 - d_i)\rho_i \frac{\partial}{\partial \rho_i} \tag{2.576}$$

Then the Callan-Symanzik equation becomes:

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i (\gamma_i(\lambda) + d_i - 4)\rho_i \frac{\partial}{\partial \rho_i} \right) G^{(n)} = 0, \tag{2.577}$$

where we can define the β function for new coupling ρ_i:

$$\beta_i(\lambda) = (\gamma_i(\lambda) + d_i - 4)\rho_i, \tag{2.578}$$

then the Callan-Symanzik equation can be reduced as:

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda)\frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i \beta_i \rho_i \frac{\partial}{\partial \rho_i} \right) G^{(n)} = 0. \tag{2.579}$$

With this, we can easily solve the running coupling ρ_i from the renormalization group equation for β_i

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)} \bar{\rho}_i = \beta_i(\bar{\rho}_i, \lambda). \tag{2.580}$$

Since β_i in the low energy limit, namely $\gamma_i(\lambda) \ll 1$, is dominated by the classical dimension $d_i - 4$, we then have

$$\frac{\mathrm{d}}{\mathrm{d}\ln(p/M)} \bar{\rho}_i \simeq (d_i - 4 + \dots).\bar{\rho}_i \tag{2.581}$$

Therefore, the solution in low energy limit reads:

$$\bar{\rho}_i = \rho \left(\frac{p}{M}\right)^{d_i - 4}. \tag{2.582}$$

Now we can clearly see that for the operators with mass dimension $d_i > 4$ will vanish as $p \to 0$, e.g., ϕ^6 theory with $d_i = 6$. These operators are called the **irrelevant operators**, which usually decrease fast until vanish in low energy limit, namely one can never feel these operators' contribution. On the contrary, the **relevant operators** are the operators with mass dimension $d_i < 4$, e.g., the mass term $\frac{1}{2}m^2\phi^2$, which will exponentially increase in low energy limit. And the operators like ϕ^4 are called the **marginal operators**, which do not changing with the energy scale due to $d_i = 4$. This type operators are only determined by the anomalous dimension $\gamma_i(\lambda)$.

This reminds us that the effective field theory truly works even though we don't know all the details in high energy level. But we still can derive the precise renormalizable Lagrangian in low energy level.

Non-Abelian gauge theories

1. Recap of gauge theory and brief look at Lie group theory

1.1 Recap: Abelian gauge theory

We firstly start with QED, which as we know is an Abelian $U(1)$ gauge theory. It start from the free fermion Lagrangian:

$$\mathcal{L} = \mathrm{i}\bar{\psi}\gamma^\mu \partial_\mu \psi - m\bar{\psi}\psi. \tag{3.1}$$

The Noether currentis is: $j^\mu = \bar{\psi}\gamma^\mu\psi$. To approach the QED Lagrangian, we introduce the (global) gauge transformation as:

$$\psi \to \psi' = e^{\mathrm{i}\alpha}\psi \tag{3.2}$$
$$\bar{\psi} \to \bar{\psi}' = e^{-\mathrm{i}\alpha}\bar{\psi} \quad , \quad \alpha \in \mathbb{R}. \tag{3.3}$$

The local gauge transformation is just changing the parameter α rely on the coordinate:

$$\alpha \to \alpha(x). \tag{3.4}$$

The gauge symmetry is saying that the Lagrangian is invariant under the gauge transformation. This requires us to introduce the covariant derivative D_μ:

$$D_\mu\psi(x) = (\partial_\mu - \mathrm{i}gA_\mu)\,\psi(x). \tag{3.5}$$

where in QED case, the charge $g = e$, and the gauge field A_μ is just the photon field, which satisfies the transformation rule:

$$A_\mu \to A'_\mu = A_\mu + \frac{1}{g}\partial_\mu\alpha(x). \tag{3.6}$$

Then we immediately have the covariant derivative transfers as

$$D_\mu\psi \to (D_\mu\psi)' = \partial_\mu\psi' - \mathrm{i}gA'_\mu\psi' = e^{\mathrm{i}\alpha}D_\mu\psi. \tag{3.7}$$

Thus the invariant of $\mathcal{L}$ requires that

$$\mathcal{L}(\psi, A_\mu) = \mathcal{L}(\psi', A'_\mu) = \mathrm{i}\bar{\psi}\gamma^\mu D_\mu\psi - m\bar{\psi}\psi \tag{3.8}$$
$$= \mathrm{i}\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi + gA_\mu j^\mu. \tag{3.9}$$

For photon dynamical, we need to add kinetic term for A_μ:

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \tag{3.10}$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

1.2 SU(N) gauge theory

Now we consider the non-Abelian $SU(N)$ gauge theory with N Dirac spinors ψ_j, where $j = 1, \dots, N$. The original Lagrangian is

$$\mathcal{L} = \sum_{j=1}^{N} \mathrm{i}\bar{\psi}_j \gamma^\mu \partial_\mu \psi_j - m\bar{\psi}_j \psi_j. \tag{3.11}$$

Under the global unitary transformation $U_{ij} \in \mathbb{C}$, $UU^\dagger = \mathbb{1}$, the spinors obey

$$\psi_j \to \psi'_j = \sum_k U_{jk}\psi_k \ , \tag{3.12}$$

$$\bar{\psi}_j \to \bar{\psi}'_j = \sum_k \bar{\psi}_k U^\dagger_{kj}, \tag{3.13}$$

thus the mass term transformation under $SU(N)$

$$\sum_j \bar{\psi}_j \psi_j \to \sum_j \bar{\psi}'_j \psi'_j = \sum_{k,l,j} \bar{\psi}_k \underbrace{U^\dagger_{kj} U_{jl}}_{\delta_{kl}} \psi_l = \sum_k \bar{\psi}_k \psi_k \tag{3.14}$$

is invariant.

Similar with the Abelian case, the local gauge transformation is $U \to U(x)$. We also require the Lagrangian invariant under the transformation. This gives coupling of fermions to gauge field in covariant derivative analogous to QED:

$$D_\mu \psi_j = \partial_\mu \psi_j - \mathrm{i}g \sum_k (A_\mu)_{jk} \psi_k. \tag{3.15}$$

This form of covariant derivative also requires the transformation rule as

$$D_\mu \psi_j \to (D_\mu \psi_j)' = \partial_\mu \psi'_j - \mathrm{i}g \sum_k (A'_\mu)_{jk} \psi'_k \tag{3.16}$$

$$= \sum_k U_{jk} (D_\mu \psi_k) \tag{3.17}$$

For convenient, we will use the short-hand notation(Matrix notation), which basically just ignore the index of the matrix element:

$$D_\mu \psi \to (D_\mu \psi)' = U D_\mu \psi = e^{\mathrm{i}\alpha(x)} D_\mu \psi. \tag{3.18}$$

We claim that the gauge field transforms following

$$A_\mu \to A'_\mu = U A_\mu U^\dagger - \frac{\mathrm{i}}{g} (\partial_\mu U) U^\dagger. \tag{3.19}$$

Although we can directly use the definition to compute this expression, I just don't want to write so much. Instead, we can have a quick check that this claim indeed satisfies our requirement:

$$D_\mu \psi \to (D_\mu \psi)' = \partial_\mu (U\psi) - \mathrm{i}g \left(U A_\mu U^\dagger - \frac{\mathrm{i}}{g} (\partial_\mu U) U^\dagger \right) U\psi \tag{3.20}$$

$$= U \partial_\mu \psi + (\partial_\mu U)\psi - \mathrm{i}g U A_\mu \psi - (\partial_\mu U)\psi \tag{3.21}$$

$$= U(\partial_\mu \psi - \mathrm{i}g A_\mu \psi) = U D_\mu \psi \ . \tag{3.22}$$

Thus, the Lagrangian is invariant under $SU(N)$ gauge transformation.

Now we need to approach the dynamical of non-Abelian gauge field, which requires a field strength. Since we keep calling it as "non-Abelian", but until now it's not quite obvious to see how it is "non-Abelian", therefore, the firld strength should give you some direct idea when we claim:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - \mathrm{i}g[A_\mu, A_\nu]. \tag{3.23}$$

We can also check it is true that this field strength indeed satisfies our requirement, namely keep the Lagrangian invariant. To show this, we firstly have the transformation rule:

$$F_{\mu\nu} \to F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu - \mathrm{i}g[A'_\mu, A'_\nu] = UF_{\mu\nu}U^\dagger. \tag{3.24}$$

The Lagrangian is proportion to the trace over $F_{\mu\nu}F^{\mu\nu} = (F_{\mu\nu})_{jk}(F^{\mu\nu})^{kj} = \mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$, due to the Lagrangian is a scalar, thus the only possible combination is the trace over $F_{\mu\nu}$. Thus $\mathrm{tr}\, F_{\mu\nu}$ transfers as

$$\begin{aligned}
\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) \to \mathrm{tr}(F'_{\mu\nu}F'^{\mu\nu}) &= \mathrm{tr}(UF_{\mu\nu}U^\dagger UF^{\mu\nu}U^\dagger) \\
&= \mathrm{tr}(UU^\dagger F_{\mu\nu}F^{\mu\nu}) \\
&= \mathrm{tr}(F_{\mu\nu}F^{\mu\nu}).
\end{aligned} \tag{3.25}$$

Then we get **the Lagrangian of Yang-Mills theory**

$$\mathcal{L} = -\kappa\, \mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) + \sum_{j,k=1}^{N} \left[\mathrm{i}\bar{\psi}_j\gamma^\mu(D_\mu)_{jk}\psi_k - m\bar{\psi}_j\psi_j\right], \tag{3.26}$$

where κ is normalized constant.

1.3 Recap for essential Lie group theory

To continue our journey, it's necessary to master some basic knowledge about Lie group and Lie algebra. So in the next, we will recall some essential knowledge about these mathematical tools and show how are they used in gauge theory.

Firstly, we recall the infinitesimal transformation of the gauge operation. For Abelian theory, e.g. QED:

$$e^{\mathrm{i}\alpha} \simeq 1 + \mathrm{i}\alpha. \tag{3.27}$$

For non-Abelian gauge theory, e.g. Yang Mills theory:

$$U = \mathbb{1} + \hat{H} + \mathcal{O}(\hat{H}^2), \tag{3.28}$$

$$U^\dagger = \mathbb{1} + \hat{H}^\dagger + \mathcal{O}(\hat{H}^2). \tag{3.29}$$

Thus we have

$$U^{-1} = U^\dagger \implies \hat{H} = -\hat{H}^\dagger \quad \text{(anti-hermitian)} \tag{3.30}$$

where we note $\hat{H} = \mathrm{i}H$ with $H = H^\dagger$. Expand $\hat{H}$ in parameters of transformation α^a

$$\hat{H}_{jk} = \mathrm{i}\sum_a \alpha^a (t^a)_{jk} \tag{3.31}$$

with $\alpha^a \in \mathbb{R}$, $t^a = (t^a)^\dagger$ is hermitian. Thus t^a span a basis in the space of hermitian $N \times N$ matrices such that we can expand $\hat{H}$ in terms of this basis. t^a are called the generators of gauge transformations.

Generally, the unitary $N \times N$ matrix has N^2 linear independent generators due to t^a is $N \times N$. However, in our case, we require $\det U = 1$, which gives one constraint. So there are $\mathbf{N^2 - 1}$ **generators in** $\mathbf{SU(N)}$. Further more, base on $\det U = 1$ and the Jacobi identity $\det\left(e^M\right) = e^{\mathrm{tr}(M)}$, we find

$$\det U = \det\left(e^{\hat{H}}\right) = e^{\mathrm{tr}\,\hat{H}} = 1, \tag{3.32}$$

thus, $\mathrm{tr}\,\hat{H} = 0$ and implies

$$\mathrm{tr}(t^a) = 0. \tag{3.33}$$

So $\mathbf{t^a}$ **are traceless generators**.

Now we discuss the algebra structure for generators t^a. The Lie algebra, namely the commutator is given by:

$$[t^a, t^b] = \mathrm{i} \sum_c f^{abc} t^c. \tag{3.34}$$

where f^{abc} is the structure constant of Lie algebra. The elements of Lie group can be expanded around $\mathbb{1}$.

$$g(\alpha) = \mathbb{1} + \mathrm{i} \sum_{a=1}^{d} \alpha_a t^a + \frac{1}{2} \sum_{a,b=1}^{d} \alpha_a \alpha_b T^{ab} + \mathcal{O}(\alpha^3), \tag{3.35}$$

where $T^{ab} = T^{ba}$ is symmetric, which can be expanded in terms of the generator t^a, t^b. Notice $g(0) = \mathbb{1}$ and because of $g(\alpha)\mathbb{1} = \mathbb{1}g(\alpha)$, we then have

$$\gamma_a(\alpha_1, \ldots, \alpha_d; 0, \ldots, 0) = \gamma_a(0, \ldots, 0; \alpha_1, \ldots, \alpha_d) = \alpha_a \tag{3.36}$$

where the γ is defined by the group multiplication:

$$g(\alpha_1, \ldots, \alpha_d) \cdot g(\beta_1, \ldots, \beta_d) = g(\gamma_1, \ldots, \gamma_d), \tag{3.37}$$

in which γ is the function of α and β:

$$\gamma_a = \gamma_a(\alpha_1, \ldots, \alpha_d; \beta_1, \ldots, \beta_d). \tag{3.38}$$

Base on the fact that $\gamma(\alpha, 0) = \alpha$ and $\gamma(0, \beta) = \beta$, then we have only one choice to expand γ_a until the second order to keep these relation, namely:

$$\gamma_a = \alpha_a + \beta_a + \sum_{b,c} C^a_{bc} \alpha_b \beta_c + \ldots, \tag{3.39}$$

where C^a_{bc} is a unknown coefficient. Thus, use $g(\alpha)g(\beta) = g(\gamma)$, we have for the left-hand side:

$$g(\alpha)g(\beta) = \mathbb{1} + \mathrm{i} \sum_a (\alpha_a + \beta_a) t^a + \frac{1}{2} \sum_{a,b} \left[(\alpha_a \alpha_b + \beta_a \beta_b) T^{ab} - 2\alpha^a \beta^b t^a t^b \right] + \ldots, \tag{3.40}$$

And for the right-hand side:

$$g(\gamma) = \mathbb{1} + \mathrm{i} \sum_a \gamma_a t^a + \frac{1}{2} \sum_{a,b} \gamma_a \gamma_b T^{ab} + \ldots. \tag{3.41}$$

Thus use (3.39), then the right-hand side becomes

$$g(\gamma(\alpha, \beta)) = \mathbb{1} + \mathrm{i} \sum_a (\alpha_a + \beta_a) t^a + \mathrm{i} \sum_{a,b,c} C^{bc}_a \alpha_b \beta_c t^a + \frac{1}{2} \sum_{a,b} (\alpha_a + \beta_a)(\alpha_b + \beta_b) T^{ab} + \ldots. \tag{3.42}$$

Compare (3.40) with (3.42), we find

$$-t^a t^b = \mathrm{i} \sum_c C_c^{ab} t^c. \tag{3.43}$$

this gives the structure constant:

$$[t^a, t^b] = \mathrm{i} \sum_{c=1}^{d} (C_c^{ba} - C_c^{ab}) t^c \tag{3.44}$$

$$= \mathrm{i} \sum_{c=1}^{d} f^{abc} t^c \tag{3.45}$$

where $f^{abc} = -f^{bac}$, and d is the dimension of Lie algebra.

Examples of Lie groups

- **GL(N)**: Group of $N \times N$ matrices in $\mathbb{R}^N$ with $\det M \neq 0$.
- **U(N)**: Group of unitary $N \times N$ matrices in $\mathbb{C}^N$, satisfies

$$U^\dagger U = U U^\dagger = \mathbb{1} \quad \text{and} \quad \det U \neq 0. \tag{3.46}$$

  Infinitesimal version:

$$U = \mathbb{1} + \mathrm{i} \sum_{a=1}^{d} \alpha_a t^a. \tag{3.47}$$

  with $d = N^2$ is the dimension of $U(N)$ Lie algebra/group.
- **SU(N)** $:= \{A \in U(N) | \det A = 1\}$. Dimension of Lie algebra $\mathfrak{su}(N)$ is $d = N^2 - 1$. Generators t^a are traceless

$$\det U = \det\left(\mathbb{1} + \mathrm{i} \sum_{a=1}^{d} \alpha_a t^a\right) = 1 + \mathrm{i} \sum_{a=1}^{d} \alpha_a \operatorname{tr} t^a = 1 \tag{3.48}$$

  thus

$$\operatorname{tr} t^a = 0 \tag{3.49}$$

- **O(N)**: group of orthogonal $N \times N$ matrices in $\mathbb{R}^N$.

$$O O^T = O^T O = \mathbb{1} \tag{3.50}$$

  Infinitesimal version:

$$O = \mathbb{1} + \mathrm{i} \sum_{a=1}^{d} \alpha_a t^a \tag{3.51}$$

  Dimension of Lie algebra $\mathfrak{o}(N)$ is $\frac{1}{2} N(N-1)$. And notice:

$$O O^T = \left(\mathbb{1} + \mathrm{i} \sum_{a=1}^{d} \alpha_a t^a\right) \left(\mathbb{1} + \mathrm{i} \sum_{b=1}^{d} \alpha_b (t^b)^T\right) = \mathbb{1}, \tag{3.52}$$

  thus,

$$\mathbb{1} = \mathbb{1} + \mathrm{i} \sum_{a=1}^{d} \alpha_a t^a + \mathrm{i} \sum_{b=1}^{d} \alpha_b (t^b)^T + \mathcal{O}(\alpha^2) \tag{3.53}$$

$$= \mathbb{1} + \mathrm{i} \sum_{a=1}^{d} \left(\alpha_a t^a + \alpha_a (t^a)^T\right), \tag{3.54}$$

which implies

$$t^a = -(t^a)^T. \tag{3.55}$$

- $\mathbf{SO(N)}$:= $\{A \in O(N) | \det A = 1\}$
- $\mathbf{SO(N, M)} = \{\Lambda \in \mathrm{Mat}(N+M, \mathbb{R}) \mid \Lambda^T \eta \Lambda = \eta, \det \Lambda = 1\}$, where

$$\eta = \mathrm{diag}(\underbrace{1, \ldots, 1}_{N}, \underbrace{-1, \ldots, -1}_{M}) \tag{3.56}$$

- **Lorentz group**: $SO(1, 3)$

1.4 Representations of Lie algebra

Def: Representation of group

A **representation of a group G on a $\mathbf{d_r}$-dimensional vector space V** is a group homomorphism

$$\rho : G \to GL(V), \tag{3.57}$$

which maps each group element $g \in G$ to an invertible linear operator $\rho(g)$ acting on V, such that:

$$\rho(g_i) \cdot \rho(g_j) = \rho(g_i g_j) \quad \forall g_i, g_j \in G. \tag{3.58}$$

The dimension d_r of the vector space V is called **the dimension of the representation**.

- A representation ρ is called **reducible** if there exists a non-trivial proper subspace $U \subset V$ (where $U \neq \{0\}$ and $U \neq V$) that is invariant under the action of all group elements, meaning:

$$\rho(g)u \in U \quad \forall g \in G, \forall u \in U. \tag{3.59}$$

- If no such invariant subspace exists, the representation is called irreducible (irrep).

For a Lie group characterized by continuous parameters α, a representation is a mapping $\rho : g(\alpha) \to M(\alpha)$, where the matrices (or operators) $M(\alpha)$ preserve the group multiplication:

$$M(\alpha) \cdot M(\beta) = M(\gamma) \quad \text{whenever} \quad g(\alpha) \cdot g(\beta) = g(\gamma). \tag{3.60}$$

In the next, we introduce some useful representations.

- **Fundamental(defining) representation**

  For a matrix Lie group, the **fundamental representation** is simply the group of matrices themselves, acting on their natural vector space.

  Example: $SU(2)$ fundamental representation

  - **Representation Space (V):** A 2-dimensional complex vector space $\mathbb{C}^2$, physically corresponding to a spin-1/2 state

  $$|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle . \tag{3.61}$$

  - **Dimension of the representation:** $d = 2$ (since the matrices are 2×2). Note that the dimension of the $SU(2)$ group/algebra itself is $2^2 - 1 = 3$ (the number of independent generators).The 3 Hermitian generators for this $d = 2$ fundamental representation are the Pauli matrices:

$$t^a = \frac{1}{2}\sigma^a \quad (a = 1, 2, 3), \tag{3.62}$$

which satisfy the Lie algebra $[t^a, t^b] = i\epsilon^{abc}t^c$, and the Pauli matrix identity:

$$\sigma^k\sigma^l = \delta^{kl}\mathbb{1} + i\epsilon^{klm}\sigma^m \tag{3.63}$$

Physically, we can use these fundamental building blocks to construct higher-dimensional representations via tensor products, such as the 3-dimensional spin-1 triplet state. In fact, $SU(2)$ has spin-j representations with dimension $2j+1$. For instance, $j = 1/2$ corresponds to the 2-dimensional fundamental representation using 2×2 Pauli matrices.

- **Adjoint representation**

  Instead of acting on physical states, the **adjoint representation** is a d-dimensional representation where the group/algebra acts on the Lie algebra space itself (spanned by the generators $\{t^a\}$ as basis vectors). The definition is as follows

Ajoint representation

We first define a automorphism map of group G:

$$\begin{aligned} \Psi : G &\to \mathrm{Aut}(G) && (3.64)\\ g &\mapsto hgh^{-1} \equiv \Psi_h(g), \quad \forall h \in G, && (3.65)\end{aligned}$$

where $\mathrm{Aut}(G)$ is the automorphism group of G. Then we can define the derivative of Φ_g at the origin noted as Ad_g:

$$\mathrm{Ad}_g = (\mathrm{d}\Phi_g)_e : T_eG \to T_eG, \tag{3.66}$$

where T_eG is the tangent space in the natural element of G. Thus the **adjoint representation** is a homomorphism:

$$\begin{aligned} \mathrm{Ad} : G &\to \mathrm{Aut}(T_eG) && (3.67)\\ g &\mapsto \mathrm{Ad}_g. && (3.68)\end{aligned}$$

To obtain the explicit matrix form of the adjoint representation, we consider an infinitesimal group element $g = e + \mathrm{i}\alpha^a t^a \in G$. Acting on a generator $t^b \in T_eG$ via the adjoint map Ad_g, we expand to the linear order of α:

$$\mathrm{Ad}_{e+\mathrm{i}\alpha^a t^a}(t^b) = (e + \mathrm{i}\alpha^a t^a)t^b(e - \mathrm{i}\alpha^a t^a) \simeq t^b + \mathrm{i}\alpha^a[t^a, t^b] + \mathcal{O}(\alpha^2) \tag{3.69}$$

This motivates us to define the infinitesimal adjoint map $\mathrm{ad}_{t^a} \equiv \mathrm{d}(\mathrm{Ad})_e(t^a)$ on the Lie algebra, which acts purely as a commutator:

$$\mathrm{ad}_{t^a}(t^b) = \lim_{\alpha^a \to 0} \frac{\left(\mathrm{Ad}_{e+\mathrm{i}\alpha^a t^a}(t^b)\right)_{kl} - (t^b)_{kl}}{\mathrm{i}\alpha^a} = \left[t^a, t^b\right]_{kl} = \mathrm{i}\sum_{c=1}^{d} f^{abc}(t^c)_{kl}. \tag{3.70}$$

Viewing ad_{t^a} as a linear transformation matrix acting on the basis vector t^b to produce t^c, the expansion coefficients (structure constants) directly define the matrix elements of the adjoint generators:

$$(t^a_{\text{adj}})_{bc} = \mathrm{i} f^{abc} \tag{3.71}$$

Hence, the structure constants f^{abc} are not just algebraic constants, they are precisely **used to construct the adjoint representation**.

In this representation, the group elements act on the Lie algebra itself. The generators are $d \times d$ matrices (where $d = N^2 - 1$), with matrix elements defined by (3.71). We can verify that these matrices satisfy the Lie algebra by using the Jacobi identity:

$$\sum_{A=1}^{d_r} (f^{aAe} f^{bcA} + f^{bAe} f^{caA} + f^{cAe} f^{abA}) = 0. \tag{3.72}$$

Writing the commutation relation

$$[t^a_{\text{adj}}, t^b_{\text{adj}}]_{de} = \mathrm{i} \sum_{c=1}^{d} f^{abc} (t^c_{\text{adj}})_{de}, \tag{3.73}$$

in terms of structure constants gives:

$$\sum_{A=1}^{d} \left[(\mathrm{i} f^{aAd})(\mathrm{i} f^{bAe}) - (\mathrm{i} f^{bAd})(\mathrm{i} f^{aAe}) \right] = \mathrm{i} \sum_{c=1}^{d} f^{abc} (\mathrm{i} f^{cde}). \tag{3.74}$$

Thus we have:

$$\sum_{A=1}^{d} \left(-f^{aAd} f^{bAe} + f^{bAd} f^{aAe} \right) = -\sum_{A=1}^{d} f^{abA} f^{Ade}. \tag{3.75}$$

Using the anti-symmetry property of structure constant: $f^{abc} = -f^{bac}$, we can reduce to the standard Jacobi identity form:

$$0 = \sum_{A=1}^{d} \left(-f^{aAd} f^{bAe} + f^{bAd} f^{aAe} + f^{Ade} f^{abA} \right) \tag{3.76}$$

$$= \sum_{A=1}^{d} \left(f^{aAd} f^{beA} + f^{bAd} f^{eaA} + f^{eAd} f^{abA} \right), \tag{3.77}$$

which is nothing but the Jacobi identity (3.72).

1.5 Inner product of generators

Now since the Lie algebra generators are all matrices, so it's natural to consider the inner product of two matrices. The most natural way is by the trace of matrix product. So we define a matrix D_r^{ab} for generators t_r^a as

$$D_r^{ab} = \mathrm{tr}\big(t_r^a t_r^b\big). \tag{3.78}$$

For $a, b = 1, \ldots, d = N^2 - 1$. Cyclicity of trace doesn't change the matrix, so it's symmetry and real-valued:

$$D_r^{ab} = D_r^{ba}, \tag{3.79}$$

and also Hermitian:

$$(D_r^{ab})^\dagger = \mathrm{tr}((t_r^a)^\dagger (t_r^b)^\dagger) = D_r^{ab}. \tag{3.80}$$

As we know from linear algebra, for any real-valued symmetry matrix, one can always diagonalize it by a orthogonal transformation. Thus, diagonalization of D_r^{ab} gives

$$\mathrm{tr}\left(t_r^a t_r^b\right) = C(r)\delta^{ab}. \tag{3.81}$$

Since the generators are diagonal, then the position of each components in the same t_r are equal, which imply that $C(r)$ is the same for any component of t_r. Thus $C(r)$ is called the **index of representation** or **Dynkin index**.

Example: $SU(N)$

The generators for $SU(2)$ is $t^a_{\text{fund}} = \frac{\sigma^a}{2} \quad (a = 1, 2, 3)$, where σ^a is the Pauli matrix:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.82}$$

For the fundamental representation matrix, we can check that

$$\mathrm{tr}(t_r^a t_r^a) = \frac{1}{2}. \tag{3.83}$$

So

$$C(r) = \frac{1}{2}. \tag{3.84}$$

More over, for general case $SU(N)$, $C(r) = \frac{1}{2}$ due to the normalization of N-dim matrices are all similar with $SU(2)$. Using the index of representation, we can find the structure constants of $SU(N)$ satisfy:

$$\begin{aligned}\mathrm{tr}([t^a, t^b]t_r^c) &= \mathrm{i}f^{abc}\mathrm{tr}(t_r^c t_r^c) \\ &= \mathrm{tr}(t_r^a t_r^b t_r^c - t_r^b t_r^a t_r^c) = \mathrm{i}f^{abc}C(r) \\ &= \mathrm{tr}(t_r^a[t_r^b, t_r^c]) = \mathrm{i}f^{abc}C(r).\end{aligned} \tag{3.85}$$

This shows that f^{abc} are totally anti-symmetric:

$$f^{abc} = -f^{bac} = f^{bca}. \tag{3.86}$$

As we all learned from quantum mechanics: $[J^2, J_i] = 0$. So similarly, one can define an operator T^2 that satisfies the similar commutation relation. This operator is called the **quadratic Casimir operator**, which is defined as:

$$T_{ij}^2 = \sum_{a=1}^{d(G)} \sum_{k=1}^{d(r)} t_{ik}^a t_{kj}^a, \tag{3.87}$$

where t_{ij}^a is fundamental representation. We can check that for any a, we all have: $[t^a, T^2] = 0$, namely T^2 commutes with all generators:

$$[t^a, T^2] = \left[t^a, \sum_b t^b t^b\right] = \sum_b \left([t^a, t^b]t^b + t^b[t^a, t^b]\right) \tag{3.88}$$

$$= \sum_b \left(\left(\mathrm{i}\sum_c f^{abc}t^c\right)t^b + t^b\left(\mathrm{i}\sum_c f^{abc}t^c\right)\right) \tag{3.89}$$

$$= \mathrm{i} \sum_{b,c} f^{abc}(t^c t^b + t^b t^c) \tag{3.90}$$

$$= \mathrm{i} \sum_{c,b} f^{acb}(t^b t^c + t^c t^b) \tag{3.91}$$

$$= \mathrm{i} \sum_{b,c} (-f^{abc})(t^c t^b + t^b t^c) \tag{3.92}$$

$$= -\mathrm{i} \sum_{b,c} f^{abc}(t^c t^b + t^b t^c) \tag{3.93}$$

$$= 0 \tag{3.94}$$

This implies that T^2 must be proportional to identity:

$$T^2_{ij} = C_2(r)\delta_{ij} \tag{3.95}$$

And $C_2(r)$ is called the **quadratic Casimir coefficient**. It can be obtained by taking the trace in both sides:

$$\mathrm{tr}\big(T^2\big) = \mathrm{tr}\left(\sum_{a=1}^{d(G)} t^a t^a\right) = \sum_{a=1}^{d(G)} \mathrm{tr}(t^a t^a) = C_2(r)\,\mathrm{tr}\,\delta_{ij}, \tag{3.96}$$

thus we have:

$$C(r)d(G) = C_2(r)d(r) \tag{3.97}$$

The identity involving two structure constants follows by taking r to be the adjoint representation. From (3.71),

$$(t^m_{\mathrm{adj}})_{an} = \mathrm{i} f^{man}. \tag{3.98}$$

Therefore the quadratic Casimir in the adjoint representation gives

$$\begin{aligned}\sum_{m,n} (t^m_{\mathrm{adj}})_{an}(t^m_{\mathrm{adj}})_{nb} &= \sum_{m,n} (\mathrm{i} f^{man})(\mathrm{i} f^{mnb}) \\ &= -\sum_{m,n} f^{man} f^{mnb} \\ &= \sum_{m,n} f^{amn} f^{bmn} = C_2(\mathrm{adj})\delta^{ab}. \end{aligned} \tag{3.99}$$

We denote

$$C_A \equiv C_2(\mathrm{adj}), \tag{3.100}$$

so

$$f^{amn} f^{bmn} = C_A \delta^{ab}. \tag{3.101}$$

1.6 Representation of Lorentz group

The Lorentz group, as we mentioned before, is $SO(1,3)$ with metric $\eta = (1,-1,-1,-1)$. The group element Λ with infinitesimal expansion can be written as

$$\Lambda_\nu{}^\mu = \delta_\nu{}^\mu + \mathrm{i} \sum_a \omega^a (t^a)_\nu{}^\mu + \mathcal{O}(\omega^2). \tag{3.102}$$

Defining relation of $SO(N, M)$ groups says

$$\Lambda^T \eta \Lambda = \eta, \tag{3.103}$$

thus the infinitesimal form implies

$$\left(\mathbb{1} + \mathrm{i}\omega^a (t^a)^T\right) \cdot \eta \cdot (\mathbb{1} + \mathrm{i}\omega^a t^a) = \eta, \tag{3.104}$$

after dropping the $\mathcal{O}(\omega^2)$, one obtains the anti-symmetric relation:

$$(t^a)^T \eta = -\eta t^a. \tag{3.105}$$

As we mentioned in the last subsection, $SO(1,3)$, as a subgroup of $O(1,3)$, has $\frac{1}{2} \times 4 \times (4-1) = 6$ generators, physically are 3 rotation + 3 boosts, which usually noted as $J^{\mu\nu}$. To limit the number of free parameters in $J^{\mu\nu}$, we require

$$J^{\rho\sigma} = -J^{\sigma\rho}. \tag{3.106}$$

So we change the note as

$$\sum_a \omega^a t^a \to \sum_{\rho\sigma} \omega_{\rho\sigma} J^{\rho\sigma}, \tag{3.107}$$

where $J^{\rho\sigma}$ fulfill $SO(1,3)$ algebra. Generally, all $SO(N, M)$ groups have also spinor representations generators

$$S^{\mu\nu} = \frac{\mathrm{i}}{4}[\gamma^\mu, \gamma^\nu], \tag{3.108}$$

fulfill $SO(1,3)$ algebra:

$$[J^{\mu\nu}, J^{\rho\sigma}] = \mathrm{i}(\eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}). \tag{3.109}$$

1.7 Integration over gauge group

Since a Lie group is also a manifold, so we can introduce the integration structure based on the group element g and function $f(g)$, the measurement $\mathrm{d}g$ is called Haar measure, which include the left and right invariant measure:

$$\text{left invariant measure}: \quad \int (\mathrm{d}g)_L f(g) = \int (\mathrm{d}g)_L f(g'g); \tag{3.110}$$

$$\text{right invariant measure}: \quad \int (\mathrm{d}g)_R f(g) = \int (\mathrm{d}g)_R f(gg'), \tag{3.111}$$

where the normalization is $\int \mathrm{d}g\, 1 = 1$, and for most of situation the left and right Haar measure are the same $\mathrm{d}g = (\mathrm{d}g)_L = (\mathrm{d}g)_R$. This measurement is just like the normal integral is invariant under the transformation $x \to x + C$, where C is a constant:

$$\int \mathrm{d}x f(x) = \int \mathrm{d}x f(x + C). \tag{3.112}$$

To obtain the computable form of the meature, we need explicit realization of $\mathrm{d}g$ with metric tensor M_{ij}:

$$M_{ij} \equiv \mathrm{tr}\left(g^{-1}\partial_i g \cdot g^{-1}\partial_j g\right), \tag{3.113}$$

where $\partial_i g = \frac{\partial}{\partial \alpha_i} g(\alpha)$. Then we can connect to the normal integral:

$$\int \mathrm{d}g\, f(g) = N \int \mathrm{d}\alpha |\det M|^{1/2} f(g(\alpha)), \tag{3.114}$$

in which the normalization is the Jacobi determinant.

Example

- $U(1) = \{e^{\mathrm{i}\theta} \mid -\pi < \theta < \pi\}$: For $g(\theta) = e^{\mathrm{i}\theta}$ we have

$$M_{\theta\theta} = \mathrm{tr}\left(g^{-1}\partial_\theta g \cdot g^{-1}\partial_\theta g\right) = \mathrm{tr}\left(\mathrm{i}\cdot\mathrm{i}\right) = -1. \tag{3.115}$$

And using the fact

$$1 = N\int_{-\pi}^{\pi} \mathrm{d}\theta \cdot 1 = N \cdot 2\pi, \tag{3.116}$$

thus $N = \frac{1}{2\pi}$. So the integral is

$$\int \mathrm{d}g\, f(g) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \mathrm{d}\theta f(e^{\mathrm{i}\theta}) \tag{3.117}$$

- $SU(2) = \{a_0 + \mathrm{i}\vec{\sigma}\cdot\vec{a} \mid a_0^2 + \vec{a}^2 = 1\}$: For the element $g(a)$ we have:

$$\partial_0 g = \frac{\partial g}{\partial a_0} = \mathbb{1}_{2\times 2} \tag{3.118}$$

$$\partial_k g = \frac{\partial g}{\partial a_k} = \mathrm{i}\sigma_k \quad (k = 1, 2, 3) \tag{3.119}$$

Thus, the metric is

$$M = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}, \tag{3.120}$$

therefore, $|\det M|^{1/2} = \sqrt{16} = 4$. The last piece is the normalization factor, for that we need to compute the integral for unity. Since the constrain require a sphere, we then can express the unity by delta function:

$$1 = N\int \mathrm{d}^4 a\, \delta(a^2 - 1) = N\int_0^\infty r^3 \mathrm{d}r \int \mathrm{d}\Omega_3\, \delta(r^2 - 1), \tag{3.121}$$

thus $N = \frac{1}{\pi^2}$. And finally:

$$\int \mathrm{d}g\, f(g) = \frac{1}{\pi^2}\int \mathrm{d}^4 a\, \delta(a^2 - 1) f(g(a)). \tag{3.122}$$

- More general case, for $SU(N)$, Schur's orthogonality relation requires

$$\int \mathrm{d}g\, g_{ij}(g^{-1})_{kl} = \frac{1}{N}\delta_{il}\delta_{jk}. \tag{3.123}$$

2. Quantization of non-Abelian gauge theory

2.1 Preparation

To quantize the non-Abelian gauge theory, we need to list what do we have now, everything begins with the Lagrangian:

$$\mathcal{L} = \sum_{j=1}^{d(r)} \left(\mathrm{i}\bar{\psi}_j \gamma^\mu (D_\mu)_{jk} \psi_k - m\bar{\psi}_j \psi_j \right) - \kappa \mathrm{tr}(F_{\mu\nu} F^{\mu\nu}), \tag{3.124}$$

where the covariant derivative is defined as:

$$(D_\mu)_{jk} = \partial_\mu \delta_{jk} - \mathrm{i}g(A_\mu)_{jk}, \tag{3.125}$$

and the field strength is:

$$(F_{\mu\nu})_{jk} = [D_\mu, D_\nu]_{jk}. \tag{3.126}$$

Under the gauge transformation, the gauge fields obey the rule:

$$A_\mu \to A'_\mu = U A_\mu U^\dagger - \frac{\mathrm{i}}{g} (\partial_\mu U) U^\dagger. \tag{3.127}$$

To work in the perturbation theory, we need to explicit show the colour indices. Thus we expand A_μ in basis of adjoint representation:

$$(A_\mu)_{jk} = \sum_{a=1}^{d(G)} A_\mu^a t_{jk}^a \tag{3.128}$$

The infinitesimal transformation with $U_{jk} = \mathbb{1}_{jk} + \mathrm{i} \sum_a \alpha^a (t^a)_{jk}$ then becomes:

$$\begin{aligned}
&(\delta A_\mu)_{jk} = (A'_\mu)_{jk} - (A_\mu)_{jk} \\
&= -(A_\mu)_{jk} + \left(\mathbb{1} + \mathrm{i} \sum_a \alpha^a (t^a) \right)_{jm} (A_\mu)_{mn} \left(\mathbb{1} - \mathrm{i} \sum_b \alpha^b (t^b) \right)_{nk} + \frac{1}{g} \sum_a (\partial_\mu \alpha^a)(t^a)_{jm} \left(\mathbb{1} - \mathrm{i} \sum_b \alpha^b (t^b) \right)_{mk} \\
&\simeq \mathrm{i} \sum_{a=1}^{d(G)} \alpha^a A_\mu^b [t^a, t^b]_{jk} + \frac{1}{g} \sum_{a=1}^{d(G)} (\partial_\mu \alpha^a) t_{jk}^a + \mathcal{O}(\alpha^2) \\
&= \frac{1}{g} \left[-g \sum_{b,c=1}^{d(G)} f^{abc} A_\mu^c \alpha^b + \partial_\mu \alpha^a \right] t_{jk}^a,
\end{aligned} \tag{3.129}$$

thus we have

$$\delta A_\mu^a = \frac{1}{g} (D_\mu \alpha)^a = \frac{1}{g} \sum_{b=1}^{d(G)} (D_\mu)^{ab} \alpha^b. \tag{3.130}$$

where $(D_\mu)^{ab}$ is the covariant derivative in adjoint representation:

$$(D_\mu)^{ab} = \partial_\mu \delta^{ab} - g \sum_{c=1}^{d(G)} f^{abc} A_\mu^c. \tag{3.131}$$

2.2 Quantization by path integral

Now we are ready to start quantization via path integral method. We have already seen that the action $S[A]$ is gauge invariant. But

$$\int \mathcal{D}A\, e^{-S[A]} \tag{3.132}$$

is not well defined, because $\mathcal{D}A$ counts infinite gauge redundant. And if one integrate two times over the pure Yang-Mills term $F^a_{\mu\nu}F^{\mu\nu a}$, one can see that the momentum $M_{\mu\nu}$ satisfies

$$M_{\mu\nu} = -k^2\eta_{\mu\nu} + k_\mu k_\nu, \tag{3.133}$$

which is degenerated (or not invertible):

$$M_{\mu\nu}k^\nu = -k^2 k_\mu + k_\mu k^2 = 0, \tag{3.134}$$

which means the determinate is 0. So we cannot define the propagator due to it's nothing but the inverse of the momentum operator, which we proved in chapter 1.

The idea to solve these problem comes from Faddeev and Popov. To divide out redundant gauge degrees of freedom, we require a gauge condition

$$G^a(A^u) = 0, \tag{3.135}$$

for a particular point A^u in an orbit of the gauge group, where u is a group element of the non-Abelian gauge group (e.g. $SU(3)$). Therefore, A^u is a gauge choice for A, in the other words, A is gauge-less. So like the way we having $\int \delta(x-a)\mathrm{d}x = 1$, we insert the gauge condition into the delta function and define a similar quantity called **Faddeev-Popov(FP) determinant $\Delta_{\mathbf{G}}[\mathbf{A}]$**:

$$\Delta_G^{-1}[A] \equiv \int \mathcal{D}u\, \delta(G^a(A^u)). \tag{3.136}$$

So by this definition, one can get a useful identity:

$$1 = \Delta_G[A] \int \mathcal{D}u\, \delta\left(G^a(A^u)\right). \tag{3.137}$$

One can check that this functional determinant is gauge invariant by continuously doing two times gauge transformation: u and u'. So the fields change to $(A^u)^{u'} = A^{u'u}$. So by definition, the new determinant is given by

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u\, \delta\Big(G^a\big((A^{u'})^u\big)\Big) = \int \mathcal{D}u\, \delta(G^a(A^{u'u})), \tag{3.138}$$

Now we introduce a new group element $u'' = u'u$, in which transformation the measure keeps invariant: $\mathcal{D}u = \mathcal{D}u''$, thus

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u''\, \delta(G^a(A^{u''})). \tag{3.139}$$

Changing the integral variable name from u'' to u, we then verify that the FP determinant is indeed invariant under gauge transformation:

$$\Delta_G^{-1}[A^{u'}] = \int \mathcal{D}u\, \delta(G^a A^u) = \Delta_G^{-1}[A] \tag{3.140}$$

Now we insert (3.137) into path integral

$$\int \mathcal{D}A\,(1)\,e^{-S[A]} = \int \mathcal{D}A\,\Delta_G[A] \int \mathcal{D}u\,\delta(G^a(A^u))e^{-S[A]}. \tag{3.141}$$

Since only the delta function in the middle integral is u dependence, and the measure $\mathcal{D}$, the determinant $\Delta_G[A]$ and the action $S[A]$ are all gauge invariant, so we can change the variable (or choose this specific gauge) $A \to A' = A^u$ without causing any difference:

$$\int \mathcal{D}A\,\Delta_G[A] \int \mathcal{D}u\,\delta(G^a(A^u))e^{-S[A]} = \int \mathcal{D}A'^{u^{-1}}\,\Delta_G\left[A'^{u^{-1}}\right] \int \mathcal{D}u\,\delta(G^a(A'))e^{-S\left[A'^{u^{-1}}\right]} \tag{3.142}$$

$$= \int \mathcal{D}A'\,\Delta_G[A'] \int \mathcal{D}u\,\delta(G^a(A'))e^{-S[A']}. \tag{3.143}$$

The trick that we are playing now can be understand with a simple comparison in calculus. For an integral in one dimsion

$$I = \int_{-\infty}^{+\infty} \mathrm{d}x \int_0^{2\pi} \mathrm{d}\theta\, f(x+\theta)g(x+\theta), \tag{3.144}$$

which obviously is θ dependent. But we can change the variable $y \equiv x + \theta$. Since θ in the second integral is a constant, thus $\mathrm{d}y = \mathrm{d}x$ with the same integration range. So the integral changes to

$$I = \int_{-\infty}^{+\infty} \mathrm{d}y \int_0^{2\pi} \mathrm{d}\theta\, f(y)g(y), \tag{3.145}$$

where the integrand is now θ independent. With the same trick, now the functional integral is u independent, so we can change the integration sequence and rename the variable from A' to A as

$$\left(\int \mathcal{D}u\right) \cdot \int \mathcal{D}A\,\Delta_G[A]\,\delta(G^a(A))\,e^{-S[A]}, \tag{3.146}$$

and since the integral over $\mathcal{D}u$ represent the "volume" of the gauge group, which is infinite. So we can just consider the rest part, which is physical and removed all the gauge redundancy as the correct definition of path integral for non-Abelian gauge theories:

$$\int \mathcal{D}A\,\Delta_G[A]\,\delta(G^a(A))\,e^{-S[A]}. \tag{3.147}$$

So far so good, it seems that we successfully eliminated the gauge redundancy. However, if we consider a specific example, say Coulomb gauge $G^a(A) = \nabla \cdot A^a = 0$, and consider the derivative of G with respect to α, so we have

$$\frac{\delta G^a_\mu}{\delta \alpha^b} = \nabla \cdot \frac{\delta A^a_\mu}{\delta \alpha^b} = \frac{1}{g}\nabla \cdot D^{ab}_\mu, \tag{3.148}$$

where δA^a_μ is given by (3.130), and (3.131) gives

$$\vec{\nabla} \cdot \vec{D}^{ab} = \nabla^2\delta^{ab} - gf^{abc}\vec{A}^c \cdot \vec{\nabla}, \tag{3.149}$$

which is the Coulomb gauge $\vec{\nabla} \cdot \vec{D}^{ab}$ operator. The problem is it can have non-trivial solution for A_i, namely,

$$(\vec{\nabla} \cdot \vec{D}^{ab})\alpha^b = 0, \tag{3.150}$$

where $\alpha^b(x) \neq 0$. It means we find a new field by the gauge transformation $\alpha^b(x)$ as $A' = A + \delta A$, with the new gauge condition:

$$G^a(A') = G^a(A) + \delta G^a = 0 + \frac{1}{g}(\vec{\nabla} \cdot \vec{D}^{ab})\alpha^b = 0, \tag{3.151}$$

which also satisfies the Coulomb gauge $\vec{\nabla} \cdot \vec{A}'^a = 0$. This problem is called the **Gribov Ambiguity**, which only troubles in non-Abelian gauge theory, e.g. QCD, due to the Abelian gauge theory have the boundary $\alpha \to 0$ at infinity and zero structure constant $f^{abc} = 0$. But for non-Abelian theory, with larger enough A in low energy confinement case, it's possible for a non-zero α satisfying

$$\nabla^2 \alpha^a = g f^{abc} \vec{A}^c \cdot \vec{\nabla} \alpha^b. \tag{3.152}$$

But at least we solved the divergent problem due to infinite gauge choice with the price of introducing a delta function and an FP determinant. These two, however also make trouble because we cannot expand perturbation series directly. The latter is limited by the delta function that only valid around $G = 0$. Therefore, to compute FP determinant, we can do the infinitesimal transformation around $G = 0$. We first consider $G(A^u)$ as function of group element $u(\alpha)$, then change the integral measure in (3.136) to: $\mathcal{D}u = \mathcal{D}G \det \left|\frac{\delta\alpha}{\delta G}\right|$, so we have:

$$\Delta_G^{-1}[A] = \int \mathcal{D}G \det \left|\frac{\delta\alpha}{\delta G}\right| \delta(G). \tag{3.153}$$

Integration over gauge group G gives the FP determinant, then we get:

$$\Delta_G[A] = \det \left|\frac{\delta G}{\delta\alpha}\right|_{G=0}. \tag{3.154}$$

So (3.147) becomes:

$$\int \mathcal{D}A\, \delta(G(A)) \det \left|\frac{\delta G}{\delta\alpha}\right| e^{-S[A]}. \tag{3.155}$$

Next step we are going to put $\delta(G)$ and FP determinant into the exponential to get a pure path integral form. For that, we consider the Gauss integral

$$\int_{-\infty}^{+\infty} \mathrm{d}c\; \mathrm{e}^{-\frac{c^2}{2\xi}} \delta(G - c) = \mathrm{e}^{-\frac{G^2}{2\xi}}. \tag{3.156}$$

Generating into the functional space, with $a = 1, \ldots, d(G)$, and change the gauge $G^a = 0$ to $G^a = C^a$ with constants C^a. We have:

$$\int \mathcal{D}C^a\, e^{-\frac{1}{2\xi}\int \mathrm{d}^4x\, C^a C^a} \delta(G^a - C^a) = e^{-\frac{1}{2\xi}\int \mathrm{d}^4x\, G^a G^a}, \tag{3.157}$$

where we successfully put the delta function into the exponential. The price is adding a **gauge fixing term** in Lagrangian:

$$S_{\text{gf}} = \int \mathrm{d}^4x\, \frac{1}{2\xi} G^a \cdot G^a. \tag{3.158}$$

Now we left a determinant needs to be eliminated :det $\left|\frac{\delta G}{\delta\alpha}\right|$. This can be done by introducing the Grassmann number in the path integral:

$$\det \left|\frac{\delta G^a(\alpha(x))}{\delta\alpha^b(y)}\right| = \int \mathcal{D}\bar{\eta}\mathcal{D}\eta\, e^{+\int \mathrm{d}^4x \mathrm{d}^4y\, \bar{\eta}^a(x) \frac{\delta G^a(x)}{\delta\alpha^b(y)} \eta^b(y)}, \tag{3.159}$$

where these two new field η and $\bar{\eta}$ is the **Faddeev-Popov Ghost fields**:

$$S_{\text{ghost}} = -\int \mathrm{d}^4x\mathrm{d}^4y\ \bar{\eta}^a(x)\frac{\delta G^a(x)}{\delta\alpha^b(y)}\eta^b(y). \tag{3.160}$$

Now we are ready to write the final form of Faddeev-Popov quantization:

$$\int \mathcal{D}A\mathcal{D}\bar{\eta}\mathcal{D}\eta\, e^{-S[A,\bar{\eta},\eta]}, \tag{3.161}$$

by inserting (3.158) and (3.159) into the Lagrangian:

$$\begin{aligned} S[A,\bar{\eta},\eta] &= S_{\text{YM}}[A] + S_{\text{gf}}[A] + S_{\text{ghost}}[A,\bar{\eta},\eta] && (3.162)\\ &= \int \mathrm{d}^4x \left[\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} - \bar{\psi}(\mathrm{i}\not{D} - m)\psi\right] + \frac{1}{2\xi}\int \mathrm{d}^4x\, G^a(x)G^a(x) - \int \mathrm{d}^4x\mathrm{d}^4y\, \bar{\eta}^a(x)\frac{\delta G^a(x)}{\delta\alpha^b(y)}\eta^b(y). && (3.163)\end{aligned}$$

It is worthy to note that we used **Euclidean convention**, which is

$$\int \mathcal{D}Ae^{-S}. \tag{3.164}$$

If we use **Minkowski convention** with the sign $(+---)$, which is

$$\int \mathcal{D}Ae^{\mathrm{i}S}. \tag{3.165}$$

So the action becomes

$$S_{\text{M}}[A,\bar{\eta},\eta] = \int \mathrm{d}^4x \left[\bar{\psi}(\mathrm{i}\not{D} - m)\psi - \frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu}\right] - \frac{1}{2\xi}\int \mathrm{d}^4x\, G^a(x)G^a(x) + \int \mathrm{d}^4x\mathrm{d}^4y\, \bar{\eta}^a(x)\frac{\delta G^a(x)}{\delta\alpha^b(y)}\eta^b(y). \tag{3.166}$$

2.3 Feynman rules for non-Abelian gauge theory

The gauge condition we choose is the **covariant gauge(Lorenz gauge)**

$$G^a = \partial^\mu A^a_\mu = 0. \tag{3.167}$$

The quadratic terms in A_μ from $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + gf^{abc}A^b_\mu A^c_\nu$ and $G^a = \partial^\mu A^a_\mu$ can be rewritten by the partial integration as:

$$\begin{aligned} S_{\text{quadratic}} &= \int \mathrm{d}^4x \left(\frac{1}{4}(\partial^\mu A^a_\nu - \partial^\nu A^a_\mu)(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu) + \frac{1}{2\xi}(\partial^\mu A^a_\mu)(\partial^\nu A^a_\nu)\right) && (3.168)\\ &= \int \mathrm{d}^4x \left(\frac{1}{2}\partial^\mu A^{\nu a}\partial_\mu A^a_\nu - \frac{1}{2}\partial^\nu A^{\mu a}\partial_\mu A^a_\nu + \frac{1}{2\xi}(\partial^\mu A^a_\mu)(\partial^\nu A^a_\nu)\right) && (3.169)\\ &= \frac{1}{2}\int \mathrm{d}^4x\, A^a_\mu \left(-\partial_\rho\partial^\rho g_{\mu\nu} + \left(1 - \frac{1}{\xi}\right)\partial_\mu\partial_\nu\right) A^{\nu a} + \text{Boundary terms}. && (3.170)\end{aligned}$$

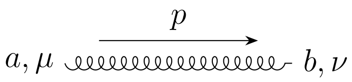
Now with this gauge fixing term, we can check that the momentum operator becomes:

$$M_{\mu\nu} = -k^2 g_{\mu\nu} + \left(1 - \frac{1}{\xi}\right)k_\mu k_\nu, \tag{3.171}$$

satisfying:

$$M_{\mu\nu}k^\nu = -k^2 k_\mu + \left(1 - \frac{1}{\xi}\right) k_\mu k^2 = -\frac{1}{\xi} k^2 k_\mu \neq 0. \tag{3.172}$$

Thus, we can then define the **propagators** legally by take inverse of momentum operator, which can be read from (3.170) as



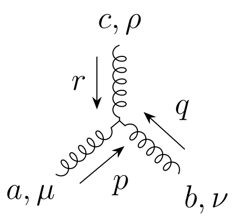
$$= \frac{-\mathrm{i}\delta^{ab}}{p^2 + \mathrm{i}\varepsilon} \left(g_{\mu\nu} - (1-\xi)\frac{p_\mu p_\nu}{p^2}\right). \tag{3.173}$$

And the **gauge boson self-interaction** is vertices is given by the non-Abelian term in $\frac{1}{4}F^2$:

$$\mathcal{L}_{3\mathrm{A}} = -g f^{abc}(\partial_\mu A_\nu^a) A^{\mu b} A^{\nu c} \tag{3.174}$$

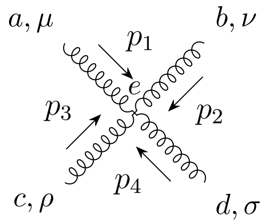
$$\mathcal{L}_{4\mathrm{A}} = \frac{1}{4} g^2 f^{abc} f^{ade} A_\mu^b A_\nu^c A^{d\mu} A^{e\nu}. \tag{3.175}$$

The first three gauge boson self interaction gives:



$$= -\mathrm{i} g f^{abc} \left[(r_\mu - q_\mu) g_{\nu\rho} + (q_\rho - p_\rho) g_{\mu\nu} + (p_\nu - r_\nu) g_{\mu\rho}\right] \tag{3.176}$$

The second four gauge boson self interaction gives:



$$= -\mathrm{i} g^2 \left[f^{abe} f^{cde}(g_{\mu\rho} g_{\nu\sigma} - g_{\mu\sigma} g_{\nu\rho}) + f^{ace} f^{bde}(g_{\mu\nu} g_{\rho\sigma} - g_{\mu\sigma} g_{\nu\rho}) + f^{ade} f^{bce}(g_{\mu\nu} g_{\rho\sigma} - g_{\mu\rho} g_{\nu\sigma})\right]. \tag{3.177}$$

Since we have introduce a new field: ghost field $\bar{\eta}, \eta$, we then can also find it's Feynman rule from the Lagrangian. For this, we need to first calculate the derivative by (3.130) and the gauge $G^a = \partial^\mu A_\mu^a$:

$$\frac{\delta G^a(x)}{\delta \alpha^b(y)} = \frac{\delta G^a(x)}{\delta A_\mu^c(x)} \frac{\delta A_\mu^c(x)}{\delta \alpha^b(y)} = \frac{1}{g} \frac{\delta G^a(x)}{\delta A_\mu^c(x)} D_\mu^{bc} \delta(x-y) = \frac{1}{g} \partial^\mu D_\mu^{bc} \delta(x-y), \tag{3.178}$$

with the form of covariant derivative, we have:

$$\frac{\delta G^a(x)}{\delta \alpha^b(y)} = \frac{1}{g} \partial^\mu (\partial_\mu \delta^{ab} - g f^{abc} A_\mu^c) \delta(x-y)。 \tag{3.179}$$

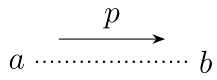
Thus the ghost action becomes:

$$S_{\text{ghost}} = \int \mathrm{d}^4 x\, \bar{\eta}^a(x) \left[\frac{1}{g} \partial^\mu (\partial_\mu \delta^{ab} - g f^{abc} A_\mu^c)\right] \eta^b(x) \tag{3.180}$$

$$= \frac{1}{g} \int \mathrm{d}^4 x\, \bar{\eta}^a \partial_\mu \partial^\mu \eta^a - \int \mathrm{d}^4 x\, f^{abc} \bar{\eta}^a \partial^\mu (A_\mu^c \eta^b) \tag{3.181}$$

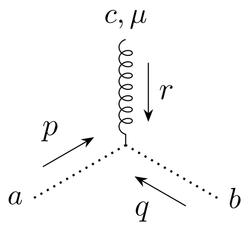
$$= \int \mathrm{d}^4 x\, \bar{\eta}^a \partial_\mu \partial^\mu \eta^a - g \int \mathrm{d}^4 x\, f^{abc} \bar{\eta}^a \partial^\mu (A_\mu^c \eta^b) \tag{3.182}$$

where we rescaled the field as $\eta \to \sqrt{g}\eta$ and $\bar{\eta} \to \sqrt{g}\bar{\eta}$ in the last step. Thus the first term is kinetic term and second is interaction term. So the kinetic term gives the **ghost propagators**:



$$= \mathrm{i}\frac{\delta^{ab}}{p^2}. \tag{3.183}$$

And second interaction term gives the **vertex of ghost-ghost-gluon:**



$$= \frac{g}{2} f^{abc}(r^\mu + p^\mu - q^\mu) = -g f^{abc} q^\mu, \tag{3.184}$$

where the vertex energy-momentum conservation requires $r^\mu + p^\mu + q^\mu = 0$.

3. BRST Symmetry

3.1 Problem of negative norm

In last section, we introduced the gauge fixing term in the action to absorbing the gauge condition $G^a = 0$:

$$S_{\mathrm{gf}} = \int \mathrm{d}^4 x \, \frac{1}{2\xi} G^a \cdot G^a, \tag{3.185}$$

which leads to the dynamical term of the gauge field as

$$\frac{1}{2\xi}(\partial^\mu A^a_\mu)(\partial^\nu A^a_\nu). \tag{3.186}$$

However, this operation breaks the local gauge symmetry because it contains $\partial_0 A_0$. The problem of this term is A_0 has a negative norm:

$$\langle 0|A_\mu(k)A^\dagger_\nu(k')|0\rangle \propto -\eta_{\mu\nu}\delta^{(4)}(k-k'), \tag{3.187}$$

thus

$$\langle 0|A_0(k)A^\dagger_0(k')|0\rangle \propto -\delta^{(4)}(k-k') < 0. \tag{3.188}$$

This should not be a problem if local gauge symmetry was not broken, because there wouldn't be the term like $\partial_0 A_0$ get into the kinematic term, i.e., propagator. So the negative norm wasn't shown when we are doing real calculations. Now everything is on the table, the optical theorem tell us we need to count all the contributions including the negative norm final states. So it may cause the sum over all physical final states (positive norm states) is greater than 1. And since the gauge choice can be changed, then the value of the probability may no longer conserved or even depends on the gauge choice, which is unacceptable.

3.2 Infinitesimal BRST transformation

To solve this problem, Becchi, Rouet, Stora and Tyutin(BRST) found that there exist a global symmetry, namely the **BRST symmetry**, which allows us to separate the non-physical states.

To introduce this symmetry, we first consider the Gaussian integral formula:

$$\int_{-\infty}^{\infty} \mathrm{d}B \exp\left\{-\frac{a}{2}B^2\right\} = \sqrt{\frac{2\pi}{a}}. \tag{3.189}$$

Change the variable $B \to B + \frac{J}{a}$:

$$\int_{-\infty}^{\infty} \mathrm{d}B \exp\left\{-\frac{a}{2}\left(B+\frac{J}{a}\right)^2\right\} = \int_{-\infty}^{\infty} \mathrm{d}B \exp\left\{-\frac{a}{2}B^2 - JB - \frac{J^2}{2a}\right\} = \text{Constant}. \tag{3.190}$$

Thus, dividing $\exp\left(\frac{J^2}{2a}\right)$ on both sides, we have

$$\exp\left\{\frac{J^2}{2a}\right\} \propto \int_{-\infty}^{\infty} \mathrm{d}B \exp\left\{-\frac{a}{2}B^2 - JB\right\}. \tag{3.191}$$

Now we can rewrite the gauge fixing term from the Faddeev-Popov Lagrangian by letting $J = \partial^\mu A^a_\mu$ and $a = -\xi$ as

$$\exp\left\{i\int \mathrm{d}^4x \left(-\frac{1}{2\xi}(\partial^\mu A^a_\mu)^2\right)\right\} \propto \int \mathcal{D}B \exp\left\{i\int \mathrm{d}^4x \left(\frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A^a_\mu)\right)\right\}, \tag{3.192}$$

where B^a is an auxiliary field, which means this field is **non-dynamical**, because it doesn't contribute any term including $\partial_\mu B^a$. So the gauge fixing Lagrangian becomes:

$$\mathcal{L}_{\text{gf}} = \frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A^a_\mu). \tag{3.193}$$

We can check this Lagrangian gives the same form with the Gaussian integration by implying the Euler-Lagrange equation:

$$\frac{\delta \mathcal{L}_{\text{gauge fix}}}{\delta B^a} = \xi B^a + \partial^\mu A^a_\mu = 0, \tag{3.194}$$

$$\implies B^a = -\frac{1}{\xi}\partial^\mu A^a_\mu. \tag{3.195}$$

So the Faddeev-Popov Lagrangian is

$$\mathcal{L}_{\text{FP}} = \bar{\psi}(\mathrm{i}\not{D} - m)\psi - \frac{1}{4}F^a_{\mu\nu}F^{\mu\nu a} + \frac{\xi}{2}(B^a)^2 + B^a(\partial^\mu A^a_\mu) + \bar{c}^a(-\partial^\mu D^{ab}_\mu)c^b. \tag{3.196}$$

We have already derived variation of A from (3.130):

$$\delta A^a_\mu = D^{ab}_\mu \alpha^b(x), \tag{3.197}$$

where we absorbed the factor $1/g$ into the arbitrary function $\alpha^b(x) \to g\alpha^b(x)$. But we now want $\alpha^b(x)$ can related to the ghost field:

$$\alpha^b(x) = g\varepsilon c^b(x), \tag{3.198}$$

where ε is a constant that non-related with the space-time. So if we also do the same thing for the other fields' infinitesimal transformations, how will the Lagrangian change? This leads us to the **BRST transformations**. BRST claimed that there is a global symmetry from ε that the transformation keep the Lagrangian invariant

$$\delta_\varepsilon \mathcal{L} = 0. \tag{3.199}$$

And the infinitesimal parameter must be Grassmann-valued:

$$\varepsilon^2 = 0, \tag{3.200}$$

because A^a_μ is a bosonic field and c^b is a fermionic field, so ε must be a ferimionic field, which fulfills the Grassmann algebra.

$$\varepsilon c^b = -c^b \varepsilon. \tag{3.201}$$

Thus the BRST transformation for A^a_μ and ψ_j can be derived by implying (3.198) into the normal infinitesimal transformation $\delta A^a_\mu = D^{ab}_\mu \alpha^b(x)$ and $\delta_\varepsilon \psi_j = \mathrm{i}\sum_a \alpha^a (t^a\psi)_j$:

$$\delta_\varepsilon A^a_\mu = \varepsilon D^{ab}_\mu c^b = \varepsilon \left(\partial_\mu c^a + g \sum_{b,c} f^{abc} A^b_\mu c^c \right), \tag{3.202}$$

$$\delta_\varepsilon \psi_j = \mathrm{i} g \varepsilon \sum_a c^a (t^a \psi)_j. \tag{3.203}$$

These two transformations obviously keep the $\mathcal{L}_{\mathrm{YM}}$ invariant, because it just changes a name of α, and since αc^a is still a local gauge transformation, then $\mathcal{L}_{\mathrm{YM}}$ is automatically invariant under this transformation. One can check that $F^a_{\mu\nu}$ follows

$$\delta_\varepsilon F^a_{\mu\nu} = g\varepsilon f^{abc} F^b_{\mu\nu} c^c, \tag{3.204}$$

thus

$$\delta_\varepsilon \left(-\frac{1}{4} F^a_{\mu\nu} F^{\mu\nu a} \right) = 0. \tag{3.205}$$

And for the fermionic field, the covariant derivative follows:

$$\delta_\varepsilon (D_\mu \psi) = ig\varepsilon c^a t^a (\partial_\mu \psi - igA^b_\mu t^b \psi) = ig\varepsilon c^a t^a (D_\mu \psi), \tag{3.206}$$

thus

$$\delta_\varepsilon \left(\overline{\psi}(i\gamma^\mu D_\mu - m)\psi \right) = (\delta_\varepsilon \overline{\psi})(i\gamma^\mu D_\mu - m)\psi + \overline{\psi}(i\gamma^\mu \delta_\varepsilon (D_\mu \psi) - m\delta_\varepsilon \psi) = 0. \tag{3.207}$$

But it's worthy to mention that **BRST transformation is not a local gauge transformation (transformation parameter ε is global)**. But it indeed **comes from a local gauge transformation (transformation parameter $\alpha^a(x) = g\varepsilon c^a(x)$ is local)**.

Now it comes to the most tedious part, the gauge fixing term and the ghost term. This is because these two terms are not gauge invariant. For convenient, we can just check the following variation of the ghost field and auxiliary field:

$$\delta_\varepsilon c^a = -\frac{1}{2} g\varepsilon \sum_{b,c} f^{abc} c^b c^c \tag{3.208}$$

$$\delta_\varepsilon \bar{c}^a = \varepsilon B^a \tag{3.209}$$

$$\delta_\varepsilon B^a = 0, \tag{3.210}$$

keep the Lagrangian of gauge fixing and ghost invariant:

$$\delta_\varepsilon (\mathcal{L}_{\mathrm{gf}} + \mathcal{L}_{\mathrm{ghost}}) = 0, \tag{3.211}$$

namely to prove:

$$\delta_\varepsilon\left(\frac{\xi}{2}(B^a)^2\right)+\underbrace{(\delta_\varepsilon B^a)}_{=0}(\partial^\mu A^a_\mu)+B^a\ \underbrace{\delta_\varepsilon(\partial^\mu A^a_\mu)}_{=\partial^\mu(\varepsilon D^{ab}_\mu c^b)}-\underbrace{(\delta_\varepsilon\bar{c}^a)}_{\varepsilon B^a}(\partial^\mu D^{ab}_\mu)c^b+\underbrace{\bar{c}^a\delta_\varepsilon(\partial^\mu D^{ab}_\mu)c^b+\bar{c}^a(\partial^\mu D^{ab}_\mu)(\delta_\varepsilon c^b)}_{\bar{c}^a\partial^\mu\delta_\varepsilon(D^{ab}_\mu c^b)}=0. \tag{3.212}$$

The first term is simply 0 because

$$\delta_\varepsilon(B^a)^2=2\delta_\varepsilon B^a=0 \tag{3.213}$$

And for the third term

$$\partial^\mu(\varepsilon D^{ab}_\mu c^b)=\underbrace{(\partial^\mu\varepsilon)}_{=0}D^{ab}_\mu c^b+\varepsilon\partial^\mu(D^{ab}_\mu c^b)=\varepsilon\partial^\mu(D^{ab}_\mu c^b), \tag{3.214}$$

due to ε is a constant. Now the expression to be proved reduces to

$$\cancel{B^a\varepsilon\partial^\mu D^{ab}_\mu c^b}-\cancel{\varepsilon B^a\partial^\mu D^{ab}_\mu c^b}-\bar{c}^a\partial^\mu(\delta_\varepsilon D^{ab}_\mu c^b) \tag{3.215}$$

$$=-\bar{c}^a\partial^\mu\left\{\partial_\mu(\delta_\varepsilon c^a)+g\sum_{b,c}f^{abc}(\delta_\varepsilon A^b_\mu)c^c+g\sum_{b,c}f^{abc}A^b_\mu(\delta_\varepsilon c^c)\right\} \tag{3.216}$$

$$=-\bar{c}^a\partial^\mu\left\{\partial_\mu\left(-\frac{g}{2}\varepsilon\sum_{b,c}f^{abc}c^bc^c\right)+g\sum_{b,c}f^{abc}\left[\varepsilon\left(\partial_\mu c^b+g\sum_{d,e}f^{bde}A^d_\mu c^e\right)\right]c^c\right.$$
$$\left.+g\sum_{b,c}f^{abc}A^b_\mu\left(-\frac{g}{2}\varepsilon\sum_{d,e}f^{cde}c^dc^e\right)\right\} \tag{3.217}$$

$$=-\bar{c}^a\partial^\mu\left\{g\varepsilon\sum_{b,c}f^{abc}\left[-\frac{1}{2}(\partial_\mu c^bc^c)+(\partial_\mu c^b)c^c\right]+g^2\varepsilon\sum_{b,c,e,f}f^{abc}\left(f^{bde}A^d_\mu c^ec^c-\frac{1}{2}f^{cde}A^b_\mu c^dc^e\right)\right\} \tag{3.218}$$

The first term will be canceled by using the property of Grassmann number $c^ac^b=-c^bc^a$ and structure constant $f^{abc}=-f^{acb}$:

$$\sum_{b,c}f^{abc}\partial_\mu(c^bc^c)=\sum_{b,c}f^{abc}\left[(\partial_\mu c^b)c^c+c^b(\partial_\mu c^c)\right] \tag{3.219}$$
$$=\sum_{b,c}\left[f^{abc}(\partial_\mu c^b)c^c+f^{acb}c^c(\partial_\mu c^b)\right] \tag{3.220}$$
$$=\sum_{b,c}\left[f^{abc}(\partial_\mu c^b)c^c-f^{abc}c^c(\partial_\mu c^b)\right] \tag{3.221}$$
$$=\sum_{b,c}\left[f^{abc}(\partial_\mu c^b)c^c+f^{abc}(\partial_\mu c^b)c^c\right] \tag{3.222}$$
$$=2\sum_{b,c}f^{abc}(\partial_\mu c^b)c^c. \tag{3.223}$$

Now the only left term is

$$-\bar{c}^a\partial^\mu g^2\varepsilon\sum_{b,c,e,f}f^{abc}\left(f^{bde}A^d_\mu c^ec^c-\frac{1}{2}f^{cde}A^b_\mu c^dc^e\right), \tag{3.224}$$

which can be canceled by Jacobi identity

$$f^{ade}f^{bcd}+f^{bde}f^{cad}+f^{cde}f^{abd}=0. \tag{3.225}$$

For that, we first relabel the indices d,e in the first term:

$$\sum_{b,c,d,e} f^{abc}f^{bde}c^ec^c=\frac{1}{2}\sum_{b,c,d,e}(f^{abc}f^{bde}-f^{abe}f^{bdc})c^ec^c. \tag{3.226}$$

Thus,

$$\begin{aligned}
&\sum_{b,c,d,e} f^{abc}\left(f^{bde}A_\mu^dc^ec^c-\frac{1}{2}f^{cde}A_\mu^bc^dc^e\right) &(3.227)\\
&=\sum_{b,c,d,e} f^{abc}f^{bde}A_\mu^dc^ec^c-\frac{1}{2}f^{abc}f^{cde}A_\mu^bc^dc^e &(3.228)\\
&=\frac{1}{2}\sum_{b,c,d,e}(f^{abc}f^{bde}-f^{abe}f^{bdc})A_\mu^dc^ec^c-f^{abc}f^{cde}A_\mu^bc^dc^e &(3.229)\\
&=\frac{1}{2}\sum_{b,c,d,e}\left(f^{ace}f^{cbd}-f^{acd}f^{cbe}-f^{abc}f^{cde}\right)A_\mu^bc^dc^e &(3.230)\\
&=\frac{1}{2}\sum_{b,c,d,e}\left(-f^{dbc}f^{ace}+f^{ebc}f^{acd}-f^{abc}f^{cde}\right)A_\mu^bc^dc^e &(3.231)\\
&=\frac{1}{2}\sum_{b,c,d,e}\left(f^{dbc}f^{aec}-f^{ebc}f^{adc}-f^{abc}f^{dec}\right)A_\mu^bc^dc^e &(3.232)\\
&=\frac{1}{2}\sum_{b,c,d,e}\left(f^{dbc}f^{aec}+f^{ebc}f^{dac}+f^{abc}f^{edc}\right)A_\mu^bc^dc^e &(3.233)\\
&=0. &(3.234)
\end{aligned}$$

Finally, with such tedious calculation, we have successfully proved (3.211) being satisfied, which means the gauge fixing term and ghost term together are BRST invariant. Thus, the Faddeev-Popov Lagrangian is BRST invariant:

$$\delta_\varepsilon\mathcal{L}_{\text{FP}}=\delta_\varepsilon\mathcal{L}_{\text{YM}}+\delta_\varepsilon\left(\mathcal{L}_{\text{gf}}+\mathcal{L}_{\text{ghost}}\right)=0. \tag{3.235}$$

3.3 Factorization of Hilbert space via BRST operator

Now since BRST symmetry is a global symmetry, we can in principle define a conservation charge Q, which should upgrade to an operator in Hilbert space since we have already quantized the classical field. This operator thus called the **BRST operator** Q, which can be easily defined from the infinitesimal transformation:

$$\delta_\varepsilon\Phi=[\varepsilon Q,\Phi]=\varepsilon(Q\Phi);\qquad \text{for bosonic field } \Phi \tag{3.236}$$

$$\delta_\varepsilon\Psi=\{\varepsilon Q,\Psi\}=-\varepsilon(Q\Psi);\qquad \text{for fermionic field } \Psi. \tag{3.237}$$

Thus Q has similar properties with δ_ε, which can be directly copy from (3.202),(3.203),(3.208),(3.209) and (3.210):

$$QA^a_\mu = D^{ab}_\mu c^b \tag{3.238}$$
$$Q\psi_j = \mathrm{i}gc^a(t^a\psi_j) \tag{3.239}$$
$$Qc^a = -\frac{1}{2}gf^{abc}c^bc^c \tag{3.240}$$
$$Q\bar{c}^a = B^a \tag{3.241}$$
$$QB^a = 0, \tag{3.242}$$

and with (3.200), one can prove that Q satisfies **Nilpotency**:

$$Q^2 = 0, \tag{3.243}$$

by acting on the gauge field A^a_μ, the ghost and anti-ghost field c^a and $\bar{c}^a$ and the auxiliary field B^a respectively to show the results are all vanished.

With BRST operator, we can rewrite the Faddeev-Popov Lagrangian as

$$\mathcal{L}_{\text{FP}} = \mathcal{L}_{\text{YM}} + Q\underbrace{\left(\bar{c}^a\partial^\mu A^a_\mu + \frac{\xi}{2}\bar{c}^aB^a\right)}_{\text{BRST ancestors}}. \tag{3.244}$$

With this form, we can easily see that it's BRST invariant by using (3.243) and $\delta_\varepsilon\mathcal{L}_{\text{YM}} = 0$:

$$Q\mathcal{L}_{\text{FP}} = 0, \quad Q\mathcal{L}_{\text{YM}} = 0 \tag{3.245}$$

Now we can use BRST operator to show how those "nonphysical" states are separated. Since BRST operator as a conserved "charge". It naturally commutes with Hamiltonian operator H:

$$[Q, H] = 0, \quad Q = Q^\dagger. \tag{3.246}$$

The Hilbert space of eigenstates of H in Heisenberg picture can be formally expressed as

$$\mathcal{H} = \mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_0, \tag{3.247}$$

where the subspace can be defined by the map:

$$Q\,:\mathcal{H} \to \mathcal{H}. \tag{3.248}$$

The kernel of Q, i.e., those states are mapped to 0, is

$$\text{Ker}(Q) = \{|\psi\rangle \in \mathcal{H} \mid Q|\psi\rangle = 0\}, \tag{3.249}$$

and the image is

$$\text{Im}(Q) = \{|\psi\rangle \in \mathcal{H} \mid |\psi\rangle = Q|\phi\rangle,\ \text{ for } |\phi\rangle \in \mathcal{H},\}. \tag{3.250}$$

And since $Q^2 = 0$, so for any states in $\text{Im}(Q)$ are all in $\text{Ker}(Q)$ too, i.e.,

$$\text{Im}(Q) \subset \text{Ker}(Q). \tag{3.251}$$

Thus, we define the three states as follows:

$$\mathcal{H}_1 = \mathcal{H} \setminus \mathrm{Ker}(Q) = \{|\psi_1\rangle \in \mathcal{H} \mid Q|\psi_1\rangle \neq 0\}, \tag{3.252}$$
$$\mathcal{H}_2 = \mathrm{Im}(Q) = \{|\psi_2\rangle \in \mathcal{H} \mid |\psi_2\rangle = Q|\phi\rangle,\ |\phi\rangle \in \mathcal{H}_1\}, \tag{3.253}$$
$$\mathcal{H}_0 = \mathrm{Ker}(Q) \setminus \mathrm{Im}(Q) = \{|\psi_0\rangle \in \mathcal{H} \mid Q|\psi_0\rangle = 0 \text{ and } \forall|\phi\rangle \in \mathcal{H},\ |\psi\rangle \neq Q|\phi\rangle\}. \tag{3.254}$$

With these definition, we can clearly see that Hilbert space is factorized into the direct sum over three subspace :

$$\mathcal{H} = \underbrace{\big(\mathcal{H} \setminus \mathrm{Ker}(Q)\big)}_{\mathcal{H}_1} \oplus \underbrace{\mathrm{Im}(Q)}_{\mathcal{H}_2} \oplus \underbrace{\big(\mathrm{Ker}(Q) \setminus \mathrm{Im}(Q)\big)}_{\mathcal{H}_0}. \tag{3.255}$$

With this, we claim the set of all physical states is the kernel of $\mathcal{H}$ mode the image:

$$\mathcal{H}_{\text{phys}} \cong \mathcal{H}_0 = \mathrm{coh}(Q) = \frac{\mathrm{Ker}(Q)}{\mathrm{Im}(Q)}, \tag{3.256}$$

which is a **cohomology space** satisfying:

$$Q\,|\psi\rangle = 0. \tag{3.257}$$

To show that it indeed make sense, we need to go back to the free theory, namely $g \to 0$, so the BRST transformation reduce to

$$QA^a_\mu = \partial_\mu c^a, \tag{3.258}$$
$$Q\bar{c}^a = B^a, \tag{3.259}$$
$$Q\psi_j = Qc^a = QB^a = 0. \tag{3.260}$$

We find this result with the definition of $\mathcal{H}_1$, $\mathcal{H}_2$ and $\mathcal{H}_0$ can help us to classify these asymptotic states:

$$A^{\mathrm{L}}_\mu \in \mathcal{H}_1, \quad \bar{c}^a \in \mathcal{H}_1; \tag{3.261}$$
$$A^{\mathrm{T}}_\mu \in \mathcal{H}_0, \quad \psi_j \in \mathcal{H}_0; \tag{3.262}$$
$$c^a \in \mathcal{H}_2, \quad B^a \in \mathcal{H}_2. \tag{3.263}$$

where A^{L}_μ represent the longitudinal polarization

$$A^{\mathrm{L}}_\mu \sim k_\mu \phi \implies Q|A^{\mathrm{L}}_\mu\rangle \sim |c^a\rangle \neq 0, \tag{3.264}$$

and A^{T}_μ represent the transversal polarization

$$k^\mu A^{\mathrm{T}}_\mu = 0 \implies QA^{\mathrm{T}}_\mu \sim k_\mu c^a = 0. \tag{3.265}$$

With this classification, we can see that for the states in $\mathcal{H}_1$, like A^{L}_μ or $\bar{c}^a$, the norm of time component are **negative**:

$$\big\langle A^{\mathrm{L}}_0 \big| A^{\mathrm{L}}_0 \big\rangle = \langle 0|a_0 a_0^\dagger|0\rangle = \langle 0|[a_0, a_0^\dagger]|0\rangle = -1 < 0, \tag{3.266}$$
$$\langle \bar{\eta}|\bar{\eta}\rangle = \langle 0|\bar{c}\,\bar{c}^\dagger|0\rangle = \langle 0|(-\bar{c}^\dagger)^\dagger \bar{c}^\dagger|0\rangle = -\langle 0|\bar{c}\,\bar{c}^\dagger|0\rangle = \langle 0|\,\{c, \bar{c}^\dagger\}\,|0\rangle = -1 < 0. \tag{3.267}$$

For the states in $\mathcal{H}_2$ are **zero norm** due to the nilpotency $Q^2 = 0$:

$$\langle \psi_2|\psi_2\rangle = \langle \psi_1|Q^\dagger Q|\psi_1\rangle = \langle \psi_1|Q^2|\psi_1\rangle \equiv 0. \tag{3.268}$$

And the states in $\mathcal{H}_0$ are **positive norm**, which is the real physical state space. So we say, physical degrees of freedom are elements of $\mathcal{H}_0$, and unphysical degrees of freedom are in $\mathcal{H}_1, \mathcal{H}_2$.

This result however, one may wondering if it is picture dependent since we assume the Heisenburg picture that the states doesn't evolve with time. The answer is lucky no, picture does not change under time evolution: for any state $|\psi_0, t\rangle$ in $\mathcal{H}$, we have

$$Q|\psi_0, t\rangle = Qe^{\mathrm{i}Ht}|\psi_0\rangle = e^{\mathrm{i}Ht}Q|\psi_0\rangle = 0, \tag{3.269}$$

which means $|\psi_0, t\rangle$ must be a linear combination of states from $\mathcal{H}_0$ and $\mathcal{H}_2$, i.e., $|\psi_0, t\rangle \in \mathcal{H}_0 \oplus \mathcal{H}_2$. But if we consider a scattering process from $|\psi_0, t\rangle$ to any $|\psi_2\rangle \in \mathcal{H}_2$ that satisfies $|\psi_2\rangle = Q\,|\psi_1\rangle$, we have

$$\langle\psi_2|\psi_0, t\rangle = \langle\psi_1|Q|\psi_0, t\rangle = 0, \tag{3.270}$$

which means the scalar product vanishes! Namely, $|\psi_0, t\rangle$ has no effective component in $\mathcal{H}_2$. So even though we mix some element of $\mathcal{H}_2$ in the combination of the construction of $|\psi_0, t\rangle$, it doesn't matter because the elements in $\mathcal{H}_2$ won't contribute to the final result.

Thus, we find that **BRST operator factorize the space based on positive norm, negative norm and zero norm**. which helps us to select and drop the negative and zero norm states.

4. Renormalization of non-Abelian gauge theory

Now we have enough foundation to consider the renormalization of Padeev–Popov theory. For that, we first recall the Feynman rules of Faddeev–Popov Lagrangian for a Yang–Mills theory coupled to fermions,

$$\mathcal{L}_{\text{FP}} = \mathcal{L}_{\text{YM}} - \frac{1}{2\xi}\left(\partial^\mu A^a_\mu\right)\left(\partial^\nu A^a_\nu\right) + \bar{c}^a\left(-\partial^\mu D^{ab}_\mu\right)c^b. \tag{3.271}$$

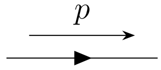
It is useful to separate it into the free part and the interaction part:

$$\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}. \tag{3.272}$$

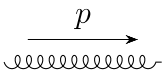
In Feynman gauge, $\xi = 1$, the free Lagrangian reads

$$\mathcal{L}_0 = \bar{\psi}\left(\mathrm{i}\not{\partial} - m\right)\psi + \frac{1}{2}A^a_\mu\left(g^{\mu\nu}\Box - \partial^\mu\partial^\nu\right)A^a_\nu. \tag{3.273}$$

The corresponding fermion and gauge boson propagators are



$$= \frac{\mathrm{i}}{\not{p} - m + \mathrm{i}\varepsilon} = \frac{\mathrm{i}(\not{p} + m)}{p^2 - m^2 + \mathrm{i}\varepsilon}, \tag{3.274}$$

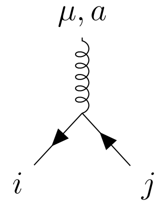
a, μ  b, ν

$$= \frac{-\mathrm{i}\,\delta^{ab}}{p^2 + \mathrm{i}\varepsilon}\left[g_{\mu\nu} - (1 - \xi)\frac{p_\mu p_\nu}{p^2 + \mathrm{i}\varepsilon}\right]. \tag{3.275}$$

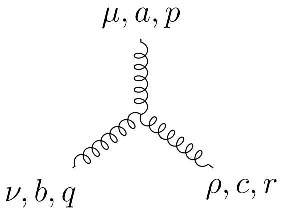
The interaction terms are

$$\begin{aligned}\mathcal{L}_{\text{int}} =& gA^a_\mu\left(\bar{\psi}\gamma^\mu t^a\psi\right) - gf^{abc}\left(\partial_\mu A^a_\nu\right)A^{b\mu}A^{c\nu} \\ & - \frac{1}{4}g^2\left(f^{abc}A^b_\mu A^c_\nu\right)\left(f^{ade}A^{d\mu}A^{e\nu}\right) + gf^{abc}\left(\partial^\mu\bar{c}^a\right)A^b_\mu c^c.\end{aligned} \tag{3.276}$$

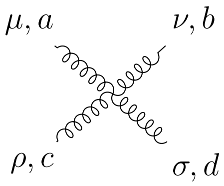
With all momenta conventionally taken incoming, the basic vertices are:



$$= \mathrm{i}g\,\gamma^\mu (t^a)_{ij}, \tag{3.277}$$

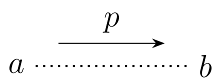


$$= g f^{abc}\left[g^{\mu\nu}(p-q)^\rho + g^{\nu\rho}(q-r)^\mu + g^{\rho\mu}(r-p)^\nu\right], \tag{3.278}$$

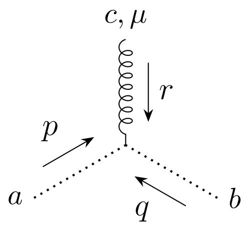


$$\begin{aligned} = -\mathrm{i}g^2\Big[& f^{abe} f^{cde}\left(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\sigma}g^{\nu\rho}\right) \\ & + f^{ace} f^{bde}\left(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\sigma}g^{\nu\rho}\right) \\ & + f^{ade} f^{bce}\left(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma}\right)\Big]. \end{aligned} \tag{3.279}$$

For the ghost propagator and ghost–gluon vertex one has



$$= \mathrm{i}\frac{\delta^{ab}}{p^2}; \tag{3.280}$$



$$= \frac{g}{2} f^{abc}(r^\mu + p^\mu - q^\mu) = -g f^{abc} q^\mu, \tag{3.281}$$

which are the result that we should be familiar with in previous section.

4.1 Why the Faddeev–Popov theory is renormalizable

Before introducing the explicit counterterms, we should explain why the Faddeev–Popov theory is expected to be renormalizable at all. The point is not to guess the counterterms first, but to prove that the ultraviolet divergent part of the theory can only have the same local form as the original Faddeev–Popov Lagrangian. This statement follows from two ingredients: **power counting** and **BRST symmetry**.

Limitation from power counting

We work in four space-time dimensions. Just as we did in Chapter 2. Since the action is dimensionless,

$$[S] = 0, \qquad [\mathrm{d}^4 x] = -4, \qquad [\mathcal{L}] = 4. \tag{3.282}$$

The dimensions of the fields are fixed by the quadratic terms. From the gauge kinetic term,

$$-\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} \supset -\frac{1}{2}(\partial_\mu A^a_\nu)(\partial^\mu A^{a\nu}), \tag{3.283}$$

we get

$$2(1+[A_\mu]) = 4 \quad \Longrightarrow \quad [A_\mu] = 1. \tag{3.284}$$

From the fermion kinetic term,

$$\bar{\psi}\, \mathrm{i}\gamma^\mu \partial_\mu \psi, \tag{3.285}$$

we get

$$[\bar{\psi}] + 1 + [\psi] = 4, \qquad [\bar{\psi}] = [\psi], \quad \Longrightarrow \quad [\psi] = [\bar{\psi}] = \frac{3}{2}. \tag{3.286}$$

Finally, from the ghost kinetic term,

$$-\bar{c}^a \partial_\mu \partial^\mu c^a, \tag{3.287}$$

we get

$$[\bar{c}] + 2 + [c] = 4. \tag{3.288}$$

The natural symmetric assignment is

$$[c] = [\bar{c}] = 1. \tag{3.289}$$

Therefore the dimensions of all interaction terms are

$$[A_\mu \bar{\psi}\gamma^\mu \psi] = 1 + \frac{3}{2} + \frac{3}{2} = 4, \tag{3.290}$$

$$[(\partial A)AA] = 1 + 1 + 1 + 1 = 4, \tag{3.291}$$

$$[AAAA] = 4, \tag{3.292}$$

$$[(\partial \bar{c})Ac] = 1 + 1 + 1 + 1 = 4. \tag{3.293}$$

Thus the coupling g is dimensionless:

$$[g] = 0. \tag{3.294}$$

This is the first indication of renormalizability just like (2.265). A negative-dimensional coupling would force higher and higher powers of momentum to appear at higher orders, producing infinitely many independent divergent structures. Here this does not happen.

Now we make the statement precise by computing the superficial degree of divergence. At large loop momentum, the propagators in (3.274), (3.275) and (3.280), behave as

$$S_\psi(k) \sim \frac{\not{k}}{k^2} \sim \frac{1}{k}, \qquad D_A(k) \sim \frac{1}{k^2}, \qquad D_c(k) \sim \frac{1}{k^2}. \tag{3.295}$$

The three-gluon vertex and the ghost-gluon vertex each contain one derivative, which in momentum space becomes a off-shell momentum that need to be integrated, so they contribute one power of loop momentum in the numerator. The fermion-gluon vertex and the four-gluon vertex contain no derivative.

Let

$$I_A,\ I_c,\ I_\psi \tag{3.296}$$

be the numbers of internal gauge-boson, ghost and fermion lines. Let

$$V_3,\ V_4,\ V_c,\ V_\psi \tag{3.297}$$

denote the numbers of AAA, $AAAA$, $A\bar{c}c$ and $A\bar{\psi}\psi$ vertices. Let the corresponding external lines be

$$E_A,\ E_c,\ E_\psi, \tag{3.298}$$

where E_c counts external ghost and anti-ghost lines together, and E_ψ counts external fermion and anti-fermion lines together. The superficial degree of divergence in $d = 4$ from (2.256) is then

$$D = 4L - 2I_A - 2I_c - I_\psi + V_3 + V_c. \tag{3.299}$$

The terms in this formula have the following origin:

- each **loop integral** contributes $+4$ (integral measure d^4k);
- each **internal gauge** or **ghost propagator** contributes -2 ($\sim \frac{1}{k^2}$);
- each **internal fermion propagator** contributes -1 ($\sim \frac{\not{k}}{k^2}$);
- each **gauge boson vertex** $(\partial A)AA$ or **ghost-boson vertex** $\partial\bar{c}Ac$ vertex contributes $+1$ from its derivative ($\mathrm{i}\partial \to k$).

The topological relation between loops, internal lines and vertices is

$$L = I_A + I_c + I_\psi - (V_3 + V_4 + V_c + V_\psi) + 1. \tag{3.300}$$

We also have the line-counting identities

$$\begin{aligned} 2I_\psi + E_\psi &= 2V_\psi, &(3.301)\\ 2I_c + E_c &= 2V_c, &(3.302)\\ 2I_A + E_A &= 3V_3 + 4V_4 + V_c + V_\psi. &(3.303)\end{aligned}$$

Substituting the loop relation into D gives

$$\begin{aligned} D &= 4(I_A + I_c + I_\psi - V_3 - V_4 - V_c - V_\psi + 1) - 2I_A - 2I_c - I_\psi + V_3 + V_c\\ &= 4 + 2I_A + 2I_c + 3I_\psi - 3V_3 - 4V_4 - 3V_c - 4V_\psi. \end{aligned} \tag{3.304}$$

Now use the three line-counting identities:

$$\begin{aligned} 2I_A &= 3V_3 + 4V_4 + V_c + V_\psi - E_A, &(3.305)\\ 2I_c &= 2V_c - E_c, &(3.306)\\ 3I_\psi &= 3V_\psi - \frac{3}{2}E_\psi. &(3.307)\end{aligned}$$

Hence

$$\begin{aligned} D &= 4 + (3V_3 + 4V_4 + V_c + V_\psi - E_A) + (2V_c - E_c) + \left(3V_\psi - \frac{3}{2}E_\psi\right)\\ &\quad - 3V_3 - 4V_4 - 3V_c - 4V_\psi\\ &= 4 - E_A - E_c - \frac{3}{2}E_\psi. \end{aligned} \tag{3.308}$$

This is the crucial power-counting result:

$$\boxed{D = 4 - E_A - E_c - \frac{3}{2}E_\psi.} \tag{3.309}$$

The important point is that D is independent of the number of vertices. Thus **going to higher loop order does not create more and more divergent structures**. Only Green functions with sufficiently few external legs can be superficially divergent. For example,

- Two gauge boson external lines (gauge 1PI):

$$AA: \qquad D = 2. \tag{3.310}$$

- Two ghost external lines (ghost 1PI):

$$\bar{c}c: \qquad D = 2. \tag{3.311}$$

- Two fermion external lines (fermion 1PI):

$$\bar{\psi}\psi: \qquad D = 1. \tag{3.312}$$

- Three gauge external lines:

$$AAA: \qquad D = 1. \tag{3.313}$$

- Three gauge-ghost external lines:

$$A\bar{c}c: \qquad D = 1. \tag{3.314}$$

- Three gauge-fermion external lines:

$$A\bar{\psi}\psi: \qquad D = 0. \tag{3.315}$$

- Four gauge external lines:

$$AAAA: \qquad D = 0. \tag{3.316}$$

For amplitudes with more external fields, $D < 0$, which is (superficially) convergent after subdivergences have been subtracted, namely we don't need to consider the renormalization of terms like $AAAAA$. So we only need to consider the divergent in $D \leq 4$

Limitation from BRST symmetry: Slavnov-Taylor identity

However, power counting alone is not yet the full proof. It tells us that the divergent terms are local and have dimension no larger than 4, but it does not by itself explain why these local terms must combine into the gauge-invariant Faddeev–Popov structure. For this we use BRST symmetry. It is most transparent to introduce the auxiliary field B^a and write

$$\mathcal{L}_{\mathrm{FP}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\mu\nu} + \bar{\psi}(\mathrm{i}\not{D} - m)\psi + \frac{\xi}{2}(B^a)^2 + B^a\partial^\mu A^a_\mu + \bar{c}^a(-\partial^\mu D^{ab}_\mu)c^b. \tag{3.317}$$

The BRST transformations are

$$QA^a_\mu = D^{ab}_\mu c^b, \tag{3.318}$$
$$Qc^a = -\frac{g}{2}f^{abc}c^bc^c, \tag{3.319}$$
$$Q\bar{c}^a = B^a, \tag{3.320}$$
$$QB^a = 0, \tag{3.321}$$
$$Q\psi = \mathrm{i}gc^at^a\psi, \tag{3.322}$$
$$Q\bar{\psi} = -\mathrm{i}g\bar{\psi}c^at^a. \tag{3.323}$$

As shown in the BRST section of the notes, these transformations are nilpotent,

$$Q^2 = 0, \tag{3.324}$$

and the Faddeev–Popov Lagrangian is BRST invariant:

$$Q\mathcal{L}_{\mathrm{FP}} = 0. \tag{3.325}$$

Next, for understanding how the BRST symmetry limits the renormalization counter terms, we need to briefly introduce the Slavnov–Taylor identity. The idea is not mysterious: it is simply the Ward identity associated with BRST symmetry. In the path integral we integrate over all fields as

$$Z[J] = \int \mathcal{D}\Phi \, \exp\left\{ \mathrm{i}S_{\mathrm{FP}}[\Phi] + \mathrm{i}\int J\Phi \right\}, \tag{3.326}$$

where Φ denotes all fields, for example $A_\mu, c, \bar{c}, B, \psi, \bar{\psi}$. And S_{FP} is the **classical action**, by contrast the **effective action** can be expressed as:

$$\Gamma[\phi_{\mathrm{cl}}] = W[J] - \int d^4x \, J(x)\phi_{\mathrm{cl}}(x), \tag{3.327}$$

where $W[J] = -i \ln Z[J]$, and $\phi_{\mathrm{cl}}(x) = \frac{\delta W[J]}{\delta J(x)} = \langle \phi(x) \rangle_J$. It can related with the source by

$$\frac{\delta \Gamma[\phi_{\mathrm{cl}}]}{\delta \phi_{\mathrm{cl}}(x)} = -J(x). \tag{3.328}$$

Thus, the effective action $\Gamma[\phi_{\mathrm{cl}}]$ gives the quantum corrected equation of motion instead of the classical one given by $S[\phi]$ by taking $J = 0$. If we make the BRST change of integration variables

$$\Phi \longrightarrow \Phi + \varepsilon Q\Phi, \tag{3.329}$$

with ε an infinitesimal Grassmann parameter, the value of the path integral cannot change. The reason is the same as in ordinary calculus: changing integration variables does not change the integral. If the action is BRST invariant and the regularization also preserves this symmetry, then both the measure and S_{FP} are unchanged. Therefore the only variation comes from the source term. This gives a Ward identity for the generating functional. After performing the Legendre transform from the connected generating functional to the quantum effective action Γ, this Ward identity is written as an equation of effective action Γ:

$$\mathcal{S}(\Gamma) = 0. \tag{3.330}$$

This equation is called the **Slavnov–Taylor identity**. Thus, in simple words, the Slavnov–Taylor identity says: **the full quantum effective action must still respect the BRST version of gauge symmetry.** Here Γ is important because it generates one-particle-irreducible amplitudes. Therefore $\mathcal{S}(\Gamma) = 0$ is not just a formal statement; it relates the possible divergent parts of propagators and vertices.

Next we explain why this constrains counterterms. Expand the effective action in loops:

$$\Gamma = S_{\mathrm{FP}} + \hbar\Gamma^{(1)} + \hbar^2\Gamma^{(2)} + \cdots . \tag{3.331}$$

At a fixed loop order, say at n loops, the contribution can be separated into a divergent part and a finite part:

$$\Gamma^{(n)} = \Gamma^{(n)}_{\mathrm{div}} + \Gamma^{(n)}_{\mathrm{fin}}. \tag{3.332}$$

In dimensional regularization the divergent part has the schematic local form

$$\Gamma^{(n)}_{\text{div}} = \sum_i \frac{a_i^{(n)}}{\epsilon} \int \mathrm{d}^4x\, \mathcal{O}_i(x), \qquad d = 4 - \epsilon. \tag{3.333}$$

The symbols $\mathcal{O}_i$ denote local operators built from the fields and their derivatives. For example, possible operators one might first write down are

$$A^a_\mu A^{a\mu}, \qquad F^a_{\mu\nu} F^{a\mu\nu}, \qquad \bar{\psi} \mathrm{i} \not{D} \psi. \tag{3.334}$$

Only some of them will actually be allowed after imposing BRST symmetry.

From now on, when we analyze one fixed loop order, we use the shorter notation

$$\boxed{\Delta \equiv \Gamma^{(n)}_{\text{div}}} \tag{3.335}$$

with the overall factor $\hbar^n$ suppressed. Thus Δ is not a new interaction invented by hand. It is the possible ultraviolet divergent part produced by loop integrals. Equivalently,

$$\Delta = \int \mathrm{d}^4x\, \mathcal{L}^{(n)}_{\text{div}}. \tag{3.336}$$

The counterterm action is introduced only after this divergence has appeared. At the same loop order we add

$$S^{(n)}_{\text{ct}} = -\Gamma^{(n)}_{\text{div}} = -\Delta, \tag{3.337}$$

up to possible finite scheme-dependent counterterms. Then the divergent part of the effective action is cancelled:

$$\Gamma^{(n)}_{\text{div}} + S^{(n)}_{\text{ct}} = \Delta - \Delta = 0. \tag{3.338}$$

This is the precise relation between Δ and the counterterm. Strictly speaking, Δ denotes the divergent local functional, while the counterterm added to the action is $-\Delta$. In common language one often calls the local structure Δ itself a "candidate counterterm", because the same operator appears in the counterterm action with the opposite coefficient. Therefore, asking whether a candidate Δ is allowed means asking whether a loop divergence of that local form can appear and, if it appears, whether it can be cancelled by a local counterterm. First introduce external sources, usually called BRST sources or antifields, which record the BRST variations of the fields. For the present discussion we write the extended classical action as

$$S_{\text{ext}} = S_{\text{FP}} + \int \mathrm{d}^4x \left[K^{a\mu}_A Q A^a_\mu + K^a_c Q c^a + K_\psi Q \psi + Q\bar{\psi}\, K_{\bar{\psi}} \right]. \tag{3.339}$$

The K's are not new physical fields. They are only a bookkeeping device. At the classical level their job is to make the following identities true:

$$\frac{\delta S_{\text{ext}}}{\delta K^{a\mu}_A} = Q A^a_\mu, \qquad \frac{\delta S_{\text{ext}}}{\delta K^a_c} = Q c^a, \qquad \frac{\delta S_{\text{ext}}}{\delta K_\psi} = Q\psi, \qquad \frac{\delta S_{\text{ext}}}{\delta K_{\bar{\psi}}} = Q\bar{\psi}. \tag{3.340}$$

Now define the path integral with two kinds of sources:

$$Z[J, K] = \int \mathcal{D}\Phi\, \exp\left\{ \mathrm{i} S_{\text{ext}}[\Phi, K] + \mathrm{i} \int \mathrm{d}^4x\, J_\Phi \Phi \right\}. \tag{3.341}$$

Here J_Φ is the ordinary source coupled to the field Φ, while K_Φ is the source coupled to the composite BRST variation $Q\Phi$. Define

$$W[J,K] = -\mathrm{i}\ln Z[J,K]. \tag{3.342}$$

Because K_A is coupled to QA, differentiating W with respect to K_A inserts QA into the path integral:

$$\frac{\delta W}{\delta K_A^{a\mu}(x)} = \left\langle QA_\mu^a(x)\right\rangle_{J,K}, \tag{3.343}$$

and similarly for $K_c, K_\psi, K_{\bar\psi}$. Then we can pass from W to the effective action. Define classical fields by

$$\Phi(x) = \frac{\delta W[J,K]}{\delta J_\Phi(x)} \tag{3.344}$$

and Legendre transform only with respect to the ordinary sources J, not with respect to the BRST sources K:

$$\Gamma[\Phi,K] = W[J,K] - \int \mathrm{d}^4x\, J_\Phi\Phi. \tag{3.345}$$

This gives

$$\frac{\delta\Gamma}{\delta\Phi} = -J_\Phi. \tag{3.346}$$

It also gives

$$\frac{\delta\Gamma}{\delta K_\Phi} = \frac{\delta W}{\delta K_\Phi}. \tag{3.347}$$

Let us check the second relation explicitly. When differentiating Γ with respect to K_Φ at fixed classical fields, the source J may depend on K:

$$\frac{\delta\Gamma}{\delta K_\Phi} = \frac{\delta W}{\delta K_\Phi} + \int \mathrm{d}^4x\,\frac{\delta W}{\delta J_\Psi}\frac{\delta J_\Psi}{\delta K_\Phi} - \int \mathrm{d}^4x\,\frac{\delta J_\Psi}{\delta K_\Phi}\Psi = \frac{\delta W}{\delta K_\Phi}, \tag{3.348}$$

because $\delta W/\delta J_\Psi = \Psi$. Therefore the two extra terms cancel.

Next perform the BRST change of variables inside the path integral,

$$\Phi \longrightarrow \Phi + \epsilon Q\Phi. \tag{3.349}$$

If the measure and the regularization preserve BRST symmetry, the integral does not change. The extended action is invariant because

$$QS_{\mathrm{FP}} = 0, \qquad Q\left(K_\Phi Q\Phi\right) = K_\Phi Q^2\Phi = 0, \tag{3.350}$$

where the K's are fixed external sources and $Q^2 = 0$. Therefore the only variation comes from the ordinary source term that changes as

$$\begin{aligned}\int \mathrm{d}^4x\, J_\Phi\Phi &\longrightarrow \int \mathrm{d}^4x\, J_\Phi(\Phi+\epsilon Q\Phi)\\ &= \int \mathrm{d}^4x\, J_\Phi\Phi + \epsilon\int \mathrm{d}^4x\, J_\Phi Q\Phi.\end{aligned} \tag{3.351}$$

Thus the exponential in $Z[J,K]$ changes, to first order in ϵ, by

$$\delta\exp\left\{\mathrm{i}S_{\mathrm{ext}} + \mathrm{i}\int \mathrm{d}^4x\, J_\Phi\Phi\right\} = \mathrm{i}\epsilon\left(\int \mathrm{d}^4x\, J_\Phi Q\Phi\right)\exp\left\{\mathrm{i}S_{\mathrm{ext}} + \mathrm{i}\int \mathrm{d}^4x\, J_\Phi\Phi\right\}. \tag{3.352}$$

But the whole path integral cannot change under a change of integration variables, so $\delta Z = 0$. Therefore

$$0 = \delta Z = \mathrm{i}\epsilon \int \mathcal{D}\Phi \left(\int \mathrm{d}^4x \, J_\Phi Q\Phi \right) \exp\left\{ \mathrm{i}S_{\text{ext}} + \mathrm{i}\int \mathrm{d}^4x \, J_\Phi \Phi \right\}. \tag{3.353}$$

Now divide by $Z[J,K]$ and remove the common factor iϵ. We get

$$0 = \int \mathrm{d}^4x \, J_\Phi \left\langle Q\Phi \right\rangle_{J,K}. \tag{3.354}$$

Expanding the shorthand over all fields gives

$$0 = \int \mathrm{d}^4x \left[J_A^{a\mu} \left\langle QA_\mu^a \right\rangle_{J,K} + J_c^a \left\langle Qc^a \right\rangle_{J,K} + J_\psi \left\langle Q\psi \right\rangle_{J,K} + J_{\bar\psi} \left\langle Q\bar\psi \right\rangle_{J,K} + J_{\bar c}^a \left\langle Q\bar c^a \right\rangle_{J,K} + J_B^a \left\langle QB^a \right\rangle_{J,K} \right]. \tag{3.355}$$

Using $Q\bar c^a = B^a$ and $QB^a = 0$, this becomes

$$0 = \int \mathrm{d}^4x \left[J_A^{a\mu} \left\langle QA_\mu^a \right\rangle_{J,K} + J_c^a \left\langle Qc^a \right\rangle_{J,K} + J_\psi \left\langle Q\psi \right\rangle_{J,K} + J_{\bar\psi} \left\langle Q\bar\psi \right\rangle_{J,K} + J_{\bar c}^a \left\langle B^a \right\rangle_{J,K} \right]. \tag{3.356}$$

Finally, the reason for introducing the K-sources was precisely that

$$\frac{\delta W}{\delta K_A^{a\mu}} = \left\langle QA_\mu^a \right\rangle_{J,K}, \qquad \frac{\delta W}{\delta K_c^a} = \left\langle Qc^a \right\rangle_{J,K}, \qquad \frac{\delta W}{\delta K_\psi} = \left\langle Q\psi \right\rangle_{J,K}, \qquad \frac{\delta W}{\delta K_{\bar\psi}} = \left\langle Q\bar\psi \right\rangle_{J,K}. \tag{3.357}$$

Substituting these identities into the previous equation gives

$$0 = \int \mathrm{d}^4x \left[J_A^{a\mu} \frac{\delta W}{\delta K_A^{a\mu}} + J_c^a \frac{\delta W}{\delta K_c^a} + J_\psi \frac{\delta W}{\delta K_\psi} + J_{\bar\psi} \frac{\delta W}{\delta K_{\bar\psi}} + J_{\bar c}^a \left\langle B^a \right\rangle_{J,K} \right], \tag{3.358}$$

up to the standard Grassmann signs. This is the BRST Ward identity for W.

Finally substitute (3.346) and (3.347) into (3.358). After multiplying the whole equation by an irrelevant minus sign, we obtain the Slavnov–Taylor identity for the effective action:

$$\mathcal{S}(\Gamma) = \int \mathrm{d}^4x \left[\frac{\delta\Gamma}{\delta K_A^{a\mu}} \frac{\delta\Gamma}{\delta A_\mu^a} + \frac{\delta\Gamma}{\delta K_c^a} \frac{\delta\Gamma}{\delta c^a} + \frac{\delta\Gamma}{\delta K_\psi} \frac{\delta\Gamma}{\delta\psi} + \frac{\delta\Gamma}{\delta K_{\bar\psi}} \frac{\delta\Gamma}{\delta\bar\psi} + B^a \frac{\delta\Gamma}{\delta\bar c^a} \right] = 0. \tag{3.359}$$

This derivation shows the important point: S_{ext} does not "turn into" Γ. Rather, S_{ext} defines the path integral, the BRST change of variables gives the identity for $W[J,K]$, and the Legendre transform rewrites that identity as an identity for $\Gamma[\Phi,K]$.

At tree level the effective action is the extended classical action,

$$\Gamma^{(0)} = S_{\text{ext}}. \tag{3.360}$$

Therefore, when we study the divergent part at one fixed loop order, we may write schematically

$$\Gamma = S_{\text{ext}} + \Delta \tag{3.361}$$

where Δ stands for the local divergent functional defined above. We have suppressed finite terms and higher-loop terms, because they are not needed for the first-order constraint on the divergent part. Now expand the Slavnov–Taylor identity to first order in Δ:

$$\mathcal{S}(S_{\text{ext}} + \Delta) = \mathcal{S}(S_{\text{ext}}) + \mathcal{S}_{S_{\text{ext}}}\Delta + \mathcal{O}(\Delta^2). \tag{3.362}$$

This first-order operator $\mathcal{S}_{S_{\rm ext}}$ is the linearized Slavnov–Taylor operator. In the notation of the discussion above we often write it as $\mathcal{S}_{S_{\rm FP}}$, because after setting the external sources K to zero it is the BRST operator associated with the Faddeev–Popov action.

To see the linearization explicitly, look only at the gauge-field part:

$$\begin{aligned}\frac{\delta\Gamma}{\delta K_A^{a\mu}}\frac{\delta\Gamma}{\delta A_\mu^a} &= \left(\frac{\delta S_{\rm ext}}{\delta K_A^{a\mu}}+\frac{\delta\Delta}{\delta K_A^{a\mu}}\right)\left(\frac{\delta S_{\rm ext}}{\delta A_\mu^a}+\frac{\delta\Delta}{\delta A_\mu^a}\right)\\ &= \underbrace{\frac{\delta S_{\rm ext}}{\delta K_A^{a\mu}}\frac{\delta S_{\rm ext}}{\delta A_\mu^a}}_{\mathcal{S}(S_{\rm ext})}+\underbrace{\frac{\delta S_{\rm ext}}{\delta K_A^{a\mu}}\frac{\delta\Delta}{\delta A_\mu^a}+\frac{\delta\Delta}{\delta K_A^{a\mu}}\frac{\delta S_{\rm ext}}{\delta A_\mu^a}}_{\mathcal{S}_{S_{\rm ext}(A)}\Delta}+\mathcal{O}(\Delta^2).\end{aligned} \tag{3.363}$$

The first term belongs to $\mathcal{S}(S_{\rm ext})$, which vanishes by classical BRST invariance. The remaining two terms are the linearized contribution. Doing the same expansion for every field gives

$$\begin{aligned}\mathcal{S}_{S_{\rm ext}}\Delta = \int \mathrm{d}^4x\, \Bigg[&\frac{\delta S_{\rm ext}}{\delta K_A^{a\mu}}\frac{\delta\Delta}{\delta A_\mu^a}+\frac{\delta\Delta}{\delta K_A^{a\mu}}\frac{\delta S_{\rm ext}}{\delta A_\mu^a}+\frac{\delta S_{\rm ext}}{\delta K_c^a}\frac{\delta\Delta}{\delta c^a}+\frac{\delta\Delta}{\delta K_c^a}\frac{\delta S_{\rm ext}}{\delta c^a}\\ &+\frac{\delta S_{\rm ext}}{\delta K_\psi}\frac{\delta\Delta}{\delta\psi}+\frac{\delta\Delta}{\delta K_\psi}\frac{\delta S_{\rm ext}}{\delta\psi}+\frac{\delta S_{\rm ext}}{\delta K_{\bar\psi}}\frac{\delta\Delta}{\delta\bar\psi}+\frac{\delta\Delta}{\delta K_{\bar\psi}}\frac{\delta S_{\rm ext}}{\delta\bar\psi}+B^a\frac{\delta\Delta}{\delta\bar c^a}\Bigg],\end{aligned} \tag{3.364}$$

again suppressing only the Grassmann sign factors. This is the explicit linearized Slavnov–Taylor operator.

For the ordinary counterterms in the Lagrangian, such as $A_\mu^a A^{a\mu}$, $F_{\mu\nu}^a F^{a\mu\nu}$, or $\bar\psi \mathrm{i}\not{D}\psi$, the candidate Δ does not contain the external sources K. Hence

$$\frac{\delta\Delta}{\delta K_A}=\frac{\delta\Delta}{\delta K_c}=\frac{\delta\Delta}{\delta K_\psi}=\frac{\delta\Delta}{\delta K_{\bar\psi}}=0. \tag{3.365}$$

Using the definition of the K-sources, (3.364) then reduces to

$$\begin{aligned}\mathcal{S}_{S_{\rm ext}}\Delta &= \int \mathrm{d}^4x\,\left[QA_\mu^a\frac{\delta\Delta}{\delta A_\mu^a}+Qc^a\frac{\delta\Delta}{\delta c^a}+Q\psi\frac{\delta\Delta}{\delta\psi}+Q\bar\psi\frac{\delta\Delta}{\delta\bar\psi}+B^a\frac{\delta\Delta}{\delta\bar c^a}\right]\\ &= \int \mathrm{d}^4x\,\left[QA_\mu^a\frac{\delta\Delta}{\delta A_\mu^a}+Qc^a\frac{\delta\Delta}{\delta c^a}+Q\bar c^a\frac{\delta\Delta}{\delta\bar c^a}+Q\psi\frac{\delta\Delta}{\delta\psi}+Q\bar\psi\frac{\delta\Delta}{\delta\bar\psi}\right],\end{aligned} \tag{3.366}$$

because $Q\bar c^a = B^a$. But the last line is exactly the BRST variation of the functional Δ:

$$Q\Delta = \int \mathrm{d}^4x \sum_{\Phi=A,c,\bar c,\psi,\bar\psi} Q\Phi\,\frac{\delta\Delta}{\delta\Phi}. \tag{3.367}$$

Therefore, for ordinary local counterterms which do not contain the external BRST sources,

$$\boxed{\mathcal{S}_{S_{\rm ext}}\Delta = Q\Delta} \tag{3.368}$$

up to total derivatives and the standard Grassmann signs. This is why, in practice, we can test whether a candidate counterterm is allowed by simply applying the BRST transformations to it. If $Q\Delta=0$, or if $Q\Delta$ is only a spacetime total derivative, then the term is compatible with the linearized Slavnov–Taylor identity. If not, it is forbidden.

Now (3.330) tells us (3.362) must satisfy

$$\mathcal{S}(S_{\rm ext})+\mathcal{S}_{S_{\rm ext}}\Delta=0. \tag{3.369}$$

Since the classical action is BRST invariant, we have

$$\mathcal{S}\left(S_{\text{ext}}\right) = 0. \tag{3.370}$$

Therefore, after lower-loop subdivergences have been subtracted, with the relation in (3.368) the remaining divergent part must satisfy

$$\mathcal{S}_{S_{\text{ext}}}\Gamma^{(n)}_{\text{div}} = Q\Gamma^{(n)}_{\text{div}} = 0. \tag{3.371}$$

This is the precise meaning of the sentence "the divergent part satisfies the linearized Slavnov–Taylor identity". Less formally, it means that the divergent counterterm cannot be an arbitrary local expression; it must be compatible with BRST symmetry.

Combining this with power counting gives the following list of restrictions:

UV divergences must be local, have dimension no larger than four, have ghost number zero, and be Lorentz invariant, color invariant, and BRST invariant.

Let us justify each phrase one by one.

- **Local.** UV divergences come from the region where loop momentum is very large. In that region the graph can only depend on external momenta through a polynomial divergent part. A polynomial in external momenta is a local operator in position space.
- **Dimension no larger than four.** From the superficial degree of divergence computed above, only operators of mass dimension ≤ 4 can be needed as counterterms in four spacetime dimensions. Operators such as A^6, $F_{\mu\nu}D^2F^{\mu\nu}$, or $(\bar{\psi}\psi)^2$ have dimension larger than four, so power counting already excludes them.
- **Ghost number zero.** The action and the effective action have ghost number zero. Hence a divergent counterterm must also have ghost number zero. For example, $\bar{c}c$ has ghost number $-1+1=0$, but a single c cannot appear by itself.
- **Lorentz invariance.** The original theory and dimensional regularization preserve Lorentz symmetry. Therefore all Lorentz indices must be properly contracted. For example, $F^a_{\mu\nu}F^{a\mu\nu}$ is allowed by Lorentz symmetry, while an expression with a free Lorentz index is not.
- **Color invariance.** The original action has global color invariance. Therefore color indices must also be contracted, for example by δ^{ab} or f^{abc} in the appropriate places.
- **BRST invariance.** This is the extra condition coming from the Slavnov–Taylor identity. It removes terms which are allowed by naive power counting but are incompatible with gauge symmetry.

For instance, a gauge-boson mass term has dimension two,

$$\frac{1}{2}M_A^2 A^a_\mu A^{a\mu}, \tag{3.372}$$

and it also has ghost number zero, Lorentz invariance, and color invariance. Thus, if we only used power counting, this term would look allowed. However, it is not BRST invariant. Indeed,

$$Q(A^a_\mu A^{a\mu}) = (QA^a_\mu)A^{a\mu} + A^a_\mu Q(A^{a\mu}) = 2A^{a\mu}D^{ab}_\mu c^b. \tag{3.373}$$

Expanding the covariant derivative makes the problem visible:

$$2A^{a\mu}D_\mu^{ab}c^b = 2A^{a\mu}\partial_\mu c^a + 2gf^{acb}A^{a\mu}A_\mu^c c^b. \tag{3.374}$$

This is not zero, and it is not merely a total derivative. Therefore a mass term for the gauge boson violates the linearized Slavnov–Taylor identity. In other words, BRST symmetry forbids it as a counterterm.

The remaining allowed divergent local terms are precisely those that can be absorbed into the normalization of the fields and parameters already present in $\mathcal{L}_{\mathrm{FP}}$:

$$A_\mu, \quad \psi, \quad c, \quad g, \quad m \tag{3.375}$$

plus the gauge-fixing parameter ξ if one does not stay in Feynman gauge.

Equivalently, the most general allowed local divergent structure has the same form as

$$-\frac{1}{4}F_{\mu\nu}^a F^{a\mu\nu}, \qquad \bar\psi \mathrm{i}\not{D}\psi, \qquad m\bar\psi\psi, \qquad \bar c^a(-\partial^\mu D_\mu^{ab})c^b, \qquad -\frac{1}{2\xi}(\partial^\mu A_\mu^a)^2. \tag{3.376}$$

Therefore no new independent interaction has to be introduced. This is the meaning of perturbative renormalizability of the Faddeev–Popov theory.

4.2 Counter terms of Faddeev–Popov Lagrangian

Now we have prepared to write the renormalization relates bare quantities to renormalized ones for the three type fields ψ, A_μ^a and c^a:

$$\psi_0 = Z_2^{1/2}\psi, \quad \bar\psi_0 = Z_2^{1/2}\bar\psi, \tag{3.377}$$
$$c_0^a = (Z_2^c)^{1/2}c^a, \bar c_0^a = (Z_2^c)^{1/2}\bar c^a, \tag{3.378}$$
$$A_{0,\mu}^a = Z_3^{1/2}A_\mu^a, \tag{3.379}$$

where fields renormalization can be expanded to get counter terms:

$$Z_2 = 1+\delta_2, \qquad Z_3 = 1+\delta_3, \qquad Z_2^c = 1+\delta_2^c. \tag{3.380}$$

The mass and the several vertex renormalizations can be seen by putting the field renormalization factors into the Lagrangian.

Fermion mass $m\bar\psi\psi$

First the bare mass term becomes

$$m_0\bar\psi_0\psi_0 = m_0(Z_2^{1/2}\bar\psi)(Z_2^{1/2}\psi) = Z_2 m_0\,\bar\psi\psi. \tag{3.381}$$

Thus the coefficient multiplying $\bar\psi\psi$ is defined as

$$Z_2 m_0 = m + \delta m, \tag{3.382}$$

so the original mass term becomes the renormalized mass term plus the mass counterterm,

$$\boxed{Z_2 m_0\,\bar\psi\psi = m\bar\psi\psi + \delta m\,\bar\psi\psi.} \tag{3.383}$$

Similarly, each interaction vertex gives one coefficient relation.

Fermion–gauge-boson vertex $A\bar{\psi}\psi$.

This vertex comes from the fermion covariant-derivative term $\bar{\psi}_0 i\not{D}_0\psi_0$. Its interaction part is

$$\begin{aligned}\mathcal{L}^{(0)}_{A\bar{\psi}\psi} &= g_0 A^a_{0,\mu}\bar{\psi}_0\gamma^\mu t^a\psi_0 \\ &= g_0 Z_2 Z_3^{1/2} A^a_\mu\bar{\psi}\gamma^\mu t^a\psi. \end{aligned} \tag{3.384}$$

We define the renormalized coefficient of the same operator by

$$\mathcal{L}_{A\bar{\psi}\psi} = g Z_1 A^a_\mu \bar{\psi}\gamma^\mu t^a\psi. \tag{3.385}$$

Therefore

$$\boxed{Z_2 Z_3^{1/2} g_0 = g Z_1 = g(1+\delta_1).} \tag{3.386}$$

Three-gauge-boson vertex AAA.

This vertex comes from the Yang–Mills kinetic term $-\frac{1}{4}F^a_{0,\mu\nu}F_0^{a\mu\nu}$. The part with three gauge fields is

$$\begin{aligned}\mathcal{L}^{(0)}_{AAA} &= -g_0 f^{abc}(\partial_\mu A^a_{0,\nu})A_0^{b\mu}A_0^{c\nu} \\ &= -g_0 Z_3^{3/2} f^{abc}(\partial_\mu A^a_\nu)A^{b\mu}A^{c\nu}. \end{aligned} \tag{3.387}$$

We write the renormalized coefficient as

$$\mathcal{L}_{AAA} = -g Z_1^{3g} f^{abc}(\partial_\mu A^a_\nu)A^{b\mu}A^{c\nu}. \tag{3.388}$$

Therefore

$$\boxed{Z_3^{3/2} g_0 = g Z_1^{3g} = g(1+\delta_1^{3g}).} \tag{3.389}$$

Four-gauge-boson vertex $AAAA$.

This vertex also comes from $-\frac{1}{4}F^a_{0,\mu\nu}F_0^{a\mu\nu}$. The part with four gauge fields is

$$\begin{aligned}\mathcal{L}^{(0)}_{AAAA} &= -\frac{1}{4}g_0^2\left(f^{abc}A^b_{0,\mu}A^c_{0,\nu}\right)\left(f^{ade}A_0^{d\mu}A_0^{e\nu}\right) \\ &= -\frac{1}{4}g_0^2 Z_3^2\left(f^{abc}A^b_\mu A^c_\nu\right)\left(f^{ade}A^{d\mu}A^{e\nu}\right). \end{aligned} \tag{3.390}$$

We write the renormalized coefficient as

$$\mathcal{L}_{AAAA} = -\frac{1}{4}g^2 Z_1^{4g}\left(f^{abc}A^b_\mu A^c_\nu\right)\left(f^{ade}A^{d\mu}A^{e\nu}\right). \tag{3.391}$$

Therefore

$$\boxed{Z_3^2 g_0^2 = g^2 Z_1^{4g} = g^2(1+\delta_1^{4g}).} \tag{3.392}$$

Ghost–gauge-boson vertex $A\bar{c}c$.

This vertex comes from the ghost term $\bar{c}^a_0(-\partial^\mu D^{ab}_{0,\mu})c^b_0$. After integrating by parts, its interaction part can be written as

$$\begin{aligned}\mathcal{L}^{(0)}_{A\bar{c}c} &= g_0 f^{abc}(\partial^\mu\bar{c}^a_0)A^b_{0,\mu}c^c_0 \\ &= g_0 Z_2^c Z_3^{1/2} f^{abc}(\partial^\mu\bar{c}^a)A^b_\mu c^c. \end{aligned} \tag{3.393}$$

We write the renormalized coefficient as

$$\mathcal{L}_{A\bar{c}c} = gZ_1^c f^{abc}(\partial^\mu \bar{c}^a) A_\mu^b c^c. \tag{3.394}$$

Therefore

$$\boxed{Z_2^c Z_3^{1/2} g_0 = gZ_1^c = g(1+\delta_1^c).} \tag{3.395}$$

Thus the counterterm Lagrangian can be written as

$$\begin{aligned}\mathcal{L}_{\mathrm{ct}} &= \bar{\psi}\left(\mathrm{i}\delta_2 \not{\partial} - \delta m\right)\psi + g\delta_1 A_\mu^a \bar{\psi}\gamma^\mu t^a \psi \\ &\quad - \frac{1}{4}\delta_3\left(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a\right)^2 - g\delta_1^{3g} f^{abc}\left(\partial_\mu A_\nu^a\right) A^{b\mu} A^{c\nu} - \frac{1}{4} g^2 \delta_1^{4g}\left(f^{abc} A_\mu^b A_\nu^c\right)\left(f^{ade} A^{d\mu} A^{e\nu}\right) \\ &\quad - \delta_2^c \, \bar{c}^a \partial_\mu \partial^\mu c^a + g\delta_1^c f^{abc}\left(\partial^\mu \bar{c}^a\right) A_\mu^b c^c. \end{aligned} \tag{3.396}$$

However there is still a problem, namely the renormalized Lagrangian also need to fulfill BRST invariant. This requires these three lines in counter term Larangian (3.396) can be also written as

$$\mathcal{L}_{\mathrm{ct}} \sim C_1 F_{\mu\nu}^2 + C_2 \bar{\psi} i \not{D} \psi + C_3 \bar{c}(-\partial D)c, \tag{3.397}$$

because we have seen that these three terms are BRST invariant by using Slavnov-Taylor identity. Therefore, although (3.396) contains **eight counterterms**, Slavnov–Taylor identity at one loop order leaves **only five independent physical renormalizations to guarantee the Lagrangian is BRST invariant**.

To fulfill this, we require the renormalized coupling constant in different vertices must be equal. First, coupling renormalization is

$$g_0 = Z_g g = (1+\delta_g) g. \tag{3.398}$$

Comparing with (3.386), and expand to the first order, we find

$$Z_2 Z_3^{1/2} Z_g g = gZ_1 \implies 1 + \delta_2 + \frac{1}{2}\delta_3 + \delta_g + \mathcal{O}(\delta^2) = 1 + \delta_1 + \mathcal{O}(\delta^2). \tag{3.399}$$

Similarly, together with (3.389), (3.392) and (3.395) we have:

$$\begin{aligned} \delta_g &= \delta_1 - \delta_2 - \frac{1}{2}\delta_3 \quad (\text{from } A\bar{\psi}\psi) && (3.400)\\ &= \delta_1^{3g} - \frac{3}{2}\delta_3 \quad (\text{from } AAA) && (3.401)\\ &= \frac{1}{2}\delta_1^{4g} - \delta_3 \quad (\text{from } AAAA) && (3.402)\\ &= \delta_1^c - \delta_2^c - \frac{1}{2}\delta_3 \quad (\text{from } A\bar{c}c). && (3.403)\end{aligned}$$

Thus this gives three constraint:

$$\delta_1 - \delta_2 = \delta_1^{3g} - \delta_3 = \delta_1^c - \delta_2^c = \frac{1}{2}\left(\delta_1^{4g} - \delta_3\right). \tag{3.404}$$

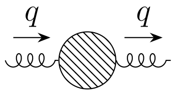
Hence the counter terms are exactly the finite set of local structures allowed by power counting and BRST symmetry.

4.3 One-loop structure of non-Abelian gauge theory

Now we can do some actual calculation. So we consider the one loop correction of Fadeev-Popov theory.

Gauge boson propagator

The one-loop vacuum polarization is defined by



$$= \Pi^{ab}_{\mu\nu}(q) = \mathrm{i}\left(g_{\mu\nu}q^2 - q_\mu q_\nu\right)\delta^{ab}\Pi(q^2), \tag{3.405}$$

because of $q^\mu \Pi^{ab}_{\mu\nu}(q) = 0$. This relation is similar with the Ward identity when we do the charge renormalization for QED. It can obtained from Slavnon-Taylor identity, the proof as follows:

Proof

We begin with the Slavnon-Taylor identity:

$$\mathcal{S}(\Gamma) = \int \mathrm{d}^4x \left[\frac{\delta\Gamma}{\delta K^{a\mu}_A}\frac{\delta\Gamma}{\delta A^a_\mu} + \frac{\delta\Gamma}{\delta K^a_c}\frac{\delta\Gamma}{\delta c^a} + B^a\frac{\delta\Gamma}{\delta\bar{c}^a} + \cdots\right] = 0. \tag{3.406}$$

Now take derivative to it:

$$\frac{\delta^2}{\delta c^b(y)\delta A^c_\nu(z)}\mathcal{S}(\Gamma) = 0. \tag{3.407}$$

And setting $A = c = \bar{c} = \psi = \bar{\psi} = B = K = 0$. So most of terms will vanish because of $\frac{\delta\Gamma}{\delta K_\Phi} = \langle Q\Phi\rangle_{J,K}$, only left

$$\int \mathrm{d}^4x \frac{\delta^2\Gamma}{\delta K^{a\mu}_A\delta c^b}\frac{\delta^2\Gamma}{\delta A^a_\mu\delta A^c_\nu} = 0. \tag{3.408}$$

Notice that for $A = 0$, we have

$$QA^a_\mu = (D_\mu c)^a = \partial_\mu c^a + \underbrace{gf^{abc}A^b_\mu c^c}_{0}. \tag{3.409}$$

Thus at tree level, we have

$$\Gamma^{(0)} = S_{\text{ext}} \supset \int d^4x\, K^{a\mu}_A(x)\left[\partial_\mu c^a(x) + gf^{ade}A^d_\mu(x)c^e(x)\right], \tag{3.410}$$

thus

$$\Gamma^{(0)}_{K^{a\mu}_A c^b}(x,y) \equiv \frac{\delta^2\Gamma}{\delta K^{a\mu}_A\delta c^b} = \delta^{ab}\partial^x_\mu\delta(x-y). \tag{3.411}$$

In the momentum space, we have

$$\Gamma^{(0)}_{K^{a\mu}_A c^b}(q) = \mathrm{i}\delta^{ab}q_\mu. \tag{3.412}$$

The loop correction is merely contribute a scalar function $G(p^2) = 1+\mathcal{O}(\hbar)$, thus it becomes

$$\Gamma_{K^{a\mu}_A c^b}(q) = iq_\mu\delta^{ab}G(q^2). \tag{3.413}$$

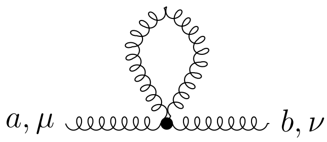
Put it back to (3.408), we then have

$$iq^\mu G(q^2)\,\Gamma_{A^b_\mu A^c_\nu}(q) = 0. \tag{3.414}$$

Because $G(p^2) \neq 0$, $\Gamma_{A^a_\mu A^b_\nu} = \Gamma^{(0)}_{A^a_\mu A^b_\nu} + \Pi^{ab}_{\mu\nu} + \cdots$ and the tree level part originally satisfies $q^\mu\Gamma^{(0)}_{A^a_\mu A^b_\nu} = 0$, thus

$$q^\mu\Pi^{ab}_{\mu\nu} = 0. \tag{3.415}$$

Base on th Feynman rules, there should be four 1-loop correction diagrams plus a counter term graph. However the massless gluon tadpole



(3.416)

vanishes in dimensional regularization since the vertex Feynman rule doesn't contain any momentum term, so the integral is:

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{g_{\mu\nu}}{k^2} \delta^{ab}. \tag{3.417}$$

This kind of integral is called **scaleless**, because it doesn't involve any external momentum or mass, so under the rescaling $k \to \lambda k$, it has

$$d^d k \to \lambda^d d^d k, \qquad \frac{1}{k^2} \to \lambda^{-2} \frac{1}{k^2}. \tag{3.418}$$

This leads to

$$I \to \lambda^{d-2} I, \tag{3.419}$$

which means the integral relies on the arbitrary rescaling parameter. This is unphysical unless $I = 0$. Or we can also explain it by direct calculation. Ignore the harmless tensor and color factors $g_{\mu\nu}\delta^{ab}$ and define the Euclidean scalar integral

$$I = \mu^{4-d} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{k^2}. \tag{3.420}$$

The factor μ^{4-d} only keeps the dimension of the integral fixed when $d \neq 4$. After angular integration,

$$I = \mu^{4-d} \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty \mathrm{d}k\, k^{d-3}, \qquad \Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(d/2)}. \tag{3.421}$$

Now introduce an arbitrary separating scale Λ and split the radial integral into an infrared part and an ultraviolet part:

$$\int_0^\infty \mathrm{d}k\, k^{d-3} = \int_0^\Lambda \mathrm{d}k\, k^{d-3} + \int_\Lambda^\infty \mathrm{d}k\, k^{d-3}. \tag{3.422}$$

The first integral is convergent for $\mathrm{Re}\, d > 2$ and gives

$$I_{\mathrm{IR}} = \int_0^\Lambda \mathrm{d}k\, k^{d-3} = \frac{\Lambda^{d-2}}{d-2}. \tag{3.423}$$

The second integral is convergent for $\mathrm{Re}\, d < 2$ and gives

$$I_{\mathrm{UV}} = \int_\Lambda^\infty \mathrm{d}k\, k^{d-3} = -\frac{\Lambda^{d-2}}{d-2}. \tag{3.424}$$

These two convergence regions do not overlap. Dimensional regularization means that we regard these answers as analytic functions of d and then continue them to the same value of d. Adding the two analytically continued pieces,

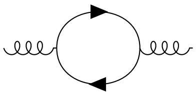
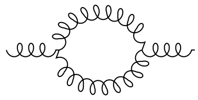
$$I = \mu^{4-d} \frac{\Omega_{d-1}}{(2\pi)^d} \left(\frac{\Lambda^{d-2}}{d-2} - \frac{\Lambda^{d-2}}{d-2} \right) = 0. \tag{3.425}$$

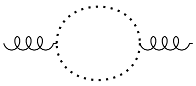
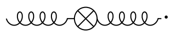
Thus the massless tadpole is zero in dimensional regularization:

$$\boxed{\int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{k^2} = 0.} \tag{3.426}$$

Generally speaking, we have the conclusion: **scaleless integral is zero in dimensional regularization**.

So non-vanishing 1-loop correction and counter term diagrams are

fermion loop:  gluon loop:  (3.427)

ghost loop:  counterterm: . (3.428)

Before the real tedious loop calculations, we may do some preparation works for them. First we derive the scalar functions $\Pi(q^2)$ appearing in these diagrams since it is the only difference structure with these diagrams. We use Feynman gauge and keep only the divergent part. Since

$$\Pi^{ab}_{\mu\nu}(q) = \mathrm{i}\left(g_{\mu\nu} q^2 - q_\mu q_\nu\right)\delta^{ab}\Pi(q^2), \tag{3.429}$$

the scalar coefficient can be extracted by the transverse projector

$$P^{\mu\nu}(q) = g^{\mu\nu} - \frac{q^\mu q^\nu}{q^2}, \qquad \Pi(q^2) = \frac{1}{\mathrm{i}(d-1)q^2\delta^{ab}} P^{\mu\nu}(q)\Pi^{ab}_{\mu\nu}(q). \tag{3.430}$$

Next preparation is to compute the common denominator for fermion loop, ghost loop and gluon loop:

$$\begin{aligned} I_0(q^2) &= \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{k^2(k+q)^2} \\ &= \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{1}{\left[\ell^2 + x(1-x)q^2\right]^2} \\ &= \frac{\mathrm{i}}{(4\pi)^2}\Gamma(\epsilon)(-q^2)^{-\epsilon} + \text{finite terms}, \qquad d = 4 - 2\epsilon, \end{aligned} \tag{3.431}$$

where $\ell = k + xq$. The logarithmically divergent part is obtained by replacing

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{k^2(k+q)^2} \longrightarrow \frac{\mathrm{i}}{(4\pi)^2}\Gamma(\epsilon)(-q^2)^{-\epsilon}. \tag{3.432}$$

Odd powers of ℓ vanish, and the tensor integrals are reduced by

$$\int \mathrm{d}^d \ell \, \ell_\mu \ell_\nu \, F(\ell^2) = \frac{g_{\mu\nu}}{d} \int \mathrm{d}^d \ell \, \ell^2 F(\ell^2). \tag{3.433}$$

We will also use the logarithmically divergent part of

$$\int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{\ell^2}{(\ell^2 - \Delta)^2} = -\frac{d}{d-2}\Delta \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta)^2} + \text{finite terms}. \tag{3.434}$$

For the present two-point graphs,

$$\Delta = -x(1-x)q^2. \tag{3.435}$$

Now we are well prepared to do the actual computation.

Fermion loop.

For one fermion flavor, the closed fermion loop gives the color trace

$$(t^a)_{ij}(t^b)_{ji} = \mathrm{tr}\big(t^a t^b\big), \tag{3.436}$$

and the closed spinor indices give the Dirac trace

$$(\gamma_\mu)_{\alpha\beta}(\not{k})_{\beta\gamma}(\gamma_\nu)_{\gamma\delta}(\not{k}+\not{q})_{\delta\alpha} = \mathrm{tr}\left[\gamma_\mu \not{k}\gamma_\nu(\not{k}+\not{q})\right]. \tag{3.437}$$

With n_f flavors,

$$\begin{aligned}\Pi^{ab}_{(f)\mu\nu}(q) &= -n_f g^2\,\mathrm{tr}\big(t^a t^b\big)\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{tr}\left[\gamma_\mu \not{k}\gamma_\nu(\not{k}+\not{q})\right]}{k^2(k+q)^2} \\ &= -n_f g^2 C(r)\delta^{ab}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{N^{(f)}_{\mu\nu}}{k^2(k+q)^2},\end{aligned} \tag{3.438}$$

where the color trace gives the fundamental representation of $SU(N)$ gauge group, namely: $\mathrm{tr}\big(t^a t^b\big) = C(r)\delta^{ab}$, and $C(r) = T_F = \frac{1}{2}$. The Dirac trace is

$$\begin{aligned}N^{(f)}_{\mu\nu} &= \mathrm{tr}\left[\gamma_\mu \not{k}\gamma_\nu(\not{k}+\not{q})\right] \\ &= 4\left[k_\mu(k+q)_\nu + k_\nu(k+q)_\mu - g_{\mu\nu}k\cdot(k+q)\right].\end{aligned} \tag{3.439}$$

Now introduce the Feynman parameter and shift

$$k = \ell - xq, \qquad k+q = \ell + (1-x)q. \tag{3.440}$$

Then

$$\begin{aligned}k_\mu(k+q)_\nu &= \ell_\mu\ell_\nu + (1-x)\ell_\mu q_\nu - xq_\mu\ell_\nu - x(1-x)q_\mu q_\nu, \\ k_\nu(k+q)_\mu &= \ell_\nu\ell_\mu + (1-x)\ell_\nu q_\mu - xq_\nu\ell_\mu - x(1-x)q_\nu q_\mu, \\ k\cdot(k+q) &= \ell^2 + (1-2x)\ell\cdot q - x(1-x)q^2.\end{aligned} \tag{3.441}$$

All terms odd in ℓ vanish after integration. Therefore the numerator can be replaced, under the integral, by

$$N^{(f)}_{\mu\nu} \longrightarrow 4\left[2\ell_\mu\ell_\nu - g_{\mu\nu}\ell^2 - 2x(1-x)q_\mu q_\nu + x(1-x)g_{\mu\nu}q^2\right]. \tag{3.442}$$

Now contract with the transverse projector. Since

$$P^{\mu\nu}q_\mu q_\nu = 0, \qquad P^{\mu\nu}g_{\mu\nu} = d-1, \qquad P^{\mu\nu}\ell_\mu\ell_\nu \to \frac{d-1}{d}\ell^2, \tag{3.443}$$

we get

$$P^{\mu\nu}N^{(f)}_{\mu\nu} \longrightarrow 4(d-1)\left[\left(\frac{2}{d}-1\right)\ell^2 + x(1-x)q^2\right]. \tag{3.444}$$

Using the ℓ^2 integral above with $\Delta = -x(1-x)q^2$, the divergent part inside the square bracket becomes

$$\left(\frac{2}{d}-1\right)\left[-\frac{d}{d-2}\Delta\right] + x(1-x)q^2 = 2x(1-x)q^2. \tag{3.445}$$

Thus

$$P^{\mu\nu} N^{(f)}_{\mu\nu} \longrightarrow 8(d-1)x(1-x)q^2. \tag{3.446}$$

After dividing by the projector normalization $\mathrm{i}(d-1)q^2\delta^{ab}$, the divergent part becomes

$$\Pi_{(f)}(q^2) = -\frac{g^2}{(4\pi)^2} n_f C(r)\,\Gamma(\epsilon)(-q^2)^{-\epsilon} \int_0^1 \mathrm{d}x\, 8x(1-x) + \text{finite terms}. \tag{3.447}$$

The remaining x integral is elementary:

$$\int_0^1 \mathrm{d}x\, 8x(1-x) = 8\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{4}{3}. \tag{3.448}$$

Therefore

$$\Pi_{(f)}(q^2) = -\frac{g^2}{(4\pi)^2} \frac{4}{3} n_f C(r)\,\Gamma(\epsilon)(-q^2)^{-\epsilon} + \text{finite terms}. \tag{3.449}$$

Ghost loop.

The ghost loop also carries a minus sign because ghosts are Grassmann fields. Using the adjoint representation relation $f^{acd} f^{bcd} = C_2(G)\delta^{ab}$, the diagram is

$$\Pi^{ab}_{(c)\mu\nu}(q) = -g^2 C_2(G)\delta^{ab} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{k_\mu (k+q)_\nu}{k^2 (k+q)^2}. \tag{3.450}$$

Again set

$$k = \ell - xq, \qquad k+q = \ell + (1-x)q. \tag{3.451}$$

Then the numerator is

$$k_\mu (k+q)_\nu = \ell_\mu \ell_\nu + (1-x)\ell_\mu q_\nu - x q_\mu \ell_\nu - x(1-x) q_\mu q_\nu. \tag{3.452}$$

The terms linear in ℓ integrate to zero, and the $q_\mu q_\nu$ term is killed by the transverse projector. Hence only the $\ell_\mu \ell_\nu$ term contributes:

$$P^{\mu\nu} k_\mu (k+q)_\nu \longrightarrow P^{\mu\nu} \ell_\mu \ell_\nu \to \frac{d-1}{d}\ell^2. \tag{3.453}$$

Using the logarithmically divergent part of the ℓ^2 integral and then dividing by $\mathrm{i}(d-1)q^2\delta^{ab}$ gives the scalar coefficient

$$\frac{1}{(d-1)q^2} \frac{d-1}{d} \left[-\frac{d}{d-2}\Delta\right] = \frac{x(1-x)}{d-2} = \frac{1}{2} x(1-x) + \mathcal{O}(\epsilon). \tag{3.454}$$

With the standard ghost-gluon vertex convention used in the Feynman rule, the two momentum factors give an extra factor 4. Thus the divergent contribution is equivalently written as

$$\Pi_{(c)}(q^2) = -\frac{g^2}{(4\pi)^2} C_2(G)\,\Gamma(\epsilon)(-q^2)^{-\epsilon} \int_0^1 \mathrm{d}x\, 2x(1-x) + \text{finite terms}. \tag{3.455}$$

Since

$$\int_0^1 \mathrm{d}x\, 2x(1-x) = 2\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{3}, \tag{3.456}$$

the ghost loop contributes

$$\Pi_{(c)}(q^2) = -\frac{g^2}{(4\pi)^2} \frac{1}{3} C_2(G)\,\Gamma(\epsilon)(-q^2)^{-\epsilon} + \text{finite terms}. \tag{3.457}$$

Gluon loop.

The gluon loop uses two three-gauge-boson vertices and has a symmetry factor 1/2. With all momenta incoming, write the three-gauge-boson vertex without the overall factor gf^{abc} as

$$V_{\alpha\beta\gamma}(p,q,r) = g_{\alpha\beta}(p-q)_\gamma + g_{\beta\gamma}(q-r)_\alpha + g_{\gamma\alpha}(r-p)_\beta. \tag{3.458}$$

The diagram is therefore

$$\begin{aligned}\Pi^{ab}_{(g)\mu\nu}(q) &= \frac{1}{2}g^2 f^{acd} f^{bcd} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{V_{\mu\rho\sigma}(q,k,-k-q)V_\nu{}^{\rho\sigma}(-q,k+q,-k)}{k^2(k+q)^2} \\ &= \frac{1}{2}g^2 C_2(G)\delta^{ab} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{N^{(g)}_{\mu\nu}}{k^2(k+q)^2}.\end{aligned} \tag{3.459}$$

For the numerator, first expand the two vertices. One finds

$$\begin{aligned}V_{\mu\rho\sigma}(q,k,-k-q) &= g_{\mu\rho}(q-k)_\sigma + g_{\rho\sigma}(2k+q)_\mu - g_{\sigma\mu}(k+2q)_\rho, \\ V_\nu{}^{\rho\sigma}(-q,k+q,-k) &= \delta^\rho_\nu(-k-2q)^\sigma + g^{\rho\sigma}(2k+q)_\nu + \delta^\sigma_\nu(q-k)^\rho.\end{aligned} \tag{3.460}$$

Therefore

$$N^{(g)}_{\mu\nu} = V_{\mu\rho\sigma}(q,k,-k-q)V_\nu{}^{\rho\sigma}(-q,k+q,-k). \tag{3.461}$$

Multiplying the two brackets and using $g_{\rho\sigma}g^{\rho\sigma} = d$ gives

$$\begin{aligned}N^{(g)}_{\mu\nu} &= (4d-6)k_\mu k_\nu + (2d-3)(k_\mu q_\nu + q_\mu k_\nu) \\ &\quad + (d+3)q_\mu q_\nu + 2g_{\mu\nu}\left(k^2 + k\cdot q - 2q^2\right).\end{aligned} \tag{3.462}$$

Now set $k = \ell - xq$. Odd powers of ℓ vanish. After the shift, the even part can be written as

$$\begin{aligned}N^{(g)}_{\mu\nu} &\longrightarrow (4d-6)\ell_\mu\ell_\nu + 2g_{\mu\nu}\ell^2 \\ &\quad + x(1-x)\left[-2(4d-6)q_\mu q_\nu + 4g_{\mu\nu}q^2\right] + \text{terms killed by } P^{\mu\nu}.\end{aligned} \tag{3.463}$$

Applying the transverse projector and reducing $\ell_\mu\ell_\nu \to g_{\mu\nu}\ell^2/d$ gives

$$P^{\mu\nu}N^{(g)}_{\mu\nu} \longrightarrow (d-1)\left[\left(\frac{4d-6}{d} + 2\right)\ell^2 + 4x(1-x)q^2\right]. \tag{3.464}$$

Using the same ℓ^2 integral relation with $\Delta = -x(1-x)q^2$, the divergent part of the bracket becomes

$$24x(1-x)q^2. \tag{3.465}$$

The diagram has the symmetry factor 1/2, so after dividing by $\mathrm{i}(d-1)q^2\delta^{ab}$ the scalar integrand is

$$\frac{1}{2}\,24x(1-x) = 12x(1-x). \tag{3.466}$$

Thus the divergent scalar part is

$$\Pi_{(g)}(q^2) = \frac{g^2}{(4\pi)^2}C_2(G)\,\Gamma(\epsilon)(-q^2)^{-\epsilon}\int_0^1 \mathrm{d}x\,12x(1-x) + \text{finite terms}. \tag{3.467}$$

The parameter integral is

$$\int_0^1 \mathrm{d}x\,12x(1-x) = 12\left(\frac{1}{2} - \frac{1}{3}\right) = 2. \tag{3.468}$$

Thus

$$\Pi_{(g)}(q^2) = \frac{g^2}{(4\pi)^2} 2C_2(G)\,\Gamma(\epsilon)(-q^2)^{-\epsilon} + \text{finite terms}. \tag{3.469}$$

Combining the gluon and ghost loops,

$$\begin{aligned} \Pi_{(g+c)}(q^2) &= \Pi_{(g)}(q^2) + \Pi_{(c)}(q^2) \\ &= \frac{g^2}{(4\pi)^2}\left(2 - \frac{1}{3}\right) C_2(G)\,\Gamma(\epsilon)(-q^2)^{-\epsilon} + \text{finite terms} \\ &= \frac{g^2}{(4\pi)^2}\frac{5}{3} C_2(G)\,\Gamma(\epsilon)(-q^2)^{-\epsilon} + \text{finite terms}. \end{aligned} \tag{3.470}$$

Counterterm graph.

The gauge-field counterterm has the quadratic part

$$\mathcal{L}_{\text{ct}} \supset -\frac{1}{4}\delta_3 \left(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu\right)^2. \tag{3.471}$$

In momentum space this gives

$$\Pi^{ab}_{\text{ct},\mu\nu}(q) = -\mathrm{i}\delta_3 \left(g_{\mu\nu}q^2 - q_\mu q_\nu\right)\delta^{ab}. \tag{3.472}$$

Comparing with $\Pi^{ab}_{\mu\nu} = \mathrm{i}(g_{\mu\nu}q^2 - q_\mu q_\nu)\delta^{ab}\Pi(q^2)$, we read off

$$\Pi_{\text{ct}}(q^2) = -\delta_3. \tag{3.473}$$

Adding the non-vanishing one-loop diagrams gives

$$\begin{aligned} \Pi_{1-\text{loop}}(q^2) &= \Pi_{(f)}(q^2) + \Pi_{(g+c)}(q^2) + \Pi_{\text{ct}}(q^2) \\ &= \frac{g^2}{(4\pi)^2}\left(\frac{5}{3}C_2(G) - \frac{4}{3}n_f C(r)\right)\Gamma(\epsilon)(-q^2)^{-\epsilon} - \delta_3 + \text{finite terms}. \end{aligned} \tag{3.474}$$

The subtraction condition at $-q^2 = \mu^2$ is that the divergent part of the renormalized one-loop vacuum polarization vanishes:

$$\Pi_{1-\text{loop}}(q^2)\big|_{-q^2=\mu^2,\ \text{div}} = 0. \tag{3.475}$$

Therefore

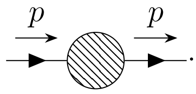
$$0 = \frac{g^2}{(4\pi)^2}\left(\frac{5}{3}C_2(G) - \frac{4}{3}n_f C(r)\right)\Gamma(\epsilon)(\mu^2)^{-\epsilon} - \delta_3. \tag{3.476}$$

Solving for the counterterm gives

$$\boxed{\delta_3 = \frac{g^2}{(4\pi)^2}\left(\frac{5}{3}C_2(G) - \frac{4}{3}n_f C(r)\right)\Gamma(\epsilon)(\mu^2)^{-\epsilon}.} \tag{3.477}$$

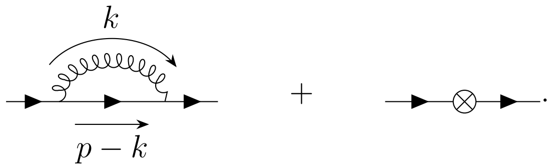
Fermion self-energy

The fermion two-point function is defined by

$\Sigma(p) =$ 

. (3.478)

At one loop the non-vanishing diagrams are the gauge-boson loop and the fermion-field counterterm:



(3.479)

We keep only the coefficient of $\not{p}$, because that is the part cancelled by the wave-function counterterm δ_2. In Feynman gauge, using $(t^a t^a)_{ij} = C_2(r)\delta_{ij}$, the loop contribution is

$$\mathrm{i}\Sigma_{\text{loop}}(p) = \mathrm{i}g^2 C_2(r) \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\gamma^\mu(\not{p}-\not{k})\gamma_\mu}{k^2(p-k)^2}. \tag{3.480}$$

The numerator is reduced by

$$\gamma^\mu\gamma^\alpha\gamma_\mu = (2-d)\gamma^\alpha, \qquad \gamma^\mu(\not{p}-\not{k})\gamma_\mu = (2-d)(\not{p}-\not{k}). \tag{3.481}$$

Combine the two denominators:

$$\begin{aligned}\frac{1}{k^2(p-k)^2} &= \int_0^1 \mathrm{d}x\, \frac{1}{[xk^2+(1-x)(p-k)^2]^2} \\ &= \int_0^1 \mathrm{d}x\, \frac{1}{[\ell^2+x(1-x)p^2]^2}, \qquad \ell = k-(1-x)p.\end{aligned} \tag{3.482}$$

Since

$$p-k = xp-\ell, \tag{3.483}$$

the odd term proportional to ℓ integrates to zero:

$$\mathrm{i}\Sigma_{\text{loop}}(p) = \mathrm{i}g^2 C_2(r)(2-d)\not{p} \int_0^1 \mathrm{d}x\, x \int \frac{\mathrm{d}^d\ell}{(2\pi)^d} \frac{1}{[\ell^2+x(1-x)p^2]^2} + \text{finite terms}. \tag{3.484}$$

Using the same logarithmically divergent scalar integral as above,

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d} \frac{1}{[\ell^2+x(1-x)p^2]^2} = \frac{\mathrm{i}}{(4\pi)^2}\Gamma(\epsilon)(-p^2)^{-\epsilon} + \text{finite terms}, \tag{3.485}$$

and the x integral comes only from the even part xp in $p-k = xp-\ell$:

$$(2-d)\int_0^1 \mathrm{d}x\, x = (2-d)\frac{1}{2} = -1+\epsilon, \tag{3.486}$$

so the divergent coefficient is simply -1 at leading order in ϵ. the divergent part has the form

$$\Sigma_{\text{loop}}(p) = \mathrm{i}\frac{g^2}{(4\pi)^2} C_2(r)\not{p}\,\Gamma(\epsilon)(-p^2)^{-\epsilon} + \text{finite terms}. \tag{3.487}$$

The counterterm $\bar{\psi}\mathrm{i}\delta_2\not{\partial}\psi$ contributes

$$\Sigma_{\text{ct}}(p) = \delta_2\not{p}. \tag{3.488}$$

Therefore

$$\Sigma(p) = \mathrm{i}\frac{g^2}{(4\pi)^2} C_2(r)\not{p}\,\Gamma(\epsilon)(-p^2)^{-\epsilon} + \delta_2\not{p} + \text{finite terms}. \tag{3.489}$$

Choosing the subtraction point $-p^2 = \mu^2$ and requiring the divergent part of the renormalized self-energy to vanish gives

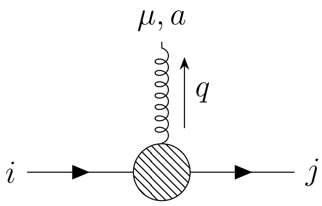
$$0 = \frac{g^2}{(4\pi)^2} C_2(r)\Gamma(\epsilon)(\mu^2)^{-\epsilon} + \delta_2. \tag{3.490}$$

Thus

$$\boxed{\delta_2 = -\frac{g^2}{(4\pi)^2} C_2(r)\,\Gamma(\epsilon)(\mu^2)^{-\epsilon}}. \tag{3.491}$$

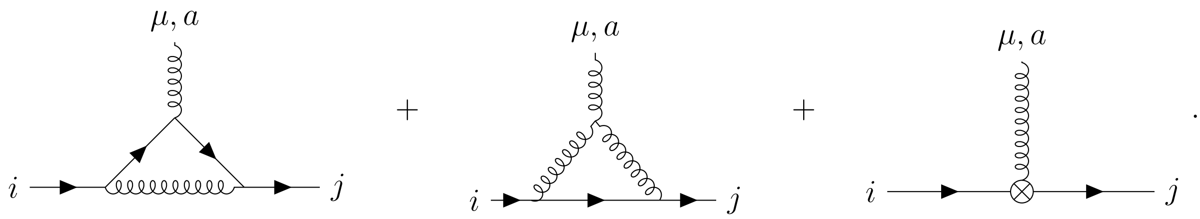
Fermion–gauge-boson vertex

The corrected fermion–gauge-boson vertex is defined by



$$= -\mathrm{i}g\,\Gamma^\mu(q)(t^a)_{ij}, \qquad \Gamma^\mu(q) = \gamma^\mu + \mathcal{O}(g^2). \tag{3.492}$$

At one loop the divergent part comes from the Abelian-like diagram, the non-Abelian diagram with a three-gauge-boson vertex, and the vertex counterterm:



. (3.493)

Abelian-like diagram

The Abelian-like diagram has color factor

$$t^b t^a t^b = \left(C_2(r) - \frac{1}{2} C_2(G) \right) t^a. \tag{3.494}$$

Its loop integral is

$$\Lambda^\mu_{\mathrm{A}} = \mathrm{i}g^3 (t^b t^a t^b)_{ij} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\gamma^\rho (\not{p}' - \not{k})\gamma^\mu(\not{p} - \not{k})\gamma_\rho}{k^2 (p' - k)^2 (p - k)^2}. \tag{3.495}$$

To see the coefficient of the logarithmic divergence explicitly, keep only the large-k part of the numerator. The terms with fewer powers of k are less divergent or finite for the vertex counterterm. Thus

$$\begin{aligned} \gamma^\rho (\not{p}' - \not{k})\gamma^\mu(\not{p} - \not{k})\gamma_\rho &\longrightarrow \gamma^\rho \not{k} \gamma^\mu \not{k} \gamma_\rho \\ &= k_\alpha k_\beta \gamma^\rho \gamma^\alpha \gamma^\mu \gamma^\beta \gamma_\rho. \end{aligned} \tag{3.496}$$

After symmetric integration,

$$k_\alpha k_\beta \longrightarrow \frac{g_{\alpha\beta}}{d} k^2, \tag{3.497}$$

so the Dirac numerator becomes

$$\begin{aligned}
k_\alpha k_\beta \gamma^\rho \gamma^\alpha \gamma^\mu \gamma^\beta \gamma_\rho &\longrightarrow \frac{k^2}{d}\gamma^\rho \gamma^\alpha \gamma^\mu \gamma_\alpha \gamma_\rho \\
&= \frac{k^2}{d}(2-d)\gamma^\rho \gamma^\mu \gamma_\rho \\
&= \frac{k^2}{d}(2-d)^2 \gamma^\mu \\
&= k^2 \gamma^\mu + \mathcal{O}(\epsilon).
\end{aligned} \tag{3.498}$$

Meanwhile the three denominators behave in the logarithmic part as

$$k^2(p'-k)^2(p-k)^2 \longrightarrow (k^2)^3. \tag{3.499}$$

Therefore the UV divergent part reduces to the standard logarithmic integral

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{k^2}{(k^2)^3} = \int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{1}{(k^2)^2} \longrightarrow \frac{\mathrm{i}}{(4\pi)^2}\Gamma(\epsilon)(-q^2)^{-\epsilon}. \tag{3.500}$$

Combining this with $t^b t^a t^b = (C_2(r) - \frac{1}{2}C_2(G))t^a$, the logarithmically divergent part is

$$\Lambda^\mu_{\mathrm{A,div}} = \mathrm{i}g\gamma^\mu (t^a)_{ij}\frac{g^2}{(4\pi)^2}\left(C_2(r) - \frac{1}{2}C_2(G)\right)\Gamma(\epsilon)(-q^2)^{-\epsilon}. \tag{3.501}$$

Non-Abelian diagram

The non-Abelian diagram has the characteristic color factor

$$\begin{aligned}
f^{abc}(t^b)_{i\ell}(t^c)_{\ell j} &= \frac{1}{2}f^{abc}\left([t^b,t^c] + \{t^b,t^c\}\right)_{ij} \\
&= \frac{1}{2}f^{abc}[t^b,t^c]_{ij} \\
&= \frac{\mathrm{i}}{2}f^{abc}f^{bcd}(t^d)_{ij} = \frac{\mathrm{i}}{2}C_2(G)(t^a)_{ij}.
\end{aligned} \tag{3.502}$$

Its loop integral is

$$\Lambda^\mu_{\mathrm{NA}} = \mathrm{i}g^3 f^{abc}(t^b t^c)_{ij}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\gamma^\rho(\not{p}'-\not{k})\gamma^\sigma V^\mu{}_{\rho\sigma}(q,k-p',p-k)}{k^2(p'-k)^2(p-k)^2}, \tag{3.503}$$

where $V^\mu{}_{\rho\sigma}$ is the three-gauge-boson vertex tensor. The same logic gives the coefficient of the logarithmic divergence. In the ultraviolet part, the three-gauge-boson vertex contributes the part linear in the loop momentum,

$$V^\mu{}_{\rho\sigma}(q,k-p',p-k) \longrightarrow 2k^\mu g_{\rho\sigma} - k_\rho \delta^\mu_\sigma - k_\sigma \delta^\mu_\rho + \text{terms independent of } k. \tag{3.504}$$

Keeping only the even powers of k after multiplication by $\gamma^\rho(\not{p}'-\not{k})\gamma^\sigma$, we get

$$\begin{aligned}
\gamma^\rho(\not{p}'-\not{k})\gamma^\sigma V^\mu{}_{\rho\sigma} &\longrightarrow -\gamma^\rho \not{k}\gamma^\sigma\left(2k^\mu g_{\rho\sigma} - k_\rho\delta^\mu_\sigma - k_\sigma\delta^\mu_\rho\right) \\
&= -2k^\mu k_\alpha \gamma^\rho\gamma^\alpha\gamma_\rho + k_\rho k_\alpha \gamma^\rho\gamma^\alpha\gamma^\mu + k_\sigma k_\alpha \gamma^\mu\gamma^\alpha\gamma^\sigma.
\end{aligned} \tag{3.505}$$

Now use symmetric integration:

$$k^\mu k_\alpha \to \frac{1}{d}\delta^\mu_\alpha k^2, \qquad k_\rho k_\alpha \to \frac{1}{d}g_{\rho\alpha}k^2, \qquad k_\sigma k_\alpha \to \frac{1}{d}g_{\sigma\alpha}k^2. \tag{3.506}$$

Then

$$\begin{aligned}\gamma^\rho(\not{p}'-\not{k})\gamma^\sigma V^\mu{}_{\rho\sigma} &\longrightarrow \frac{k^2}{d}\left[-2\gamma^\rho\gamma^\mu\gamma_\rho+\gamma^\rho\gamma_\rho\gamma^\mu+\gamma^\mu\gamma^\sigma\gamma_\sigma\right]\\ &=\frac{k^2}{d}\left[-2(2-d)+d+d\right]\gamma^\mu\\ &=\frac{4(d-1)}{d}k^2\gamma^\mu=3k^2\gamma^\mu+\mathcal{O}(\epsilon).\end{aligned} \tag{3.507}$$

The same logarithmic integral then appears:

$$\int\frac{\mathrm{d}^dk}{(2\pi)^d}\frac{1}{(k^2)^2}\longrightarrow\frac{\mathrm{i}}{(4\pi)^2}\Gamma(\epsilon)(-q^2)^{-\epsilon}. \tag{3.508}$$

Together with the color factor $\frac{1}{2}C_2(G)t^a$ obtained above, this gives

$$\Lambda^\mu_{\mathrm{NA,div}}=\mathrm{i}g\gamma^\mu(t^a)_{ij}\frac{g^2}{(4\pi)^2}\frac{3}{2}C_2(G)\,\Gamma(\epsilon)(-q^2)^{-\epsilon}. \tag{3.509}$$

Counterterm and subtraction

The vertex counterterm comes from $g\delta_1A^a_\mu\bar\psi\gamma^\mu t^a\psi$, hence

$$\Lambda^\mu_{\mathrm{ct}}=\mathrm{i}g\gamma^\mu(t^a)_{ij}\delta_1. \tag{3.510}$$

Adding the three pieces, and notice that only the coefficient of $\gamma^\mu(t^a)_{ij}$ is needed for Z_1.

$$\begin{aligned}\Lambda^\mu_{\mathrm{div}}&=\mathrm{i}g\gamma^\mu(t^a)_{ij}\left[\frac{g^2}{(4\pi)^2}\left(C_2(r)-\frac{1}{2}C_2(G)+\frac{3}{2}C_2(G)\right)\Gamma(\epsilon)(-q^2)^{-\epsilon}+\delta_1\right]\\ &=\mathrm{i}g\gamma^\mu(t^a)_{ij}\left[\frac{g^2}{(4\pi)^2}(C_2(r)+C_2(G))\Gamma(\epsilon)(-q^2)^{-\epsilon}+\delta_1\right].\end{aligned} \tag{3.511}$$

At the subtraction point $-q^2=\mu^2$, the divergent part is cancelled by choosing

$$0=\frac{g^2}{(4\pi)^2}(C_2(r)+C_2(G))\Gamma(\epsilon)(\mu^2)^{-\epsilon}+\delta_1. \tag{3.512}$$

Therefore

$$\boxed{\delta_1=-\frac{g^2}{(4\pi)^2}(C_2(r)+C_2(G))\Gamma(\epsilon)(\mu^2)^{-\epsilon}}. \tag{3.513}$$

4.4 Callan–Symanzik equation in non-Abelian gauge theory

Now we can try to write down the Callan–Symanzik equation and analyse the running coupling. For the fermion–gauge-boson three-point Green's function define

$$G_{A\bar\psi\psi}=\left\langle A^a_\mu(x)\psi_i(y)\bar\psi_j(z)\right\rangle. \tag{3.514}$$

The bare and renormalized fields are related by

$$A^a_{0,\mu}=Z_3^{1/2}A^a_\mu,\qquad\psi_0=Z_2^{1/2}\psi,\qquad\bar\psi_0=Z_2^{1/2}\bar\psi. \tag{3.515}$$

Therefore the bare three-point Green's function is

$$\begin{aligned} G_{0,A\bar{\psi}\psi} &= \left\langle A^a_{0,\mu}\psi_{0,i}\bar{\psi}_{0,j}\right\rangle \\ &= Z_3^{1/2}Z_2G_{A\bar{\psi}\psi}. \end{aligned} \tag{3.516}$$

Here the factor Z_2 appears because there are two fermion fields, and the factor $Z_3^{1/2}$ appears because there is one gauge field.

Now act with $\mu\mathrm{d}/\mathrm{d}\mu$ while keeping all bare quantities fixed, and μ is the arbitrary subtraction scale. Since $G_{0,A\bar{\psi}\psi}$ is a bare quantity,

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu}G_{0,A\bar{\psi}\psi} = 0. \tag{3.517}$$

Using $G_{0,A\bar{\psi}\psi} = Z_2Z_3^{1/2}G_{A\bar{\psi}\psi}$, this gives

$$\begin{aligned} 0 &= \mu\frac{\mathrm{d}}{\mathrm{d}\mu}\left(Z_2Z_3^{1/2}G_{A\bar{\psi}\psi}\right) \\ &= Z_2Z_3^{1/2}\left[\mu\frac{\mathrm{d}}{\mathrm{d}\mu} + \mu\frac{\mathrm{d}\ln Z_2}{\mathrm{d}\mu} + \frac{1}{2}\mu\frac{\mathrm{d}\ln Z_3}{\mathrm{d}\mu}\right]G_{A\bar{\psi}\psi}. \end{aligned} \tag{3.518}$$

The total μ derivative acting on the renormalized Green's function contains the explicit scale dependence and the implicit dependence through the running coupling:

$$\mu\frac{\mathrm{d}}{\mathrm{d}\mu} = \mu\frac{\partial}{\partial\mu} + \mu\frac{\mathrm{d}g}{\mathrm{d}\mu}\frac{\partial}{\partial g} = \mu\frac{\partial}{\partial\mu} + \beta(g)\frac{\partial}{\partial g}. \tag{3.519}$$

We define the field anomalous dimensions by

$$\gamma_2 = \frac{1}{2}\mu\frac{\mathrm{d}\ln Z_2}{\mathrm{d}\mu}, \qquad \gamma_3 = \frac{1}{2}\mu\frac{\mathrm{d}\ln Z_3}{\mathrm{d}\mu}. \tag{3.520}$$

Then

$$\mu\frac{\mathrm{d}\ln Z_2}{\mathrm{d}\mu} = 2\gamma_2, \qquad \frac{1}{2}\mu\frac{\mathrm{d}\ln Z_3}{\mathrm{d}\mu} = \gamma_3. \tag{3.521}$$

Thus the Callan–Symanzik equation for this three-point function is

$$\boxed{\left(\mu\frac{\partial}{\partial\mu} + \beta(g)\frac{\partial}{\partial g} + 2\gamma_2 + \gamma_3\right)G_{A\bar{\psi}\psi} = 0}. \tag{3.522}$$

In a massive theory one should also add the mass-running term $-\gamma_m m\partial/\partial m$. Here we focus on the massless or high-energy part relevant for the gauge coupling running.

Anomalous dimensions.

At one loop,

$$Z_i = 1 + \delta_i, \qquad \ln Z_i = \delta_i + \mathcal{O}(g^4). \tag{3.523}$$

Therefore

$$\gamma_i = \frac{1}{2}\mu\frac{\mathrm{d}\delta_i}{\mathrm{d}\mu} + \mathcal{O}(g^4). \tag{3.524}$$

The divergent factors found above all have the form

$$F_\epsilon(\mu) = \Gamma(\epsilon)(\mu^2)^{-\epsilon}. \tag{3.525}$$

Since

$$\begin{aligned}\mu\frac{\mathrm{d}}{\mathrm{d}\mu}F_\epsilon(\mu) &= \Gamma(\epsilon)\,\mu\frac{\mathrm{d}}{\mathrm{d}\mu}(\mu^2)^{-\epsilon}\\ &= -2\epsilon\,\Gamma(\epsilon)(\mu^2)^{-\epsilon}\\ &= -2+\mathcal{O}(\epsilon),\end{aligned} \tag{3.526}$$

we will use

$$\frac{1}{2}\mu\frac{\mathrm{d}}{\mathrm{d}\mu}F_\epsilon(\mu) = -1+\mathcal{O}(\epsilon). \tag{3.527}$$

From the one-loop self-energy and vacuum-polarization counterterms,

$$\begin{aligned}\delta_2 &= -\frac{g^2}{(4\pi)^2}C_2(r)F_\epsilon(\mu),\\ \delta_3 &= \frac{g^2}{(4\pi)^2}\left(\frac{5}{3}C_2(G)-\frac{4}{3}n_fC(r)\right)F_\epsilon(\mu).\end{aligned} \tag{3.528}$$

Thus

$$\begin{aligned}\gamma_2 = \frac{1}{2}\mu\frac{\mathrm{d}\delta_2}{\mathrm{d}\mu} &= -\frac{g^2}{(4\pi)^2}C_2(r)\left[\frac{1}{2}\mu\frac{\mathrm{d}F_\epsilon}{\mathrm{d}\mu}\right]\\ &= \frac{g^2}{(4\pi)^2}C_2(r),\end{aligned} \tag{3.529}$$

$$\gamma_3 = \frac{1}{2}\mu\frac{\mathrm{d}\delta_3}{\mathrm{d}\mu} = -\frac{g^2}{(4\pi)^2}\left(\frac{5}{3}C_2(G)-\frac{4}{3}n_fC(r)\right). \tag{3.530}$$

The sign of γ_3 here follows from the convention $F_\epsilon(\mu)=\Gamma(\epsilon)(\mu^2)^{-\epsilon}$. Some books absorb the opposite sign into the definition of the subtraction factor; then the displayed intermediate sign changes, while the beta function below is unchanged.

Gauge coupling running.

The gauge coupling as we computed in (3.400) at 1-loop level is

$$\delta_g = \delta_1-\delta_2-\frac{1}{2}\delta_3, \tag{3.531}$$

where $g_0 = Z_2^{-1}Z_3^{-1/2}Z_1 g$. Because g_0 is bare, it is independent of μ:

$$\begin{aligned}0 = \mu\frac{\mathrm{d}g_0}{\mathrm{d}\mu} &= \mu\frac{\mathrm{d}}{\mathrm{d}\mu}(Z_g g)\\ &= Z_g\beta(g)+g\,\mu\frac{\mathrm{d}Z_g}{\mathrm{d}\mu}.\end{aligned} \tag{3.532}$$

To one-loop order $Z_g = 1+\delta_g+\mathcal{O}(g^2)$, so

$$\beta(g) = -g\,\mu\frac{\mathrm{d}\delta_g}{\mathrm{d}\mu}+\mathcal{O}(g^5). \tag{3.533}$$

Now insert the one-loop counterterms

$$\begin{aligned}\delta_1 &= -\frac{g^2}{(4\pi)^2}\left(C_2(r)+C_2(G)\right)F_\epsilon(\mu),\\ \delta_2 &= -\frac{g^2}{(4\pi)^2}C_2(r)F_\epsilon(\mu),\\ \delta_3 &= \frac{g^2}{(4\pi)^2}\left(\frac{5}{3}C_2(G)-\frac{4}{3}n_fC(r)\right)F_\epsilon(\mu).\end{aligned} \tag{3.534}$$

Then

$$\begin{aligned}\delta_g &= -\frac{g^2}{(4\pi)^2}\left[\left(C_2(r)+C_2(G)\right)-C_2(r)+\frac{1}{2}\left(\frac{5}{3}C_2(G)-\frac{4}{3}n_fC(r)\right)\right]F_\epsilon(\mu)\\ &= -\frac{g^2}{(4\pi)^2}\left(\frac{11}{6}C_2(G)-\frac{2}{3}n_fC(r)\right)F_\epsilon(\mu)\\ &= -\frac{1}{2}\frac{g^2}{(4\pi)^2}\left(\frac{11}{3}C_2(G)-\frac{4}{3}n_fC(r)\right)F_\epsilon(\mu).\end{aligned} \tag{3.535}$$

Define

$$\beta_0 = \frac{11}{3}C_2(G)-\frac{4}{3}n_fC(r). \tag{3.536}$$

Therefore

$$\delta_g = -\frac{1}{2}\frac{g^2}{(4\pi)^2}\beta_0F_\epsilon(\mu). \tag{3.537}$$

Taking the scale derivative,

$$\begin{aligned}\mu\frac{\mathrm{d}\delta_g}{\mathrm{d}\mu} &= -\frac{1}{2}\frac{g^2}{(4\pi)^2}\beta_0\mu\frac{\mathrm{d}F_\epsilon}{\mathrm{d}\mu}+\mathcal{O}(g^4)\\ &= -\frac{1}{2}\frac{g^2}{(4\pi)^2}\beta_0\left(-2+\mathcal{O}(\epsilon)\right)\\ &= \frac{g^2}{(4\pi)^2}\beta_0.\end{aligned} \tag{3.538}$$

Finally,

$$\boxed{\beta(g) = -g\frac{g^2}{(4\pi)^2}\left(\frac{11}{3}C_2(G)-\frac{4}{3}n_fC(r)\right)+\mathcal{O}(g^4)} \tag{3.539}$$

or, equivalently,

$$\beta(g) = -g\left[\frac{g^2}{(4\pi)^2}\beta_0+\frac{g^4}{(4\pi)^4}\beta_1+\cdots\right]. \tag{3.540}$$

The positive value of β_0 for sufficiently small n_f means

$$\mu\frac{\mathrm{d}g}{\mathrm{d}\mu} < 0, \tag{3.541}$$

so the gauge coupling becomes smaller at higher energy. This is the one-loop statement of asymptotic freedom in non-Abelian gauge theory.

5. Quantum Chromodynamics: QCD

This section organizes the basic physics of quantum chromodynamics (QCD): why the strong interaction needs color, why the coupling becomes weak at high energies, why perturbation theory breaks down in the infrared, and why apparently infrared-divergent partonic cross sections can still lead to finite physical predictions.

5.1 What QCD is

QCD is the non-Abelian gauge theory of the strong interaction. Its gauge group is

$$SU(3)_{\text{color}}. \tag{3.542}$$

The word "color" does not refer to visible color. It is an internal quantum number carried by quarks. A quark field has a color index

$$q_i(x), \qquad i = 1, 2, 3, \tag{3.543}$$

and the three basis states are often called red, green, and blue. The gluon field is written as

$$A^a_\mu(x), \qquad a = 1, \dots, 8, \tag{3.544}$$

because gluons transform in the adjoint representation of $SU(3)$, whose dimension is $3^2 - 1 = 8$.

The classical QCD Lagrangian is

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} + \sum_f \bar{q}_f \left(\mathrm{i}\gamma^\mu D_\mu - m_f \right) q_f, \tag{3.545}$$

where

$$D_\mu = \partial_\mu + \mathrm{i} g_s A^a_\mu T^a, \tag{3.546}$$

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g_s f^{abc} A^b_\mu A^c_\nu. \tag{3.547}$$

The matrices T^a are the generators of $SU(3)$ in the fundamental representation. We use the normalization

$$\operatorname{tr}(T^a T^b) = T_F \delta^{ab}, \qquad T_F = \frac{1}{2}. \tag{3.548}$$

For $SU(N_c)$, the standard group factors are

$$C_2(G) = N_c, \qquad C_2(r) = \frac{N_c^2 - 1}{2N_c}, \qquad T_F = \frac{1}{2}. \tag{3.549}$$

For real QCD, $N_c = 3$, so $C_2(G) = 3$ and $C_2(r) = 4/3$.

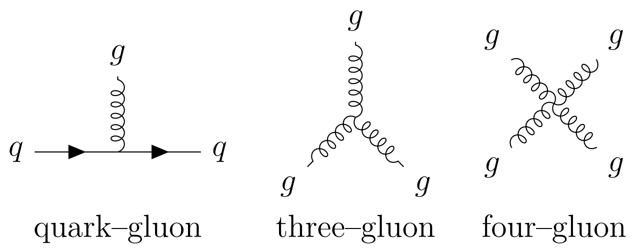


Figure 3.1: The elementary QCD interactions drawn from the QCD Lagrangian. The last two vertices are absent in QED and are responsible for much of the non-Abelian physics of QCD.

5.2 Why color is needed

Color is not an artificial decoration. It is required by both theory and experiment. Three classic pieces of evidence come from baryon spectroscopy, $\pi^0 \to \gamma\gamma$, and e^+e^- annihilation.

1. The Δ^{++} baryon and Fermi statistics

The quark content of the Δ^{++} baryon is

$$\Delta^{++} = uuu. \tag{3.550}$$

Its spin is 3/2. In the state with maximal spin projection, the three u quarks have parallel spins:

$$|s_z = 3/2\rangle = |u\uparrow\, u\uparrow\, u\uparrow\rangle. \tag{3.551}$$

Thus the spin part of the wave function is completely symmetric. Since all three quarks are u quarks, the flavour part is also symmetric. The ground-state baryon has orbital angular momentum $L = 0$, so the spatial wave function is symmetric as well.

Quarks are fermions. The total wave function must be antisymmetric under the exchange of any two quarks:

$$P_{ij}\Psi_{\text{total}} = -\Psi_{\text{total}}. \tag{3.552}$$

Writing

$$\Psi_{\text{total}} = \Psi_{\text{space}}\Psi_{\text{spin}}\Psi_{\text{flavour}}\Psi_{\text{color}}, \tag{3.553}$$

the first three factors are symmetric for the ground-state Δ^{++}. Therefore the missing antisymmetry must come from the color wave function. The color-singlet baryon wave function is

$$\Psi_{\text{color}} = \frac{1}{\sqrt{6}}\,\epsilon_{ijk}|i\,j\,k\rangle, \qquad i, j, k \in \{r, g, b\}. \tag{3.554}$$

For example, exchanging the first two color labels gives

$$\epsilon_{jik} = -\epsilon_{ijk}, \tag{3.555}$$

so the color wave function changes sign. With color included, the full Δ^{++} wave function obeys the Pauli principle. Without color, the observed Δ^{++} state would be inconsistent with Fermi statistics.

2. The decay $\pi^0 \to \gamma\gamma$

The flavour wave function of the neutral pion is

$$|\pi^0\rangle = \frac{1}{\sqrt{2}}\left(|u\bar{u}\rangle - |d\bar{d}\rangle\right). \tag{3.556}$$

The two photons couple to the quarks through a triangle diagram. Since photons couple to

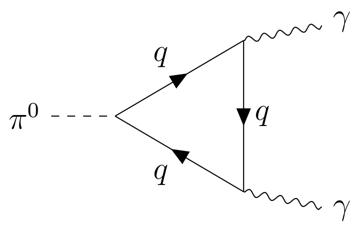


Figure 3.2: Triangle topology for $\pi^0 \to \gamma\gamma$. The loop is summed over quark flavours and colors.

electric charge, each color gives the same contribution, and summing over color gives a factor of N_c. Up to a common kinematic factor, the charge part of the amplitude is

$$\begin{aligned}\mathcal{A}(\pi^0 \to \gamma\gamma) &\propto \frac{N_c}{\sqrt{2}}\left(Q_u^2 - Q_d^2\right)\\ &= \frac{N_c}{\sqrt{2}}\left[\left(\frac{2}{3}\right)^2 - \left(-\frac{1}{3}\right)^2\right] = \frac{N_c}{3\sqrt{2}}.\end{aligned} \tag{3.557}$$

More completely, the chiral anomaly gives

$$\Gamma(\pi^0 \to \gamma\gamma) = \frac{m_\pi^3 \alpha^2}{64\pi^3 f_\pi^2}\left[N_c\left(Q_u^2 - Q_d^2\right)\right]^2. \tag{3.558}$$

Since $Q_u = 2/3$ and $Q_d = -1/3$, for $N_c = 3$ the factor in brackets is

$$3\left(\frac{4}{9} - \frac{1}{9}\right) = 1. \tag{3.559}$$

If there were no color, namely $N_c = 1$, the amplitude would be smaller by a factor of 3, and the decay width would be smaller by a factor of 9. The observed size of $\pi^0 \to \gamma\gamma$ supports $N_c = 3$.

3. The ratio $R(s)$ in $e^+e^- \to$ hadrons

Define

$$R(s) = \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)}. \tag{3.560}$$

At leading order, and neglecting the lepton mass,

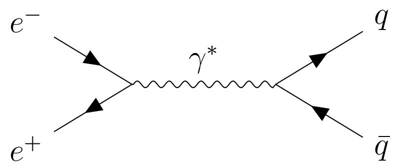


Figure 3.3: Leading-order partonic process behind $e^+e^- \to$ hadrons. At this order the hadronic final state is approximated by $q\bar{q}$, followed by nonperturbative hadronization.

$$\sigma(e^+e^- \to \mu^+\mu^-) = \frac{4\pi\alpha^2}{3s}. \tag{3.561}$$

For a final-state quark pair $q\bar{q}$, the photon–quark vertex contains the electric charge Q_q, so the squared amplitude receives a factor Q_q^2. The final-state quark can also have any of N_c colors, so

$$\sigma(e^+e^- \to q\bar{q}) = N_c Q_q^2 \sigma(e^+e^- \to \mu^+\mu^-)\left[1 + \mathcal{O}\left(\frac{m_q^2}{s}\right)\right]. \tag{3.562}$$

Therefore, when the energy is high enough to produce a set of light quark flavours,

$$R(s) = N_c \sum_{q\ \text{open}} Q_q^2 + \mathcal{O}(\alpha_s). \tag{3.563}$$

In the energy region where only u, d, s are open,

$$\sum_{q=u,d,s} Q_q^2 = \left(\frac{2}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 + \left(-\frac{1}{3}\right)^2 = \frac{2}{3}. \tag{3.564}$$

Experimentally the corresponding plateau in $R(s)$ is approximately 2, hence

$$2 \simeq N_c \frac{2}{3} \qquad \Longrightarrow \qquad N_c = 3. \tag{3.565}$$

When $\sqrt{s}$ crosses the charm and bottom thresholds, $R(s)$ increases. Near narrow resonances

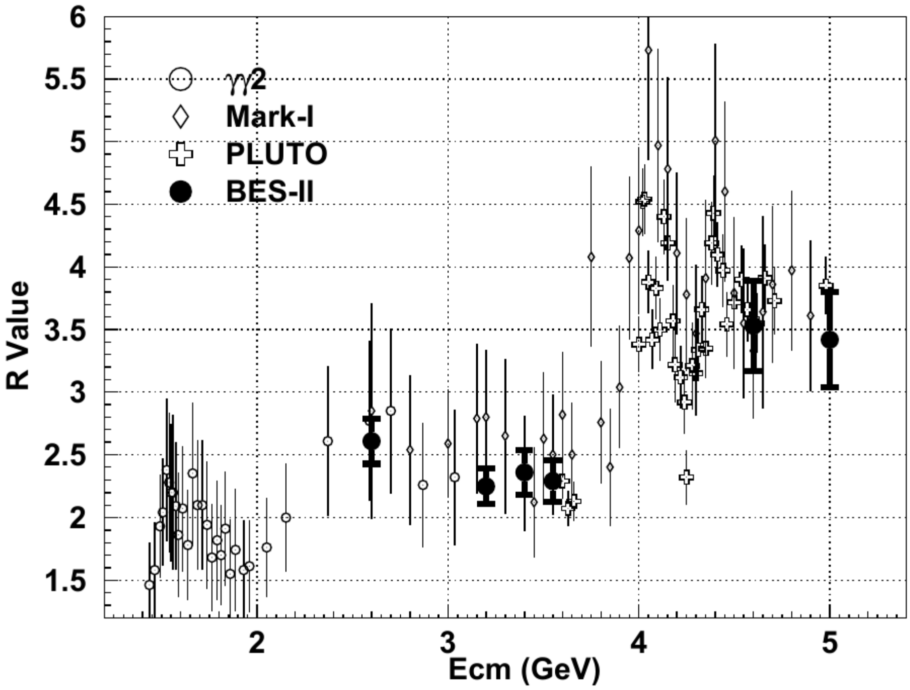


Figure 3.4: Measured values of R in the few-GeV region. The visible step-like structure is the experimental counterpart of changing open quark flavours. Image cropped from BES Collaboration, *Measurements of the cross-section for* $e^+e^- \to$ *hadrons at center-of-mass energies from 2 to 5 GeV*, arXiv:hep-ex/9908046, Fig. 2.

such as J/ψ and Υ, sharp peaks appear.

5.3 No free quarks and confinement

Isolated quarks have not been observed. Quarks and gluons appear only inside color-singlet hadrons. This is called confinement. From the viewpoint of perturbative QCD, confinement cannot be proven by summing a finite number of Feynman diagrams. However, the running of the strong coupling gives an important hint: the coupling is small at short distances and large at long distances, so perturbation theory eventually breaks down in the infrared.

A useful qualitative picture is the static quark potential

$$V(r) \simeq -C_2(r)\frac{\alpha_s(r)}{r} + \sigma r. \tag{3.566}$$

The first term is Coulomb-like at short distances. The linear term at large distances means that separating color charges costs an energy that grows with distance.

5.4 Renormalization constants and the running coupling

The strong coupling is defined by

$$\alpha_s = \frac{g_s^2}{4\pi}. \tag{3.567}$$

In dimensional regularization,

$$d = 4 - 2\epsilon. \tag{3.568}$$

Because g_s has a mass dimension away from four dimensions, the bare coupling is written as

$$g_{s,0} = \mu^\epsilon Z_g g_s(\mu). \tag{3.569}$$

Equivalently, if

$$a_s \equiv \frac{\alpha_s}{4\pi} = \frac{g_s^2}{(4\pi)^2}, \tag{3.570}$$

then

$$a_{s,0} = \mu^{2\epsilon} Z_{a_s} a_s(\mu), \qquad Z_{a_s} = Z_g^2. \tag{3.571}$$

At one loop,

$$Z_g = 1 - \frac{1}{2}\frac{g_s^2}{(4\pi)^2}\frac{\beta_0}{\epsilon} + \mathcal{O}(g_s^4), \tag{3.572}$$

where

$$\beta_0 = \frac{11}{3}C_2(G) - \frac{4}{3}T_F n_f = \frac{11}{3}C_2(G) - \frac{2}{3}n_f. \tag{3.573}$$

Here n_f is the number of light quark flavours participating in the running. The first term comes from gluon and ghost effects, especially the non-Abelian self-interaction of gluons. The second term comes from quark loops. Physically, quark loops behave like the screening effect in QED, while gluon self-interactions produce antiscreening. Asymptotic freedom occurs because the gluon contribution dominates.

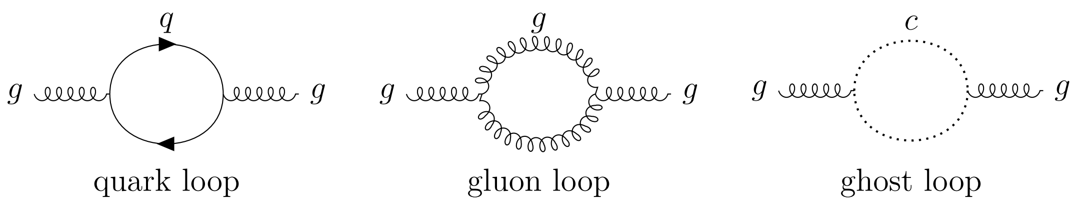


Figure 3.5: One-loop vacuum-polarization topologies contributing to the QCD beta function. The quark loop gives the $-\frac{2}{3}n_f$ part of β_0 for $SU(3)$; the gluon and ghost sectors together give the positive 11 term.

In the four-dimensional limit, the beta function is defined by

$$\frac{\mathrm{d}a_s}{\mathrm{d}\ln\mu^2} = \beta(a_s). \tag{3.574}$$

Its perturbative expansion is

$$\beta(a_s) = -\beta_0 a_s^2 - \beta_1 a_s^3 + \mathcal{O}(a_s^4), \tag{3.575}$$

with

$$\beta_1 = \frac{34}{3}C_2(G)^2 - \frac{20}{3}C_2(G)T_F n_f - 4C_2(r)T_F n_f. \tag{3.576}$$

For $SU(3)$,

$$\beta_0 = 11 - \frac{2}{3}n_f, \qquad \beta_1 = 102 - \frac{38}{3}n_f. \tag{3.577}$$

If one uses α_s rather than $a_s = \alpha_s/(4\pi)$, then

$$\frac{\mathrm{d}\alpha_s}{\mathrm{d}\ln\mu^2} = -\frac{\beta_0}{4\pi}\alpha_s^2 - \frac{\beta_1}{(4\pi)^2}\alpha_s^3 + \cdots . \tag{3.578}$$

5.5 Solving the one-loop running explicitly

Keeping only the one-loop term gives

$$\frac{\mathrm{d}a_s}{\mathrm{d}\ln Q^2} = -\beta_0 a_s^2. \tag{3.579}$$

Separate variables:

$$\frac{\mathrm{d}a_s}{a_s^2} = -\beta_0\,\mathrm{d}\ln Q^2. \tag{3.580}$$

Integrating from a reference scale μ to a scale Q gives

$$\int_{a_s(\mu^2)}^{a_s(Q^2)} \frac{\mathrm{d}a}{a^2} = -\beta_0 \int_{\ln\mu^2}^{\ln Q^2} \mathrm{d}L, \tag{3.581}$$

$$-\frac{1}{a_s(Q^2)} + \frac{1}{a_s(\mu^2)} = -\beta_0 \ln\frac{Q^2}{\mu^2}. \tag{3.582}$$

Therefore

$$\frac{1}{a_s(Q^2)} = \frac{1}{a_s(\mu^2)} + \beta_0 \ln\frac{Q^2}{\mu^2}. \tag{3.583}$$

Writing the integration constant as Λ_{QCD}, one obtains

$$a_s(Q^2) = \frac{1}{\beta_0 \ln\left(Q^2/\Lambda_{\mathrm{QCD}}^2\right)}. \tag{3.584}$$

Thus

$$\boxed{\alpha_s(Q^2) = \frac{4\pi}{\beta_0 \ln\left(Q^2/\Lambda_{\mathrm{QCD}}^2\right)}} \qquad \text{one-loop}. \tag{3.585}$$

If an experimental input $\alpha_s(\mu^2)$ is given, then

$$\Lambda_{\mathrm{QCD}} = \mu \exp\left[-\frac{1}{2\beta_0 a_s(\mu^2)}\right] = \mu \exp\left[-\frac{2\pi}{\beta_0 \alpha_s(\mu^2)}\right]. \tag{3.586}$$

In precision applications, one uses higher-loop running and threshold matching. A common reference input is

$$\alpha_s(M_Z) \simeq 0.1180. \tag{3.587}$$

For large Q,

$$\ln\frac{Q^2}{\Lambda_{\mathrm{QCD}}^2} \to \infty, \qquad \alpha_s(Q^2) \to 0. \tag{3.588}$$

This is asymptotic freedom: high-energy, short-distance processes are weakly coupled and can be treated perturbatively. When Q approaches Λ_{QCD}, the one-loop expression has a Landau-pole-like singularity. This does not mean that a physical observable literally diverges; it means

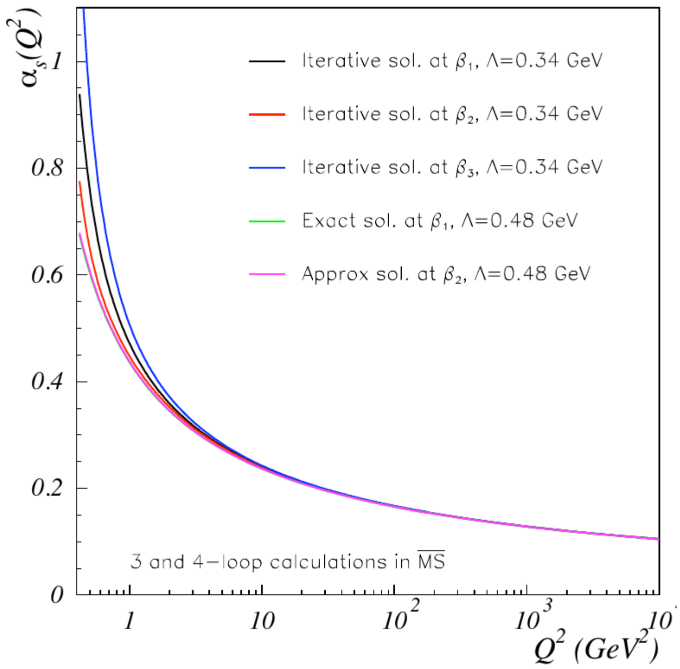


Figure 3.6: Perturbative running of $\alpha_s(Q^2)$ at different loop orders. The coupling decreases at large momentum transfer. Image cropped from A. Deur, S. J. Brodsky, and G. F. de Téramond, *The QCD Running Coupling*, arXiv:1604.08082, Fig. 3.1.

that perturbation theory is no longer reliable and nonperturbative confinement physics becomes important.

Asymptotic freedom requires

$$\beta_0 > 0. \tag{3.589}$$

For $SU(3)$,

$$11 - \frac{2}{3}n_f > 0 \qquad \Longrightarrow \qquad n_f < \frac{33}{2} = 16.5. \tag{3.590}$$

Thus integer flavour numbers obeying

$$n_f \leq 16 \tag{3.591}$$

lead to one-loop asymptotic freedom. The real world has at most six quark flavours, so real QCD is safely in the asymptotically free regime.

5.6 The basic idea of a conformal window

Including the two-loop term,

$$\beta(a_s) = -\beta_0 a_s^2 - \beta_1 a_s^3 + \cdots . \tag{3.592}$$

A nonzero fixed point satisfies

$$\beta(a_s^*) = 0 \qquad \Longrightarrow \qquad a_s^* = -\frac{\beta_0}{\beta_1}. \tag{3.593}$$

For this fixed point to occur at positive coupling, β_0 and β_1 must have opposite signs. For $SU(3)$,

$$\beta_0 > 0 \quad \Longleftrightarrow \quad n_f < 16.5, \tag{3.594}$$

while

$$\beta_1 < 0 \quad \iff \quad 102 - \frac{38}{3}n_f < 0 \quad \iff \quad n_f > \frac{306}{38} \simeq 8.05. \tag{3.595}$$

Therefore two-loop perturbation theory suggests that, roughly for $9 \leq n_f \leq 16$, an infrared fixed point may exist. When n_f is close to 16.5, β_0 is small and a_s^* is also small. This is the Banks–Zaks fixed point, which is perturbatively controlled. The exact lower edge of the conformal window is a nonperturbative question and cannot be fixed reliably by the two-loop formula alone.

5.7 Classical running in dimensional regularization

In dimensional regularization, the coupling already has a mass dimension. This can be seen from the interaction term

$$g_s \bar{q} \gamma^\mu A_\mu q. \tag{3.596}$$

The field dimensions in d dimensions are

$$[q] = \frac{d-1}{2}, \qquad [A_\mu] = \frac{d-2}{2}. \tag{3.597}$$

Since the Lagrangian density has mass dimension d,

$$[g_s] + 2\frac{d-1}{2} + \frac{d-2}{2} = d, \tag{3.598}$$

$$[g_s] = \frac{4-d}{2} = \epsilon. \tag{3.599}$$

Therefore a dimensionless renormalized coupling requires the factor μ^ϵ:

$$g_{s,0} = \mu^\epsilon Z_g g_s(\mu). \tag{3.600}$$

This produces a classical contribution to the d-dimensional beta function:

$$\boxed{\frac{\mathrm{d}a_s}{\mathrm{d}\ln\mu^2} = -\epsilon a_s - \beta_0 a_s^2 - \beta_1 a_s^3 + \cdots .} \tag{3.601}$$

The term $-\epsilon a_s$ is not a quantum loop effect. It comes from the engineering dimension of the coupling. Only after taking $\epsilon \to 0$ does one recover the usual four-dimensional QCD running.

5.8 Heavy-quark thresholds and matching of the running coupling

The number n_f in the running coupling is the number of quark flavours that are effectively light at the scale Q. For example,

$$Q \lesssim m_c : \quad n_f = 3 \quad (u, d, s), \tag{3.602}$$

while

$$m_c \lesssim Q \lesssim m_b : \quad n_f = 4 \quad (u, d, s, c). \tag{3.603}$$

When the scale crosses a heavy-quark mass m_h, one must match the coupling in the n_f-flavour effective theory to the coupling in the $(n_f + 1)$-flavour theory.

At one loop, the matching is determined by the heavy-quark loop contribution to the gluon propagator. Gauge invariance forces the gluon vacuum polarization to be transverse:

$$\mathrm{i}\Pi^{ab}_{\mu\nu}(q) = \mathrm{i}\delta^{ab}\left(g_{\mu\nu}q^2 - q_\mu q_\nu\right)\Pi(q^2). \tag{3.604}$$

First consider n_f massless light quarks. Their loop gives

$$\mathrm{i}\Pi^{ab}_{\mu\nu}(q) = -(-1)g_s^2 T_F n_f \delta^{ab}\mu^{2\epsilon}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\frac{\mathrm{tr}\left[\gamma_\mu \not{k}\gamma_\nu(\not{k}+\not{q})\right]}{k^2(k+q)^2}. \tag{3.605}$$

The fermion trace is

$$\mathrm{tr}\left[\gamma_\mu \not{k}\gamma_\nu(\not{k}+\not{q})\right] = 4\left[k_\mu(k+q)_\nu + k_\nu(k+q)_\mu - g_{\mu\nu}k\cdot(k+q)\right]. \tag{3.606}$$

Introduce a Feynman parameter:

$$\frac{1}{k^2(k+q)^2} = \int_0^1 \mathrm{d}x\,\frac{1}{\left[(k+xq)^2 + x(1-x)(-q^2)\right]^2}. \tag{3.607}$$

After shifting $\ell = k + xq$, all integrals odd in ℓ vanish. Even tensor integrals reduce according to

$$\int \mathrm{d}^d\ell\,\ell_\mu\ell_\nu F(\ell^2) = \frac{g_{\mu\nu}}{d}\int \mathrm{d}^d\ell\,\ell^2 F(\ell^2). \tag{3.608}$$

Only the transverse structure remains. The divergent and logarithmic part of the scalar polarization is

$$\Pi^{(n_f)}_{\text{light}}(q^2) = -\frac{\alpha_s}{4\pi}\frac{4}{3}T_F n_f\left[\frac{1}{\bar\epsilon} + \ln\frac{\mu^2}{-q^2} + \text{finite const.}\right], \tag{3.609}$$

or, using $T_F = 1/2$,

$$\Pi^{(n_f)}_{\text{light}}(q^2) = -\frac{\alpha_s}{4\pi}\frac{2}{3}n_f\left[\frac{1}{\bar\epsilon} + \ln\frac{\mu^2}{-q^2} + \text{finite const.}\right]. \tag{3.610}$$

Here

$$\frac{1}{\bar\epsilon} = \frac{1}{\epsilon} - \gamma_E + \ln 4\pi. \tag{3.611}$$

In handwritten notes one often suppresses scheme-dependent constants and keeps only the pole and logarithm, since these are the terms that determine running and matching.

Now suppose the quark in the loop has mass m_h, and the external momentum is much smaller than that mass:

$$q^2 \ll m_h^2. \tag{3.612}$$

The same Feynman-parameter calculation gives

$$\Pi_h(q^2) = -\frac{\alpha_s}{4\pi}\frac{2}{3}\left[\frac{1}{\bar\epsilon} + \ln\frac{\mu^2}{m_h^2} + \mathcal{O}\left(\frac{q^2}{m_h^2}\right)\right]. \tag{3.613}$$

In the $\overline{\text{MS}}$ scheme, after subtracting $1/\bar\epsilon$, the heavy-quark loop leaves only a local low-energy correction. It can be absorbed into the definition of the coupling in the low-energy theory. The one-loop decoupling relation is

$$\boxed{\alpha_s^{(n_f)}(\mu) = \alpha_s^{(n_f+1)}(\mu)\left[1 + \frac{\alpha_s^{(n_f+1)}(\mu)}{6\pi}\ln\frac{\mu^2}{m_h^2}\right] + \mathcal{O}(\alpha_s^3)}. \tag{3.614}$$

If the matching scale is chosen to be $\mu = m_h$, then at one loop

$$\alpha_s^{(n_f)}(m_h) = \alpha_s^{(n_f+1)}(m_h) + \mathcal{O}(\alpha_s^3). \tag{3.615}$$

Thus the running coupling is continuous across a heavy-quark threshold at one-loop order. At higher orders finite matching corrections remain, so precision QCD requires threshold matching at each heavy-quark mass.

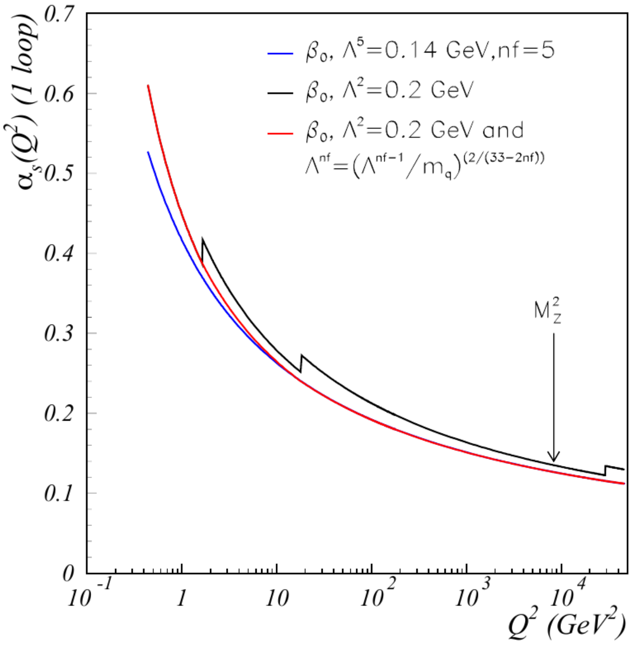


Figure 3.7: Effect of quark thresholds on the running coupling. The red curve illustrates threshold matching, while the black curve shows the discontinuous behaviour that one would obtain from changing the active flavour number without matching. Image cropped from Deur, Brodsky, and de Téramond, arXiv:1604.08082, Fig. 3.2.

5.9 Cross-section predictions and infrared safety

QCD predicts many collider observables, for example

- hadron production in e^+e^- annihilation: LEP, Belle II, BES;
- deep inelastic scattering: HERA, EIC;
- the Drell–Yan process: $pp \to \gamma^*/Z \to \ell^+\ell^-$;
- jets, heavy flavour, and vector-boson production at the LHC.

The key requirement is infrared safety: the observable must not change under the emission of a soft gluon or under a collinear splitting.

Consider inclusive $e^+e^- \to$ hadrons. At leading order,

$$R_0(s) = N_c \sum_q Q_q^2. \tag{3.616}$$

At next-to-leading order there are two classes of corrections:

$$\text{real emission:} \qquad e^+e^- \to q\bar{q}g, \tag{3.617}$$
$$\text{virtual correction:} \qquad e^+e^- \to q\bar{q} \quad \text{with one gluon loop.} \tag{3.618}$$

Each class is infrared divergent by itself. But the inclusive hadronic cross section counts both $q\bar{q}$ and $q\bar{q}g$ final states as hadrons, and the soft and collinear singularities cancel in the sum.

Let a massless quark have momentum

$$p = (E_q, \vec{p}), \qquad p^2 = 0, \tag{3.619}$$

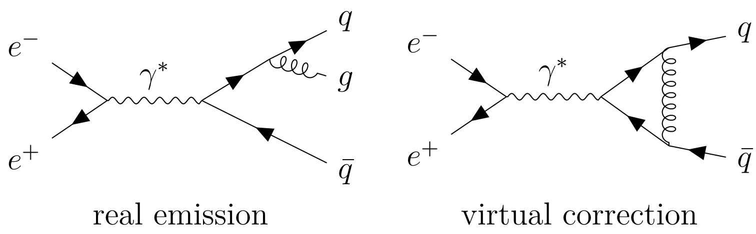


Figure 3.8: The two NLO ingredients in inclusive $e^+e^- \to$ hadrons. Each piece is infrared divergent separately, but the divergences cancel in the inclusive sum.

and let it radiate a gluon

$$k = (E_g, \vec{k}), \qquad k^2 = 0. \tag{3.620}$$

If the angle between them is θ_{qg}, then since $|\vec{p}| = E_q$ and $|\vec{k}| = E_g$,

$$2p \cdot k = 2E_q E_g (1 - \cos\theta_{qg}). \tag{3.621}$$

When the gluon is soft,

$$E_g \to 0, \tag{3.622}$$

this denominator becomes small. When the gluon is collinear to the quark,

$$\theta_{qg} \to 0, \qquad 1 - \cos\theta_{qg} \simeq \frac{\theta_{qg}^2}{2}, \tag{3.623}$$

the denominator also becomes small.

In the soft limit, the squared matrix element factorizes:

$$|\mathcal{M}_{q\bar{q}g}|^2 \simeq |\mathcal{M}_{q\bar{q}}|^2 \, 4\pi\alpha_s C_2(r) \frac{2p_q \cdot p_{\bar{q}}}{(p_q \cdot k)(p_{\bar{q}} \cdot k)}. \tag{3.624}$$

This eikonal factor contains two denominators of the form $p \cdot k$, so it generates soft singularities. In the collinear limit, the propagator

$$(p + k)^2 = p^2 + k^2 + 2p \cdot k = 2p \cdot k \tag{3.625}$$

approaches zero, generating a collinear singularity.

The phase-space measure makes the logarithmic structure explicit. In $d = 4 - 2\epsilon$, a soft gluon has approximately

$$\frac{\mathrm{d}^{d-1}k}{2E_g} \sim \mathrm{d}E_g \, E_g^{1-2\epsilon} \, \mathrm{d}\theta \, \theta^{1-2\epsilon}. \tag{3.626}$$

Multiplying by the soft-collinear eikonal factor

$$\frac{1}{E_g^2 \theta^2}, \tag{3.627}$$

one obtains

$$\int_0 \mathrm{d}E_g \, E_g^{-1-2\epsilon} \int_0 \mathrm{d}\theta \, \theta^{-1-2\epsilon}. \tag{3.628}$$

Therefore

$$\int_0^{\Lambda} \mathrm{d}E_g \, E_g^{-1-2\epsilon} = -\frac{1}{2\epsilon_{\mathrm{IR}}} + \ln\Lambda + \cdots, \tag{3.629}$$

$$\int_0^{\delta} \mathrm{d}\theta \, \theta^{-1-2\epsilon} = -\frac{1}{2\epsilon_{\mathrm{IR}}} + \ln\delta + \cdots. \tag{3.630}$$

The energy integral gives a soft pole, and the angular integral gives a collinear pole. Their overlap gives a double pole $1/\epsilon_{\mathrm{IR}}^2$.

The one-loop structure can be summarized as

$$\sigma_{\mathrm{real}} = \sigma_0 \frac{\alpha_s}{4\pi} C_2(r) \left[\frac{2}{\epsilon_{\mathrm{IR}}^2} + \frac{3}{\epsilon_{\mathrm{IR}}} + \mathrm{finite}_{\mathrm{real}} \right], \tag{3.631}$$

$$\sigma_{\mathrm{virtual}} = \sigma_0 \frac{\alpha_s}{4\pi} C_2(r) \left[-\frac{2}{\epsilon_{\mathrm{IR}}^2} - \frac{3}{\epsilon_{\mathrm{IR}}} + \mathrm{finite}_{\mathrm{virtual}} \right]. \tag{3.632}$$

The infrared poles cancel in the sum:

$$\sigma_{\mathrm{NLO}} = \sigma_0 + \sigma_{\mathrm{real}} + \sigma_{\mathrm{virtual}} \quad \text{is finite.} \tag{3.633}$$

This is the KLN theorem at work in the inclusive e^+e^- hadronic cross section.

The finite part gives the famous QCD correction

$$R(s) = N_c \sum_q Q_q^2 \left[1 + \frac{\alpha_s(s)}{4\pi} 3C_2(r) + \mathcal{O}(\alpha_s^2) \right]. \tag{3.634}$$

For QCD, $C_2(r) = 4/3$, so

$$\frac{\alpha_s}{4\pi} 3C_2(r) = \frac{\alpha_s}{4\pi} \cdot 4 = \frac{\alpha_s}{\pi}. \tag{3.635}$$

Thus

$$\boxed{R(s) = N_c \sum_q Q_q^2 \left[1 + \frac{\alpha_s(s)}{\pi} + \mathcal{O}(\alpha_s^2) \right]}. \tag{3.636}$$

This formula ties together the main themes of this section: the number of colors fixes the leading normalization, the coupling $\alpha_s(s)$ runs with the energy scale, and the real and virtual infrared singularities cancel in an inclusive observable.

Standard Model

1. Spontaneous Symmetry Breaking

Spontaneous symmetry breaking is the situation in which the equations and the Lagrangian have a symmetry, but the vacuum chosen by the theory does not. The symmetry is still present in the theory; it is not a mistake in the Lagrangian. The point is that the set of lowest-energy field configurations contains more than one equivalent choice, and choosing one of them hides part of the symmetry.

This section develops the idea in three steps. First we study a real scalar with a discrete symmetry, where there is symmetry breaking but no Goldstone boson. Then we study a complex scalar with a continuous $U(1)$ symmetry, where a massless Goldstone mode appears. Finally we generalize to N complex scalars and prove the Goldstone theorem by counting flat directions of the potential.

1.1 A real scalar with a discrete symmetry

Consider one real scalar field ϕ with Lagrangian

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V(\phi), \tag{4.1}$$

and potential

$$V(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{\lambda}{4}\phi^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{4.2}$$

This is often called the "wrong-sign mass" potential. Around $\phi = 0$ the quadratic term in the potential has negative curvature, so $\phi = 0$ is not a stable vacuum. This corrects a common misleading phrase: the theory does have a quadratic term; what is wrong is the sign of the quadratic term around the origin.

The potential is invariant under the discrete transformation

$$\phi \longrightarrow -\phi, \qquad V(\phi) = V(-\phi). \tag{4.3}$$

This is a $\mathbb{Z}_2$ symmetry.

To find the extrema, compute

$$\frac{\mathrm{d}V}{\mathrm{d}\phi} = -\mu^2\phi + \lambda\phi^3 = \phi\left(-\mu^2 + \lambda\phi^2\right). \tag{4.4}$$

Thus

$$\frac{\mathrm{d}V}{\mathrm{d}\phi} = 0 \qquad \Longrightarrow \qquad \phi = 0 \quad \text{or} \quad \phi^2 = \frac{\mu^2}{\lambda}. \tag{4.5}$$

Define

$$v = \frac{\mu}{\sqrt{\lambda}}. \tag{4.6}$$

The extrema are therefore

$$\phi = 0, \qquad \phi = \pm v. \tag{4.7}$$

To decide which are minima, compute the second derivative:

$$\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2} = -\mu^2 + 3\lambda\phi^2. \tag{4.8}$$

At the origin,

$$\left.\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2}\right|_{\phi=0} = -\mu^2 < 0, \tag{4.9}$$

so $\phi = 0$ is a local maximum. At $\phi = \pm v$,

$$\left.\frac{\mathrm{d}^2 V}{\mathrm{d}\phi^2}\right|_{\phi=\pm v} = -\mu^2 + 3\lambda\frac{\mu^2}{\lambda} = 2\mu^2 > 0, \tag{4.10}$$

so $\phi = \pm v$ are true minima.

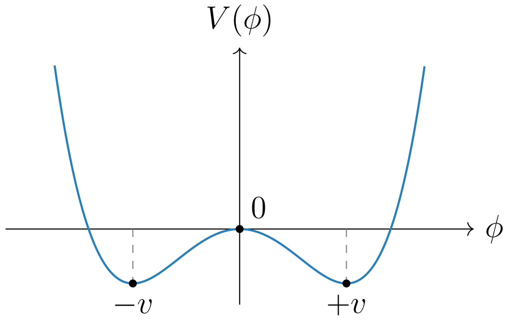


Figure 4.1: The double-well potential. The Lagrangian is symmetric under $\phi \to -\phi$, but either vacuum $\phi = +v$ or $\phi = -v$ chooses one side.

The theory has the symmetry $\phi \to -\phi$, but the vacuum does not. If the system chooses

$$\langle 0|\phi|0\rangle = +v, \tag{4.11}$$

then the transformed configuration has

$$\langle 0|\phi|0\rangle \longrightarrow -v, \tag{4.12}$$

which is a different vacuum. The two vacua are physically equivalent, but once one is chosen the symmetry is no longer manifest. This is spontaneous breaking of a discrete symmetry.

There is no Goldstone boson in this example. Goldstone bosons arise from broken continuous symmetries, where moving along the vacuum manifold can be done infinitesimally with no energy cost. A discrete symmetry does not provide such a continuous flat direction.

1.2 A complex scalar and global U(1) breaking

Now consider a complex scalar field

$$\phi = \frac{1}{\sqrt{2}}\left(\phi_1 + \mathrm{i}\phi_2\right), \tag{4.13}$$

with Lagrangian

$$\mathcal{L} = \partial_\mu\phi^*\,\partial^\mu\phi - V(\phi,\phi^*), \tag{4.14}$$

and potential

$$V(\phi,\phi^*) = -\mu^2\phi^*\phi + \frac{\lambda}{2}(\phi^*\phi)^2, \qquad \mu^2>0, \quad \lambda>0. \tag{4.15}$$

The Lagrangian is invariant under the global $U(1)$ transformation

$$\phi(x) \longrightarrow \phi'(x) = e^{\mathrm{i}\alpha}\phi(x), \qquad \alpha\in\mathbb{R}. \tag{4.16}$$

This is a global symmetry because α is constant in spacetime.

The potential depends only on the radial variable $\phi^*\phi$. Minimizing with respect to ϕ^* gives

$$\frac{\partial V}{\partial\phi^*} = \phi\left(-\mu^2+\lambda\phi^*\phi\right) = 0. \tag{4.17}$$

Therefore

$$\phi = 0 \quad \text{or} \quad \phi^*\phi = \frac{\mu^2}{\lambda}. \tag{4.18}$$

Again $\phi = 0$ is the unstable point. The true vacua satisfy

$$|\phi|^2 = \frac{\mu^2}{\lambda}. \tag{4.19}$$

In the complex plane this is a circle. Since

$$|\phi|^2 = \frac{1}{2}\left(\phi_1^2+\phi_2^2\right), \tag{4.20}$$

the vacuum circle can also be written as

$$\phi_1^2+\phi_2^2 = \frac{2\mu^2}{\lambda}. \tag{4.21}$$

It is conventional to define

$$v^2 = \frac{2\mu^2}{\lambda}. \tag{4.22}$$

Then the vacuum condition is

$$|\phi|^2 = \frac{v^2}{2}. \tag{4.23}$$

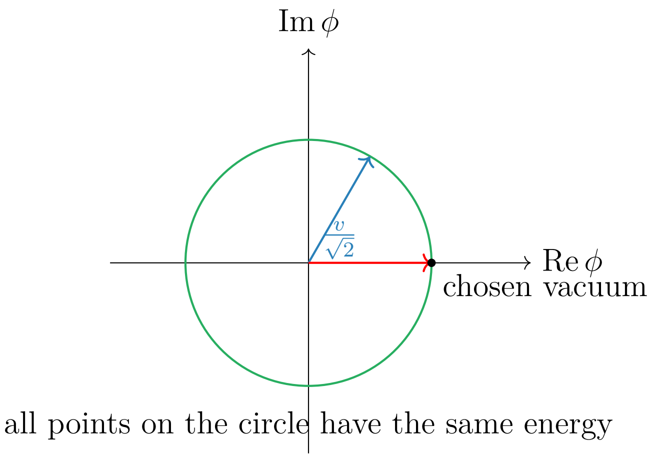


Figure 4.2: The vacuum manifold for the broken global $U(1)$ model. Choosing one point on the circle breaks the manifest $U(1)$ symmetry.

Every point on this circle is a possible ground state. Choosing one point breaks the global $U(1)$ symmetry. We may choose the vacuum to lie along the real direction:

$$\phi_0 = \frac{v}{\sqrt{2}}, \qquad \phi_{1,0} = v, \quad \phi_{2,0} = 0. \tag{4.24}$$

This choice is not a loss of generality; any other point on the circle can be rotated into this one by a $U(1)$ transformation.

1.3 Expansion around the chosen vacuum

To find the physical particles, expand the field around the chosen vacuum:

$$\phi(x) = \frac{1}{\sqrt{2}} \left[v + h(x) + \mathrm{i}\sigma(x) \right]. \tag{4.25}$$

Here $h(x)$ is the radial fluctuation and $\sigma(x)$ is the angular fluctuation. At the vacuum,

$$h = 0, \qquad \sigma = 0. \tag{4.26}$$

The kinetic term becomes

$$\begin{aligned} \partial_\mu \phi^* \, \partial^\mu \phi &= \frac{1}{2} \partial_\mu (v + h - \mathrm{i}\sigma) \partial^\mu (v + h + \mathrm{i}\sigma) \\ &= \frac{1}{2} \partial_\mu h \, \partial^\mu h + \frac{1}{2} \partial_\mu \sigma \, \partial^\mu \sigma. \end{aligned} \tag{4.27}$$

Since v is constant, its derivative vanishes.

Now compute the potential in terms of h and σ. First,

$$\begin{aligned} \phi^* \phi &= \frac{1}{2} \left[(v + h)^2 + \sigma^2 \right] \\ &= \frac{1}{2} \left[v^2 + 2vh + h^2 + \sigma^2 \right]. \end{aligned} \tag{4.28}$$

Therefore

$$V(h, \sigma) = -\frac{\mu^2}{2} \left[(v + h)^2 + \sigma^2 \right] + \frac{\lambda}{8} \left[(v + h)^2 + \sigma^2 \right]^2. \tag{4.29}$$

Expanding the square,

$$\begin{aligned} \left[(v + h)^2 + \sigma^2 \right]^2 &= \left(v^2 + 2vh + h^2 + \sigma^2 \right)^2 \\ &= v^4 + 4v^3 h + 6v^2 h^2 + 2v^2 \sigma^2 + \mathcal{O}(\text{cubic}). \end{aligned} \tag{4.30}$$

Hence the terms up to quadratic order are

$$V(h, \sigma) = -\frac{\mu^2}{2} \left(v^2 + 2vh + h^2 + \sigma^2 \right) + \frac{\lambda}{8} \left(v^4 + 4v^3 h + 6v^2 h^2 + 2v^2 \sigma^2 \right) + \cdots. \tag{4.31}$$

Separate constant, linear, and quadratic terms:

$$\begin{aligned} V(h, \sigma) &= \left(-\frac{\mu^2 v^2}{2} + \frac{\lambda v^4}{8} \right) + \left(-\mu^2 v + \frac{\lambda v^3}{2} \right) h \\ &\quad + \left(-\frac{\mu^2}{2} + \frac{3\lambda v^2}{4} \right) h^2 + \left(-\frac{\mu^2}{2} + \frac{\lambda v^2}{4} \right) \sigma^2 + \cdots. \end{aligned} \tag{4.32}$$

Using

$$v^2 = \frac{2\mu^2}{\lambda}, \tag{4.33}$$

the linear term vanishes:

$$-\mu^2 v + \frac{\lambda v^3}{2} = v \left(-\mu^2 + \frac{\lambda v^2}{2} \right) = 0. \tag{4.34}$$

This must happen because we expanded around a minimum. The quadratic terms become

$$-\frac{\mu^2}{2} + \frac{3\lambda v^2}{4} = -\frac{\mu^2}{2} + \frac{3\mu^2}{2} = \mu^2, \tag{4.35}$$

$$-\frac{\mu^2}{2} + \frac{\lambda v^2}{4} = -\frac{\mu^2}{2} + \frac{\mu^2}{2} = 0. \tag{4.36}$$

Thus

$$V(h, \sigma) = V(v) + \mu^2 h^2 + \mathcal{O}(h^3, h^2\sigma, h\sigma^2, \sigma^4). \tag{4.37}$$

The Lagrangian contains $-\mu^2 h^2$. Comparing with the standard mass term

$$-\frac{1}{2} m_h^2 h^2, \tag{4.38}$$

we find

$$m_h^2 = 2\mu^2. \tag{4.39}$$

There is no quadratic term for σ, so

$$m_\sigma^2 = 0. \tag{4.40}$$

The radial field h is massive. The angular field σ is massless and is called the Goldstone boson.

The reason is geometric. The h direction moves away from the vacuum circle, so the potential increases. The σ direction moves along the vacuum circle, where the potential is flat. A flat direction has zero curvature, and zero curvature means zero mass squared.

1.4 N complex scalar fields

Now consider N complex scalar fields ϕ^i, $i = 1, \ldots, N$, with

$$\mathcal{L} = \sum_{i=1}^{N} \partial_\mu \phi^{i*} \partial^\mu \phi^i - V(\phi, \phi^*), \tag{4.41}$$

and

$$V(\phi, \phi^*) = -\mu^2 \sum_{i=1}^{N} \phi^{i*}\phi^i + \frac{\lambda}{2}\left(\sum_{i=1}^{N} \phi^{i*}\phi^i\right)^2. \tag{4.42}$$

This potential is invariant under global $U(N)$ transformations:

$$\phi^i \longrightarrow \phi'^i = \sum_{j=1}^{N} U^{ij}\phi^j, \qquad U^\dagger U = \mathbf{1}. \tag{4.43}$$

The quantity

$$\sum_i \phi^{i*}\phi^i \tag{4.44}$$

is just the squared length of the complex vector ϕ, so it is unchanged by unitary rotations.

The minimum satisfies

$$\sum_{i=1}^{N} \phi^{i*}\phi^i = \frac{\mu^2}{\lambda}. \tag{4.45}$$

This is a sphere in the $2N$ real-dimensional field space. Define again

$$\frac{v^2}{2} = \frac{\mu^2}{\lambda}, \qquad v^2 = \frac{2\mu^2}{\lambda}. \tag{4.46}$$

Using the $U(N)$ symmetry, choose the vacuum to point in the first complex direction:

$$\phi_0^1 = \frac{v}{\sqrt{2}}, \qquad \phi_0^2 = \cdots = \phi_0^N = 0. \tag{4.47}$$

Then write the fields as

$$\phi^1 = \frac{1}{\sqrt{2}} \left[v + h(x) + \mathrm{i}\sigma(x) \right], \tag{4.48}$$

$$\phi^k = \frac{1}{\sqrt{2}} \left[\pi_{2k-2}(x) + \mathrm{i}\pi_{2k-1}(x) \right], \qquad k = 2, \dots, N. \tag{4.49}$$

There are $2N$ real fields in total:

$$h, \quad \sigma, \quad \pi_2, \dots, \pi_{2N-1}. \tag{4.50}$$

The radial field h changes the length of the vector ϕ and is massive. All other fields change only the direction of ϕ in the vacuum manifold and are massless at tree level.

Equivalently, the chosen vacuum leaves a residual $U(N-1)$ symmetry acting on the fields

$$\phi^2, \dots, \phi^N. \tag{4.51}$$

Thus the symmetry-breaking pattern is

$$U(N) \longrightarrow U(N-1). \tag{4.52}$$

The number of broken generators is

$$\dim U(N) - \dim U(N-1) = N^2 - (N-1)^2 = 2N - 1. \tag{4.53}$$

Therefore there are

$$2N - 1 \tag{4.54}$$

massless Goldstone bosons. This agrees with the field decomposition above: σ plus the $2N-2$ real components of $\phi^2, \dots, \phi^N$.

1.5 Goldstone theorem

The examples above illustrate the general result.

Goldstone theorem. In a relativistic theory with a continuous global internal symmetry, each spontaneously broken generator gives one massless real scalar excitation, called a Goldstone boson.

There are important assumptions in this statement. The symmetry must be global, not gauged. If the symmetry is gauged, the massless scalar is removed from the physical spectrum by the Higgs mechanism. Also, the symmetry must be exact. If it is explicitly broken, the would-be Goldstone boson generally becomes light but not massless.

1.6 Proof by the mass matrix

Consider N real scalar fields ϕ^i with Lagrangian

$$\mathcal{L} = \frac{1}{2}\sum_{i=1}^{N} \partial_\mu\phi^i\,\partial^\mu\phi^i - V(\phi). \tag{4.55}$$

Suppose the potential is invariant under a continuous symmetry group G. For an infinitesimal transformation,

$$\phi^i \longrightarrow \phi^i + \delta_a\phi^i, \tag{4.56}$$

where

$$\delta_a\phi^i = \alpha^a (T_a)^i{}_j \phi^j. \tag{4.57}$$

Here T_a is a generator of the symmetry and α^a is an infinitesimal real parameter. Since V is invariant,

$$\delta_a V = \sum_i \frac{\partial V}{\partial\phi^i}\delta_a\phi^i = 0. \tag{4.58}$$

Substitute the infinitesimal transformation:

$$\sum_i \frac{\partial V}{\partial\phi^i}(T_a)^i{}_j\phi^j = 0. \tag{4.59}$$

This identity holds for every field configuration ϕ, not only at the minimum.

Now differentiate this identity with respect to ϕ^k:

$$\sum_i \frac{\partial^2 V}{\partial\phi^k\partial\phi^i}(T_a)^i{}_j\phi^j + \sum_i \frac{\partial V}{\partial\phi^i}(T_a)^i{}_k = 0. \tag{4.60}$$

Evaluate at a vacuum ϕ_0. Since ϕ_0 is a minimum,

$$\left.\frac{\partial V}{\partial\phi^i}\right|_{\phi_0} = 0. \tag{4.61}$$

The second term therefore vanishes. Define the mass matrix

$$(M^2)_{ki} = \left.\frac{\partial^2 V}{\partial\phi^k\partial\phi^i}\right|_{\phi_0}. \tag{4.62}$$

Then

$$\sum_i (M^2)_{ki}(T_a)^i{}_j\phi_0^j = 0. \tag{4.63}$$

In vector notation this is

$$M^2\,(T_a\phi_0) = 0. \tag{4.64}$$

Now distinguish two cases.

Unbroken generator.

If T_a is unbroken, it leaves the vacuum invariant:

$$T_a\phi_0 = 0. \tag{4.65}$$

Then the equation above is trivial and gives no new massless field.

Broken generator.

If T_a is broken, then it moves the vacuum:

$$T_a\phi_0 \neq 0. \tag{4.66}$$

But the mass-matrix equation says

$$M^2\left(T_a\phi_0\right) = 0. \tag{4.67}$$

Thus $T_a\phi_0$ is a nonzero eigenvector of the mass matrix with eigenvalue zero. A zero eigenvalue of the scalar mass matrix means a massless scalar mode. Therefore each broken generator gives a massless scalar field.

This is the mathematical form of the geometric picture. Broken generators move the vacuum along the manifold of degenerate minima. Motion along that manifold costs no potential energy, so the corresponding direction has zero curvature. Zero curvature of the potential is precisely zero mass squared.

2. The Higgs Mechanism

2.1 Difference between Higgs mechanism and Goldstone's theorem

As we have discussed in last section, Goldstone's theorem should be applied directly only to a *global* continuous symmetry. Suppose a continuous global symmetry is a true physical symmetry of the Lagrangian, but the vacuum does not respect it. Then each broken generator produces one physical massless scalar particle, called a Goldstone boson.

The Higgs mechanism uses a scalar potential with the same geometric shape, but now the symmetry is *local*. A local gauge symmetry is not an ordinary physical symmetry that relates two different states; it is a redundancy in our description of the same physical state. For this reason, Goldstone's theorem does not say that a broken gauge symmetry must produce physical massless particles.

The correct way to think about the Higgs mechanism is the following. First, temporarily ignore the gauge field. The scalar potential then has flat angular directions, and fluctuations along those directions would be Goldstone bosons in the corresponding global theory. After the symmetry is gauged, those same angular fields are no longer physical particles. They can be removed by a gauge choice and become the longitudinal polarizations of the massive gauge bosons. This is why they are called *would-be Goldstone bosons*.

Thus, when we later write a phase field such as $\theta(x)$, we are not claiming that θ is a physical massless Goldstone particle in the gauge theory. We are identifying the angular scalar mode which would have been a Goldstone boson before gauging, and then showing explicitly how the gauge field absorbs it.

In short,

$$\begin{aligned} \text{global spontaneous symmetry breaking} &\Longrightarrow \text{physical massless Goldstone bosons,} \\ \text{gauged version of the same scalar theory} &\Longrightarrow \text{would-be Goldstone modes are eaten.} \end{aligned} \tag{4.68}$$

The result is a massive gauge field. Its extra longitudinal polarization comes from the scalar mode which would have been a Goldstone boson in the ungauged global theory.

2.2 The Abelian Higgs model

We first study the simplest example: a complex scalar field coupled to a $U(1)$ gauge field. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^*(D^\mu\phi) - V(\phi), \qquad D_\mu = \partial_\mu + \mathrm{i}eA_\mu, \tag{4.69}$$

with

$$V(\phi) = -\mu^2|\phi|^2 + \frac{\lambda}{2}|\phi|^4, \qquad \mu^2 > 0, \quad \lambda > 0. \tag{4.70}$$

Equivalently, the scalar part of the Lagrangian contains

$$\mathcal{L} \supset (D_\mu\phi)^*(D^\mu\phi) + \mu^2|\phi|^2 - \frac{\lambda}{2}|\phi|^4. \tag{4.71}$$

Notice that the sign of quadratic term is positive, which means this is not the mass term yet. The Lagrangian is invariant under the local transformation

$$\phi(x) \longrightarrow e^{\mathrm{i}\alpha(x)}\phi(x), \qquad A_\mu(x) \longrightarrow A_\mu(x) - \frac{1}{e}\partial_\mu\alpha(x). \tag{4.72}$$

The reason for introducing A_μ is exactly to make derivatives compatible with a position-dependent phase. Indeed,

$$D_\mu\phi = (\partial_\mu + \mathrm{i}eA_\mu)\phi \quad \Longrightarrow \quad D_\mu\phi \longrightarrow e^{\mathrm{i}\alpha(x)}D_\mu\phi, \tag{4.73}$$

so $(D_\mu\phi)^*(D^\mu\phi)$ is gauge invariant.

The vacuum

The vacuum is found by minimizing V. Since V depends only on $|\phi|$, write $r = |\phi|$. Then

$$V(r) = -\mu^2 r^2 + \frac{\lambda}{2}r^4. \tag{4.74}$$

The stationary condition is

$$\frac{\mathrm{d}V}{\mathrm{d}r} = -2\mu^2 r + 2\lambda r^3 = 2r(-\mu^2 + \lambda r^2) = 0. \tag{4.75}$$

Thus either $r = 0$, or

$$r^2 = \frac{\mu^2}{\lambda}. \tag{4.76}$$

The point $r = 0$ is a local maximum, because

$$\left.\frac{\mathrm{d}^2V}{\mathrm{d}r^2}\right|_{r=0} = -2\mu^2 < 0. \tag{4.77}$$

Therefore the minima lie on the circle

$$|\phi| = v, \qquad v^2 = \frac{\mu^2}{\lambda}. \tag{4.78}$$

This circle of minima is the familiar “Mexican-hat” vacuum manifold. The radial direction costs energy, while moving around the circle costs no potential energy before the field is coupled to a gauge field.

Polar form and unitary gauge

Near the vacuum it is useful to separate radial and angular fluctuations:

$$\phi(x) = \left(v + \frac{h(x)}{\sqrt{2}}\right) e^{\mathrm{i}\theta(x)}. \tag{4.79}$$

Here $h(x)$ is the radial fluctuation, while $\theta(x)$ is the angular coordinate along the circle of classical minima. It is important not to misread this sentence: in the local $U(1)$ theory, $\theta(x)$ is not a physical Goldstone particle. Rather, it is the field that *would have been* the Goldstone boson if the same scalar potential had only a global $U(1)$ symmetry. Because the $U(1)$ symmetry is gauged here, $\theta(x)$ is a would-be Goldstone mode and can be removed by a gauge transformation.

To see this explicitly, substitute (4.79) into the covariant derivative:

$$D_\mu\phi = e^{\mathrm{i}\theta}\left[\frac{1}{\sqrt{2}}\partial_\mu h + \mathrm{i}\left(v + \frac{h}{\sqrt{2}}\right)(\partial_\mu\theta + eA_\mu)\right]. \tag{4.80}$$

The phase $e^{\mathrm{i}\theta}$ drops out of the absolute value, so

$$|D_\mu\phi|^2 = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + e^2\left(v + \frac{h}{\sqrt{2}}\right)^2 B_\mu B^\mu, \qquad B_\mu \equiv A_\mu + \frac{1}{e}\partial_\mu\theta. \tag{4.81}$$

The combination B_μ is what remains after choosing unitary gauge, where the scalar is made real:

$$\phi(x) \longrightarrow v + \frac{h(x)}{\sqrt{2}}, \qquad A_\mu(x) \longrightarrow B_\mu(x). \tag{4.82}$$

Thus the angular field θ has not disappeared mysteriously; it has become part of the gauge field.

Expanding the second term in (4.81) gives

$$e^2\left(v + \frac{h}{\sqrt{2}}\right)^2 B_\mu B^\mu = e^2v^2 B_\mu B^\mu + \sqrt{2}e^2 v\, hB_\mu B^\mu + \frac{e^2}{2}h^2 B_\mu B^\mu. \tag{4.83}$$

The first term is a mass term for the vector field. A massive vector mass term is conventionally written as

$$\frac{1}{2}m_A^2 B_\mu B^\mu. \tag{4.84}$$

Comparing this with $e^2v^2B_\mu B^\mu$, we obtain

$$m_A^2 = 2e^2v^2. \tag{4.85}$$

The remaining terms in (4.83) are interactions between the Higgs fluctuation h and the massive vector boson.

The radial scalar mass follows from the potential. Put

$$r = v + \frac{h}{\sqrt{2}}. \tag{4.86}$$

Since $V'(v) = 0$, the leading nonconstant term is

$$V(r) = V(v) + \frac{1}{2}V''(v)\left(\frac{h}{\sqrt{2}}\right)^2 + \cdots. \tag{4.87}$$

Now

$$V''(r) = -2\mu^2 + 6\lambda r^2, \qquad V''(v) = -2\mu^2 + 6\lambda v^2 = 4\mu^2, \tag{4.88}$$

where $v^2 = \mu^2/\lambda$ has been used. Hence

$$V(r) = V(v) + \mu^2 h^2 + \cdots . \tag{4.89}$$

Because the Lagrangian contains $-V$, the quadratic scalar term is

$$-\mu^2 h^2 = -\frac{1}{2}(2\mu^2)h^2, \tag{4.90}$$

so

$$m_h^2 = 2\mu^2. \tag{4.91}$$

Counting degrees of freedom

Before the Higgs mechanism, the complex scalar has two real degrees of freedom and the massless gauge boson has two transverse polarizations:

$$2 + 2 = 4. \tag{4.92}$$

After choosing the vacuum, only the radial scalar h remains as a physical scalar. The gauge boson is now massive, so it has three polarizations:

$$1 + 3 = 4. \tag{4.93}$$

No physical degree of freedom is lost. The would-be Goldstone mode supplies the longitudinal polarization of the massive vector boson.

	scalar degrees of freedom	vector degrees of freedom	total
before	2	2	4
after	1	3	4

2.3 The same physics in Cartesian fields

The polar form makes the Higgs mechanism transparent, but it is also useful to see how the result appears if we expand the complex scalar into two real fields:

$$\phi(x) = v + \frac{1}{\sqrt{2}}\big(\phi_1(x) + \mathrm{i}\phi_2(x)\big), \qquad \phi_1, \phi_2 \in \mathbb{R}. \tag{4.94}$$

Here ϕ_1 is the radial fluctuation and ϕ_2 is the angular fluctuation to leading order.

To derive the quadratic part of the potential, first compute

$$|\phi|^2 = \left(v + \frac{\phi_1}{\sqrt{2}}\right)^2 + \left(\frac{\phi_2}{\sqrt{2}}\right)^2 = v^2 + \sqrt{2}v\phi_1 + \frac{1}{2}(\phi_1^2 + \phi_2^2). \tag{4.95}$$

It is useful to write this as

$$|\phi|^2 = v^2 + \Delta, \qquad \Delta = \sqrt{2}v\phi_1 + \frac{1}{2}(\phi_1^2 + \phi_2^2). \tag{4.96}$$

Substituting this into $V = -\mu^2|\phi|^2 + \frac{\lambda}{2}|\phi|^4$ gives

$$\begin{aligned} V &= -\mu^2(v^2+\Delta) + \frac{\lambda}{2}(v^2+\Delta)^2 \\ &= V(v) + (-\mu^2+\lambda v^2)\Delta + \frac{\lambda}{2}\Delta^2. \end{aligned} \tag{4.97}$$

The coefficient of Δ vanishes because the vacuum satisfies $v^2 = \mu^2/\lambda$. Therefore there is no linear term, and the quadratic part comes only from Δ^2. To second order in the fluctuating fields,

$$\Delta^2 = \left(\sqrt{2}v\phi_1\right)^2 + \text{terms of cubic or higher order} = 2v^2\phi_1^2 + \cdots . \tag{4.98}$$

Using $v^2 = \mu^2/\lambda$, the potential therefore expands as

$$V(\phi) = V(v) + \mu^2\phi_1^2 + \text{interaction terms.} \tag{4.99}$$

Therefore

$$-V(\phi) = -V(v) - \mu^2\phi_1^2 + \text{interaction terms.} \tag{4.100}$$

This gives

$$m_{\phi_1}^2 = 2\mu^2, \qquad m_{\phi_2}^2 = 0 \tag{4.101}$$

before gauge fixing. The absence of a ϕ_2^2 term is the usual Goldstone-boson result for the scalar potential. In a gauge theory this field will mix with the gauge field and is not a physical massless particle.

Now expand the covariant derivative:

$$\begin{aligned} D_\mu\phi &= \partial_\mu\phi + \mathrm{i}eA_\mu\phi \\ &= \frac{1}{\sqrt{2}}\left(\partial_\mu\phi_1 + \mathrm{i}\partial_\mu\phi_2\right) + \mathrm{i}evA_\mu + \frac{\mathrm{i}e}{\sqrt{2}}A_\mu(\phi_1+\mathrm{i}\phi_2). \end{aligned} \tag{4.102}$$

Keeping only terms quadratic in the fields gives

$$|D_\mu\phi|^2 = \frac{1}{2}(\partial_\mu\phi_1)^2 + \frac{1}{2}(\partial_\mu\phi_2)^2 + \sqrt{2}ev\,A^\mu\partial_\mu\phi_2 + e^2v^2A_\mu A^\mu + \cdots . \tag{4.103}$$

The last term is the same gauge-boson mass term as before:

$$e^2v^2A_\mu A^\mu = \frac{1}{2}m_A^2A_\mu A^\mu, \qquad m_A^2 = 2e^2v^2. \tag{4.104}$$

The mixed term

$$\sqrt{2}ev\,A^\mu\partial_\mu\phi_2 \tag{4.105}$$

is the Cartesian-field signal that ϕ_2 is not an independent physical particle. In unitary gauge it is removed. In a covariant R_ξ gauge it is cancelled by a gauge-fixing term, leaving a gauge-dependent unphysical would-be Goldstone field whose effects cancel from physical amplitudes.

2.4 What the propagator is telling us

Now we shall top for a moment and discuss how "the gauge field eats the would-be Goldstone mode" . And what really shows up in the propagator of the gauge field.

Recall the degree-of-freedom count:

$$\text{massless vector: 2 polarizations,} \qquad \text{massive vector: 3 polarizations.} \tag{4.106}$$

The new third polarization is the longitudinal polarization. The question is: where does it come from? The answer is that it comes from the angular scalar mode, namely the would-be Goldstone field.

In unitary gauge the would-be Goldstone field has been removed from the scalar field, so it is hidden inside the gauge field. The massive-vector propagator then takes the form

$$D_{\mu\nu}^{\text{unitary}}(k) = \frac{-\mathrm{i}}{k^2 - m_A^2 + \mathrm{i}\epsilon}\left(g_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2}\right). \tag{4.107}$$

The term $k_\mu k_\nu/m_A^2$ is the part associated with the longitudinal polarization. Therefore this propagator is the unitary-gauge way of saying: the vector field now contains a longitudinal mode.

The same physics can be seen before choosing unitary gauge. In the Cartesian expansion, the quadratic Lagrangian contains the mixing term

$$\sqrt{2}ev\, A^\mu \partial_\mu \phi_2. \tag{4.108}$$

Using $m_A = \sqrt{2}ev$, this is

$$m_A A^\mu \partial_\mu \phi_2. \tag{4.109}$$

The derivative gives a momentum factor k_μ, so the mixing between A_μ and the would-be Goldstone field carries a factor schematically like $m_A k_\mu$. This is why a would-be Goldstone exchange naturally produces terms proportional to $k_\mu k_\nu$, exactly the tensor structure needed for the longitudinal part of a vector propagator.

Schematically, the gauge-boson mass insertion gives a contribution

$$\mathrm{i}m_A^2 g_{\mu\nu}. \tag{4.110}$$

The exchange of the would-be Goldstone mode gives a longitudinal contribution of the form

$$m_A k_\mu \frac{\mathrm{i}}{k^2}(-m_A k_\nu) = -\mathrm{i}m_A^2 \frac{k_\mu k_\nu}{k^2}. \tag{4.111}$$

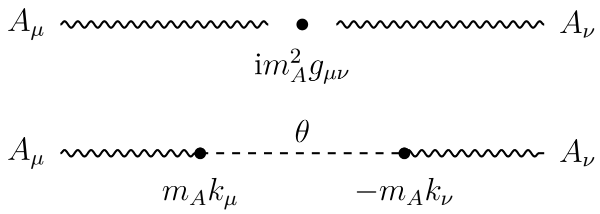


Figure 4.3: The mass insertion and the would-be Goldstone exchange combine to produce the longitudinal structure of the massive gauge boson. This is a schematic diagram; the precise signs depend on the gauge-fixing convention.

Together they form the transverse structure

$$\mathrm{i}m_A^2\left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}\right). \tag{4.112}$$

This equation should be read only as a schematic explanation, not as a complete propagator formula. The lesson is simple: the same scalar mode that disappears in unitary gauge is precisely the mode that supplies the longitudinal polarization of the massive gauge boson. Thus the phrase "the gauge boson eats the Goldstone boson" means:

$$A_\mu \text{ with 2 polarizations} + 1 \text{ would-be Goldstone mode} \implies \text{massive } A_\mu \text{ with 3 polarizations.} \tag{4.113}$$

2.5 $SU(2)$ gauge symmetry breaking

The non-Abelian version uses a complex scalar doublet

$$\Phi = \begin{pmatrix} \Phi_1 \\ \Phi_2 \end{pmatrix}, \qquad \Phi_1, \Phi_2 \in \mathbb{C}, \tag{4.114}$$

coupled to an $SU(2)$ gauge field. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} + (D_\mu \Phi)^\dagger (D^\mu \Phi) - V(\Phi), \tag{4.115}$$

where

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2, \qquad \mu^2 > 0, \quad \lambda > 0, \tag{4.116}$$

and

$$D_\mu = \partial_\mu - \mathrm{i} g \frac{\sigma^a}{2} W^a_\mu. \tag{4.117}$$

The σ^a are the Pauli matrices, and repeated adjoint indices $a = 1, 2, 3$ are summed.

The potential depends only on $\Phi^\dagger \Phi$. Let

$$\rho^2 = \Phi^\dagger \Phi. \tag{4.118}$$

Then

$$V(\rho) = -\mu^2 \rho^2 + \lambda \rho^4. \tag{4.119}$$

The minimum satisfies

$$\frac{\mathrm{d}V}{\mathrm{d}\rho} = -2\mu^2 \rho + 4\lambda \rho^3 = 2\rho(-\mu^2 + 2\lambda \rho^2) = 0. \tag{4.120}$$

Thus the nonzero vacuum satisfies

$$\Phi^\dagger \Phi = \rho^2 = \frac{\mu^2}{2\lambda}. \tag{4.121}$$

It is conventional to write

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \qquad v^2 = \frac{\mu^2}{\lambda}. \tag{4.122}$$

Indeed,

$$\langle \Phi \rangle^\dagger \langle \Phi \rangle = \frac{v^2}{2} = \frac{\mu^2}{2\lambda}. \tag{4.123}$$

Notice the dimensions: $\Phi^\dagger \Phi$ has dimension two, so it is proportional to v^2, not to v.

Which generators are broken?

For this pure $SU(2)$ theory, all three generators are broken by the vacuum. Acting with the Pauli matrices gives

$$\sigma^1 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} v \\ 0 \end{pmatrix}, \qquad \sigma^2 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} -\mathrm{i}v \\ 0 \end{pmatrix}, \qquad \sigma^3 \begin{pmatrix} 0 \\ v \end{pmatrix} = \begin{pmatrix} 0 \\ -v \end{pmatrix}. \tag{4.124}$$

None of these is zero, so the vacuum is moved by every generator. Hence the three would-be Goldstone fields are absorbed by the three $SU(2)$ gauge bosons.

This statement is specific to a pure $SU(2)$ gauge theory with one scalar doublet. In the Standard Model the gauge group is $SU(2)_L \times U(1)_Y$, and one linear combination remains unbroken; that unbroken combination is electromagnetism.

Unitary gauge and the gauge-boson masses

The most general fluctuation around the chosen vacuum may be written as

$$\Phi(x) = \exp\left[\frac{\mathrm{i}}{2}\sigma^a\theta^a(x)\right]\frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v+h(x)\end{pmatrix}. \tag{4.125}$$

The three fields $\theta^a(x)$ are the would-be Goldstone fields. An $SU(2)$ gauge transformation can remove them, leaving unitary gauge:

$$\Phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v+h(x)\end{pmatrix}. \tag{4.126}$$

In matrix notation,

$$W_\mu = W_\mu^a\frac{\sigma^a}{2}. \tag{4.127}$$

Under a local $SU(2)$ transformation $U(x)$, the gauge field transforms as

$$W_\mu \longrightarrow UW_\mu U^\dagger - \frac{\mathrm{i}}{g}U\partial_\mu U^\dagger, \tag{4.128}$$

with the sign matching the convention

$$D_\mu = \partial_\mu - \mathrm{i}gW_\mu. \tag{4.129}$$

Now compute the covariant derivative in unitary gauge. Since

$$\sigma^1\begin{pmatrix}0\\ v+h\end{pmatrix} = \begin{pmatrix}v+h\\ 0\end{pmatrix}, \quad \sigma^2\begin{pmatrix}0\\ v+h\end{pmatrix} = \begin{pmatrix}-\mathrm{i}(v+h)\\ 0\end{pmatrix}, \quad \sigma^3\begin{pmatrix}0\\ v+h\end{pmatrix} = \begin{pmatrix}0\\ -(v+h)\end{pmatrix}, \tag{4.130}$$

we find

$$D_\mu\Phi = \frac{1}{\sqrt{2}}\begin{pmatrix}-\frac{\mathrm{i}g}{2}(v+h)(W_\mu^1 - \mathrm{i}W_\mu^2)\\ \partial_\mu h + \frac{\mathrm{i}g}{2}(v+h)W_\mu^3\end{pmatrix}. \tag{4.131}$$

Taking the Hermitian norm gives

$$(D_\mu\Phi)^\dagger(D^\mu\Phi) = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{g^2}{8}(v+h)^2 W_\mu^a W^{a\mu}. \tag{4.132}$$

The gauge-boson mass term is therefore

$$\frac{g^2v^2}{8}W_\mu^a W^{a\mu}. \tag{4.133}$$

Since the canonical mass term for each real vector boson is

$$\frac{1}{2}m_W^2 W_\mu^a W^{a\mu}, \tag{4.134}$$

we obtain

$$m_W^2 = \frac{g^2v^2}{4}. \tag{4.135}$$

All three gauge bosons have the same mass because the original theory has an $SU(2)$ symmetry and the scalar representation treats the three generators symmetrically after all of them are broken.

Degree-of-freedom count for the $SU(2)$ model

The complex doublet contains four real scalar degrees of freedom. Before the Higgs mechanism, the three massless gauge bosons each have two transverse polarizations:

$$4 + 3 \times 2 = 10. \tag{4.136}$$

After the Higgs mechanism, one scalar h remains, and the three gauge bosons are massive, each with three polarizations:

$$1 + 3 \times 3 = 10. \tag{4.137}$$

Again the number of physical degrees of freedom is unchanged. The three would-be Goldstone modes have become the longitudinal polarizations of the three massive gauge bosons.

	scalar degrees of freedom	vector degrees of freedom	total
before	4	$3 \times 2 = 6$	10
after	1	$3 \times 3 = 9$	10

3. Chiral Gauge Theories and Fermion Masses

The purpose of this section is to explain why chiral gauge theories are special. The main point is simple:

A chiral gauge symmetry can forbid an ordinary fermion mass term.

This statement is the reason the Higgs mechanism is needed not only for gauge boson masses, but also for fermion masses. A bare Dirac mass would directly couple a left-handed fermion to a right-handed fermion. If those two fields carry different gauge quantum numbers, that coupling is not gauge invariant. The Higgs field supplies exactly the missing gauge quantum numbers. After the Higgs field develops a vacuum expectation value, the Yukawa interaction becomes an ordinary fermion mass term.

3.1 Recap for Dirac spinors and Weyl spinors

A four-component Dirac spinor can be written in terms of two two-component Weyl spinors. In the chiral representation we write

$$\Psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}. \tag{4.138}$$

Here ψ_L is the left-handed Weyl spinor and ψ_R is the right-handed Weyl spinor. The labels L and R refer to chirality. For massless fermions chirality is the same as helicity, but for massive fermions they are not the same concept.

The Dirac adjoint is

$$\bar{\Psi} = \Psi^\dagger \gamma^0. \tag{4.139}$$

With the chiral gamma matrices used below, this becomes

$$\bar{\Psi} = \begin{pmatrix} \psi_R^\dagger & \psi_L^\dagger \end{pmatrix}. \tag{4.140}$$

This reversal of L and R inside $\bar{\Psi}$ is a common source of confusion. It happens because γ^0 is off diagonal in the chiral basis.

We use

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \qquad \sigma^\mu = (1, \vec{\sigma}), \qquad \bar{\sigma}^\mu = (1, -\vec{\sigma}), \tag{4.141}$$

where $\vec{\sigma}$ are the Pauli matrices. In this convention

$$\gamma^5 = \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{4.142}$$

Therefore the chiral projection operators are

$$P_L = \frac{1-\gamma^5}{2}, \qquad P_R = \frac{1+\gamma^5}{2}. \tag{4.143}$$

They obey

$$P_L^2 = P_L, \qquad P_R^2 = P_R, \qquad P_L P_R = 0, \qquad P_L + P_R = 1. \tag{4.144}$$

Thus

$$\Psi_L = P_L \Psi, \qquad \Psi_R = P_R \Psi. \tag{4.145}$$

Some books define γ^5 with the opposite sign. Then the displayed formulas for P_L and P_R are interchanged. This is only a convention; one must simply use it consistently.

Now we can write the free Dirac Lagrangian is

$$\mathcal{L} = \bar{\Psi}\left(\mathrm{i}\gamma^\mu\partial_\mu - m\right)\Psi. \tag{4.146}$$

Using the block form of the gamma matrices, the kinetic term splits into a left-handed part and a right-handed part:

$$\begin{aligned} \bar{\Psi}\,\mathrm{i}\gamma^\mu\partial_\mu\Psi &= \begin{pmatrix} \psi_R^\dagger & \psi_L^\dagger \end{pmatrix} \mathrm{i} \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} \partial_\mu \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \\ &= \mathrm{i}\psi_R^\dagger\sigma^\mu\partial_\mu\psi_R + \mathrm{i}\psi_L^\dagger\bar{\sigma}^\mu\partial_\mu\psi_L. \end{aligned} \tag{4.147}$$

The mass term, however, mixes left and right:

$$\begin{aligned} -m\bar{\Psi}\Psi &= -m \begin{pmatrix} \psi_R^\dagger & \psi_L^\dagger \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \\ &= -m\left(\psi_R^\dagger\psi_L + \psi_L^\dagger\psi_R\right). \end{aligned} \tag{4.148}$$

Equivalently, in four-component notation,

$$-m\bar{\Psi}\Psi = -m\left(\bar{\Psi}_L\Psi_R + \bar{\Psi}_R\Psi_L\right). \tag{4.149}$$

This is the key algebraic fact. A fermion mass term is not a purely left-handed or purely right-handed object. It couples the two chiralities.

3.2 Chiral gauge theory

A gauge theory is called *chiral* when the left-handed and right-handed fermions transform differently under the gauge group. The simplest example is a $U(1)$ gauge symmetry. Let

$$\psi_L(x) \longrightarrow \psi'_L(x) = e^{\mathrm{i}y_L\alpha(x)}\psi_L(x), \qquad \psi_R(x) \longrightarrow \psi'_R(x) = e^{\mathrm{i}y_R\alpha(x)}\psi_R(x). \tag{4.150}$$

The function $\alpha(x)$ is the same local gauge parameter, but the charges y_L and y_R may be different. If

$$y_L = y_R, \tag{4.151}$$

then the theory is vector-like. Ordinary QED is of this type: the left- and right-handed parts of the electron carry the same electric charge. If

$$y_L \neq y_R, \tag{4.152}$$

then the theory is chiral.

With the convention

$$\psi(x) \longrightarrow e^{\mathrm{i}y\alpha(x)}\psi(x), \tag{4.153}$$

the covariant derivative is

$$D_\mu\psi = (\partial_\mu - \mathrm{i}gyA_\mu)\,\psi, \tag{4.154}$$

provided the gauge field transforms as

$$A_\mu(x) \longrightarrow A'_\mu(x) = A_\mu(x) + \frac{1}{g}\partial_\mu\alpha(x). \tag{4.155}$$

Indeed,

$$\begin{aligned} D'_\mu\psi' &= \left(\partial_\mu - \mathrm{i}gyA'_\mu\right) e^{\mathrm{i}y\alpha}\psi \\ &= e^{\mathrm{i}y\alpha}\left[\partial_\mu + \mathrm{i}y\partial_\mu\alpha - \mathrm{i}gy\left(A_\mu + \frac{1}{g}\partial_\mu\alpha\right)\right]\psi \\ &= e^{\mathrm{i}y\alpha}\left(\partial_\mu - \mathrm{i}gyA_\mu\right)\psi \\ &= e^{\mathrm{i}y\alpha}D_\mu\psi. \end{aligned} \tag{4.156}$$

Thus $D_\mu\psi$ transforms in the same way as ψ, which is precisely why the kinetic term is gauge invariant.

Now examine the mass term in a chiral $U(1)$ theory. Since

$$\bar{\psi}_L \longrightarrow e^{-\mathrm{i}y_L\alpha(x)}\bar{\psi}_L, \qquad \psi_R \longrightarrow e^{\mathrm{i}y_R\alpha(x)}\psi_R, \tag{4.157}$$

the bilinear transforms as

$$\bar{\psi}_L\psi_R \longrightarrow e^{\mathrm{i}(y_R-y_L)\alpha(x)}\bar{\psi}_L\psi_R. \tag{4.158}$$

Similarly,

$$\bar{\psi}_R\psi_L \longrightarrow e^{\mathrm{i}(y_L-y_R)\alpha(x)}\bar{\psi}_R\psi_L. \tag{4.159}$$

Therefore

$$m\left(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L\right) \tag{4.160}$$

is gauge invariant only if

$$y_L = y_R. \tag{4.161}$$

If $y_L \neq y_R$, an explicit Dirac mass is forbidden by gauge invariance. This is what the short phrase "a chiral gauge theory requires $m = 0$" really means: the Lagrangian cannot contain a bare Dirac mass before symmetry breaking. It does *not* mean that the physical fermion must remain massless forever.

3.3 The Higgs-Yukawa solution in a $U(1)$ toy model

The Higgs field can repair the gauge transformation of the forbidden mass term. Introduce a complex scalar field ϕ and write the Yukawa interaction

$$\mathcal{L}_{\text{Yuk}} = -\lambda \left(\phi\, \bar{\psi}_L \psi_R + \phi^*\, \bar{\psi}_R \psi_L \right). \tag{4.162}$$

For this to be gauge invariant, ϕ must transform as

$$\phi \longrightarrow \phi' = e^{\mathrm{i}(y_L - y_R)\alpha(x)} \phi. \tag{4.163}$$

Then

$$\begin{aligned} \phi\, \bar{\psi}_L \psi_R &\longrightarrow e^{\mathrm{i}(y_L - y_R)\alpha} \phi\, e^{\mathrm{i}(y_R - y_L)\alpha} \bar{\psi}_L \psi_R \\ &= \phi\, \bar{\psi}_L \psi_R. \end{aligned} \tag{4.164}$$

The Higgs field carries exactly the difference between the left-handed and right-handed charges.

Choose the usual symmetry-breaking potential

$$V(\phi) = -\mu^2 \phi^* \phi + \frac{\lambda_\phi}{2} \left(\phi^* \phi \right)^2, \qquad \mu^2 > 0, \qquad \lambda_\phi > 0. \tag{4.165}$$

The minimum satisfies

$$\begin{aligned} \frac{\partial V}{\partial(\phi^* \phi)} &= -\mu^2 + \lambda_\phi (\phi^* \phi) = 0, \\ \langle \phi^* \phi \rangle &= \frac{\mu^2}{\lambda_\phi}. \end{aligned} \tag{4.166}$$

Writing

$$\phi(x) = \frac{1}{\sqrt{2}} \left(v + h(x) \right) e^{\mathrm{i}\beta(x)/v}, \tag{4.167}$$

we have

$$\langle \phi^* \phi \rangle = \frac{v^2}{2}. \tag{4.168}$$

Hence

$$\frac{v^2}{2} = \frac{\mu^2}{\lambda_\phi}, \qquad v^2 = \frac{2\mu^2}{\lambda_\phi}. \tag{4.169}$$

In unitary gauge the phase $\beta(x)$ is removed, and the Yukawa term becomes

$$\begin{aligned} \mathcal{L}_{\text{Yuk}} &= -\lambda \frac{v + h(x)}{\sqrt{2}} \left(\bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L \right) \\ &= -\frac{\lambda v}{\sqrt{2}} \bar{\psi}\psi - \frac{\lambda}{\sqrt{2}} h \bar{\psi}\psi. \end{aligned} \tag{4.170}$$

Therefore the fermion mass is

$$m_\psi = \frac{\lambda v}{\sqrt{2}}. \tag{4.171}$$

The same interaction also predicts a Higgs-fermion coupling:

$$\mathcal{L}_{h\bar{\psi}\psi} = -\frac{m_\psi}{v} h \bar{\psi}\psi. \tag{4.172}$$

This relation is conceptually important. In a Higgs theory, fermion masses and Higgs-fermion couplings have the same origin.

3.4 Non-Abelian chiral gauge theories

The same idea works for a non-Abelian gauge group. Let the left-handed and right-handed fermions transform in possibly different representations:

$$\psi_L \longrightarrow U_L(x)\psi_L, \qquad \psi_R \longrightarrow U_R(x)\psi_R. \tag{4.173}$$

For infinitesimal transformations,

$$U_L(x) = 1 + \mathrm{i}\alpha^a(x)t^a_L, \qquad U_R(x) = 1 + \mathrm{i}\alpha^a(x)t^a_R, \tag{4.174}$$

so

$$\delta\psi_L = \mathrm{i}\alpha^a t^a_L\psi_L, \qquad \delta\psi_R = \mathrm{i}\alpha^a t^a_R\psi_R. \tag{4.175}$$

The covariant derivatives are

$$D_\mu\psi_L = \left(\partial_\mu - \mathrm{i}gA^a_\mu t^a_L\right)\psi_L, \qquad D_\mu\psi_R = \left(\partial_\mu - \mathrm{i}gA^a_\mu t^a_R\right)\psi_R. \tag{4.176}$$

Here t^a_L and t^a_R are matrices appropriate to the representations of ψ_L and ψ_R. They need not be the same matrices.

A bare mass term transforms schematically as

$$\bar\psi_L\psi_R \longrightarrow \bar\psi_L U^\dagger_L U_R\psi_R. \tag{4.177}$$

For this to be invariant for every gauge transformation, one needs

$$U^\dagger_L U_R = 1 \tag{4.178}$$

on the relevant fermion indices. This happens when the two chiralities are in equivalent representations. If they are in different representations, the bare mass term is not a gauge singlet and is forbidden.

The Higgs field must be chosen so that the Yukawa product is a gauge singlet:

$$\mathcal{L}_{\text{Yuk}} = -y\left(\bar\psi_L\,\Phi\,\psi_R + \text{h.c.}\right). \tag{4.179}$$

In representation language, this means that the product

$$R^*_L \otimes R_\Phi \otimes R_R \tag{4.180}$$

must contain the singlet representation. This is the non-Abelian version of the statement that the Higgs field carries the missing gauge charge.

There is one further consistency condition that is often not written in short notes: a quantum chiral gauge theory must also be anomaly free. Anomaly cancellation is separate from the classical gauge invariance discussed here. The Standard Model has precisely arranged fermion quantum numbers so that the gauge anomalies cancel.

3.5 The Glashow-Salam-Weinberg model

The electroweak theory is the most important example. Its gauge group is

$$G_{\text{EW}} = SU(2)_L \times U(1)_Y. \tag{4.181}$$

The Higgs mechanism breaks it to electromagnetism:

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}. \tag{4.182}$$

The unbroken electric charge generator is

$$Q = T^3 + Y, \tag{4.183}$$

where $T^3 = \sigma^3/2$ for an $SU(2)$ doublet and Y is hypercharge.

One lepton family

For one lepton family, the field content is

field	$SU(2)_L$ representation	Y	electric charge Q
W^a_μ, $a = 1,2,3$	$\mathbf{3}$	0	$0, \pm 1$ after $W^\pm$
B_μ	$\mathbf{1}$	0	0
$L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L$	$\mathbf{2}$	$-\frac{1}{2}$	$\begin{pmatrix} 0 \\ -1 \end{pmatrix}$
e_R	$\mathbf{1}$	-1	-1
$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}$	$\mathbf{2}$	$\frac{1}{2}$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

The charges in the last column follow from $Q = T^3 + Y$. For example, for the lepton doublet

$$Q_L = \frac{\sigma^3}{2} - \frac{1}{2}\mathbf{1} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.184}$$

For the Higgs doublet,

$$Q_\Phi = \frac{\sigma^3}{2} + \frac{1}{2}\mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{4.185}$$

Thus the lower component ϕ^0 is electrically neutral. This is why it is allowed to develop a vacuum expectation value without breaking $U(1)_{\text{em}}$.

The electroweak Lagrangian

For the lepton-Higgs sector, the Lagrangian is

$$\begin{aligned} \mathcal{L} = &-\frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + (D_\mu\Phi)^\dagger(D^\mu\Phi) - V(\Phi) \\ &+ \mathrm{i}\bar{L}\bar{\sigma}^\mu D_\mu L + \mathrm{i}\bar{e}_R\sigma^\mu D_\mu e_R - y_e\left(\bar{L}\Phi e_R + \bar{e}_R\Phi^\dagger L\right). \end{aligned} \tag{4.186}$$

The covariant derivative acting on a field with weak isospin generators T^a and hypercharge Y is

$$D_\mu = \partial_\mu - \mathrm{i}gW^a_\mu T^a - \mathrm{i}g'YB_\mu. \tag{4.187}$$

For an $SU(2)$ doublet,

$$T^a = \frac{\sigma^a}{2}. \tag{4.188}$$

For a singlet, $T^a = 0$.

The electron Yukawa term is gauge invariant for two independent reasons. First, the $SU(2)$ indices can be contracted:

$$\bar{L}\Phi = \bar{L}_i \Phi^i. \tag{4.189}$$

Second, the hypercharges add to zero:

$$Y(\bar{L}) + Y(\Phi) + Y(e_R) = \frac{1}{2} + \frac{1}{2} - 1 = 0. \tag{4.190}$$

By contrast, the bare mass term $\bar{L}e_R$ is not gauge invariant:

$$\bar{L}e_R \quad \text{is an } SU(2)_L \text{ doublet, not a singlet.} \tag{4.191}$$

This is the electroweak version of the earlier $U(1)$ lesson.

Gauge boson masses

The Higgs potential is

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda \left(\Phi^\dagger \Phi\right)^2. \tag{4.192}$$

The vacuum can be chosen as

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \tag{4.193}$$

Since the lower component has $Q = 0$, this vacuum preserves electromagnetism.

Insert the vacuum into the Higgs kinetic term:

$$\begin{aligned} D_\mu \langle \Phi \rangle &= \left(-\mathrm{i} g W_\mu^a \frac{\sigma^a}{2} - \mathrm{i} g' \frac{1}{2} B_\mu \right) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} \\ &= -\frac{\mathrm{i} v}{2\sqrt{2}} \begin{pmatrix} g(W_\mu^1 - \mathrm{i} W_\mu^2) \\ -g W_\mu^3 + g' B_\mu \end{pmatrix}. \end{aligned} \tag{4.194}$$

Therefore

$$(D_\mu \langle \Phi \rangle)^\dagger (D^\mu \langle \Phi \rangle) = \frac{g^2 v^2}{8} \left[(W_\mu^1)^2 + (W_\mu^2)^2 \right] + \frac{v^2}{8} \left(g W_\mu^3 - g' B_\mu \right)^2. \tag{4.195}$$

Define

$$W_\mu^\pm = \frac{1}{\sqrt{2}} \left(W_\mu^1 \mp \mathrm{i} W_\mu^2 \right). \tag{4.196}$$

Then

$$\frac{g^2 v^2}{8} \left[(W_\mu^1)^2 + (W_\mu^2)^2 \right] = \frac{g^2 v^2}{4} W_\mu^+ W^{-\mu}. \tag{4.197}$$

For a charged complex vector field the mass term has the form

$$m_W^2 W_\mu^+ W^{-\mu}. \tag{4.198}$$

Thus

$$m_W = \frac{g v}{2}. \tag{4.199}$$

For the neutral gauge bosons, define the weak mixing angle by

$$\tan \theta_W = \frac{g'}{g}. \tag{4.200}$$

The massive and massless neutral fields are

$$Z_\mu = \cos\theta_W\, W^3_\mu - \sin\theta_W\, B_\mu, \qquad A_\mu = \sin\theta_W\, W^3_\mu + \cos\theta_W\, B_\mu. \tag{4.201}$$

The mass term becomes

$$\frac{v^2}{8}\left(gW^3_\mu - g'B_\mu\right)^2 = \frac{v^2}{8}(g^2+g'^2)Z_\mu Z^\mu. \tag{4.202}$$

For a real vector field the mass term has the form

$$\frac{1}{2}m_Z^2 Z_\mu Z^\mu. \tag{4.203}$$

Hence

$$m_Z = \frac{v}{2}\sqrt{g^2+g'^2}, \qquad m_A = 0. \tag{4.204}$$

The photon remains massless because it is the gauge boson of the unbroken $U(1)_{\text{em}}$.

The weak mixing angle also satisfies

$$\cos\theta_W = \frac{g}{\sqrt{g^2+g'^2}} = \frac{m_W}{m_Z}, \qquad \sin\theta_W = \frac{g'}{\sqrt{g^2+g'^2}}. \tag{4.205}$$

The electric charge is

$$e = g\sin\theta_W = g'\cos\theta_W = \frac{gg'}{\sqrt{g^2+g'^2}}. \tag{4.206}$$

Covariant derivatives after mixing

It is useful to rewrite the neutral part of the covariant derivative in terms of A_μ and Z_μ. Starting from

$$D_\mu = \partial_\mu - \mathrm{i}gW^a_\mu T^a - \mathrm{i}g'YB_\mu, \tag{4.207}$$

use

$$W^3_\mu = \cos\theta_W Z_\mu + \sin\theta_W A_\mu, \qquad B_\mu = -\sin\theta_W Z_\mu + \cos\theta_W A_\mu. \tag{4.208}$$

The A_μ coefficient is

$$\begin{aligned} -\mathrm{i}\left(g\sin\theta_W T^3 + g'\cos\theta_W Y\right)A_\mu &= -\mathrm{i}e(T^3+Y)A_\mu \\ &= -\mathrm{i}eQA_\mu. \end{aligned} \tag{4.209}$$

This proves that the massless field A_μ couples to electric charge.

The Z_μ coefficient is

$$-\mathrm{i}\left(g\cos\theta_W T^3 - g'\sin\theta_W Y\right)Z_\mu = -\mathrm{i}\frac{g}{\cos\theta_W}\left(T^3 - \sin^2\theta_W Q\right)Z_\mu. \tag{4.210}$$

Thus, for a weak doublet,

$$\begin{aligned} D_\mu L = \Bigg[&\partial_\mu - \mathrm{i}\frac{g}{\sqrt{2}}\left(W^+_\mu\sigma^+ + W^-_\mu\sigma^-\right) \\ &- \mathrm{i}\frac{g}{\cos\theta_W}Z_\mu\left(T^3 - \sin^2\theta_W Q\right) - \mathrm{i}eA_\mu Q\Bigg]L, \end{aligned} \tag{4.211}$$

where

$$\sigma^\pm = \frac{1}{2}\left(\sigma^1 \pm \mathrm{i}\sigma^2\right). \tag{4.212}$$

For the right-handed electron, $T^3 = 0$ and $Q = -1$, so

$$D_\mu e_R = \left[\partial_\mu - \mathrm{i}\frac{g}{\cos\theta_W}\sin^2\theta_W Z_\mu + \mathrm{i}eA_\mu\right]e_R. \tag{4.213}$$

Electron mass from the Higgs field

In unitary gauge,

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}. \tag{4.214}$$

Then the electron Yukawa term becomes

$$\begin{aligned} -y_e \left(\bar{L}\Phi e_R + \bar{e}_R \Phi^\dagger L \right) &= -y_e \frac{v+h}{\sqrt{2}} \left(\bar{e}_L e_R + \bar{e}_R e_L \right) \\ &= -m_e \bar{e} e - \frac{m_e}{v} h \bar{e} e. \end{aligned} \tag{4.215}$$

Therefore

$$m_e = \frac{y_e v}{\sqrt{2}}. \tag{4.216}$$

The neutrino remains massless in this minimal lepton model because there is no right-handed neutrino field and therefore no analogous renormalizable Dirac Yukawa term.

3.6 Summary of the logic

The whole construction can be compressed into the following chain:

- different gauge quantum numbers for $\psi_L, \psi_R \Longrightarrow$ no gauge-invariant bare Dirac mass,
- Higgs field carries the missing quantum numbers $\Longrightarrow$ gauge-invariant Yukawa interaction,
- $\langle \Phi \rangle \neq 0 \Longrightarrow$ fermion mass after symmetry breaking.

This is why the Higgs mechanism is tied so deeply to chiral gauge theories. It does not merely give masses to $W^\pm$ and Z; it also allows fermion masses without explicitly violating the gauge symmetry.

4. The Standard Model

The Standard Model is the quantum field theory of the strong, weak, and electromagnetic interactions. Its starting point is the gauge group

$$G_{\mathrm{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y. \tag{4.217}$$

Each factor has a distinct physical meaning. The group $SU(3)_c$ is the color gauge group of QCD. The group $SU(2)_L$ acts on left-handed weak doublets. The group $U(1)_Y$ is the hypercharge gauge group. The subscript L in $SU(2)_L$ is not decorative: it tells us that the weak interaction distinguishes left-handed and right-handed fermions.

The Higgs field breaks only the electroweak part of the gauge group:

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\mathrm{em}}. \tag{4.218}$$

Color is not broken, so after electroweak symmetry breaking the unbroken gauge symmetry is

$$SU(3)_c \times U(1)_{\mathrm{em}}. \tag{4.219}$$

The generator of the remaining electromagnetic gauge group is

$$Q = T^3 + Y. \tag{4.220}$$

Here T^3 is the third weak-isospin generator and Y is hypercharge. For an $SU(2)_L$ doublet,

$$T^3 = \frac{\sigma^3}{2} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}. \tag{4.221}$$

For an $SU(2)_L$ singlet, $T^3 = 0$, so its electric charge is simply equal to its hypercharge.

4.1 Field content

Representations will be denoted by their dimensions:

$$\mathbf{1}, \qquad \mathbf{2}, \qquad \mathbf{3}, \qquad \mathbf{8}. \tag{4.222}$$

For $SU(3)_c$, $\mathbf{3}$ is the fundamental color representation, $\overline{\mathbf{3}}$ is the anti-fundamental representation, and $\mathbf{8}$ is the adjoint representation. For $SU(2)_L$, $\mathbf{2}$ is a weak doublet, $\mathbf{1}$ is a weak singlet, and $\mathbf{3}$ is the adjoint representation.

The gauge bosons are fixed by the gauge group. The gluons G^a_μ are the gauge bosons of $SU(3)_c$, the fields W^a_μ are the gauge bosons of $SU(2)_L$, and B_μ is the gauge boson of $U(1)_Y$:

field	$SU(3)_c$	$SU(2)_L$	Y	electric charge
G^a_μ, $a = 1, \ldots, 8$	$\mathbf{8}$	$\mathbf{1}$	0	0
W^a_μ, $a = 1, 2, 3$	$\mathbf{1}$	$\mathbf{3}$	0	$0, \pm 1$ after mixing
B_μ	$\mathbf{1}$	$\mathbf{1}$	0	0

The charged weak bosons are not W^1_μ and W^2_μ separately. They are the complex combinations

$$W^\pm_\mu = \frac{1}{\sqrt{2}} \left(W^1_\mu \mp \mathrm{i} W^2_\mu \right). \tag{4.223}$$

The remaining neutral fields W^3_μ and B_μ later combine into the photon and the Z boson.

There are three families of fermions:

$$I = 1, 2, 3. \tag{4.224}$$

For leptons these are the e, μ, τ families. For quarks they are

$$(u, d), \qquad (c, s), \qquad (t, b). \tag{4.225}$$

In each family the left-handed lepton fields form a weak doublet,

$$L^I_L = \begin{pmatrix} \nu^I_L \\ e^I_L \end{pmatrix}, \qquad L^I_L \sim (\mathbf{1}, \mathbf{2})_{-\frac{1}{2}}, \tag{4.226}$$

while the right-handed charged lepton is a weak singlet,

$$e^I_R \sim (\mathbf{1}, \mathbf{1})_{-1}. \tag{4.227}$$

The electric charges follow from $Q = T^3 + Y$. For the lepton doublet,

$$Q(L_L) = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}. \tag{4.228}$$

Thus $Q(\nu_L) = 0$ and $Q(e_L) = -1$. Since e_R is a weak singlet,

$$Q(e_R) = Y(e_R) = -1. \tag{4.229}$$

For quarks, the left-handed fields form a color triplet and weak doublet:

$$Q_L^I = \begin{pmatrix} u_L^I \\ d_L^I \end{pmatrix}, \qquad Q_L^I \sim (\mathbf{3}, \mathbf{2})_{\frac{1}{6}}. \tag{4.230}$$

The right-handed quarks are weak singlets:

$$u_R^I \sim (\mathbf{3}, \mathbf{1})_{\frac{2}{3}}, \qquad d_R^I \sim (\mathbf{3}, \mathbf{1})_{-\frac{1}{3}}. \tag{4.231}$$

For the left-handed quark doublet,

$$Q(Q_L) = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} + \frac{1}{6}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & 0 \\ 0 & -\frac{1}{3} \end{pmatrix}. \tag{4.232}$$

Therefore u-type quarks have electric charge $2/3$, and d-type quarks have electric charge $-1/3$.

The scalar sector contains one complex Higgs doublet,

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad \Phi \sim (\mathbf{1}, \mathbf{2})_{\frac{1}{2}}. \tag{4.233}$$

Again using $Q = T^3 + Y$,

$$Q_\Phi = \frac{\sigma^3}{2} + \frac{1}{2}\mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \tag{4.234}$$

The lower component ϕ^0 is electrically neutral. This is essential: only a neutral field can acquire a vacuum expectation value without breaking electromagnetism.

sector	field	$SU(3)_c$	$SU(2)_L$	Y	Q
gauge bosons	$G_\mu^a,\ a = 1, \dots, 8$	$\mathbf{8}$	$\mathbf{1}$	0	0
	$W_\mu^a,\ a = 1, 2, 3$	$\mathbf{1}$	$\mathbf{3}$	0	$W^\pm : \pm 1,\ W^3 : 0$
	B_μ	$\mathbf{1}$	$\mathbf{1}$	0	0
leptons	$L_L^I = (\nu_L^I, e_L^I)^T$	$\mathbf{1}$	$\mathbf{2}$	$-\frac{1}{2}$	$(0, -1)$
	e_R^I	$\mathbf{1}$	$\mathbf{1}$	-1	-1
quarks	$Q_L^I = (u_L^I, d_L^I)^T$	$\mathbf{3}$	$\mathbf{2}$	$\frac{1}{6}$	$(\frac{2}{3}, -\frac{1}{3})$
	u_R^I	$\mathbf{3}$	$\mathbf{1}$	$\frac{2}{3}$	$\frac{2}{3}$
	d_R^I	$\mathbf{3}$	$\mathbf{1}$	$-\frac{1}{3}$	$-\frac{1}{3}$
Higgs	$\Phi = (\phi^+, \phi^0)^T$	$\mathbf{1}$	$\mathbf{2}$	$\frac{1}{2}$	$(1, 0)$

表 4.1: Minimal Standard Model field content. The family index is $I = 1, 2, 3$. The electric charges in the last column are obtained from $Q = T^3 + Y$.

4.2 The Lagrangian before symmetry breaking

The Standard Model Lagrangian is built from gauge kinetic terms, fermion kinetic terms, the Higgs kinetic and potential terms, and Yukawa interactions:

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}. \tag{4.235}$$

The gauge kinetic terms are

$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu} - \frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu}. \tag{4.236}$$

The indices are $a = 1, \ldots, 8$ for $SU(3)_c$ and $a = 1, 2, 3$ for $SU(2)_L$.

The covariant derivative tells us how every field couples to the gauge bosons:

$$D_\mu = \partial_\mu - \mathrm{i}g_s G^a_\mu T^a_c - \mathrm{i}g W^a_\mu T^a_L - \mathrm{i}g' Y B_\mu. \tag{4.237}$$

If a field is a singlet under a group, the corresponding generator is zero. For example, e_R is a singlet under $SU(3)_c$ and $SU(2)_L$, so only the hypercharge term acts on e_R.

In four-component notation the fermion kinetic terms are

$$\begin{aligned}\mathcal{L}_{\text{fermion}} = \mathrm{i}\sum_{I=1}^{3}\left[\bar{L}^I_L\gamma^\mu D_\mu L^I_L + \bar{e}^I_R\gamma^\mu D_\mu e^I_R\right] \\ + \mathrm{i}\sum_{I=1}^{3}\left[\bar{Q}^I_L\gamma^\mu D_\mu Q^I_L + \bar{u}^I_R\gamma^\mu D_\mu u^I_R + \bar{d}^I_R\gamma^\mu D_\mu d^I_R\right].\end{aligned} \tag{4.238}$$

This form is compact, but the left-handed and right-handed labels should not be ignored: the two chiralities carry different electroweak quantum numbers.

The Higgs part is

$$\mathcal{L}_{\text{Higgs}} = (D_\mu\Phi)^\dagger(D^\mu\Phi) - V(\Phi), \tag{4.239}$$

with

$$V(\Phi) = -\mu^2\Phi^\dagger\Phi + \lambda\left(\Phi^\dagger\Phi\right)^2. \tag{4.240}$$

The Yukawa interactions are the gauge-invariant substitutes for ordinary fermion mass terms. The Standard Model does not allow a bare term such as $\bar{L}_L e_R$, because L_L is a weak doublet and e_R is a weak singlet. The Higgs doublet supplies the missing weak-isospin and hypercharge quantum numbers.

For up-type masses one needs the conjugate Higgs doublet

$$\widetilde{\Phi} = \mathrm{i}\sigma^2\Phi^* = \begin{pmatrix}\phi^{0*} \\ -\phi^-\end{pmatrix}, \qquad \widetilde{\Phi} \sim (\mathbf{1}, \mathbf{2})_{-\frac{1}{2}}. \tag{4.241}$$

The full Yukawa interaction is

$$\begin{aligned}\mathcal{L}_{\text{Yukawa}} = -(Y_e)_{IJ}\bar{L}^I_L\Phi e^J_R - (Y_d)_{IJ}\bar{Q}^I_L\Phi d^J_R \\ - (Y_u)_{IJ}\bar{Q}^I_L\widetilde{\Phi}u^J_R + \text{h.c.}\end{aligned} \tag{4.242}$$

The matrices Y_e, Y_d, Y_u act in family space.

The hypercharge checks show why these terms have exactly this form. Since $Y(\bar{\psi}) = -Y(\psi)$, the electron Yukawa term has

$$Y(\bar{L}_L) + Y(\Phi) + Y(e_R) = \frac{1}{2} + \frac{1}{2} - 1 = 0. \tag{4.243}$$

The down-quark Yukawa term has

$$Y(\bar{Q}_L) + Y(\Phi) + Y(d_R) = -\frac{1}{6} + \frac{1}{2} - \frac{1}{3} = 0, \tag{4.244}$$

and the up-quark Yukawa term has

$$Y(\bar{Q}_L) + Y(\widetilde{\Phi}) + Y(u_R) = -\frac{1}{6} - \frac{1}{2} + \frac{2}{3} = 0. \tag{4.245}$$

Thus every Yukawa term is a gauge singlet.

4.3 Electroweak symmetry breaking and currents

The Higgs vacuum can be chosen as

$$\langle\Phi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix}. \tag{4.246}$$

In unitary gauge,

$$\Phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v+h(x)\end{pmatrix}, \qquad \widetilde{\Phi}(x) = \frac{1}{\sqrt{2}}\begin{pmatrix}v+h(x)\\ 0\end{pmatrix}. \tag{4.247}$$

The neutral gauge bosons mix according to

$$A_\mu = \sin\theta_W W^3_\mu + \cos\theta_W B_\mu, \qquad Z_\mu = \cos\theta_W W^3_\mu - \sin\theta_W B_\mu, \tag{4.248}$$

where

$$\tan\theta_W = \frac{g'}{g}, \qquad e = g\sin\theta_W = g'\cos\theta_W. \tag{4.249}$$

The field A_μ is the photon and remains massless; Z_μ is massive.

For a weak doublet, the broken-phase covariant derivative can be written as

$$\begin{aligned} D_\mu = \bar{D}_\mu &- \mathrm{i}\frac{g}{\sqrt{2}}\left(W^+_\mu T^+ + W^-_\mu T^-\right) \\ &- \mathrm{i}\frac{g}{\cos\theta_W} Z_\mu\left(T^3 - \sin^2\theta_W Q\right) - \mathrm{i}eA_\mu Q. \end{aligned} \tag{4.250}$$

Here $\bar{D}_\mu$ contains the color part when the field is a quark, and

$$T^\pm = \frac{\sigma^1 \pm \mathrm{i}\sigma^2}{2}. \tag{4.251}$$

The $W^\pm$ terms change the upper component of a weak doublet into the lower component or vice versa. The photon term couples to electric charge. The Z term couples to the combination $T^3 - \sin^2\theta_W Q$.

After symmetry breaking, the fermion-gauge interactions can be summarized by

$$\begin{aligned} \mathcal{L}_{\text{int}} = \frac{g}{\sqrt{2}}&\left(W^+_\mu J^\mu_+ + W^-_\mu J^\mu_-\right) + \frac{g}{\cos\theta_W} Z_\mu J^\mu_Z \\ &+ eA_\mu J^\mu_{\text{em}} + g_s G^a_\mu J^\mu_a. \end{aligned} \tag{4.252}$$

Ignoring quark mixing, the charged currents are

$$J^\mu_+ = \sum_{I=1}^{3}\left(\bar{\nu}^I_L\gamma^\mu e^I_L + \bar{u}^I_L\gamma^\mu d^I_L\right), \tag{4.253}$$

and

$$J^\mu_- = \sum_{I=1}^{3}\left(\bar{e}^I_L\gamma^\mu \nu^I_L + \bar{d}^I_L\gamma^\mu u^I_L\right). \tag{4.254}$$

The charged current is purely left-handed. In the quark sector, the realistic charged current contains the CKM matrix:

$$\bar{u}^I_L\gamma^\mu d^I_L \quad\longrightarrow\quad \bar{u}^I_L\gamma^\mu V_{IJ} d^J_L. \tag{4.255}$$

The electromagnetic current is vector-like:

$$J^\mu_{\rm em} = \sum_f Q_f \bar f \gamma^\mu f. \tag{4.256}$$

For one family,

$$J^\mu_{\rm em} = -\bar e \gamma^\mu e + \frac{2}{3}\bar u \gamma^\mu u - \frac{1}{3}\bar d \gamma^\mu d + \cdots, \tag{4.257}$$

where the dots indicate the remaining families.

The neutral weak current is chiral:

$$J^\mu_Z = \sum_f \bar f \gamma^\mu \left(g^f_L P_L + g^f_R P_R \right) f, \tag{4.258}$$

with

$$g^f_L = T^3_f - Q_f \sin^2\theta_W, \qquad g^f_R = -Q_f \sin^2\theta_W. \tag{4.259}$$

For one family this gives

$$\begin{aligned} J^\mu_Z = {} & \frac{1}{2}\bar\nu_L\gamma^\mu\nu_L + \left(-\frac{1}{2} + \sin^2\theta_W\right)\bar e_L\gamma^\mu e_L + \sin^2\theta_W \bar e_R \gamma^\mu e_R \\ & + \left(\frac{1}{2} - \frac{2}{3}\sin^2\theta_W\right)\bar u_L \gamma^\mu u_L - \frac{2}{3}\sin^2\theta_W \bar u_R\gamma^\mu u_R \\ & + \left(-\frac{1}{2} + \frac{1}{3}\sin^2\theta_W\right)\bar d_L\gamma^\mu d_L + \frac{1}{3}\sin^2\theta_W \bar d_R \gamma^\mu d_R. \end{aligned} \tag{4.260}$$

This expression is long, but its logic is simple: for each fermion, insert its T^3 and Q into Eq. (4.259).

One direct experimental consequence is seen in

$$e^+e^- \longrightarrow \gamma^*, Z^* \longrightarrow f\bar f. \tag{4.261}$$

The amplitude has a photon part and a Z-boson part:

$$\mathcal{M} = \mathcal{M}_\gamma + \mathcal{M}_Z. \tag{4.262}$$

The photon propagator is proportional to $1/s$, while the Z-exchange piece contains the resonance factor

$$\frac{1}{s - m_Z^2 + i m_Z \Gamma_Z}. \tag{4.263}$$

Near $s = m_Z^2$, the Z contribution dominates. Since $g^f_L \neq g^f_R$, the Z boson couples differently to left- and right-handed fermions. A useful measure of this difference is

$$A_f = \frac{\Gamma(Z\to f_L\bar f_R) - \Gamma(Z\to f_R \bar f_L)}{\Gamma(Z\to f_L\bar f_R) + \Gamma(Z\to f_R\bar f_L)} = \frac{(g^f_L)^2 - (g^f_R)^2}{(g^f_L)^2 + (g^f_R)^2}. \tag{4.264}$$

For a charged lepton,

$$g^\ell_L = -\frac{1}{2} + \sin^2\theta_W, \qquad g^\ell_R = \sin^2\theta_W. \tag{4.265}$$

Therefore

$$A_\ell = \frac{\left(-\frac{1}{2} + \sin^2\theta_W\right)^2 - \left(\sin^2\theta_W\right)^2}{\left(-\frac{1}{2} + \sin^2\theta_W\right)^2 + \left(\sin^2\theta_W\right)^2} \neq 0. \tag{4.266}$$

A nonzero asymmetry is a direct sign that the weak neutral current violates parity.

4.4 Masses from the Higgs field

Now substitute Eq. (4.247) into the Yukawa interactions. For charged leptons,

$$\begin{aligned}-(Y_e)_{IJ}\bar{L}_L^I\Phi e_R^J + \text{h.c.} &= -(Y_e)_{IJ}\frac{v+h}{\sqrt{2}}\bar{e}_L^I e_R^J + \text{h.c.} \\ &= -(M_e)_{IJ}\bar{e}_L^I e_R^J - \frac{h}{v}(M_e)_{IJ}\bar{e}_L^I e_R^J + \text{h.c.},\end{aligned} \tag{4.267}$$

where

$$M_e = \frac{v}{\sqrt{2}}Y_e. \tag{4.268}$$

Similarly,

$$M_d = \frac{v}{\sqrt{2}}Y_d, \qquad M_u = \frac{v}{\sqrt{2}}Y_u. \tag{4.269}$$

The Yukawa matrices are not generally diagonal. The physical masses are found by unitary rotations of the left- and right-handed fermion fields. For charged leptons,

$$U_{eL}^\dagger M_e U_{eR} = \text{diag}(m_e, m_\mu, m_\tau). \tag{4.270}$$

The quark mass matrices are diagonalized in the same way. The mismatch between the left-handed rotations for up-type and down-type quarks produces the CKM matrix in the charged current.

In the minimal Standard Model field content listed above, there is no right-handed neutrino. Hence there is no renormalizable Dirac Yukawa mass term for neutrinos, and the minimal theory predicts massless neutrinos. Neutrino oscillations show that this cannot be the full story.

A simple extension is to add right-handed neutrinos

$$\nu_R^I = (\nu_{eR}, \nu_{\mu R}, \nu_{\tau R}), \qquad \nu_R^I \sim (\mathbf{1}, \mathbf{1})_0\,. \tag{4.271}$$

They are singlets under all Standard Model gauge groups, so they are called sterile neutrinos. Once they are included, the Dirac neutrino Yukawa interaction is allowed:

$$\mathcal{L}_{\nu\text{Yuk}} = -(Y_\nu)_{IJ}\bar{L}_L^I\widetilde{\Phi}\nu_R^J + \text{h.c.} \tag{4.272}$$

Using explicit $SU(2)$ indices,

$$\bar{L}_L\widetilde{\Phi} = \epsilon_{ij}\Phi^{*i}\bar{L}_L^j. \tag{4.273}$$

After electroweak symmetry breaking,

$$\begin{aligned}\mathcal{L}_{\nu\text{Yuk}} &= -(Y_\nu)_{IJ}\frac{v+h}{\sqrt{2}}\bar{\nu}_L^I\nu_R^J + \text{h.c.}, \\ M_\nu &= \frac{v}{\sqrt{2}}Y_\nu.\end{aligned} \tag{4.274}$$

Because ν_R is a complete gauge singlet, a Majorana mass term is also allowed:

$$\mathcal{L}_{M_R} = -\frac{1}{2}(M_R)_{IJ}\overline{\nu_R^{cI}}\nu_R^J + \text{h.c.} \tag{4.275}$$

This is the starting point of the seesaw mechanism. The important lesson here is that neutrino masses require physics beyond the minimal Standard Model field content, even though they can be incorporated in a gauge-invariant way.

5. Quark Mixing and the CKM Matrix

The goal of this section is to explain why the Standard Model contains a mixing matrix in the quark charged current. The short version is simple: the weak interaction is written in a basis fixed by the gauge group, while particle masses are found by diagonalizing the Yukawa matrices. These two bases do not have to be the same. Their mismatch is the Cabibbo-Kobayashi-Maskawa matrix, usually called the CKM matrix.

5.1 Yukawa Couplings Before Electroweak Symmetry Breaking

For each generation $I = 1, 2, 3$, the Standard Model quarks are

$$Q_L^I = \begin{pmatrix} u_L^I \\ d_L^I \end{pmatrix}, \qquad u_R^I, \qquad d_R^I. \tag{4.276}$$

Here Q_L^I is an $SU(2)_L$ doublet, while u_R^I and d_R^I are $SU(2)_L$ singlets. The index I labels families. Before we diagonalize any mass matrix, it is better to call these fields *weak-interaction eigenstates*, not mass eigenstates.

The Higgs doublet is

$$\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad \widetilde{\phi} = \mathrm{i}\sigma^2\phi^* = \begin{pmatrix} \phi^{0*} \\ -\phi^- \end{pmatrix}. \tag{4.277}$$

The object $\widetilde{\phi}$ is needed because the up-type quark Yukawa term must be made gauge invariant with the conjugate Higgs doublet. With generation indices shown explicitly, the quark Yukawa Lagrangian is

$$\mathcal{L}_Y^q = -\overline{Q}_L^I (Y_d)_{IJ} \phi \, d_R^J - \overline{Q}_L^I (Y_u)_{IJ} \widetilde{\phi} \, u_R^J + \text{h.c.} \tag{4.278}$$

The two matrices Y_u and Y_d are arbitrary complex 3×3 matrices. They are not assumed to be diagonal. Therefore the fields (u^I, d^I) appearing here are generally not the physical quarks (u, c, t, d, s, b).

For leptons, the analogous charged-lepton Yukawa term is

$$\mathcal{L}_Y^e = -\overline{L}_L^I (Y_e)_{IJ} \phi \, e_R^J + \text{h.c.} \tag{4.279}$$

where

$$L_L^I = \begin{pmatrix} \nu_L^I \\ e_L^I \end{pmatrix}. \tag{4.280}$$

The minimal Standard Model has no right-handed neutrino field and therefore no neutrino Yukawa mass term. If right-handed neutrinos ν_R^I are added, then one may also write

$$\mathcal{L}_Y^\nu = -\overline{L}_L^I (Y_\nu)_{IJ} \widetilde{\phi} \, \nu_R^J + \text{h.c.}, \tag{4.281}$$

but this is an extension of the minimal Standard Model. This distinction is important: quark mixing is already present in the minimal Standard Model, while neutrino mixing requires neutrino masses.

5.2 Mass Matrices After the Higgs Gets a Vacuum Expectation Value

In unitary gauge the Higgs field may be written as

$$\phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \qquad \widetilde{\phi}(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} v + h(x) \\ 0 \end{pmatrix}. \tag{4.282}$$

Substituting this into the Yukawa Lagrangian gives the quark mass terms

$$\mathcal{L}^q_{\text{mass}} = -\overline{d}^I_L (M_d)_{IJ} d^J_R - \overline{u}^I_L (M_u)_{IJ} u^J_R + \text{h.c.}, \tag{4.283}$$

$$M_d = \frac{v}{\sqrt{2}} Y_d, \qquad M_u = \frac{v}{\sqrt{2}} Y_u. \tag{4.284}$$

The matrices M_u and M_d are complex 3×3 matrices. A generic complex matrix is not diagonal and not Hermitian. Therefore we do not diagonalize it by a single unitary transformation. Instead, we use a bi-unitary transformation, also called singular value decomposition.

For any complex matrix M, there exist unitary matrices V_L and V_R such that

$$V_L^\dagger M V_R = D, \tag{4.285}$$

where D is diagonal with non-negative real entries. In the quark sector we choose

$$V_{uL}^\dagger M_u V_{uR} = D_u = \text{diag}(m_u, m_c, m_t), \tag{4.286}$$

$$V_{dL}^\dagger M_d V_{dR} = D_d = \text{diag}(m_d, m_s, m_b). \tag{4.287}$$

The positive numbers on the diagonal are the physical quark masses.

The corresponding change of basis is

$$u^I_L = (V_{uL})_{Ii} u^i_L, \qquad u^I_R = (V_{uR})_{Ii} u^i_R, \tag{4.288}$$

$$d^I_L = (V_{dL})_{Ij} d^j_L, \qquad d^I_R = (V_{dR})_{Ij} d^j_R. \tag{4.289}$$

Uppercase indices I, J label the original weak basis, while lowercase indices i, j label the mass basis. To see explicitly why this works, look at the up-quark mass term:

$$-\overline{u}^I_L (M_u)_{IJ} u^J_R = -\overline{u}^i_L (V_{uL}^\dagger)_{iI} (M_u)_{IJ} (V_{uR})_{Jj} u^j_R \tag{4.290}$$

$$= -\overline{u}^i_L (D_u)_{ij} u^j_R \tag{4.291}$$

$$= -m_u \overline{u}_L u_R - m_c \overline{c}_L c_R - m_t \overline{t}_L t_R. \tag{4.292}$$

Adding the Hermitian conjugate gives the usual Dirac mass terms:

$$-m_i \overline{q}_{Li} q_{Ri} - m_i \overline{q}_{Ri} q_{Li} = -m_i \overline{q}_i q_i. \tag{4.293}$$

5.3 Why Most Gauge Interactions Do Not Notice the Basis Change

The kinetic terms are invariant under these constant unitary rotations. For example,

$$\mathrm{i}\overline{u}^I_L \gamma^\mu \partial_\mu u^I_L = \mathrm{i}\overline{u}^i_L (V_{uL}^\dagger)_{iI} \gamma^\mu \partial_\mu (V_{uL})_{Ij} u^j_L \tag{4.294}$$

$$= \mathrm{i}\overline{u}^i_L (V_{uL}^\dagger V_{uL})_{ij} \gamma^\mu \partial_\mu u^j_L \tag{4.295}$$

$$= \mathrm{i}\overline{u}^i_L \gamma^\mu \partial_\mu u^i_L. \tag{4.296}$$

The derivative does not act on V_{uL}, because this is a constant matrix in generation space, not a spacetime-dependent gauge transformation.

Neutral gauge currents are also unchanged. A typical neutral up-quark current becomes

$$\overline{u}^I_L \gamma^\mu u^I_L = \overline{u}^i_L (V^\dagger_{uL} V_{uL})_{ij} \gamma^\mu u^j_L \tag{4.297}$$
$$= \overline{u}^i_L \gamma^\mu u^i_L. \tag{4.298}$$

The same argument applies to the photon, gluon, and Z-boson neutral currents. This is why tree-level flavor-changing neutral currents are absent in the Standard Model. The important exception is the charged weak current.

5.4 The Charged Current and the CKM Matrix

In the weak basis, the charged-current interaction is diagonal in family space:

$$\mathcal{L}_{\text{cc}} = -\frac{g}{\sqrt{2}} \overline{u}^I_L \gamma^\mu d^I_L W^+_\mu - \frac{g}{\sqrt{2}} \overline{d}^I_L \gamma^\mu u^I_L W^-_\mu. \tag{4.299}$$

Now substitute the mass-basis rotations:

$$\overline{u}^I_L \gamma^\mu d^I_L = \overline{u}^i_L (V^\dagger_{uL})_{iI} \gamma^\mu (V_{dL})_{Ij} d^j_L \tag{4.300}$$
$$= \overline{u}^i_L \gamma^\mu (V^\dagger_{uL} V_{dL})_{ij} d^j_L. \tag{4.301}$$

This motivates the definition

$$V_{\text{CKM}} \equiv V^\dagger_{uL} V_{dL}. \tag{4.302}$$

The charged-current Lagrangian in the mass basis is therefore

$$\mathcal{L}_{\text{cc}} = -\frac{g}{\sqrt{2}} \overline{u}^i_L \gamma^\mu (V_{\text{CKM}})_{ij} d^j_L W^+_\mu + \text{h.c.} \tag{4.303}$$

$$= -\frac{g}{\sqrt{2}} \begin{pmatrix} \overline{u}_L & \overline{c}_L & \overline{t}_L \end{pmatrix} \gamma^\mu V_{\text{CKM}} \begin{pmatrix} d_L \\ s_L \\ b_L \end{pmatrix} W^+_\mu + \text{h.c.} \tag{4.304}$$

Written out as a matrix,

$$V_{\text{CKM}} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}. \tag{4.305}$$

Each entry tells us the amplitude for an up-type quark to couple to a down-type quark through a W boson. For example, V_{us} controls the strength of $u \leftrightarrow s$ charged-current transitions.

Because V_{uL} and V_{dL} are unitary, the CKM matrix is unitary:

$$V_{\text{CKM}} V^\dagger_{\text{CKM}} = V^\dagger_{uL} V_{dL} V^\dagger_{dL} V_{uL} = \mathbb{1}, \tag{4.306}$$
$$V^\dagger_{\text{CKM}} V_{\text{CKM}} = V^\dagger_{dL} V_{uL} V^\dagger_{uL} V_{dL} = \mathbb{1}. \tag{4.307}$$

5.5 Discrete Symmetries and Why the Weak Current Is Special

The electromagnetic and strong interactions are invariant under C, P, and T, and hence also under CPT. The weak charged current is different because it only contains left-handed fermion fields:

$$J^{+\mu}_W = \overline{u}^i_L \gamma^\mu (V_{\text{CKM}})_{ij} d^j_L. \tag{4.308}$$

Parity exchanges left-handed and right-handed chiral fields:

$$P\psi_L(t,\mathbf{x})P^{-1} = \eta_P \gamma^0 \psi_R(t,-\mathbf{x}), \tag{4.309}$$

where η_P is an irrelevant phase. Therefore a current made only from left-handed fields cannot be invariant under parity by itself. This is the origin of maximal parity violation in the charged weak interaction.

Charge conjugation changes particles into antiparticles. With a charge conjugation matrix C satisfying

$$C^{-1}\gamma^\mu C = -(\gamma^\mu)^T, \tag{4.310}$$

one defines

$$\psi^C = C\overline{\psi}^T. \tag{4.311}$$

The charged weak current is not invariant under C alone either. The combined transformation CP is more subtle: a left-handed particle is mapped into the corresponding right-handed antiparticle, which can still participate in weak interactions. Thus the chiral structure alone does not force CP violation.

The CKM matrix is where CP violation enters. Under CP, the charged-current coupling is complex conjugated:

$$V_{\text{CKM}} \longrightarrow V_{\text{CKM}}^*. \tag{4.312}$$

If all phases in V_{CKM} can be removed by redefining quark fields, then CP is conserved. If one complex phase remains, then CP is violated.

5.6 Counting the Physical Parameters in the CKM Matrix

A general 3×3 unitary matrix has nine real parameters. It is useful to think of them as three rotation angles and six phases:

$$9 = 3 \text{ angles} + 6 \text{ phases}. \tag{4.313}$$

However, not all six phases are physical. The six quark mass eigenstate fields can be rephased:

$$u_i \longrightarrow \mathrm{e}^{\mathrm{i}\alpha_i} u_i, \qquad d_j \longrightarrow \mathrm{e}^{\mathrm{i}\beta_j} d_j. \tag{4.314}$$

Under this change,

$$(V_{\text{CKM}})_{ij} \longrightarrow \mathrm{e}^{-\mathrm{i}\alpha_i}(V_{\text{CKM}})_{ij}\mathrm{e}^{\mathrm{i}\beta_j}. \tag{4.315}$$

There are six phases α_i, β_j, but one common phase changes nothing:

$$\alpha_1 = \alpha_2 = \alpha_3 = \beta_1 = \beta_2 = \beta_3 \quad \Longrightarrow \quad V_{\text{CKM}} \text{ unchanged}. \tag{4.316}$$

Therefore only five phases can be removed. The physical CKM matrix has

$$3 \text{ angles} + (6-5) \text{ phase} = 3 \text{ angles} + 1 \text{ CP-violating phase}. \tag{4.317}$$

For n generations, the general result is

$$N_{\text{angles}} = \frac{n(n-1)}{2}, \tag{4.318}$$

$$N_{\text{phases}} = \frac{(n-1)(n-2)}{2}. \tag{4.319}$$

For $n = 2$, there is one mixing angle and no physical complex phase. This is why two generations are not enough for CKM-type CP violation. With three generations, one irreducible complex phase remains.

5.7 Useful Parameterizations

The standard parameterization writes the CKM matrix in terms of three angles $\theta_{12}, \theta_{23}, \theta_{13}$ and one phase δ. Using $s_{ij} = \sin\theta_{ij}$ and $c_{ij} = \cos\theta_{ij}$,

$$V_{\mathrm{CKM}} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}. \tag{4.320}$$

This form makes the single physical phase explicit. If $\delta = 0$ or $\delta = \pi$, then the matrix can be chosen real and there is no CKM-type CP violation.

Phenomenologically, the CKM matrix is close to the identity matrix. The Wolfenstein parameterization makes this hierarchy transparent:

$$V_{\mathrm{CKM}} = \begin{pmatrix} 1 - \frac{\lambda^2}{2} & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \frac{\lambda^2}{2} & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + O(\lambda^4), \qquad \lambda \simeq 0.22. \tag{4.321}$$

Here λ measures the Cabibbo mixing between the first two generations, A controls the size of mixing with the third generation, and η is responsible for CP violation.

5.8 Unitarity Triangles and the Jarlskog Invariant

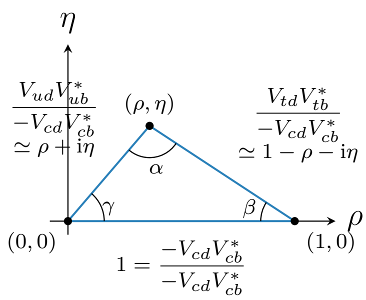
The unitarity of V_{CKM} gives orthogonality relations among different rows and columns. One important relation is

$$V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0. \tag{4.322}$$

This is the sum of three complex numbers. Since the sum is zero, the three numbers form a triangle in the complex plane. Dividing by $-V_{cd}V_{cb}^*$ gives the conventional normalized triangle:

$$\frac{V_{ud}V_{ub}^*}{-V_{cd}V_{cb}^*} + \frac{V_{td}V_{tb}^*}{-V_{cd}V_{cb}^*} = 1. \tag{4.323}$$

At leading order in the Wolfenstein expansion, the apex is approximately (ρ, η):



The angles α, β, γ are measured in many flavor-physics processes. The fact that the triangle has nonzero area is equivalent to the presence of CP violation.

A basis-independent measure of CKM CP violation is the Jarlskog invariant:

$$J = \mathrm{Im}\left(V_{ij}V_{kl}V_{il}^*V_{kj}^*\right), \qquad i \neq k, \quad j \neq l. \tag{4.324}$$

For a unitary 3×3 CKM matrix, every nonzero choice gives the same magnitude of J. In the Wolfenstein expansion,

$$J = A^2\lambda^6\eta + O(\lambda^8). \tag{4.325}$$

Thus $J = 0$ if the physical phase vanishes, if one of the mixing angles vanishes, or if any two quarks with the same charge are exactly degenerate in mass. A compact invariant way to state the last point is

$$\mathrm{Im}\det\left[M_u M_u^\dagger, M_d M_d^\dagger\right] \propto J \prod_{i<j}(m_{u_i}^2 - m_{u_j}^2) \prod_{k<l}(m_{d_k}^2 - m_{d_l}^2). \tag{4.326}$$

This formula explains why CP violation needs both a complex CKM phase and a genuinely three-generation mass spectrum.

5.9 Parameter Count in the Flavor Sector

If neutrino masses are ignored, the usual Standard Model parameter count can be summarized as

$$\text{gauge couplings} : 3, \tag{4.327}$$
$$\text{Higgs potential parameters} : 2 \quad (v, \lambda), \tag{4.328}$$
$$\text{quark masses} : 6, \tag{4.329}$$
$$\text{charged-lepton masses} : 3, \tag{4.330}$$
$$\text{CKM parameters} : 4. \tag{4.331}$$

This gives

$$3 + 2 + 6 + 3 + 4 = 18 \tag{4.332}$$

parameters, if the QCD theta angle is not included in the count.

If neutrinos are massive, the lepton sector contains additional physical parameters. For three light Majorana neutrinos one commonly adds three neutrino masses, three PMNS mixing angles, one Dirac phase, and two Majorana phases. If one counts only oscillation physics, the two Majorana phases do not appear, and the extra oscillation-relevant parameters are

$$3 \text{ neutrino masses} + 4 \text{ PMNS mixing parameters} = 7. \tag{4.333}$$

The moral is that flavor physics is mostly the study of how these mass matrices are diagonalized and how the mismatch between different diagonalizations appears in weak interactions.

6. Anomalies in Quantum Field Theory

A symmetry is said to be anomalous if it is a symmetry of the classical action but fails to be a symmetry of the quantum theory. Equivalently, a current which is conserved by the classical equations of motion is not conserved after the path integral has been defined carefully. The key point is that quantization does not only use the action. It also uses the functional integration measure. A classical transformation may leave the action invariant while changing the measure by a nontrivial Jacobian.

This distinction is especially important for gauge theories. If the anomalous symmetry is a global symmetry, the theory can still be perfectly consistent. The anomaly then has physical consequences, such as $\pi^0 \to \gamma\gamma$, the breaking of classical scale invariance, or the running of a coupling. If the anomalous symmetry is a local gauge symmetry, the situation is much more serious. Gauge symmetry is the redundancy which removes unphysical polarizations and gives the Ward or Slavnov-Taylor identities. A gauge anomaly destroys those identities and usually makes the theory inconsistent.

Classically a conserved current satisfies

$$\partial_\mu j^\mu = 0. \tag{4.334}$$

In an anomalous quantum theory the same current instead obeys

$$\partial_\mu j^\mu = \hbar\,\mathcal{A}, \tag{4.335}$$

where $\mathcal{A}$ is a local expression built from the background fields. The explicit factor of $\hbar$ is only a reminder that the effect is quantum mechanical. In practice we usually set $\hbar = 1$.

6.1 Vector and Axial Symmetries

Consider a Dirac fermion coupled to a $U(1)$ gauge field,

$$S = \int \mathrm{d}^4x\, \bar\psi(\mathrm{i}\gamma^\mu D_\mu - m)\psi, \qquad D_\mu = \partial_\mu + \mathrm{i}gA_\mu. \tag{4.336}$$

The ordinary vector transformation is

$$\psi(x) \to \psi'(x) = \exp\left[\mathrm{i}\alpha(x)\right]\psi(x), \tag{4.337}$$

$$A_\mu(x) \to A'_\mu(x) = A_\mu(x) - \frac{1}{g}\partial_\mu\alpha(x). \tag{4.338}$$

The action is invariant. For a constant parameter α, Noether's theorem gives the vector current

$$j^\mu = \bar\psi\gamma^\mu\psi, \qquad \partial_\mu j^\mu = 0. \tag{4.339}$$

This is the current coupled to the gauge field. In QED its conservation is not optional; it is part of gauge consistency.

For a massless Dirac fermion there is another classical symmetry, the axial or chiral rotation,

$$\psi(x) \to \psi'(x) = \exp\left[\mathrm{i}\beta(x)\gamma^5\right]\psi(x), \tag{4.340}$$

$$\bar\psi(x) \to \bar\psi'(x) = \bar\psi(x)\exp\left[\mathrm{i}\beta(x)\gamma^5\right]. \tag{4.341}$$

The associated axial current is

$$j_5^\mu = \bar\psi\gamma^\mu\gamma^5\psi. \tag{4.342}$$

To see what the classical action says, expand the transformation to first order in $\beta(x)$. The variation of the action is

$$\delta S = \int \mathrm{d}^4x\, \beta(x)\left[-\partial_\mu j_5^\mu + 2\mathrm{i}m\,\bar\psi\gamma^5\psi\right]. \tag{4.343}$$

Since $\beta(x)$ is arbitrary, classical invariance for $m = 0$ gives

$$\partial_\mu j_5^\mu = 0. \tag{4.344}$$

For $m \neq 0$, the mass term explicitly breaks the axial symmetry:

$$\partial_\mu j_5^\mu = 2\mathrm{i}m\,\bar\psi\gamma^5\psi. \tag{4.345}$$

The anomaly is the statement that even when $m = 0$, the quantum path integral adds another term to this equation.

6.2 The Fujikawa Idea

The path integral is

$$Z = \int \mathcal{D}\psi\,\mathcal{D}\bar{\psi}\,\exp\bigl(\mathrm{i}S[\psi,\bar{\psi},A]\bigr). \tag{4.346}$$

The classical calculation only tested the exponential of the action. The Fujikawa method instead asks whether the measure $\mathcal{D}\psi\,\mathcal{D}\bar{\psi}$ is invariant.

It is cleanest to define the measure in Euclidean spacetime. Let $\not{D}_E$ be the Euclidean Dirac operator and choose a complete orthonormal set of eigenfunctions,

$$\not{D}_E\phi_n(x) = \lambda_n\phi_n(x), \qquad \int \mathrm{d}^4x\,\phi_m^\dagger(x)\phi_n(x) = \delta_{mn}. \tag{4.347}$$

The eigenvalues λ_n are real in the standard Euclidean convention, and $\lambda_n^2 \geq 0$. Expand the fields in this basis:

$$\psi(x) = \sum_n a_n\phi_n(x), \tag{4.348}$$

$$\bar{\psi}(x) = \sum_n \bar{b}_n\phi_n^\dagger(x). \tag{4.349}$$

The coefficients a_n and $\bar{b}_n$ are Grassmann variables. This turns the formal measure into the precise object

$$\mathcal{D}\psi\,\mathcal{D}\bar{\psi} = \prod_n \mathrm{d}a_n\,\mathrm{d}\bar{b}_n. \tag{4.350}$$

Under an infinitesimal axial rotation

$$\psi'(x) = \bigl[1 + \mathrm{i}\beta(x)\gamma^5\bigr]\psi(x), \tag{4.351}$$

the new coefficients satisfy

$$a'_m = \int \mathrm{d}^4x\,\phi_m^\dagger(x)\psi'(x) \tag{4.352}$$

$$= \sum_n (\delta_{mn} + C_{mn})a_n, \tag{4.353}$$

where

$$C_{mn} = \mathrm{i}\int \mathrm{d}^4x\,\beta(x)\phi_m^\dagger(x)\gamma^5\phi_n(x). \tag{4.354}$$

For ordinary commuting variables, a change of variables by a matrix $1 + C$ gives a factor $\det(1 + C)$. For Grassmann variables it gives the inverse factor. The same factor appears for $\bar{\psi}$. Therefore

$$\mathcal{D}\psi'\,\mathcal{D}\bar{\psi}' = J^{-1}\mathcal{D}\psi\,\mathcal{D}\bar{\psi}, \qquad J = \det(1 + C)^2. \tag{4.355}$$

To first order,

$$\ln J = 2\,\mathrm{Tr}\,C \tag{4.356}$$

$$= 2\mathrm{i}\int \mathrm{d}^4x\,\beta(x)\sum_n \phi_n^\dagger(x)\gamma^5\phi_n(x). \tag{4.357}$$

This is the place where the anomaly lives. The expression $\sum_n \phi_n^\dagger \gamma^5 \phi_n$ is divergent and must be regulated. Different regulators would be dangerous if they broke gauge invariance. The natural gauge-invariant regulator is built from the Dirac operator itself:

$$\sum_n \phi_n^\dagger(x)\gamma^5\phi_n(x) \longrightarrow \lim_{M\to\infty} \sum_n \phi_n^\dagger(x)\gamma^5 \exp\left(-\frac{\lambda_n^2}{M^2}\right)\phi_n(x) \tag{4.358}$$

$$= \lim_{M\to\infty} \mathrm{tr}\,\langle x|\gamma^5 \exp\left(-\frac{\not{D}_E^2}{M^2}\right)|x\rangle\,. \tag{4.359}$$

Here tr means a trace over spinor indices, and over any internal representation indices if the fermion is non-Abelian.

6.3 Evaluating the Regulated Trace

The square of the Dirac operator contains the field strength. With

$$\sigma^{\mu\nu} = \frac{\mathrm{i}}{2}[\gamma^\mu, \gamma^\nu], \tag{4.360}$$

one may write, up to convention-dependent signs between Minkowski and Euclidean notation,

$$\not{D}_E^2 = D_E^2 + \frac{g}{2}\sigma^{\mu\nu}F_{\mu\nu}. \tag{4.361}$$

Only the term quadratic in $F_{\mu\nu}$ contributes to the trace with γ^5. The reason is simple:

$$\mathrm{tr}_D(\gamma^5) = 0, \qquad \mathrm{tr}_D(\gamma^5\sigma^{\mu\nu}) = 0, \tag{4.362}$$

while

$$\mathrm{tr}_D\left(\gamma^5\sigma^{\mu\nu}\sigma^{\rho\sigma}\right) = 4\mathrm{i}\,\epsilon^{\mu\nu\rho\sigma} \tag{4.363}$$

in the convention used here. The free heat-kernel factor is

$$\langle x|\exp\left(\frac{\partial^2}{M^2}\right)|x\rangle = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\exp\left(-\frac{k^2}{M^2}\right) \tag{4.364}$$

$$= \frac{M^4}{16\pi^2}. \tag{4.365}$$

The factor M^4 cancels the M^{-4} from the second-order expansion of the exponential. Hence the regulated local trace is finite:

$$\mathcal{A}(x) \equiv \lim_{M\to\infty} \mathrm{tr}\,\langle x|\gamma^5 \exp\left(-\frac{\not{D}_E^2}{M^2}\right)|x\rangle = \frac{g^2}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{4.366}$$

The overall sign can change if one changes the convention for γ^5, ϵ^{0123}, or the definition of D_μ. The important invariant statement is that the anomaly is proportional to $F\widetilde{F}$, where

$$\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}. \tag{4.367}$$

Combining the variation of the action with the variation of the measure gives the anomalous Ward identity

$$\partial_\mu j_5^\mu = 2\mathrm{i}m\,\bar{\psi}\gamma^5\psi + \frac{g^2}{16\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{4.368}$$

For a massless fermion, the first term vanishes, but the second term remains. This is the Adler-Bell-Jackiw anomaly.

The triangle diagram gives the same result as the Fujikawa computation. The Fujikawa method shows more directly why the result is unavoidable: the measure cannot preserve both the axial symmetry and the gauge-invariant definition of the fermion determinant.

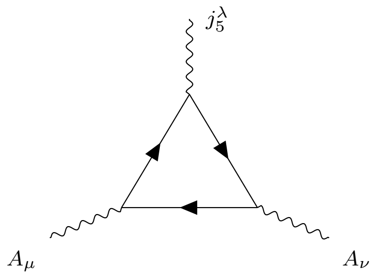


Figure 4.4: The perturbative representation of the axial anomaly. The axial current insertion and two gauge-current insertions form a triangle graph.

6.4 Non-Abelian Anomalies and Their Cancellation

For a non-Abelian gauge field

$$A_\mu = A_\mu^a T^a, \qquad F_{\mu\nu} = F_{\mu\nu}^a T^a, \tag{4.369}$$

the anomaly of a chiral gauge current is controlled by the group-theory factor

$$\mathcal{A}^{abc}(R) = \mathrm{tr}_R\left[T^a\{T^b, T^c\}\right], \tag{4.370}$$

where R is the representation carried by the left-handed Weyl fermion. Schematically,

$$(D_\mu J^\mu)^a \propto \frac{g^2}{16\pi^2}\epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}^b F_{\rho\sigma}^c \mathcal{A}^{abc}(R). \tag{4.371}$$

The exact numerical coefficient depends on the normalization of the generators, but the cancellation condition does not:

$$\sum_{\text{left-handed Weyl fermions } i} \mathcal{A}^{abc}(R_i) = 0. \tag{4.372}$$

This formula also explains why vectorlike theories are safe. A right-handed fermion in representation R can be rewritten as a left-handed antifermion in the complex conjugate representation $\bar{R}$. For complex representations,

$$\mathcal{A}^{abc}(\bar{R}) = -\mathcal{A}^{abc}(R). \tag{4.373}$$

Thus a vectorlike pair $R \oplus \bar{R}$ contributes

$$\mathcal{A}^{abc}(R) + \mathcal{A}^{abc}(\bar{R}) = 0. \tag{4.374}$$

This is why QED and QCD with ordinary Dirac fermions do not have gauge anomalies.

Real and pseudoreal representations are also anomaly-free. If $R \simeq \bar{R}$, then

$$\mathcal{A}^{abc}(R) = \mathcal{A}^{abc}(\bar{R}) = -\mathcal{A}^{abc}(R), \tag{4.375}$$

so

$$\mathcal{A}^{abc}(R) = 0. \tag{4.376}$$

This is the group-theoretic reason that many real representations cannot produce a perturbative gauge anomaly.

The Adler-Bardeen theorem says that the anomaly coefficient is one-loop exact. Higher-loop diagrams can renormalize many quantities, but they do not change the coefficient of the anomaly. Therefore, if the one-loop gauge anomaly cancels, it remains cancelled to all orders in perturbation theory.

The lesson is the following. A global anomaly can be a real physical effect. A local gauge anomaly is a consistency condition. When building a chiral gauge theory, one must check the representation content before doing any detailed dynamics:

$$\boxed{\sum_i \mathrm{tr}_{R_i}\left[T^a\{T^b,T^c\}\right]=0}. \tag{4.377}$$

Only after this condition is satisfied does the quantum theory have a chance to be a consistent gauge theory.